

AK Higher-Dimensional Projective Structure Theory

Problem Interface Appendices MS / NS

Version 21.0.0

Wide-Pipeline, Application-Capability, Arithmetic, and PDE Edition

This v21 synchronization layer strengthens the arithmetic and Navier–Stokes problem interfaces while preserving strict theorem boundaries and widening the reusable source-adaptor, intermediate-packet, alternate-route, fallback, fan-out, and application lanes. It separates source semantics, Interface realization, Core-P, Core-DG, descent, generic Bridge composition, source-specific Return, and independent validation. The complete v20 MS/NS corpus follows unchanged as immutable predecessor ancestry. No Claim Register entry is generated in this document.

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Part IV v21 Synchronization and Strengthening Layer

V21.1. Canonical dependency and ownership boundary

v21 Definition 1 (Problem-Interface route record). A v21 problem-facing route is a typed record

$$\mathfrak{R}_{\text{PI}} = (\mathcal{S}, \mathcal{R}, \mathcal{C}, \mathcal{T}, \Sigma, \Gamma, \mathfrak{D}, \mathcal{A}),$$

where $\mathcal{S}$ is the source semantics, $\mathcal{R}$ the realization and variance data, $\mathcal{C}$ the Core-facing predicate or diagram, $\mathcal{T}$ the exact external target, Σ the selected route class, Γ the propagated hypotheses, $\mathfrak{D}$ the defect and non-scalar obligation record, and $\mathcal{A}$ the immutable artifact bundle. Two routes may be composed only when the displayed codomain, domain, variance, coefficient, target, and hypothesis records agree or are connected by a proved conversion theorem.

v21 Definition 2 (Bridge-territory separation). The following ownership classes are disjoint.

- (i) A *source theorem* lives entirely in arithmetic or PDE semantics.
- (ii) An *Interface theorem* constructs a typed Core-P object, a typed Core-DG diagram, or an admitted readiness input from source data.
- (iii) A *P2DG or descent theorem* constructs actual diagrammatic data, cocycles, coherence, or an effective descended object.
- (iv) Generic Bridge-component composition and target-relative Return calculus are owned by Main Chapter 9 and Appendices HT/UB.
- (v) Appendices MS and NS may own only source-specific Interface components, source-specific Bridge components, source-specific Return theorems, benchmark recovery tracks, and explicit non-go theorems.
- (vi) Independent source comparison validates strength after the theorem-bearing chain closes; it supplies no missing arrow.

v21 Proposition 3 (Readiness does not promote a Bridge). *The statuses ready, descent_input_ready, and bridge_input_ready assert only that the corresponding input record is complete and consistent. None implies a Bridge theorem, descent effectivity, a Return theorem, or an external conclusion.*

Proof. Each readiness predicate is defined on the domain of the next construction. The claimed conclusions are values of later theorem nodes. Domain membership cannot substitute for evaluation of those nodes. $\square$

v21 Proposition 4 (Source-specific components do not acquire generic ownership). *A proved MS or NS component remains source-bound. Reuse in another arithmetic family, another PDE class, another coefficient system, or another target requires an explicit substitution theorem preserving every field of Definition 1.*

Proof. The source class, semantics, hypotheses, realization, and target are part of the theorem statement. Replacing any of them changes the typed implication unless a theorem proves invariance under that replacement. $\square$

v21 Definition 5 (Atomic and target-result sorts). Atomic verifier status is one of

pass, reject, undefined, not_invoked.

A target-indexed Bridge result is one of

reject, undefined, obstructed, partial_return, regularized_return, return.

Identical character strings in different fields do not identify the sorts.

v21 Definition 6 (Canonical descent lane). If a problem route invokes descent, exactly one active lane is selected per proof DAG:

strict_alexandrov_sheaf, strict_bounded_complex, model_certified_derived.

The canonical DescentInput contains local objects, overlap maps, cocycles, higher cells when required, window data, comparison data, a coherence model, an effectivity theorem, and artifacts. No MS or NS source observable is renamed into a descent field without construction of the actual datum.

v21 Theorem 7 (Typed Bridge non-compensation). *Let a target route require components $C_1, \dots, C_m$. If one required component is missing, ill-typed, rejected, or obstructed, then favorable metric reserve, successful components in another source domain, or a completed recovery track for another target cannot yield return for the original target.*

Proof. The route obligation is a typed conjunction. A missing conjunct is neither a metric quantity nor an alternative proof of itself. Cross-domain successes have different source and target types and therefore cannot inhabit the missing component slot. $\square$

v21 Definition 8 (Target-relative strength and independent products). For a target T , the supported strength set S_T is downward closed in its declared strength poset. For independent targets $T_1 \in C_1$ and $T_2 \in C_2$, the combined result is a product record in $C_1 \times C_2$, with supported set $S_{T_1} \times S_{T_2}$. It is not an intersection unless both sets are explicitly embedded in one common poset.

v21 Proposition 9 (MS and NS routes are noninterchangeable). *An exact arithmetic recovery and an exact Serrin-criterion recovery may coexist in one portfolio, but neither supplies a component, hypothesis, metric allowance, or promotion witness for the other.*

Proof. Their source categories, realizations, predicates, Return consequents, and source bindings are disjoint. The portfolio is a product of records, not a theorem connecting the factors. $\square$

V21.2. Arithmetic target stratification and exact finite p -group calculus

v21 Definition 10 (Arithmetic target family). The arithmetic targets

$$T_{\text{group}}, \quad T_{\text{growth}}, \quad T_{\text{control}}, \quad T_{\text{class}}, \quad T_{\text{main}}, \quad T_{pL}$$

refer respectively to finite p -group structure, eventual Iwasawa growth, bounded control defects, class-number specialization, a main-conjecture statement, and a p -adic analytic statement. The family is discrete until a proved source-specific implication supplies an order edge. In particular, T_{growth} does not automatically imply T_{class} , and neither implies T_{main} .

v21 Theorem 11 (Hilbert-profile inversion for finite abelian p -groups). *Let*

$$G \cong \bigoplus_{i=1}^s \mathbb{Z}/p^{a_i}\mathbb{Z}, \quad a_i \geq 1,$$

and define

$$h_G(j) := \dim_{\mathbb{F}_p}(p^{j-1}G/p^jG), \quad j \geq 1.$$

Then

$$h_G(j) = \#\{i : a_i \geq j\}, \quad m_G(j) := h_G(j) - h_G(j+1) = \#\{i : a_i = j\}.$$

Consequently the finite sequence $(h_G(j))_{j \geq 1}$ determines the isomorphism class of G .

Proof. For one summand $\mathbb{Z}/p^a\mathbb{Z}$, the quotient $p^{j-1}G/p^jG$ is one-dimensional exactly for $j \leq a$. Direct sums add dimensions. The first difference therefore counts summands whose exponent length is exactly j , recovering the elementary divisors. $\square$

v21 Corollary 12 (Exact arithmetic invariants from the length profile). *For nonzero G ,*

$$\begin{aligned}\dim_{\mathbb{F}_p}(G/pG) &= h_G(1), \\ \log_p \#G &= \sum_{j \geq 1} h_G(j), \\ \log_p \exp(G) &= \max\{j : h_G(j) > 0\}.\end{aligned}$$

Under the existing realization $\mathcal{L}_p(G)$, these are respectively the number of bars, total barcode length, and maximum bar length.

Proof. The first and third identities follow directly from the elementary divisors. The second is $\sum_i a_i = \sum_j \#\{i : a_i \geq j\}$. The barcode translation follows from the interval decomposition $[1, a_i + 1)$. $\square$

v21 Theorem 13 (Symmetric-monoidal object calculus). *For finite abelian p -primary groups,*

$$\mathcal{L}_p(G \oplus H) \cong \mathcal{L}_p(G) \oplus \mathcal{L}_p(H).$$

Hence Hilbert profiles add, barcodes form multiset unions, and total barcode length is additive.

Proof. The p -power filtration and its graded quotients commute with finite direct sums. All assertions follow degree-wise and then from interval decomposition. $\square$

v21 Theorem 14 (Strict filtered exactness and graded additivity). *Let*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of finite abelian p -primary groups that is strict for the intrinsic p -power filtrations. Then for every $j \geq 1$,

$$0 \longrightarrow p^{j-1}A/p^jA \longrightarrow p^{j-1}B/p^jB \longrightarrow p^{j-1}C/p^jC \longrightarrow 0$$

is exact. Therefore $h_B(j) = h_A(j) + h_C(j)$ and $\log_p \#B = \log_p \#A + \log_p \#C$.

Proof. Strictness identifies the induced subobject and quotient filtrations with the intrinsic filtrations. Associated graded is exact for a strict filtered short exact sequence. Taking dimensions and summing over j gives the assertions. $\square$

v21 Remark 15 (Strictness boundary). The sequence $0 \rightarrow \mathbb{Z}/p \rightarrow \mathbb{Z}/p^2 \rightarrow \mathbb{Z}/p \rightarrow 0$ is not strict for the intrinsic p -power filtrations. Thus Theorem 14 does not silently split arbitrary extensions or assert exactness of associated graded without the strictness record.

V21.3. Iwasawa-growth difference calculus and algebraic extension footholds

v21 Definition 16 (Tail shift operator). For a sequence $c = (c_n)$ over a characteristic-zero field, let $(Ec)_n = c_{n+1}$, and put

$$\mathcal{L}_p := (E - p)(E - 1)^2.$$

The residual is

$$(\mathcal{L}_p c)_n = c_{n+3} - (p+2)c_{n+2} + (2p+1)c_{n+1} - pc_n.$$

v21 Theorem 17 (Exact recurrence characterization of Iwasawa-form tails). *For a sequence c_n over a characteristic-zero field, the following are equivalent.*

(i) *There exist N and scalars μ, λ, ν such that*

$$c_n = \mu p^n + \lambda n + \nu \quad (n \geq N).$$

(ii) *$(\mathcal{L}_p c)_n = 0$ for every sufficiently large n .*

Proof. The forward implication follows because p^n , n , and 1 are killed by $(E - p)(E - 1)^2$. Conversely, write $z_n = (E - 1)^2 c_n$. On a tail, $z_{n+1} = pz_n$, hence $z_n = Ap^n$. Integrating the first-difference equation twice gives $c_n = A(p - 1)^{-2} p^n + Bn + C$ on that tail. $\square$

v21 Corollary 18 (Finite-support residual criterion). *A sequence has an eventual Iwasawa-form tail if and only if the residual sequence $\mathcal{L}_p c$ has finite support.*

Proof. Finite support means precisely that the residual vanishes from some index onward. Apply Theorem 17. $\square$

v21 Theorem 19 (Three-value coefficient extraction on a certified tail). *Assume the Iwasawa-form identity holds at the three consecutive indices $N, N + 1, N + 2$. Then its coefficients are uniquely determined by*

$$\mu = \frac{c_{N+2} - 2c_{N+1} + c_N}{(p - 1)^2 p^N},$$

$$\lambda = (c_{N+1} - c_N) - \mu(p - 1)p^N, \quad \nu = c_N - \mu p^N - \lambda N.$$

If the recurrence of Theorem 17 is certified for every $n \geq N$, these three values determine the entire tail.

Proof. Take two forward differences of the displayed Iwasawa form and solve successively for μ , λ , and ν . A third-order recurrence with leading coefficient one propagates from three consecutive values. $\square$

v21 Remark 20 (No three-sample promotion). Three numerical values alone do not prove that the recurrence holds on the remaining tail. Theorem 19 is a coefficient extractor after a tail theorem or recurrence certificate has been supplied, not a finite-sampling proof of eventual growth.

v21 Theorem 21 (Rational generating-series characterization). *Let $C(T) = \sum_{n \geq 0} c_n T^n$ be a formal power series over a characteristic-zero field. The sequence has an Iwasawa-form tail from N if and only if there is a polynomial $P(T)$ of degree less than N and scalars μ, λ, ν such that*

$$C(T) = P(T) + \mu \frac{(pT)^N}{1 - pT} + \lambda \frac{T^N (N - (N - 1)T)}{(1 - T)^2} + \nu \frac{T^N}{1 - T}.$$

In particular, modulo a polynomial correction, the denominator divides $(1 - pT)(1 - T)^2$.

Proof. Sum the three tail series $\sum_{n \geq N} p^n T^n$, $\sum_{n \geq N} n T^n$, and $\sum_{n \geq N} T^n$. Conversely, coefficient extraction from the displayed rational expression gives the stated tail formula. $\square$

v21 Theorem 22 (Finite-family common-tail closure). *For finitely many sequences $c^{(1)}, \dots, c^{(m)}$, if each has an Iwasawa-form tail from some witness N_i , then all formulas hold simultaneously from $N_* = \max_i N_i$. The common witness is independent of the order in which the tracks are assembled.*

Proof. Every $n \geq N_*$ lies in every individual tail. The maximum of a finite set is invariant under reordering. $\square$

v21 Theorem 23 (Multivariable exponential-polynomial annihilator). *Let K have characteristic zero and let*

$$f(\mathbf{n}) = \sum_{\alpha=1}^m P_\alpha(\mathbf{n}) \prod_{i=1}^d a_{\alpha i}^{n_i}, \quad \mathbf{n} \in \mathbb{N}^d,$$

where $a_{\alpha i} \in K^\times$, and the degree of P_α in n_i is at most $e_{\alpha i}$. If E_i shifts the i -th coordinate, then

$$\prod_{\alpha=1}^m \prod_{i=1}^d (E_i - a_{\alpha i})^{e_{\alpha i} + 1} f = 0.$$

Proof. For one term, $E_i - a_{\alpha i}$ multiplies the exponential factor by $a_{\alpha i}$ and replaces the polynomial by its forward difference in n_i , lowering its degree in that variable. The indicated power kills that term. All shifts commute, and the product kills every summand. $\square$

v21 Remark 24 (Multivariable arithmetic boundary). Theorem 23 is an algebraic foothold for future multivariable Iwasawa interfaces. It does not prove that a multivariable Iwasawa module has such a growth function, does not address pseudo-null corrections, and does not extend to noncommutative Iwasawa algebras without new source mathematics.

V21.4. Arithmetic Bridge and Return disposition

v21 Theorem 25 (v21 arithmetic route decomposition). *The exact Track G route decomposes into the following independently typed components:*

$$\begin{aligned} \text{precursor source package} &\longrightarrow \text{signed finite defect realization} \\ &\longrightarrow \text{bounded-window Core zero} \longrightarrow T_{\text{growth}}. \end{aligned}$$

The final source comparison validates exact benchmark strength. The class-number target T_{class} requires a separate class-field source-to-module Interface, and the main-conjecture target T_{main} requires an independent analytic–algebraic source theorem and Return chain.

Proof. The displayed arrows have distinct source and target types. Existing Track G closes exactly the first chain. Neither the class-number identification nor a main-conjecture implication is among its premises or conclusions. $\square$

v21 Proposition 26 (Characteristic and Fitting data remain source-side). *An equality of characteristic or Fitting ideals, a pseudo-isomorphism, or equality of Iwasawa invariants does not determine an actual Core-DG morphism, a kernel/cokernel persistence profile, or a Bridge result without a source-specific realization and transport theorem.*

Proof. Ideal-theoretic invariants can forget pseudo-null or finite subquotients, whereas a diagram morphism and its defects are actual categorical data. Equality of invariants therefore does not construct the required map or identify its kernel and cokernel. $\square$

v21 Proposition 27 (Corrected source-binding identity). *The content identity of the local Iwasawa source-statement binding consumed by this release is*

$$\text{a5f0a65a73c802492167676aba024b01651cd855bbdc00bc3d3c48d46cd6f0d7}.$$

Any registry entry carrying a different hash is stale and may be used only as repair ancestry until regenerated.

Proof. The displayed value is the SHA-256 digest of the bound local statement artifact used by the present build. Content-addressed identity, not filename or a predecessor registry string, controls the binding. $\square$

V21.5. Navier–Stokes target stratification and scaling calculus

v21 Definition 28 (NS target family). The targets

$$T_{\text{norm}}, \quad T_{\text{criterion}}, \quad T_{\text{continuation}}, \quad T_{\text{unconditional}}, \quad T_{\text{Clay}}$$

refer respectively to a certified mixed-norm bound, the fixed source regularity implication, a continuation theorem, unconditional regularity on a declared solution class, and a Clay-level conclusion. An implication edge must be proved separately; completion of $T_{\text{criterion}}$ does not promote any higher target.

v21 Theorem 29 (Mixed-norm scaling law). *For the Navier–Stokes scaling*

$$u_{\lambda}(x, t) = \lambda u(\lambda x, \lambda^2 t),$$

and finite $q, r \in [1, \infty)$,

$$\|u_{\lambda}\|_{L^q(0, T/\lambda^2; L^r(\mathbb{R}^3))} = \lambda^{1-2/q-3/r} \|u\|_{L^q(0, T; L^r(\mathbb{R}^3))}.$$

Thus the Serrin index

$$\mathfrak{s}(q, r) := \frac{2}{q} + \frac{3}{r}$$

is critical at 1.

Proof. Change variables $y = \lambda x$ in the spatial norm and $s = \lambda^2 t$ in the temporal norm. The amplitude contributes λ , spatial measure contributes $\lambda^{-3/r}$, and time contributes $\lambda^{-2/q}$. $\square$

v21 Corollary 30 (Strict-subcritical reserve of the frozen Serrin track). *For $(q, r) = (6, 6)$,*

$$\mathfrak{s}(6, 6) = \frac{5}{6}, \quad 1 - \mathfrak{s}(6, 6) = \frac{1}{6}.$$

The frozen source track is therefore strictly subcritical and does not invoke the endpoint $\mathfrak{s} = 1$.

v21 Theorem 31 (Energy interpolation line and no-go boundary). *Assume only*

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; L^6(\mathbb{R}^3)).$$

For $0 \leq \theta \leq 1$, interpolation gives

$$\frac{1}{q} = \frac{\theta}{2}, \quad \frac{1}{r} = \frac{1-\theta}{2} + \frac{\theta}{6},$$

and every such pair satisfies

$$\frac{2}{q} + \frac{3}{r} = \frac{3}{2}.$$

Consequently interpolation of the standard energy endpoints alone cannot reach the Serrin range $2/q + 3/r \leq 1$.

Proof. Spatial interpolation gives $\|u(t)\|_r \leq \|u(t)\|_2^{1-\theta} \|u(t)\|_6^\theta$. Taking $q = 2/\theta$, integrating, and using the two endpoint norms yields the mixed-norm estimate. Substitution gives the constant index $3/2$. $\square$

v21 Theorem 32 (Interpolation corridor generated by the $L_t^6 L_x^6$ input). *If*

$$u \in L_t^\infty L_x^2 \cap L_t^6 L_x^6,$$

then for $0 \leq \theta \leq 1$,

$$\frac{1}{q} = \frac{\theta}{6}, \quad \frac{1}{r} = \frac{1-\theta}{2} + \frac{\theta}{6},$$

and

$$\mathfrak{s}(q, r) = \frac{3}{2} - \frac{2\theta}{3}.$$

For $3/4 \leq \theta \leq 1$, this produces the Serrin corridor from $(q, r) = (8, 4)$ to $(6, 6)$.

Proof. Apply the same pointwise spatial interpolation and choose $q = 6/\theta$. The $L_t^\infty L_x^2$ factor is taken outside the time integral and the remaining exponent on the L_x^6 norm is six. The index formula is immediate. $\square$

v21 Remark 33 (Corridor boundary). Theorem 32 propagates an already available $L_t^6 L_x^6$ hypothesis to nearby exponent pairs. It does not derive that hypothesis from the energy inequality and therefore does not advance an unconditional regularity route by itself.

V21.6. Finite certificates, compactness, and analytic stitching

v21 Theorem 34 (Finite time partition with local L^6 budget). *If $u \in L^6(0, T; L^6(\mathbb{R}^3))$ and $\eta > 0$, there is a finite partition*

$$0 = t_0 < t_1 < \cdots < t_m = T$$

such that

$$\int_{t_{j-1}}^{t_j} \|u(t)\|_6^6 dt \leq \eta \quad \text{for every } j,$$

with

$$m \leq \max \left\{ 1, \left\lceil \frac{\|u\|_{L_t^6 L_x^6}^6}{\eta} \right\rceil \right\}.$$

Proof. The cumulative integral $F(t) = \int_0^t \|u(s)\|_6^6 ds$ is continuous and nondecreasing. Divide the interval $[0, F(T)]$ into increments of length at most η and take corresponding level-set times of F . $\square$

v21 Theorem 35 (Composition of local rational norm certificates). *For a finite partition as above, suppose rational numbers $a_j \geq 0$ are certified to satisfy*

$$\int_{t_{j-1}}^{t_j} \|u(t)\|_6^6 dt \leq a_j.$$

Then

$$\|u\|_{L^6(0,T;L^6)} \leq \left(\sum_{j=1}^m a_j \right)^{1/6}.$$

The certificate is exact at the norm-predicate level and does not by itself certify regularity.

Proof. The sixth power of the global norm is the sum of the sixth-power integrals on the disjoint partition intervals. $\square$

v21 Theorem 36 (Weak-limit preservation of a certified Serrin bound). *Let $u_n \rightharpoonup u$ weakly in $L^6(0, T; L^6(\mathbb{R}^3))$. If $\|u_n\|_{L_t^6 L_x^6} \leq M$ for every n , then*

$$\|u\|_{L_t^6 L_x^6} \leq M.$$

Proof. The Bochner space $L^6(0, T; L^6)$ is reflexive, and its norm is weakly lower semicontinuous. $\square$

v21 Remark 37 (Compactness burden). Theorem 36 consumes an actual weak-convergence theorem in the exact Bochner space and a uniform bound. A numerical plateau, distributional convergence without the mixed-norm bound, or convergence in a weaker topology does not supply those premises.

v21 Theorem 38 (Conditional interval stitching). *Let $0 = t_0 < \dots < t_m = T$. Suppose the same weak solution is regular on each $\mathbb{R}^3 \times (t_{j-1}, t_j]$, and the regular representatives and traces agree at every interior partition time. Then it is regular on $\mathbb{R}^3 \times (0, T]$.*

Proof. The regular pieces agree on their common traces and therefore glue to one regular representative on the union of the intervals, which is $(0, T]$. $\square$

v21 Remark 39 (Restart is a separate component). Theorem 38 does not prove existence of compatible traces, restart admissibility, or persistence of the hypotheses across a singular endpoint. Those are separate analytic and proof-DAG components.

v21 Theorem 40 (General fixed-exponent norm-defect zero reflection). *Fix finite $q, r > 1$, a source object $u \in L_t^q L_x^r$, and $M \geq 0$. Define*

$$a_{q,r,M}(u) := (\|u\|_{L_t^q L_x^r}^q - M^q)_+.$$

Any exact one-bar realization whose bar length is $a_{q,r,M}(u)$ satisfies

$$\text{repaired profile} = 0 \iff \|u\|_{L_t^q L_x^r} \leq M.$$

This is predicate-level zero reflection; it is not a theorem computing the norm or converting the norm bound into a PDE conclusion.

Proof. The one-bar profile is zero exactly when its nonnegative length is zero, which is equivalent to the displayed norm inequality. $\square$

V21.7. NS Bridge cuts and exact criterion-level scope

v21 Theorem 41 (Exact v21 decomposition of the frozen Serrin route). *The frozen fixed-exponent route consists of:*

$$\text{source norm and rational witness} \rightarrow \text{exact norm-defect profile} \rightarrow T_{\text{norm}} \rightarrow T_{\text{criterion}},$$

where the final arrow is the quarantined Serrin source theorem. No detector, Type IV, higher edge, descent, or finite-to-infinite tower theorem is required for this route unless separately invoked by a different construction.

Proof. The exact one-bar zero-reflection identifies the Core predicate with the mixed-norm bound. The source theorem maps that bound to interior regularity. The omitted component types are not used in either implication. $\square$

v21 Theorem 42 (Unconditional-regularity residual cut). *For the frozen Leray–Hopf source class, a theorem proving*

$$\|u\|_{L^6(0,T;L^6(\mathbb{R}^3))} < \infty$$

for every solution would compose with the exact criterion route to yield interior regularity on $(0, T]$. The existing MS/NS interface stack does not prove this universal norm bound; therefore the unconditional and Clay-level targets remain obstructed at that source-side analytic edge.

Proof. The first assertion is composition with the frozen Serrin implication. The second follows from the source and target inventory: the current recovery assumes or certifies the norm predicate for a supplied object but contains no theorem deriving it for all Leray–Hopf solutions from the energy data. $\square$

v21 Proposition 43 (Energy data cannot fill the residual cut by interpolation alone). *The standard energy endpoints cannot discharge the missing universal $L_t^6 L_x^6$ edge by interpolation alone.*

Proof. Theorem 31 confines all such interpolated pairs to Serrin index $3/2$, whereas $L_t^6 L_x^6$ has index $5/6$. $\square$

v21 Proposition 44 (Type IV and PDE regularity remain separate). *A Type IV zero result for any discretization, refinement, scaling, or time-window tower does not imply $T_{\text{criterion}}$, $T_{\text{unconditional}}$, or T_{Clay} without a source-specific soundness theorem identifying the actual kernel/cokernel predicates with the corresponding analytic obstruction.*

Proof. Type IV is a morphism-defect predicate in the selected profile category. PDE regularity is a source-domain analytic predicate. Their identification is exactly the missing source-specific Bridge component. $\square$

v21 Proposition 45 (NS source-binding identity). *The content identity of the local fixed-exponent Serrin source-statement binding consumed by this release is*

$$608216fda60bd3fdc055243c0169bdb31b53c337bb99d9b7c9cb43a215996423.$$

Proof. This is the SHA-256 digest of the local source-statement binding used by the present build. $\square$

V21.8. Pipeline capacity, alternate routes, and failure isolation

v21 Definition 46 (Problem-Interface intermediate packet). *A Problem-Interface intermediate packet is a loss-less typed record*

$$\text{PIPacket} = (\text{SourceRef}, \text{AdapterRef}, C_{\text{src}}, C_{\text{tgt}}, \text{var}, \text{coeff}, \text{scope}, \text{predicate}, \Gamma, \text{status}, \mathcal{A}).$$

It is an interchange object between source adapters and downstream consumers. A packet is reusable only when the consumer accepts its exact source and target categories, variance, coefficient convention, scope, predicate, propagated hypotheses, lifecycle status, and artifact identities. A shorter tuple, a visually identical value, or a filename alias is not an equivalent packet.

v21 Definition 47 (Route portfolio and capability set). A *route portfolio* is a finite typed DAG whose nodes are source statements, adapters, intermediate packets, Core certificates, diagrammatic or descent records, source-specific Bridge components, Return theorems, and target results. Every edge is a proved conversion or implication. Its capability set is

$$\text{Cap}(\mathfrak{P}) := \{(\mathcal{S}, T, \sigma) : \text{some closed route in } \mathfrak{P} \text{ supports strength } \sigma \text{ for target } T\}.$$

The capability set measures available typed routes, not the truth of unproved targets and not the importance of the source problem.

v21 Theorem 48 (Alternative-route support union). *Fix one target strength poset and finitely many independently auditable routes $R_1, \dots, R_m$ to the same target. Let S_i be the downward-closed strength set actually supported by R_i . Under disjunctive route selection, the portfolio-supported set is*

$$\text{Down}\left(\bigcup_{i=1}^m S_i\right).$$

No hypothesis, partial component, metric reserve, or failed edge from one route may be spliced into another unless a separate composition theorem constructs the hybrid route.

Proof. A strength is supported by the portfolio exactly when at least one complete selected route supports it. This gives the union, and the target strength convention gives downward closure. A hybrid assembled from proper fragments is not one of the closed routes and has no theorem-bearing status without an additional typed composition result. $\square$

v21 Theorem 49 (Independent-route failure isolation). *Let R_1 and R_2 be routes whose required non-source nodes, mutable execution states, and artifact outputs are disjoint. A reject, undefined, stale, or obstructed node in R_1 does not change the mathematical or lifecycle status of R_2 . More generally, a node failure propagates only to descendants that consume that node.*

Proof. The status of a node is a function of its declared dependencies. A node outside the descendant closure is not among those dependencies. Disjoint execution and artifact states exclude hidden mutation channels, so the second route retains the same inputs and result. $\square$

v21 Definition 50 (Path-disjunctive portfolio and target cut). A route portfolio is *path-disjunctive* for a target when each admissible proof route is represented by one complete source-to-target path and the target accepts any one complete path. A set of internal nodes is a *target cut* when deleting it removes every admissible source-to-target path.

v21 Theorem 51 (Finite target-cut characterization). *In a finite path-disjunctive portfolio, a set C blocks the target if and only if C meets every admissible source-to-target path. Consequently, inclusion-minimal target cuts form an antichain and identify the irreducible pipeline bottlenecks.*

Proof. If C misses some admissible path, that path survives and the target is not blocked. If C meets every path, no complete route survives. Two distinct inclusion-minimal cuts cannot properly contain one another, so they form an antichain. $\square$

v21 Theorem 52 (Content-addressed verified-node reuse). *Assume deterministic semantic replay. Two route nodes may be represented by one verified node when their complete typed records, theorem versions, dependency hashes, source bindings, execution environments, and output artifact hashes are identical. Reuse preserves the verifier result and avoids duplicate defect charging. Equality of titles, displayed values, or basenames is insufficient.*

Proof. Under deterministic replay the result is determined by the complete input and environment record. Identical content-addressed records therefore replay to the same output. The two references denote one semantic occurrence; charging it twice would count an alias rather than a second traversal. $\square$

v21 Theorem 53 (Optional-lane capability monotonicity). *Let $\mathfrak{P}'$ be obtained from $\mathfrak{P}$ by adding optional routes without mutating any existing node, dependency, artifact, or selection rule. Then*

$$\text{Cap}(\mathfrak{P}) \subseteq \text{Cap}(\mathfrak{P}'),$$

and every pre-existing route keeps its former status and strength. The inclusion does not strengthen any individual theorem.

Proof. Every old closed route remains present with identical inputs and semantics. New routes may add capability triples but cannot alter the proofs or supported strengths of the old routes. $\square$

V21.9. Arithmetic adapter bus and widened exact lanes

v21 Definition 54 (MS arithmetic adapter bus). The MS adapter bus admits separately registered source lanes for finite-length discrete-valuation-ring modules, finite abelian p -groups, Iwasawa growth sequences, strict bounded-control diagrams, congruence systems, characteristic or Fitting data, and p -adic analytic profiles. Its reusable intermediate packet classes are:

LengthPacket, RecurrencePacket, ControlPacket, DiagramPacket, AnalyticPacket.

Only the first three receive the exact theorems below. The latter two remain conditional until their source-specific realization and Return components are supplied.

v21 Theorem 55 (Exact finite-length DVR persistence realization). *Let R be a discrete valuation ring with uniformizer π and residue field $k = R/(\pi)$. For a finite-length finitely generated R -module M , define*

$$\mathcal{L}_\pi(M)_j := \pi^{j-1}M/\pi^jM \quad (j \geq 1),$$

with transition induced by multiplication by π , and extend it constantly over $[j, j+1)$. Then $\mathcal{L}_\pi$ is functorial. If

$$M \cong \bigoplus_{i=1}^s R/(\pi^{a_i}),$$

then

$$\text{Barcode}(\mathcal{L}_\pi(M)) \cong \bigsqcup_{i=1}^s \{[1, a_i + 1)\}.$$

Hence the complete barcode determines the isomorphism class of M , and its total bar length is $\text{length}_R(M)$.

Proof. An R -linear map commutes with multiplication by π , so it induces compatible maps on all graded quotients. The structure theorem for finite-length modules over a DVR gives the displayed decomposition. Each cyclic summand contributes one one-dimensional chain through levels $1, \dots, a_i$, hence one interval $[1, a_i + 1)$. The interval multiset recovers the exponents and its total length is their sum. $\square$

v21 Corollary 56 (DVR Hilbert-profile inversion). *For*

$$h_M(j) := \dim_k(\pi^{j-1}M/\pi^jM),$$

one has

$$h_M(j) - h_M(j+1) = \#\{i : a_i = j\}.$$

Thus h_M , the complete barcode, and the elementary-divisor multiset carry equivalent object-level information on this source class.

Proof. The value $h_M(j)$ counts summands with $a_i \geq j$. Taking consecutive differences counts summands of exact exponent j . $\square$

v21 Theorem 57 (Strict DVR exact-sequence transport). *Let*

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

be a short exact sequence of finite-length R -modules that is strict for the intrinsic π -power filtrations. Then the associated graded sequence is exact at every level, the resulting persistence sequence is objectwise exact, and

$$\text{length}_R(M) = \text{length}_R(M') + \text{length}_R(M'').$$

Without strictness, the associated graded exactness and kernel/cokernel profile identifications are not asserted.

Proof. Strictness is precisely the hypothesis under which passage to the associated graded preserves the short exact sequence. Summing the finite-dimensional graded dimensions yields the standard additivity of module length. $\square$

v21 Theorem 58 (Annihilator closure under sums and shifts). *Let E be the forward shift. Suppose polynomial shift operators $P(E)$ and $Q(E)$ annihilate sequences a and b , respectively, on a common tail. Then*

$$P(E)Q(E)(a + b) = 0$$

on a tail. Moreover, every finite shift $E^h a$ is annihilated by $P(E)$ on the corresponding shifted tail.

Proof. Polynomial expressions in the same shift commute. Thus $PQa = QPa = 0$ and $PQb = PQb = P(Qb) = 0$ on a common tail. Also $P(E)E^h = E^h P(E)$. $\square$

v21 Theorem 59 (Residue-class branch annihilator). *Fix $m \geq 1$ and set $c_k^{(r)} = c_{mk+r}$ for $0 \leq r < m$. If each branch is eventually annihilated by a polynomial $P_r(S)$ in the branch shift S , then the full sequence is eventually annihilated by*

$$\prod_{r=0}^{m-1} P_r(E^m).$$

This supplies an exact finite operator packet for periodic, signed, or branch-dependent growth laws once the branch recurrences are independently proved.

Proof. On indices congruent to r modulo m , the operator $P_r(E^m)$ is exactly the branch operator $P_r(S)$ and vanishes eventually. All factors commute, so the product vanishes on every residue class beyond one common cutoff. $\square$

v21 Theorem 60 (Finite determination under a certified monic recurrence). *Suppose a sequence satisfies a fixed monic recurrence of order d ,*

$$c_{n+d} + a_{d-1}c_{n+d-1} + \cdots + a_0c_n = 0 \quad (n \geq N).$$

Then $c_N, \dots, c_{N+d-1}$ determine the complete tail uniquely. Two sequences satisfying the same certified recurrence and agreeing on those d values agree for every $n \geq N$.

Proof. The recurrence solves successively for $c_{N+d}, c_{N+d+1}, \dots$. Induction gives uniqueness and equality. $\square$

v21 Theorem 61 (Finite direct-sum bounded-control portfolio). *For a finite family of strict bounded-control diagrams of finite-length DVR modules, their direct sum is again a strict bounded-control diagram. Kernels and cokernels are the direct sums of the component kernels and cokernels, and the total length bounds are the sums of the component bounds. Consequently, finitely many independent control lanes may be aggregated without identifying their source statements or Return targets.*

Proof. Finite direct sums preserve exact sequences, kernels, cokernels, and strictness of the component filtrations. Length is additive on finite direct sums. The final sentence follows because aggregation occurs in the common length packet, while source and Return records remain component-indexed. $\square$

v21 Remark 62 (Arithmetic widening boundary). The DVR lane strictly enlarges the finite p -group implementation class, and the recurrence lanes admit finite sums, shifts, and residue-class branches. None of these results identifies a general Iwasawa module over $R[[T_1, \dots, T_d]]$, a pseudo-isomorphism class, a characteristic ideal, or a p -adic L -function with a Core defect profile. Those remain source-specific Bridge programs.

V21.10. NS analytic adapter bus and widened certificate lanes

v21 Definition 63 (NS analytic adapter bus). The NS adapter bus has separate lanes for global Serrin mixed norms, parabolic-cylinder local norms, finite time partitions, finite spatial covers, approximation or Galerkin limits, interval restart and stitching, and source-side mild-solution, vorticity, pressure, or epsilon-regularity criteria. The last four source criteria remain conditional imports unless their exact theorem statements and hypotheses are bound. A lane may feed the common packet classes

MixedNormPacket, LocalScalePacket, ApproximationPacket, RegularityTargetPacket.

v21 Theorem 64 (General finite time-budget partition). *Let $1 \leq q < \infty$, $1 \leq r \leq \infty$, and $u \in L^q(0, T; L^r(\Omega))$. For every $\eta > 0$, there is a finite partition $0 = t_0 < \dots < t_m = T$ such that*

$$\int_{t_{j-1}}^{t_j} \|u(t)\|_{L^r(\Omega)}^q dt \leq \eta$$

for every j .

Proof. The nonnegative function $f(t) = \|u(t)\|_{L^r}^q$ belongs to $L^1(0, T)$. Partition by successive level times of its absolutely continuous cumulative integral. $\square$

v21 Theorem 65 (General local-certificate composition). *For a finite time partition, if certified numbers $a_j \geq 0$ satisfy*

$$\int_{t_{j-1}}^{t_j} \|u(t)\|_{L^r}^q dt \leq a_j,$$

then

$$\|u\|_{L^q(0, T; L^r)} \leq \left(\sum_j a_j \right)^{1/q}.$$

This constructs one global norm packet from finitely many local proof objects.

Proof. The q -th power of the global mixed norm is the sum of the interval integrals. $\square$

v21 Theorem 66 (Finite spatial-cover globalization). *Let $\Omega \subseteq \bigcup_{\ell=1}^L \Omega_\ell$ be a finite measurable cover and $q, r \geq 1$. Then*

$$\|u\|_{L^q(I; L^r(\Omega))} \leq \sum_{\ell=1}^L \|u\|_{L^q(I; L^r(\Omega_\ell))}.$$

Thus certified local mixed-norm packets on a finite cover yield a coarse but rigorous global packet. Overlap does not improve the bound unless an overlap or partition-of-unity theorem is separately used.

Proof. Pointwise, $|u| \mathbf{1}_\Omega \leq \sum_\ell |u| \mathbf{1}_{\Omega_\ell}$. Apply the triangle inequality first in L_x^r and then in L_t^q . $\square$

v21 Theorem 67 (General weak-limit transport of mixed-norm bounds). *Let $1 < q, r < \infty$ and suppose $u_n \rightharpoonup u$ weakly in $L^q(I; L^r(\Omega))$. If $\|u_n\|_{L^q L^r} \leq M$ for all n , then*

$$\|u\|_{L^q L^r} \leq M.$$

Proof. The mixed-norm Bochner space is reflexive in the displayed range and its norm is weakly lower semi-continuous. $\square$

v21 Theorem 68 (Mixed-norm interpolation fan-out). *Suppose $u \in L^{q_0}(I; L^{r_0}) \cap L^{q_1}(I; L^{r_1})$, with admissible exponents in $[1, \infty]$. For $0 \leq \theta \leq 1$, define*

$$\frac{1}{q_\theta} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}, \quad \frac{1}{r_\theta} = \frac{1-\theta}{r_0} + \frac{\theta}{r_1}.$$

Then

$$\|u\|_{L^{q_\theta} L^{r_\theta}} \leq \|u\|_{L^{q_0} L^{r_0}}^{1-\theta} \|u\|_{L^{q_1} L^{r_1}}^\theta.$$

One pair of certified endpoint packets may therefore fan out to a continuum of intermediate norm targets. Each PDE conclusion still requires its own source theorem at the selected exponent.

Proof. Apply interpolation to the spatial norms and then Hölder interpolation in time, with the usual endpoint interpretation. $\square$

v21 Definition 69 (Scale-normalized parabolic-cylinder packet). For a parabolic cylinder $Q_\rho(z_0) = B_\rho(x_0) \times (t_0 - \rho^2, t_0)$, define

$$\mathcal{N}_{q,r}(u; Q_\rho) := \rho^{1-\frac{2}{q}-\frac{3}{r}} \|u\|_{L_t^q L_x^r(Q_\rho)}.$$

It is a local analytic packet. Smallness, regularity, or singular-set consequences are not attached unless a source-specific epsilon-regularity or local Serrin theorem is supplied.

v21 Theorem 70 (Parabolic scaling invariance of the local packet). *Under the centered Navier–Stokes rescaling*

$$u^{(\rho)}(y, s) = \rho u(x_0 + \rho y, t_0 + \rho^2 s),$$

one has

$$\mathcal{N}_{q,r}(u; Q_\rho(z_0)) = \|u^{(\rho)}\|_{L^q(-1,0;L^r(B_1))}.$$

Hence $\mathcal{N}_{q,r}$ is dimensionless for every exponent pair, while the sign of $1 - 2/q - 3/r$ records subcritical, critical, or supercritical behavior of the unnormalized norm.

Proof. The change of variables $x = x_0 + \rho y$, $t = t_0 + \rho^2 s$ gives the factor $\rho^{1-3/r-2/q}$, which is exactly removed by the prefactor. $\square$

v21 Theorem 71 (Multi-criterion interval portfolio). *Let $0 = t_0 < \dots < t_m = T$. On each interval $(t_{j-1}, t_j]$, suppose a possibly different source-bound criterion C_j is proved and its exact source theorem yields regularity of the same weak solution on that interval. If the resulting regular representatives have compatible traces at all interior partition times, then the solution is regular on $(0, T]$. No common exponent pair is required.*

Proof. Each source theorem gives a regular representative on its interval. Trace compatibility identifies adjacent representatives, so they glue to one regular representative on the finite union. $\square$

v21 Remark 72 (NS widening boundary). The widened bus supports global, local, cover-based, approximation, interpolation, and mixed-criterion assembly. It does not prove a new epsilon-regularity theorem, a pressure or vorticity criterion, endpoint $L_t^\infty L_x^3$ regularity, a universal Serrin bound, or unconditional smoothness. Such statements enter only as separately bound source theorems and cannot be inferred from pipeline breadth.

V21.11. Application capability, fan-out, fallback, and implementation backends

v21 Definition 73 (Consumer capability signature). A consumer declares the set of accepted packet classes together with exact category, variance, coefficient, scope, hypothesis, evidence, and artifact requirements. A source adapter and consumer are connectable precisely when their signatures match or a proved converter mediates between them.

v21 Theorem 74 (Immutable packet fan-out). *One immutable intermediate packet may feed finitely many consumers in parallel. If each consumer route closes independently, the outputs form a target-product record. Failure of one consumer does not invalidate the packet or the other consumers, unless the failing execution mutates shared state in violation of immutability.*

Proof. Each consumer reads the same immutable packet and has a separate dependency subgraph. Apply independent-route failure isolation and the target-product rule. $\square$

v21 Theorem 75 (Typed fallback correctness). *Suppose a primary and a fallback route have the same exact target and the fallback is a complete route with its own discharged hypotheses and artifacts. If the primary route is undefined, rejected, stale, or obstructed, the target may still be returned through the fallback at the fallback's supported strength. No failed-primary component is imported into the fallback.*

Proof. Alternative-route semantics are disjunctive. The fallback route supplies a complete independent witness for its supported target strength; the failed route contributes nothing. $\square$

v21 Definition 76 (Evidence backend). An evidence backend is one of: human mathematical proof, formally checked proof, certified symbolic computation, certified numerical bound, or quarantined external theorem import. The backend field describes how evidence is produced and replayed. It does not change the theorem statement, target strength, source hypotheses, or logical status.

v21 Proposition 77 (Backend diversity does not collapse evidence roles). *A successful build, numerical enclosure, formal type check, or replay may discharge only the obligation for which it is sound. It does not promote advisory output to theorem evidence, replace source authority, or supply a missing Return theorem.*

Proof. The soundness domain of each backend is part of its contract. An obligation outside that domain remains a distinct unproved node. $\square$

v21 Definition 78 (Operational breadth vector). For engineering review, define the operational breadth vector

$$\mathbf{b} = (n_{\text{source}}, n_{\text{packet}}, n_{\text{exact}}, n_{\text{conditional}}, n_{\text{target}}, n_{\text{fallback}}, n_{\text{backend}}).$$

It counts registered source lanes, intermediate packet classes, exact routes, conditional routes, target classes, fallback routes, and evidence backends. It is not a theorem-strength scalar and is not entered into the metric defect ledger.

v21 Theorem 79 (Pipeline width and theorem strength are orthogonal). *Increasing $\mathbf{b}$ by adding adapters, packet classes, consumers, fallback routes, or backends can enlarge the capability set, but it does not strengthen the conclusion of any route whose theorem statement and hypotheses are unchanged. Conversely, a stronger theorem may increase target strength without increasing the number of lanes.*

Proof. Capability is a set of available typed triples, whereas theorem strength is ordered inside a fixed target specification. The two coordinates change under different operations and neither determines the other. $\square$

V21.12. Newness, extension status, and closure

v21 Definition 80 (Newness vocabulary). The label v21-local-new means newly extracted, proved, typed, and registered within the AK-HPDST v21 artifact set. It does not assert priority over the mathematical literature. Standard algebraic, functional-analytic, or interpolation facts remain standard facts even when their role in the AK architecture is new.

v21 Theorem 81 (MS/NS v21 reinforced closure). *The v21 synchronization layer establishes the following at exactly the displayed strength.*

- (i) *Source, Interface, Core-P, Core-DG, descent, generic Bridge composition, source-specific Return, and validation are separately owned and typed.*
- (ii) *Readiness does not imply effectivity or Return, and cross-domain components do not compensate.*
- (iii) *Finite abelian p -group structure is exactly recoverable from its Hilbert/length persistence profile.*
- (iv) *Eventual Iwasawa-form growth is exactly characterized by a third-order constant-coefficient recurrence, finite-support residual, and a rational generating-series tail.*
- (v) *The multivariable exponential-polynomial annihilator is available as an algebraic extension foothold, without an arithmetic realization claim.*
- (vi) *The frozen Track G remains exact only for its registered characteristic-growth benchmark relative to its precursor package; class-number and main-conjecture targets remain separate.*
- (vii) *The Navier–Stokes mixed-norm scaling law, the strict-subcritical $L_t^6 L_x^6$ reserve, and the energy-interpolation no-go line are explicit.*
- (viii) *Finite time-budget certificates, weak-limit transfer, and conditional interval stitching are available with their analytic premises displayed.*

- (ix) *The fixed Serrin criterion route remains an exact one-way source-specific recovery, while the universal $L_t^6 L_x^6$ estimate is the exposed residual cut for stronger targets.*
- (x) *Finite route portfolios support disjunctive alternative routes, failure isolation, target-cut analysis, content-addressed node reuse, immutable fan-out, and typed fallback without cross-route theorem inflation.*
- (xi) *The arithmetic adapter bus now includes an exact finite-length DVR lane, strict exact-sequence transport, recurrence closure under sums and shifts, residue-class branch annihilators, finite recurrence determination, and finite direct-sum control aggregation.*
- (xii) *The NS adapter bus now includes general mixed-norm time certificates, finite spatial-cover globalization, weak-limit transport, interpolation fan-out, scale-normalized parabolic-cylinder packets, and interval-wise heterogeneous criterion stitching.*
- (xiii) *Pipeline width and theorem strength remain orthogonal: broader applicability cannot promote an unproved arithmetic or PDE conclusion.*
- (xiv) *No unconditional three-dimensional regularity theorem, Clay-level theorem, Iwasawa main conjecture, Mazur specialization, or new p -adic L -function theorem is claimed.*

Proof. Items (i)–(ii) are the governance theorems of V21.1. Items (iii)–(vi) follow from V21.2–V21.4. Items (vii)–(ix) follow from V21.5–V21.7. The pipeline, arithmetic-bus, NS-bus, and application-capability conclusions follow from V21.8–V21.11. The final item is the explicit target boundary maintained throughout the layer. $\square$

v21 Remark 82 (Claim Register deferral). All v21-local theorem identities in this synchronization layer remain pending `_final_` CR. This document neither allocates final Claim UIDs nor declares release-registry pass.

Immutable v20 MS/NS Predecessor Corpus

The following corpus is retained verbatim as repair ancestry. Its local v20 labels and statements remain immutable; v21 current use is governed by the synchronization layer above.

1 Appendix MS-A: Arithmetic Interface Meta-Theory

MS-A.1. Scope, dependency position, and interface status

This appendix defines the arithmetic Problem-Interface meta-theory of AK-HPDST v20.0.0. It specifies how arithmetic source data may be projected into Core-readable objects, how finite arithmetic evidence may become a certified Core predicate, and what additional theorems are required before any certified Core statement may return to arithmetic semantics. It belongs to the Problem Interface layer and does not enlarge the Core mathematical package of Chapters 1–8 or Foundation Appendices A–G.

The central rule is

Arithmetic projection supplies a typed shadow;
only a proved Return Theorem carries a certified statement back.

Accordingly, this appendix proves no Birch–Swinnerton-Dyer, ABC, Riemann-hypothesis, Iwasawa-main-conjecture, modularity, Selmer-finiteness, characteristic-ideal, or p -adic L -function theorem. Its theorem-bearing content is limited to conditional interface statements whose source, target, hypotheses, artifacts, direction, and failure boundary are displayed.

For an arithmetic source object X , every Core-relevant evaluation follows the fixed path

$$\begin{aligned} X &\mapsto F_X = \mathfrak{R}_{\text{arith}}(X) \mapsto \mathbf{P}_n(F_X), \\ \mathbf{P}_n(F_X) &\mapsto B_{n,W}(F_X) := W_W \mathbf{P}_n(F_X) \mapsto \text{Eval}_{n,W,\tau}(F_X) \\ &:= T_\tau^{\text{bar}}(B_{n,W}(F_X)). \end{aligned}$$

No raw arithmetic invariant replaces any arrow in this path.

Remark 1.1 (v20 dependency position). The normative dependency order is:

- (i) the immutable repaired v19 lower baseline, used only at its retained strength;
- (ii) AK-HPDST v20.0.0 Main Chapters 1–8;
- (iii) v20 Foundation Appendices A–G;
- (iv) CR-v20 baseline governance for registered Main/Foundation claims;
- (v) CM-v20, TB-v20, and frozen Technical Extension Appendices H–N, each only at its reviewed local strength and immutable artifact identity;
- (vi) this Problem-Interface appendix.

The current CR-v20 baseline does not yet promote Part IV. Therefore a local MS-A label is not a canonical Claim UID, and appearance in this appendix is not claim promotion.

Remark 1.2 (Relation to Chapter 8). Every theorem-bearing arithmetic route must keep the following components separate:

- (i) external arithmetic source semantics;
- (ii) purpose-faithful arithmetic-to-AK realization;
- (iii) one-parameter and coefficient extraction, when needed;
- (iv) repaired window-first evaluation;
- (v) direct proof or admitted finite detector certificate;
- (vi) an n -eligible higher obstruction package, when Ext^n is invoked;
- (vii) actual-morphism Type IV and transient-defect records, when tower semantics are invoked;

- (viii) a finite-to-infinite reduction, when a finite witness is used for an infinite target;
- (ix) the typed Core audit verdict;
- (x) a separately proved arithmetic Return Theorem.

No component may be omitted, reversed, merged, or replaced by arithmetic plausibility.

Remark 1.3 (No shortcut through arithmetic meaning). Arithmetic meaning may motivate a realization, detector, window, threshold, tower, or bridge. It cannot replace

$$B_{n,W}(F_X), \quad \text{Eval}_{n,W,\tau}(F_X), \quad C_{\tau,W}F_X, \quad \phi_{i,W,\tau}, \quad \text{val}_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau(\mathcal{B}),$$

nor can it replace detector completeness, n -eligibility, finite-to-infinite reduction, claim-use preflight, artifact identity, or a Return Theorem.

Remark 1.4 (Window-first and metric guard-rails). Window restriction precedes hard deletion. No unconditional commutation

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is assumed. Whenever a metric inference crosses hard deletion, both compared pre-threshold profiles must satisfy the two-sided closed-band exclusion

$$[\tau - \varepsilon, \tau + \varepsilon],$$

and exact zero transport additionally uses $2\varepsilon \leq \tau$, with $2\varepsilon < \tau$ as the operational reserve. Raw detector ranks are not inferred to be bottleneck-stable.

MS-A.1bis. Arithmetic interface sovereignty

Definition 1.5 (Arithmetic interface sovereignty). *Arithmetic interface sovereignty* is the rule that arithmetic source data, source-domain theorems, conjectures, computations, literature, and semantic interpretations remain outside the AK Core unless they re-enter through declared verifier-side records. Admissible re-entry objects include, as invoked:

- (i) constructible one-parameter persistence objects in $\text{Pers}_k^{\text{cons}}$;
- (ii) finite-type filtered objects in $\text{FiltCh}(k)$;
- (iii) repaired pre-threshold and post-threshold profile records;
- (iv) admitted finite detector certificates with separate certificate and value statuses;
- (v) admissible representatives $C_{\tau,W}F$;
- (vi) standard n -eligible realized objects and edge records;
- (vii) actual tower comparison morphisms and typed terminal Type IV records;
- (viii) full kernel/cokernel defect profiles and complete transient-defect detector records;
- (ix) certified metric-ledger records tied to actual profile pairs;
- (x) non-scalar obligation records;
- (xi) finite-to-infinite reduction records;
- (xii) canonical manifest, claim-use, dependency, and artifact records.

A Return Theorem is not a Core re-entry object; it is the separate theorem carrying an audited AK-side conclusion back to arithmetic semantics.

Proposition 1.6 (No arithmetic sovereignty override). *No arithmetic theorem, conjecture, invariant, computation, literature artifact, expert judgment, or AI-generated arithmetic analysis overrides a missing or failed verifier-side requirement.*

Proof. The verifier consumes typed objects and evidence at their declared roles. Source-domain authority and mathematical interest do not manufacture a constructible profile, detector completeness theorem, representative, edge isomorphism, actual comparison morphism, reduction theorem, or immutable artifact. A missing required object therefore remains undefined; an evidenced false invoked condition rejects at its typed route. $\square$

Remark 1.7 (Arithmetic meaning is proposer-side by default). Arithmetic meaning may guide proposal, search, and calibration. It becomes theorem evidence only through a proved Interface or Bridge Theorem and successful claim-use preflight.

MS-A.2. Arithmetic source data, semantics, variance, and coefficients

Definition 1.8 (Arithmetic source datum). An *arithmetic source datum* is a mathematical object or structured family arising from arithmetic geometry, number theory, or arithmetic representation theory. Examples include elliptic curves, abelian varieties, number fields, Galois representations, Selmer groups or complexes, Iwasawa modules and towers, congruence or Hecke modules, automorphic data, classical or p -adic L -functions, Newton polygons, characteristic ideals, motivic or cohomological complexes, and height, conductor, regulator, Tamagawa, or local-correction data.

Definition 1.9 (Arithmetic source category). An *arithmetic source category* is a declared category, groupoid, indexed family, tower, diagram, or typed collection

$$\mathcal{A}_{\text{arith}}$$

whose objects are arithmetic source data and whose morphisms encode exactly the source-domain relations consumed by the proposed interface.

Definition 1.10 (Arithmetic source-semantics record). An *arithmetic source-semantics record* contains:

- (i) the source class and object identity;
- (ii) the ambient arithmetic category or theory;
- (iii) the precise external property, obstruction, or conclusion in scope;
- (iv) coefficient rings, fields, completions, local conditions, and topology;
- (v) every finiteness, exactness, compactness, control, interpolation, or convergence hypothesis used;
- (vi) the source theorem or proof artifact when an arithmetic fact is consumed;
- (vii) the permitted and prohibited readings of the source datum.

Definition 1.11 (Arithmetic interface family). An *arithmetic interface family* is a declared source class together with a common realization policy, coefficient and parameter extraction policy, monitored Core scope, claim role, and artifact regime. Family membership is a non-scalar obligation whenever a family-relative detector or uniform theorem is invoked.

Definition 1.12 (Variance record). A *variance record* declares whether the arithmetic-to-Core construction is covariant, contravariant, mixed-variance, pseudofunctorial, diagrammatic, or merely objectwise. It identifies the direction of every source morphism and the induced Core morphism or comparison cell.

Proposition 1.13 (Object assignment is not functorial realization). *An objectwise rule $X \mapsto F_X$ does not establish a functor*

$$\mathcal{A}_{\text{arith}} \longrightarrow \text{FiltCh}(k) \quad \text{or} \quad \mathcal{A}_{\text{arith}} \longrightarrow \text{Pers}_k^{\text{cons}}.$$

Proof. Functoriality additionally requires a typed action on morphisms, preservation of identities and composition, and the declared variance. These data are not supplied by an object assignment. $\square$

Definition 1.14 (Coefficient-passage record). A *coefficient-passage record* declares how arithmetic objects over a ring such as $\mathbb{Z}$, $\mathbb{Z}_p$, an Iwasawa algebra Λ , a Hecke algebra, or a local coefficient ring are converted to finite-dimensional k -linear data. It records:

- (i) the source coefficient object and target field k ;
- (ii) the operation used, such as tensor product, derived tensor product, reduction, localization, dualization, residue-field passage, or cohomology;
- (iii) exactness, flatness, Tor, derived-limit, completion, and finite-dimensionality hypotheses;
- (iv) the information discarded or changed;
- (v) whether the passage is exact, metric-controlled, theorem-controlled but nonmetric, advisory, or unavailable;
- (vi) immutable proof and computation artifacts.

Proposition 1.15 (Coefficient reduction is not lossless by default). *Passage from arithmetic coefficients to a field k is not assumed exact, conservative, faithful, or obstruction-reflecting.*

Proof. Tensoring, reduction, completion, taking invariants, coinvariants, duals, or cohomology may kill torsion, create Tor terms, fail to commute with limits, or forget module structure. Each required preservation or reflection property therefore needs its own theorem and artifact. $\square$

Remark 1.16 (Arithmetic data are not automatically Core data). An arithmetic object becomes Core-readable only after the source semantics, variance, coefficients, parameter extraction, constructibility, and all invoked downstream records have been supplied.

MS-A.3. Purpose-faithful arithmetic realization

Definition 1.17 (Arithmetic realization datum). An *arithmetic realization datum* is a declared objectwise procedure or functorial route

$$\mathfrak{R}_{\text{arith}} : \mathcal{A}_{\text{arith}} \rightsquigarrow \text{FiltCh}(k) \quad \text{or} \quad \mathfrak{R}_{\text{arith}} : \mathcal{A}_{\text{arith}} \rightsquigarrow \text{Pers}_k^{\text{cons}},$$

where $\rightsquigarrow$ deliberately permits an objectwise or partially functorial construction. A genuine functor arrow may be written only after the variance and composition laws are proved.

Definition 1.18 (Purpose-faithful arithmetic realization). An arithmetic realization is *purpose-faithful* for a declared claim when it identifies:

- (i) the external arithmetic feature;
- (ii) the Core-visible feature intended to represent it;
- (iii) the direction of use;
- (iv) the source class and all hypotheses;
- (v) the theorem or controlled-mismatch statement connecting the two;
- (vi) the exact failure boundary.

Purpose-faithfulness is feature-relative and direction-relative. It is not categorical faithfulness, conservativity, full faithfulness, or equivalence unless separately proved.

Definition 1.19 (Arithmetic realization target). An arithmetic interface declares one or more targets, each with its role:

- (i) a filtered object in $\text{FiltCh}(k)$;
- (ii) a constructible persistence object in $\text{Pers}_k^{\text{cons}}$;
- (iii) a windowed pre-threshold profile $B_{n,w}(F)$;
- (iv) a repaired profile $\text{Eval}_{n,w,\tau}(F)$;
- (v) an admitted detector object or certificate;
- (vi) an admissible representative $C_{\tau,w}F$;
- (vii) a standard n -eligible realized object

$$A_{n;F,w,\tau} \in D^{[-n,0]}(k\text{-mod})$$

or a separately proved admissible replacement;

- (viii) an actual tower comparison artifact $\phi_{i,w,\tau}$;
- (ix) a finite-to-infinite reduction record;
- (x) a certificate or manifest component;
- (xi) an advisory, Search, calibration, or explanatory object.

Targets in the last item are non-verdict by default.

Definition 1.20 (Purpose profile). *A purpose profile records the source feature, target feature, direction, monitored degrees, windows, thresholds, detector stage, expected Core clause, required Return conclusion, source class, and prohibited stronger reading.*

Proposition 1.21 (No arithmetic interface without declared realization). *No arithmetic source datum has theorem-bearing access to the Core until its realization datum, target, purpose profile, source semantics, coefficient policy, parameter policy, and invoked artifacts are declared.*

Proof. Without those records the verifier cannot identify the Core input, the intended external meaning, or the direction and scope of transport. The entry is therefore undefined rather than passing. $\square$

Remark 1.22 (Purpose-faithful is not globally faithful). *A realization may preserve one obstruction predicate while forgetting rank, torsion, characteristic-ideal, analytic, or functorial information. No unlisted arithmetic feature is inferred to survive.*

MS-A.3bis. Purpose-faithfulness burden and claim typing

Definition 1.23 (Purpose-faithfulness burden). *The burden consists of:*

- (i) source and target types;
- (ii) variance and coefficient conventions;
- (iii) parameter extraction and filtration orientation;
- (iv) the exact preserved or reflected predicate;
- (v) direction, naturality scope, and representative dependence;
- (vi) hypotheses and source class;
- (vii) proof or controlled-defect artifact;
- (viii) failure routing and forbidden converse.

Definition 1.24 (Purpose-faithfulness status). A purpose-faithfulness claim has evidence status

proved, conditional, operational, spec, rejected, unknown,

and a separate invocation role. Only proved, or conditional with every registered hypothesis discharged, may be consumed as theorem evidence.

Proposition 1.25 (Declared purpose is not purpose-faithfulness). *Declaring what a realization is intended to preserve does not prove that it preserves it.*

Proposition 1.26 (Unproved purpose-faithfulness blocks transport). *A purpose-faithfulness claim with status spec, rejected, unknown, or unresolved conditional cannot discharge an arithmetic transport or Return obligation.*

Proof. The missing correspondence is itself the cross-layer theorem required for the proposed implication. Neither Core certification nor arithmetic plausibility supplies it. $\square$

Definition 1.27 (Arithmetic interface claim class). Every arithmetic-facing cross-layer statement is classified as exactly one of:

interface_definition, interface_theorem, bridge_candidate, bridge_program,
bridge_theorem, toy_bridge_theorem, spec, non_claim.

Operational policies, Search artifacts, Companion templates, successor-release candidates, deprecated items, and rejected items retain their separate global roles and are not silently folded into this cross-layer list. Document role, evidence status, and invocation role remain independent.

Remark 1.28 (Feature-relative and direction-relative). A route may be faithful for

arithmetic obstruction $\implies$ Core nonzero

while failing for the converse, or faithful on a finite benchmark class while failing on an infinite tower. Each direction and class is registered separately.

MS-A.4. Core-readable arithmetic objects

Definition 1.29 (Core-readable arithmetic object). An arithmetic source datum X is *Core-readable* on a declared scope (n, W, τ) when the interface supplies:

- (i) a typed realized object $F_X \in \text{FiltCh}(k)$ or $P_X \in \text{Pers}_k^{\text{cons}}$;
- (ii) constructibility and finite-dimensionality evidence on the consumed scope;
- (iii) the exact pre-threshold profile
$$B_{n,W}(F_X) = W_W \mathbf{P}_n(F_X);$$
- (iv) the repaired profile
$$\text{Eval}_{n,W,\tau}(F_X) = T_\tau^{\text{bar}}(B_{n,W}(F_X));$$
- (v) a valid comparison mode and evidence for every consumed comparison;
- (vi) every detector, representative, higher-edge, Type IV, transient-defect, atlas, Restart, reduction, claim, and manifest record that is invoked;
- (vii) a complete and consistent canonical manifest permitting the Chapter 7 verifier to evaluate the run.

Core-readability means typed auditability of the projected data. It does not prove purpose-faithfulness or any arithmetic conclusion.

Definition 1.30 (Arithmetic certificate input). An *arithmetic certificate input* is a Core-readable arithmetic object together with immutable scope, coefficient, parameter, evaluation, detector, comparison, representative, eligibility, tower, defect, ledger, non-scalar-obligation, claim-use, and artifact records required by the intended Core clause.

Definition 1.31 (Arithmetic Core-readability failure). Core-readability is:

- (i) pass only when every invoked entry obligation is supplied and typed;
- (ii) undefined when required evidence is missing, unresolved, stale, or ill-typed;
- (iii) reject when a well-typed invoked condition is evidenced false or the manifest is inconsistent.

A clause may be marked not `_invoked` only under an immutable scope exclusion; the top-level Interface outcome itself is then not generated for that excluded route. An Interface failure is not automatically a Core mathematical rejection.

Remark 1.32 (Constructibility requirement). Wild, uncontrolled multiparameter, infinite-rank, or nonconstructible arithmetic output is outside the Core until a separate Interface theorem reduces it to the declared constructible one-parameter regime.

Remark 1.33 (No raw arithmetic-to-Eval shortcut). A rank, valuation, Newton polygon, Selmer dimension, L -value, or characteristic ideal may motivate the expected barcode. It does not define $B_{n,W}$ or $\text{Eval}_{n,W,\tau}$ without a proved typed identification.

MS-A.4bis. One-parameter extraction and constructibility discipline

Definition 1.34 (Arithmetic one-parameter extraction). An *arithmetic one-parameter extraction* converts arithmetic data with one or more natural parameters into a single ordered parameter compatible with $\text{Pers}_k^{\text{cons}}$. It records:

- (i) all source parameters and their orders;
- (ii) the selected persistence direction and orientation;
- (iii) normalization, ties, incomparable elements, branches, skipped layers, and boundary conventions;
- (iv) transition morphisms or an explicit objectwise-only restriction;
- (v) information discarded by projection;
- (vi) constructibility, local finiteness, and pointwise finite-dimensionality proofs;
- (vii) family membership and non-adaptivity when a family-relative detector is planned;
- (viii) metric and nonmetric effects separately.

Definition 1.35 (Extraction discrepancy and extraction obligation). An extraction effect is classified as:

- (i) an actual metric profile discrepancy $\mathbf{d}_{\text{arith-extract}}$;
- (ii) a theorem-controlled nonmetric semantic loss;
- (iii) a missing or failed non-scalar extraction obligation;
- (iv) an advisory description only.

Only the first class may enter a Core metric ledger, and only with an actual bottleneck-support theorem for the consumed profile pair.

Proposition 1.36 (No multiparameter shortcut). A multiparameter arithmetic object is not Core-readable merely because every source parameter is mathematically meaningful.

Proof. The Core consumes constructible one-parameter persistence. A parameter projection, order, transition system, and constructibility proof are independent requirements. $\square$

Proposition 1.37 (Extraction loss is non-scalar unless metrically represented). *Discarded branches, changed variance, loss of module structure, failure of constructibility, and missing transition morphisms are non-scalar obligations unless a separate theorem constructs an actual profile discrepancy bound representing the relevant loss.*

Remark 1.38 (Common arithmetic extraction risks). Typical risks include collapsing conductor and level, selecting one branch of a multivariable p -adic L -function, choosing only the cyclotomic direction, or identifying congruence depth with valuation depth. No such choice is neutral by default.

Proposition 1.39 (Right-unbounded detector boundary). *A right-unbounded arithmetic parameter range with unbounded possible births has no uniform complete finite detector merely because finitely many arithmetic layers were sampled. A finite certificate requires an additional finite-birth, stabilization, global-bound, family, or finite-to-infinite reduction theorem defeating the Core no-go boundary.*

MS-A.5. Arithmetic pre-gates

Definition 1.40 (Arithmetic pre-gate). An *arithmetic pre-gate* is a source-domain screening or preparation condition applied before Core verification. Examples include conductor bounds, reduction type, local conditions, torsion or rank prechecks, congruence compatibility, Selmer control hypotheses, tower admissibility, automorphic availability, branch selection, or finite-layer adequacy.

Definition 1.41 (Arithmetic pre-gate record). A pre-gate record contains the source condition, source object, theorem or computation used, lifecycle result, requiredness, artifacts, and the explicit statement that the result is not a Core atomic verdict.

Proposition 1.42 (Arithmetic pre-gates do not replace Core gates). *Passing arithmetic pre-gates does not discharge repaired profile vanishing, detector certification, representative existence, n -eligibility, actual-morphism Type IV, transient-defect absence, metric reserve, finite-to-infinite reduction, manifest closure, or Return.*

Proof. Pre-gates select or prepare source objects. The listed obligations have different types and are evaluated by their own theorems and artifacts. $\square$

Remark 1.43 (Pre-gate failure versus Core reject). Failure of a pre-gate usually means that the proposed Interface route is unavailable for that object. It is a Core rejection only when a corresponding Core clause has been properly invoked and evidenced false.

MS-A.5bis. Arithmetic finite-detector interface

Definition 1.44 (Arithmetic detector realization record). An *arithmetic detector realization record* identifies a finite arithmetic observable Q_X with a declared Core detector on exactly one source stage:

windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect.

It records the monitored degree, W , τ , endpoint convention, detector mode, sample or critical set, source profile identity, and the theorem relating Q_X to the detector.

Definition 1.45 (Arithmetic finite detector certificate). An arithmetic finite detector certificate contains, as applicable:

- (i) the exact profile stage and immutable profile identity;
- (ii) detector soundness evidence;
- (iii) detector zero-completeness or characterization evidence;
- (iv) birth/germ exhaustiveness, stencil coverage, or family-mesh evidence;

- (v) endpoint and window-cofinal policy;
- (vi) family membership and the source of every uniform threshold-excess bound;
- (vii) exact or metric-safe transport records;
- (viii) non-adaptive selection evidence;
- (ix) immutable computation and proof artifacts;
- (x) separate detector certificate status and mathematical value status.

Proposition 1.46 (Finite arithmetic observation is not finite proof). *A finite list of arithmetic ranks, valuations, layers, special values, or sampled maps is not a finite proof certificate without either a direct theorem covering the exact finite predicate or the complete detector package.*

Proof. Finiteness of the observation does not establish coverage of all retained bars or defects. Missing completeness leaves a zero-looking result undefined. $\square$

Proposition 1.47 (Soundness and completeness do not collapse). *A certified nonzero arithmetic detector witness may support only the direction proved by its soundness theorem. A zero detector value proves repaired-profile zero only under the exact completeness theorem and scope.*

Proposition 1.48 (Arithmetic detector value is not an external conclusion). *Even a complete detector certificate establishes only its registered Core predicate. An arithmetic zero or nonzero conclusion additionally requires the applicable Return or detector-soundness Bridge.*

Remark 1.49 (Metric detector transport). Metric transport preserves the certified zero predicate only through the pre-threshold bottleneck bound, two-sided threshold-gap theorem, and $2\varepsilon < \tau$ operational reserve. It does not transport arbitrary detector ranks, birth sets, certificate sizes, or arithmetic observables.

MS-A.6. Higher eligible regime for arithmetic Ext^n use

Definition 1.50 (Arithmetic standard n -eligible regime). Fix $n \geq 1$. An arithmetic interface is in the *standard n -eligible regime* on $(F_X; W, \tau)$ when:

- (i) an admissible representative $C_{\tau, W}F_X$ exists;
- (ii) an actual degree- n profile isomorphism identifies

$$\mathbf{P}_n(C_{\tau, W}F_X) \cong \text{Eval}_{n, W, \tau}(F_X);$$

- (iii) a realization

$$A_{n; X, W, \tau} := \mathcal{R}_n(C_{\tau, W}F_X) \in D^b(k\text{-mod})$$

is independently meaningful and realization-compatible;

- (iv) the standard amplitude condition

$$A_{n; X, W, \tau} \in D^{[-n, 0]}(k\text{-mod})$$

holds, or a separately registered admissible replacement is proved;

- (v) a fixed edge extractor E_n is declared on a stated repaired-profile class and satisfies $E_n(0) = 0$;
- (vi) an artifact-backed edge identification

$$H^{-n}(A_{n; X, W, \tau}) \cong E_n(\text{Eval}_{n, W, \tau}(F_X))$$

is supplied, with objectwise or natural-family scope declared;

- (vii) coefficient field, family membership, representative-change compatibility, realization-change compatibility, non-adaptivity, claim status, and artifacts are complete.

For $n = 1$, this specializes to the inherited two-term eligible regime.

Proposition 1.51 (Higher one-way arithmetic discharge). *In the standard n -eligible regime,*

$$\text{Eval}_{n,W,\tau}(F_X) = 0 \implies \text{Ext}^n(A_{n;X,W,\tau}, k) = 0.$$

Proof. Zero compatibility gives $H^{-n}(A_{n;X,W,\tau}) = 0$. The field-coefficient higher Ext edge of Appendix C then yields the stated vanishing at exactly its registered strength. $\square$

Proposition 1.52 (No higher Ext discharge without eligibility). *Missing representative, realization, amplitude, profile-class, edge-extractor, zero-compatibility, edge-identification, coefficient, non-adaptivity, or artifact evidence leaves an invoked arithmetic Ext^n clause undefined.*

Definition 1.53 (Arithmetic zero-reflection package). *An arithmetic zero-reflection package is a separately proved natural theorem on a declared profile class establishing the converse needed for*

$$\text{Ext}^n(A_{n;X,W,\tau}, k) = 0 \implies \text{Eval}_{n,W,\tau}(F_X) = 0.$$

It records the exact extractor, class, naturality, faithfulness or zero-reflection theorem, representative independence, and artifacts.

Proposition 1.54 (No converse from eligibility alone). *Standard n -eligibility and Ext^n -vanishing do not imply repaired-profile vanishing without the arithmetic zero-reflection package.*

Remark 1.55 (Arithmetic Ext is not automatically Core Ext). Galois-cohomological, motivic, deformation, Selmer, Iwasawa, or module-theoretic Ext groups are not automatically

$$\text{Ext}^n(A_{n;X,W,\tau}, k).$$

Any identification is itself an Interface theorem.

Remark 1.56 (No synthetic promotion). Encoding a detector vector space as a shifted complex may prove a toy or synthetic identity. It does not supply independent arithmetic meaning, a natural realization theorem, or a principal arithmetic PH-to-Ext bridge.

MS-A.7. Arithmetic transport, Bridge, and Return statements

Definition 1.57 (Arithmetic transport statement). *An arithmetic transport statement is a directional implication between a precisely typed arithmetic predicate and a precisely typed Core-visible predicate. The forward and reverse directions are distinct claims.*

Definition 1.58 (Arithmetic Interface theorem). *An arithmetic Interface theorem is a proved conditional theorem connecting declared arithmetic source data to a Core-readable object or predicate, or an audited Core predicate to an arithmetic conclusion. It records:*

- (i) source and target semantics;
- (ii) direction and variance;
- (iii) source class and hypotheses;
- (iv) realization, coefficient, and parameter records;
- (v) exact Core predicate and profile stage;
- (vi) detector, representative, higher-edge, tower, reduction, or Return obligations invoked;
- (vii) metric and nonmetric effects;
- (viii) immutable proof and artifact identities;

(ix) failure routes and prohibited stronger readings.

Definition 1.59 (Default status of arithmetic cross-layer implications). An unproved arithmetic cross-layer implication is a `bridge_candidate`, `bridge_program`, or `spec`, not theorem evidence. Promotion requires a new proof, exact role/status registration, and artifact binding.

Proposition 1.60 (Arithmetic transport does not enlarge the Core). *A proved arithmetic Interface or Bridge Theorem remains Problem-Interface material unless a later release explicitly imports and reproves it as Core.*

Remark 1.61 (Two directions must be separated). The claims

$$\mathcal{P}_{\text{arith}}(X) \implies \mathcal{Q}_{\text{Core}}(F_X) \quad \text{and} \quad \mathcal{Q}_{\text{Core}}(F_X) \implies \mathcal{P}_{\text{arith}}(X)$$

have different semantic burdens. The second is a Return direction and is never inferred from the first.

Definition 1.62 (Arithmetic Return Theorem). An *arithmetic Return Theorem* is a proved theorem of the form

$$\mathcal{Q}_{\text{AK}}(X) \implies \mathcal{C}_{\text{arith}}(X),$$

where $\mathcal{Q}_{\text{AK}}(X)$ is the exact audited AK-side conclusion actually established and $\mathcal{C}_{\text{arith}}(X)$ is the exact arithmetic conclusion. It records the source class, all hypotheses, every intermediate semantic identification, finite-to-infinite scope, direction, proof, artifacts, and failure boundary.

MS-A.7bis. Arithmetic comparison modes

Definition 1.63 (Arithmetic comparison). An *arithmetic comparison* may occur in the source domain, between realizations, between Core-readable profiles, or between finite and limiting objects. The comparison level is always declared.

Definition 1.64 (Arithmetic comparison classification). A source-side comparison may be classified locally as

`source_exact`, `realization_exact`, `metric_candidate`, `advisory`, `unavailable`.

Every consumed Core comparison nevertheless resolves to exactly one canonical runtime mode:

`exact`, `metric_gap_certified`, `not_compared`.

The projections are:

- (i) `realization_exact` becomes `exact` only with an actual equality or profile isomorphism of the declared Core-readable profiles;
- (ii) `metric_candidate` becomes `metric_gap_certified` only with a certified bottleneck upper bound on the actual pre-threshold profile pair, a typed scalarization, the two-sided gap record, and every required zero-separation condition;
- (iii) `source_exactness` remains source-side until transported by a theorem;
- (iv) `advisory` or `unavailable` comparisons become `not_compared`, which cannot discharge an invoked comparison obligation.

Failure is a status, not a comparison mode.

Proposition 1.65 (Arithmetic equality is not Core exactness). *An equality of arithmetic invariants, congruence, control theorem, or source isomorphism is not a Core exact comparison without an actual exact comparison of the realized profiles.*

Proposition 1.66 (Arithmetic closeness is not metric-gap certification). *A valuation estimate, congruence depth, finite-layer approximation, or numerical error bound is not metric-gap certification unless it upper-bounds the bottleneck distance of the exact profiles consumed by hard deletion and satisfies the full threshold-gap record.*

Proposition 1.67 (Object distance does not control defect profiles automatically). *A bottleneck bound between source and target tower profiles does not automatically bound the kernel or cokernel defect profiles of an actual comparison morphism. A separate morphism-level support theorem is required.*

MS-A.8. Arithmetic effects, metric ledger, and non-scalar obligations

Definition 1.68 (Arithmetic interface effect). An *arithmetic interface effect* is any discrepancy, information loss, hypothesis, or blocker introduced by realization, coefficient passage, parameter extraction, truncation, approximation, comparison, tower formation, or Return. Each effect is classified before aggregation.

Definition 1.69 (Arithmetic effect classification). An arithmetic interface effect has exactly one primary class:

- (i) `metric_profile_discrepancy`: an actual discrepancy between declared Core-readable profiles;
- (ii) `morphism_profile_discrepancy`: an actual discrepancy between declared morphism-derived defect profiles;
- (iii) `theorem_controlled_nonmetric`: semantic loss controlled by a theorem but not represented by a Core metric;
- (iv) `non_scalar_obligation`: existence, typing, naturality, completeness, coherence, reduction, Return, status, or identity requirement;
- (v) `advisory`: a non-verdict quantity;
- (vi) `unavailable`: no admissible control is supplied.

Definition 1.70 (Local arithmetic metric aliases). Local symbols such as

$$\mathfrak{d}_{\text{arith}}, \quad \mathfrak{d}_{\text{arith-extract}}, \quad \mathfrak{d}_{\text{arith-trunc}}, \quad \mathfrak{d}_{\text{arith-comp}}$$

are gate-scoped aliases for metric-bearing contributions. They are not independent Core ledgers and do not acquire a numerical value merely by being named.

Definition 1.71 (Arithmetic metric-ledger entry). A metric-ledger entry records:

- (i) the exact source and target profile identities;
- (ii) the construction path and dependency graph;
- (iii) the metric effect type and canonical Appendix M slot;
- (iv) its element in the declared commutative quantale $(V, \oplus, 0_V, \preceq)$;
- (v) the scalarization val_B ;
- (vi) a theorem or verified artifact proving

$$d_B(B, C) \leq \text{val}_B(\mathfrak{d}_{\text{met}});$$

- (vii) the protected pre-threshold family $\mathcal{B}$, its threshold clearances, and the strict reserve;
- (viii) immutable provenance and replay artifacts.

Definition 1.72 (Arithmetic non-scalar obligation record). Non-scalar arithmetic obligations include, as invoked:

`source_semantics`, `variance`, `coefficient_passage`,
`constructibility`, `purpose_faithfulness`, `parameter_extraction`,
`detector_completeness`, `representative`, `n_eligibility`, `edge_naturality`,
`actual_comparison_morphism`, `terminal_tameness`,
`transient_coverage`, `finite_to_infinite_reduction`,
`apex_agreement`, `return_theorem`, `claim_permission`,
`manifest_consistency`, `artifact_identity`.

Each has its own atomic status and evidence.

Proposition 1.73 (Only supported metric discrepancies enter the ledger). *Only effects in the first two classes of Definition 1.69 may enter the strict Core metric budget, and only through an actual support theorem bound to the consumed profile path.*

Proof. The budget is meaningful only as an upper bound on an identified bottleneck discrepancy. Semantic, existential, naturality, completeness, governance, and Return obligations are not numbers in that metric. $\square$

Proposition 1.74 (Non-scalar obligations are non-compensating). *No amount of metric slack, favorable arithmetic structure, detector score, or pre-gate success repairs a missing or failed non-scalar arithmetic obligation.*

Proposition 1.75 (Arithmetic metric budget is path-bound). *A local arithmetic error estimate may be consumed only when the complete realization-to-profile path, aggregation theorem, actual compared profiles, and strict gate-scoped gap comparison are declared.*

Remark 1.76 (Arithmetic elegance is not zero discrepancy). Canonicity, natural language, symmetry, or conceptual elegance does not prove exactness or zero metric discrepancy.

MS-A.8bis. Finite approximation, finite proof, and finite-to-infinite reduction

Definition 1.77 (Arithmetic finite approximation). An *arithmetic finite approximation* is a finite-layer, finite-precision, finite-rank, finite-window, sampled, truncated, or computational replacement for an arithmetic object, tower, series, module, or analytic profile.

Definition 1.78 (Approximation lifecycle status). An approximation has lifecycle status

sample, calibration, bounded, certified, failed, undefined.

These are not Core atomic statuses. A certified approximation may support a Core comparison only when its exact metric or theorem-controlled nonmetric role is declared.

Proposition 1.79 (Finite evidence is not an infinite arithmetic theorem). *Finite-layer, finite-precision, sampled, or numerical evidence does not establish an infinite tower statement, unbounded-window statement, completed-module statement, p -adic analytic identity, or external arithmetic theorem without a separately proved reduction and Return chain.*

Definition 1.80 (Arithmetic finite-to-infinite Interface record). Whenever finite evidence is used for an infinite arithmetic target, the record contains:

- (i) the exact finite witness and its certified predicate;
- (ii) the actual directed-system morphisms;
- (iii) the declared colimit, limit, completion, or source semantics;
- (iv) the registered finite-to-infinite reduction theorem and all hypotheses;
- (v) a compatible apex cocone or cone;
- (vi) the induced comparison morphism;
- (vii) apex agreement;
- (viii) any compactness, control, Mittag–Leffler, derived-limit, convergence, continuation, or stabilization hypotheses;
- (ix) dependency-closed immutable artifacts.

Proposition 1.81 (Finite prefix and plateau are not reduction). *Finite-prefix agreement, numerical plateau, scalar convergence, Cauchy behavior, or observed eventual constancy does not pass a finite-to-infinite clause without the registered theorem and apex-agreement artifacts.*

Proposition 1.82 (Unledged or untyped approximation is non-verdict). *If an approximation changes a consumed Core profile or comparison, then either an admissible metric support record, an exact theorem-controlled transport, or an explicit non-invocation is required. Otherwise the affected clause is undefined.*

Remark 1.83 (Derived-limit obligations). Inverse limits, completions, and nonexact limit functors may generate $\varprojlim^1$, Tor, or other derived terms. These are independent obligations and are not suppressed by finite-stage stabilization plots.

MS-A.9. Arithmetic towers and actual-morphism Type IV

Definition 1.84 (Arithmetic tower). An *arithmetic tower* is a directed or inverse system of arithmetic source data with declared index category, transition morphisms, variance, source semantics, and candidate apex. Examples include cyclotomic, anticyclotomic, congruence, level, Selmer, or local-condition towers.

Definition 1.85 (Core-readable arithmetic tower). An arithmetic tower is Core-readable for Type IV only when its realization supplies:

- (i) actual filtered or persistence objects and transition morphisms;
- (ii) repaired stagewise profiles on a fixed (i, W, τ) ;
- (iii) source provenance and a semantic-authority verdict;
- (iv) a declared apex identity and repaired apex profile;
- (v) a compatible colimit or source-profile system;
- (vi) an actual comparison morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty);$$

- (vii) an ambient abelian profile category and typing proof;
- (viii) terminal-tameness and defect-profile artifacts.

Object-level barcode agreement does not supply these data.

Definition 1.86 (Arithmetic Type IV diagnostic record). For a valid artifact, define

$$\mu_{i,W,\tau} := \text{tgdim}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} := \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}).$$

For one fixed pair (W, τ) and one finite monitored degree set I_{mon} , the defined degreewise values may be aggregated as

$$\mu_{\text{Collapse}}(W, \tau) := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \quad \nu_{\text{Collapse}}(W, \tau) := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

Aggregation across several windows, thresholds, or infinitely many degrees requires a separate typed aggregation or summability theorem. The record declares whether W is bounded right-open, where tgdim_W counts right-endpoint-cofinal defect bars, or right-unbounded, where it counts infinite defect bars.

Definition 1.87 (Arithmetic Type IV proxy). A source-domain signal such as failure of control, anomalous Iwasawa growth, congruence failure, or unstable limit behavior is an *arithmetic Type IV proxy*. It is advisory or Interface-side until a theorem identifies it with the canonical kernel/cokernel defect profiles.

Proposition 1.88 (Arithmetic Type IV proxy is not Type IV). *An arithmetic proxy, source invariant, barcode matching, finite-stage agreement, or equality of Betti numbers does not determine the canonical Type IV diagnostics without the actual comparison morphism and defect profiles.*

Proposition 1.89 (Terminal zero is not full comparison isomorphism).

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

does not exclude finite interior kernel or cokernel bars and therefore does not imply that $\phi_{i,W,\tau}$ is an isomorphism.

Remark 1.90 (Arithmetic μ versus Collapse μ). The Iwasawa invariant μ_{Iw} and the Collapse diagnostic μ_{Collapse} have different definitions and source types. Their identification requires a separate arithmetic Interface theorem.

MS-A.9bis. Arithmetic invariant namespace separation

Definition 1.91 (Arithmetic invariant namespace). Arithmetic invariants retain decorated source-domain names, for example

$$\mu_{Iw}, \quad \lambda_{Iw}, \quad r_{\text{Sel}}, \quad \text{ord}_{s=s_0} L, \quad v_p, \quad \text{char}_\Lambda.$$

They are not Core diagnostic symbols without a proved transport theorem.

Proposition 1.92 (Namespace collision is invalid). *A theorem-facing record is ill-typed if the same undecorated symbol is used for an arithmetic invariant and a Core diagnostic in a way affecting a conclusion.*

Proposition 1.93 (Arithmetic invariants are not Core obstructions by default). *Selmer rank, analytic order, Iwasawa λ - or μ -growth, valuation growth, and characteristic-ideal behavior are not automatically residual persistent homology, a higher Ext^n obstruction, terminal Type IV, or transient defect.*

Remark 1.94 (μ -collision warning). When both invariants occur, the subscripts Iw and Collapse are mandatory. The cokernel-side Collapse diagnostic is always v , never a Latin u .

MS-A.9ter. Arithmetic transient-defect discipline

Definition 1.95 (Arithmetic transient-defect interface). An arithmetic transient-defect interface uses the full profiles

$$\ker \phi_{i,W,\tau}, \quad \text{coker } \phi_{i,W,\tau},$$

together with a declared defect threshold τ_{def} , component, detector stage, endpoint convention, information class, and complete detector certificate.

Definition 1.96 (Arithmetic defect conclusion levels). The following conclusion levels are independent:

- (i) terminal Type IV status;
- (ii) long-transient kernel status;
- (iii) long-transient cokernel status;
- (iv) short-defect exclusion;
- (v) full kernel/cokernel vanishing;
- (vi) comparison-map isomorphism.

No lower level imputes a higher one.

Proposition 1.97 (Terminal and transient zero do not collapse). *Terminal certified_zero does not discharge long-transient absence, and long-transient certified_absent does not prove full defect-profile zero unless a separate short-defect-exclusion theorem is supplied.*

Proposition 1.98 (Arithmetic transient proxy is non-verdict). *Observed finite-layer fluctuations or temporary arithmetic growth are not transient-defect certificates without identification with the actual kernel or cokernel defect profile and a complete detector theorem.*

MS-A.10. Selmer, Iwasawa, and p -adic L -profile placeholders

This section introduces meta-theoretic source types only. It proves no Selmer, Iwasawa, or p -adic L -function theorem.

Definition 1.99 (Selmer interface datum). A *Selmer interface datum* consists of Selmer groups, Selmer complexes, local conditions, or Pontryagin-dual modules together with source semantics, coefficients, variance, parameter extraction, and a proposed Core-readable realization.

Definition 1.100 (Iwasawa interface datum). An *Iwasawa interface datum* consists of a tower, Iwasawa algebra or module, characteristic ideal, arithmetic invariants, control statements, and finite-layer-to-limit semantics together with a proposed Core-readable tower realization.

Definition 1.101 (p -adic L -profile interface datum). A *p -adic L -profile interface datum* consists of a p -adic L -function, branch, valuation sequence, Newton polygon, Weierstrass data, zero structure, or special-value profile together with a proposed one-parameter and coefficient extraction.

Proposition 1.102 (Placeholders do not prove transport). *The existence of any placeholder datum proves neither purpose-faithfulness nor a detector, higher-edge, tower, reduction, or Return theorem.*

Remark 1.103 (Successor appendices). Appendix MS-B may instantiate cyclotomic, Iwasawa, congruence, and p -adic L -profile routes. Its instances remain subordinate to the meta-theory and cannot weaken these obligations.

MS-A.11. Arithmetic Interface and certificate records

Definition 1.104 (Arithmetic Interface entry). An *arithmetic Interface entry* records at least:

- (i) external source object, source semantics, source class, and immutable source artifact;
- (ii) document role, evidence status, invocation role, permitted use, and forbidden stronger reading;
- (iii) realization datum, purpose profile, variance, coefficients, parameter extraction, and constructibility;
- (iv) exact windows, thresholds, monitored degrees, endpoint convention, and window-first evaluation order;
- (v) pre-threshold and repaired profile identities;
- (vi) comparison mode and exact or metric evidence for every consumed comparison;
- (vii) detector stage, mode, soundness, completeness, coverage, family, non-adaptivity, certificate status, and value status when invoked;
- (viii) representative and actual profile isomorphisms when invoked;
- (ix) n -eligible realization, edge extractor, edge identification, naturality, and zero-reflection records when invoked;
- (x) tower source semantics, transition morphisms, apex identity, actual comparison morphism, terminal tameness, terminal Type IV, and transient-defect records when invoked;
- (xi) metric ledger, scalarization, actual discrepancy support, protected family, clearance, and strict reserve;
- (xii) every non-scalar obligation and atomic status;
- (xiii) finite-to-infinite reduction, apex agreement, and derived-limit records when invoked;
- (xiv) the exact arithmetic Return claim and Return Theorem status when an external conclusion is claimed;
- (xv) known-theorem-recovery track identifiers, benchmark-class binding, source-proof quarantine, non-circularity audit, component status vector, recovered-strength relation, and independent validation artifacts when recovery is advertised;
- (xvi) portfolio shared-infrastructure, cross-track contamination, non-compensation, and closure records when several recoveries are advertised;
- (xvii) claim-use preflight, dependency closure, repair ancestry, manifest status, and immutable artifact identities.

Definition 1.105 (Arithmetic certificate record). An *arithmetic certificate record* is the canonical v20 proof-object projection of an arithmetic Interface entry together with all verifier-side evidence needed to compute the declared Core clauses. It does not include an arithmetic theorem merely because the source object is arithmetic.

Proposition 1.106 (Canonical manifest is necessary but not sufficient). *A complete and consistent arithmetic manifest is necessary for theorem-bearing use, but manifest completeness does not prove the mathematical truth of the recorded claims.*

Proof. The manifest establishes reconstructibility, identity, scope, dependency, and status semantics. Truth still depends on the proofs and artifacts referenced by the record. $\square$

Proposition 1.107 (Claim-use preflight is mandatory). *Every arithmetic claim consumed as theorem evidence must pass identity, source binding, role, evidence status, invocation role, hypotheses, dependencies, comparison policy, permitted-use, and artifact checks. Missing preflight data yield undefined.*

Remark 1.108 (Artifact identity). Basename, path, title, upload date, or literature citation alone is not immutable artifact identity. Content-addressed identity and exact locators control.

MS-A.12. Distinct arithmetic conclusion levels

Definition 1.109 (Arithmetic Interface outcome). The Interface entry outcome is one of pass, reject, or undefined. A subordinate clause may be not invoked only when it is immutably outside the declared claim scope. Interface pass means only that the arithmetic source has been projected into a complete, typed Core-readable entry at the declared scope.

Definition 1.110 (Core outcome for arithmetic data). A *Core outcome* is the Chapter 7 audit result for the invoked AK clauses. It may include repaired-profile zero/nonzero, detector, higher-edge, Type IV, transient-defect, ledger, atlas, Restart, or finite-to-infinite clauses. It is not an arithmetic theorem.

Definition 1.111 (Arithmetic external conclusion). An *arithmetic external conclusion* is a source-domain statement such as a rank, finiteness, vanishing, control, characteristic-ideal, special-value, interpolation, or L -function assertion.

Proposition 1.112 (Core pass is not arithmetic conclusion). *A Core pass for arithmetic data does not imply an arithmetic external conclusion without a proved Return Theorem whose antecedent is exactly the audited AK-side conclusion.*

Remark 1.113 (Nine-level distinction). The following levels remain distinct:

source pre-gate, Interface entry, finite certificate,
Core audit, Bridge/transport, finite-to-infinite reduction,
Return, external arithmetic conclusion,
independent recovery validation.

A success, proof, or validation at one level does not impute the corresponding status at another.

MS-A.12bis. Complete arithmetic projection-and-return chain

Definition 1.114 (Arithmetic theorem-bearing chain). A complete arithmetic AK theorem has the form

$$\begin{aligned} X &\xrightarrow{\mathfrak{R}_{\text{arith}}} F_X \xrightarrow{\mathbf{P}_n} \mathbf{P}_n(F_X) \xrightarrow{\mathbf{W}_W} B_{n,W}(F_X), \\ B_{n,W}(F_X) &\xrightarrow{T_{\tau}^{\text{bar}}} \text{Eval}_{n,W,\tau}(F_X) \xrightarrow{\text{direct proof/detector}} Q_{\text{fin}}(X), \\ Q_{\text{fin}}(X) &\xrightarrow{\text{reduction}} Q_{\text{AK}}(X) \xrightarrow{\text{typed audit}} \text{Verdict}_{\text{AK}}, \\ \text{Verdict}_{\text{AK}} &\xrightarrow{\text{Return}} C_{\text{arith}}(X). \end{aligned}$$

A higher Ext^n edge or tower-defect branch may be inserted only through its own eligible or actual-morphism package. Every arrow records source, target, direction, hypotheses, role, evidence status, artifacts, and failure boundary.

Definition 1.115 (Arithmetic theorem promotion). Promotion from any of the following statuses to theorem evidence requires:

spec, bridge_candidate, bridge_program.

- (i) a complete proof of every invoked arrow;
- (ii) exact claim typing and permitted use;
- (iii) dependency and artifact closure;
- (iv) a successful claim-use preflight;
- (v) a new or updated Claim Register entry after Part IV freeze.

Registration without proof does not promote.

Proposition 1.116 (No promotion by certification alone). *Neither Interface pass, Core pass, higher Extⁿ-vanishing, terminal Type IV zero, transient-defect absence, finite-to-infinite reduction, nor manifest closure alone promotes an arithmetic conclusion to theorem status.*

Proof. Each establishes at most one proper segment of the theorem-bearing chain. The arithmetic conclusion is licensed only by the full composition ending in Return. □

Proposition 1.117 (Missing arrow blocks external conclusion). *If any required realization, finite-certificate, higher-edge, tower, reduction, audit, or Return arrow is absent or undefined, the arithmetic external conclusion is unlicensed and remains undefined.*

Proposition 1.118 (Transport hypotheses are non-scalar). *Missing hypotheses of the theorem-bearing chain cannot be repaired by metric slack, favorable advisory data, arithmetic elegance, or successful pre-gates.*

MS-A.12ter. Arithmetic known-theorem-recovery portfolios

Definition 1.119 (Arithmetic known-theorem-recovery track). An *arithmetic known-theorem-recovery track* is a proof-bearing record for one independently proved external theorem on one immutable benchmark class. It contains:

- (i) the exact external theorem statement, hypotheses, benchmark class, and immutable source locator;
- (ii) the admissible precursor mathematics available before the benchmark conclusion is used;
- (iii) a non-circular purpose-faithful realization defined only from those admissible inputs;
- (iv) the exact finite or directly proved Core predicate;
- (v) every detector, actual-morphism, metric, higher-edge, or transient-defect record consumed;
- (vi) the finite-to-infinite reduction when finite or bounded-window predicates are used for an infinite target, and an apex-agreement record only when the chosen route invokes tower, limit, colimit, or apex semantics;
- (vii) the exact audited AK-side conclusion;
- (viii) a separately proved Return Theorem whose consequent is the advertised external conclusion;
- (ix) an independent validation comparison with the external theorem;
- (x) the exact recovered strength and all stronger readings that remain prohibited.

The independent benchmark theorem is a validation target, not an admissible proof oracle for a missing AK arrow.

Definition 1.120 (Benchmark-class immutability). A recovery benchmark class fixes the source objects, coefficient and local hypotheses, variance, topology, parameter range, exceptional cases, and conclusion strength before the AK run begins. Expanding the class, weakening hypotheses, or replacing the conclusion after inspecting the outcome creates a new recovery track and requires new artifacts.

Definition 1.121 (Source-proof quarantine). The independent proof of the benchmark conclusion is *quarantined* from the theorem-bearing AK chain. It may identify the benchmark, verify the final recovered conclusion, and compare theorem strength. It may not discharge a missing realization, detector-completeness, actual-morphism, reduction, apex-agreement, audit, or Return obligation. A precursor theorem may be consumed only when it is separately bound, logically weaker than the benchmark conclusion on the declared class, and listed in the admissible-input graph.

Definition 1.122 (Non-circular recovery realization). A recovery realization is *non-circular* when its construction does not consume the benchmark conclusion, the target constant or exceptional bound whose existence is to be recovered, or an equivalent oracle. Candidate invariants may be obtained from an independently specified presentation, algebraic decomposition, exact source map, or algorithm that is available before validation against the benchmark.

Proposition 1.123 (Encoding the answer is not recovery). *A persistence object or certificate obtained by inserting the benchmark conclusion, its target constants, its eventual-stability index, or an equivalent truth table into the realization does not constitute known-theorem recovery.*

Proof. The missing cross-layer mathematics has been assumed as input rather than reconstructed. Such an encoding may be a replay or visualization artifact, but it does not validate the advertised Interface–Core–Reduction–Return chain. $\square$

Definition 1.124 (Recovery-track status vector). Each track stores separately the statuses of

source binding, Interface, finite certificate/direct proof,
Core audit, reduction, Return, validation.

The exact recovered-strength relation is one of

exact, strictly_weaker, incomparable, mismatch, not_yet_recovered.

A source theorem may be independently proved while the AK recovery remains pending or undefined. No headline status overwrites the component vector.

Definition 1.125 (Exact recovery witness). An *exact recovery witness* is a recovery track for which every required component is proved, the recovered-strength relation is exact, the benchmark proof was not used to fill an AK arrow, and the complete dependency/artifact graph is closed. A strictly weaker or incomparable conclusion is a calibration result, not recovery of the named benchmark theorem.

Definition 1.126 (Arithmetic recovery portfolio). An *arithmetic recovery portfolio* is a finite family of recovery tracks together with:

- (i) a shared-infrastructure record for genuinely identical coefficient, realization, detector, ledger, or replay artifacts;
- (ii) a track-specific assumption and artifact graph;
- (iii) a cross-track contamination audit;
- (iv) non-compensation rules;
- (v) the release-gate interpretation of each exact witness;
- (vi) the portfolio closure status.

Shared infrastructure may avoid duplication but does not transfer a theorem conclusion between tracks.

Definition 1.127 (Portfolio closure status). A portfolio has one closure status:

empty, schema_ready, tracks_declared, partially_recovered,
dual_recovery_certified, blocked, inconsistent.

The status

partially_recovered

requires at least one exact recovery witness. The status

dual_recovery_certified

is available only when both advertised tracks independently possess exact recovery witnesses and the cross-track audit passes. A weaker calibration cannot satisfy either exact-recovery condition unless the weaker theorem is frozen as a separate benchmark track.

Definition 1.128 (Mathematical witness versus release registration). Recovery mathematics and release governance are separate fields:

mathematically_exact, ready_for_registration, registered_pass.

The first means that the exact recovery witness is proved. The second adds local manifest and artifact closure. The third additionally requires the canonical Claim Register and release-registry action demanded by the current release contract. A Problem-Interface appendix may establish the first two but must not fabricate the third.

Proposition 1.129 (Recovery tracks do not compensate). *A completed recovery track cannot discharge a missing arrow, failed hypothesis, or stronger recovered-strength claim in another track.*

Proof. The tracks have different source theorems, benchmark classes, predicates, and Return consequents. Their obligations form a typed conjunction, not a numerical balance. $\square$

Proposition 1.130 (One recovery may satisfy an existential release gate). *When a release clause requires at least one known-theorem recovery, one independently exact and registered track may satisfy that existential clause. It does not establish a portfolio claim advertising two completed recoveries.*

Proposition 1.131 (Dual recovery is calibration, not theorem promotion). *Even two exact completed recoveries validate only their two immutable benchmark chains. They do not prove a main conjecture, extend either benchmark class, identify unrelated arithmetic invariants with Collapse diagnostics, or enlarge the Core.*

Theorem 1.132 (Exact track composition). *If a recovery track has a proved non-circular Interface, proved finite/direct Core predicate, every invoked reduction and apex-agreement theorem, a proved Return Theorem, and closed artifacts, then its advertised external conclusion follows at the strength proved by the Return Theorem. It is an exact recovery witness precisely when independent validation shows that this strength equals the immutable benchmark conclusion.*

Proof. The theorem-bearing arrows compose directionally. Source-proof quarantine ensures that the benchmark conclusion was not used in that composition. The independent comparison establishes exactness of the recovered strength but contributes no missing arrow. $\square$

Theorem 1.133 (Conditional portfolio composition). *A portfolio is dual_recovery_certified precisely when both advertised tracks satisfy Theorem 1.132 at exact strength and the shared-infrastructure and contamination audits pass.*

Proof. This is the definition of dual closure together with trackwise non-compensation. $\square$

MS-A.13. Arithmetic interface package

Theorem 1.134 (Appendix MS-A v20 arithmetic interface package). *Within the Problem Interface layer, the following package is available at exactly the displayed strength:*

- (i) *arithmetic source semantics, variance, coefficient passage, and parameter extraction are explicit non-interchangeable records;*
- (ii) *purpose-faithfulness is feature-relative, direction-relative, and theorem-bearing only when proved;*
- (iii) *Core readability requires constructible window-first profiles and every invoked verifier-side record;*
- (iv) *finite arithmetic observation becomes finite proof only through an admitted detector certificate with separate soundness and completeness evidence;*
- (v) *higher arithmetic Extⁿ use requires the complete standard n-eligible package, and the Core direction is one-way unless natural zero-reflection is separately proved;*
- (vi) *source exactness and source closeness do not become Core exact or metric-gap-certified comparison without the canonical comparison evidence;*
- (vii) *only actual supported metric profile discrepancies enter the strict ledger; semantic and logical obligations remain non-scalar;*
- (viii) *an arithmetic tower becomes Type IV-readable only through actual transition and comparison morphisms, apex semantics, terminal tameness, and defect artifacts;*
- (ix) *terminal Type IV, long-transient absence, full defect zero, and comparison isomorphism are distinct conclusion levels;*
- (x) *finite evidence reaches an infinite arithmetic target only through a registered finite-to-infinite theorem and apex agreement;*
- (xi) *Interface pass, Core pass, reduction pass, arithmetic Return, and independent recovery validation remain distinct;*
- (xii) *known-theorem recovery requires benchmark immutability, source-proof quarantine, non-circular realization, exact recovered-strength accounting, and the complete projection-and-return chain;*
- (xiii) *strictly weaker or incomparable calibration does not count as recovery of the named theorem;*
- (xiv) *mathematical exactness, readiness for registration, and canonical release-registry pass remain distinct;*
- (xv) *multiple recovery tracks form a non-compensating portfolio, and dual recovery calibrates only the two registered benchmark chains;*
- (xvi) *no arithmetic conclusion is licensed without the complete projection-and-return chain.*

Proof. Items (i)–(iii) follow from Sections MS-A.2–MS-A.4bis. Item (iv) follows from Section MS-A.5bis. Item (v) follows from Section MS-A.6 and the v20 Appendix C higher-edge package. Item (vi) follows from Section MS-A.7bis and Appendix A. Item (vii) follows from Section MS-A.8 and the Appendix F/G non-compensation discipline. Items (viii)–(ix) follow from Sections MS-A.9 and MS-A.9ter and Appendix D. Item (x) follows from Section MS-A.8bis and Chapter 8. Items (xi) and the final non-shortcut conclusion follow from Sections MS-A.12–MS-A.12bis. The recovery exactness, registration separation, and portfolio conclusions follow from Section MS-A.12ter. □

Remark 1.135 (Package role). Theorem 1.134 is a meta-theoretic interface package. It proves no arithmetic object-level statement and supplies no missing realization or Return theorem.

MS-A.14. Explicit exclusions

Appendix MS-A asserts none of the following.

- No Birch–Swinnerton-Dyer, ABC, Riemann-hypothesis, Iwasawa-main-conjecture, modularity, Selmer-finiteness, rank, regulator, characteristic-ideal, or p -adic L -function theorem is proved.
- No arithmetic object, group, complex, tower, L -function, Newton polygon, or invariant is automatically Core-readable.
- No objectwise realization is automatically functorial, natural, faithful, conservative, exact, or full.
- No coefficient reduction to k is automatically exact, lossless, Tor-free, limit-compatible, or obstruction-reflecting.
- No multiparameter arithmetic object is automatically one-parameter constructible persistence.
- No parameter extraction is assumed lossless or non-adaptive.
- No raw arithmetic invariant replaces $B_{n,W}(F)$ or $\text{Eval}_{n,W,\tau}(F)$.
- No finite sample, finite layer, finite precision, numerical plateau, or computed rank is a finite detector certificate by itself.
- No detector soundness theorem is read as zero completeness.
- No zero-looking detector output is read as repaired-profile zero without complete coverage.
- No raw detector rank is assumed bottleneck-stable.
- No metric transport through hard deletion is used without the two-sided threshold-gap chain.
- No purpose declaration establishes purpose-faithfulness.
- No arithmetic pre-gate replaces any Core, detector, higher-edge, Type IV, reduction, manifest, or Return obligation.
- No arithmetic Ext group is automatically the Core Ext^n group.
- No synthetic or shifted detector encoding establishes a natural arithmetic PH-to-Ext bridge.
- No converse $\text{Ext}^n = 0 \Rightarrow \text{Eval}_n = 0$ is asserted without natural zero-reflection.
- No arithmetic equality or congruence is automatically Core exactness.
- No valuation or approximation bound is automatically a bottleneck bound.
- No object-distance bound automatically controls kernel or cokernel defect profiles.
- No semantic, naturality, completeness, morphism, reduction, Return, claim, or artifact obligation enters a scalar ledger by name.
- No metric slack repairs a non-scalar obligation.
- No arithmetic tower supplies Type IV without actual transition and source-to-apex comparison morphisms.
- No barcode matching, object isomorphism, equal Betti numbers, or finite-stage agreement supplies the Type IV morphism.
- No terminal Type IV zero implies transient-defect absence, full defect zero, or comparison isomorphism.
- No arithmetic μ -invariant is identified with μ_{Collapse} by notation.
- No Latin u is used for the cokernel-side Collapse diagnostic.

- No finite prefix, Cauchy estimate, observed stabilization, colimit existence, or formal apex proves apex agreement.
- No finite-to-infinite reduction is treated as an arithmetic Return Theorem.
- No Interface pass or Core pass is an arithmetic theorem.
- No Bridge candidate, Bridge Program, Spec, Search artifact, Companion template, AI output, or Claim Register entry is theorem evidence by itself.
- No complete manifest is automatically a true or passing proof.
- No external theorem conclusion, target invariant, eventual-stability bound, or equivalent truth table is inserted into a realization and then advertised as recovery.
- No independent source proof fills a missing AK Interface, detector, actual-morphism, reduction, apex-agreement, audit, or Return arrow.
- No completed recovery track compensates for another track that is pending, undefined, mismatched, or rejected.
- No dual recovery implies an Iwasawa main conjecture, a stronger control theorem, or an extension beyond the two immutable benchmark classes.
- No current CR-v20 baseline registration is fabricated for Part IV.

MS-A.15. Provenance, conformance, and status

This v20 reconstruction depends on:

- (i) Chapters 1–2 and Appendix A for claim discipline, window-first evaluation, threshold-gap repair, and repaired-zero transport;
- (ii) Chapter 3 and Appendix B for finite detector certificates, proof compression, and admissible representatives;
- (iii) Chapter 4 and Appendix C for standard n -eligible higher Ext edges and the zero-reflection boundary;
- (iv) Chapter 5 and Appendix D for actual-morphism Type IV, terminal tameness, and transient-defect separation;
- (v) Chapter 6 and Appendix E for atlas, overlap, and Restart obligations when invoked;
- (vi) Chapter 7 and Appendices F–G for atomic status, non-compensation, claim-use preflight, canonical manifests, and replay;
- (vii) Chapter 8 for Interface entry, finite-to-infinite reduction, Known-Theorem Recovery, and Return architecture;
- (viii) CR-v20 baseline only for currently registered Main/Foundation claim governance;
- (ix) CM-v20 and TB-v20 only at their reviewed immutable local strengths;
- (x) frozen Technical Extension Appendices H–N, especially Appendix M ledger semantics and Appendix N higher-enhancement boundaries, without silent Core import.

Its conformance status is

v20_mathematical_conformant;
 arithmetic recovery portfolio exactness schema closed;
 Track G exact witness supplied in Appendix MS-B;
 canonical G16/Part IV registration pending.

Remark 1.136 (Canonical Type IV notation). The canonical diagnostics are μ_{Collapse} and ν_{Collapse} . No Latin-letter cokernel diagnostic is introduced or revived.

Local role classification. Definitions in this appendix are `interface_definition` or `operational_policy`. Propositions and Theorem 1.134 are Problem-Interface statements proved only from displayed typing rules and inherited v20 theorems. Any problem-specific arithmetic implication not proved here remains `bridge_candidate`, `bridge_program`, `spec`, or `non_claim`. No local label is a canonical v20 Claim UID before final Part IV freeze, cross-audit, and Claim Register extraction.

Repair ancestry. This text conservatively replaces the v19 MS-A interface architecture. The v19 artifact remains immutable repair ancestry. No v19 verdict or source statement is mutated in place.

2 Appendix MS-B: Cyclotomic / Iwasawa / Congruence Interfaces

MS-B.1. Scope, dependency position, and interface status

This appendix specializes Appendix MS-A to cyclotomic towers, Iwasawa modules, congruence systems, control statements, and p -adic L -profiles. It belongs to the Problem Interface layer and does not enlarge the theorem-bearing Core of Chapters 1–8.

The governing rule is

Arithmetic tower structure supplies source semantics; it does not supply a Core theorem.

All Core-visible persistence data are read through the window-first repaired profile

$$B_{i,W}(F) := W_W \mathbf{P}_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

No argument in this appendix interchanges window restriction and hard threshold deletion without a separately proved commutation statement. Whenever metric transport through hard deletion is consumed, the actual pre-threshold profile pair must satisfy the repaired two-sided threshold-gap hypotheses. The sharp mathematical zero-transport boundary is $2\varepsilon \leq \tau$; a strict reserve $2\varepsilon < \tau$ may be imposed operationally.

Whenever Type IV is invoked, the required datum is an actual morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty)$$

in the declared abelian profile category. Objectwise similarity, common barcodes, equal ranks, a control heuristic, or an arithmetic limit symbol does not create this morphism.

Remark 2.1 (Interface status). Every cyclotomic, Iwasawa, congruence, control, main-conjecture, and p -adic analytic implication is [Spec] by default. The specific recovery theorems in Sections MS-B.15ter–MS-B.15quater are exceptions only at their displayed benchmark strength and only after their proof-bearing chains close. No exact recovery status propagates to a stronger arithmetic statement.

Remark 2.2 (Relation to Appendix MS-A). Appendix MS-A governs arithmetic sovereignty, purpose-faithfulness, coefficient passage, one-parameter extraction, finite detectors, higher Ext^n -eligibility, actual-morphism Type IV, transient defects, finite-to-infinite reduction, non-scalar obligations, and the complete Projection–Return chain. All such requirements are inherited here without weakening.

Remark 2.3 (No arithmetic tower shortcut). A number-field tower, a tower of Selmer groups, an inverse system of class groups, or a module over an Iwasawa algebra is not automatically a directed constructible persistence tower. Variance, coefficients, topology, transition morphisms, one-parameter extraction, constructibility, and apex semantics must all be resolved first.

MS-B.1bis. v20 dependency baseline and claim typing

Definition 2.4 (MS-B v20 dependency baseline). Appendix MS-B is interpreted relative to:

- (i) AK-HPDST v20.0.0 Main Chapters 1–8;
- (ii) Foundation Appendices A–G;

- (iii) the frozen Technical Extension Appendices H–N, at their individually proved strengths;
- (iv) the reviewed CM-v20 and TB-v20 statements at their exact theorem, conditional, operational, toy, or non-claim status;
- (v) Appendix MS-A v20.0.0;
- (vi) the CR-v20 baseline as governance, without inferring unrecorded Problem-Interface promotion.

No v19 wording, classical theorem name, literature citation, implementation result, or AI-produced text strengthens a v20 claim beyond this dependency closure.

Definition 2.5 (MS-B source-side status). Cyclotomic towers, Iwasawa modules, characteristic ideals, pseudo-isomorphisms, congruences, control theorems, finite-layer computations, and p -adic L -functions are source-side data until a proved typed Interface produces verifier-side records. Their local evidence class is one of

proved, conditional, operational, spec, deprecated, rejected, unknown,

and their invocation role is separately one of

theorem_evidence, operational_rule, specification_target, provenance_only.

Definition 2.6 (MS-B local claim roles). A cross-layer statement introduced here has one local semantic role:

interface_definition, interface_theorem, bridge_candidate,
bridge_program, bridge_theorem, toy_bridge_theorem,
spec, non_claim.

A local label is not a canonical Claim UID. Formal CR promotion is deferred until the full Problem Interface sequence and its artifact closure are frozen.

Proposition 2.7 (Named arithmetic structure does not strengthen Core status). *Naming a datum cyclotomic, ordinary, supersingular, Iwasawa-theoretic, modular, congruential, controlled, main-conjectural, or p -adic analytic does not strengthen any Core verdict.*

Proof. These adjectives describe source semantics. A Core verdict consumes typed profiles, representatives, eligible higher-edge records when invoked, actual morphisms, detector certificates, ledgers, manifests, and status-valid artifacts. Source terminology creates none of those records. $\square$

Remark 2.8 (MS-B hardening scope). The purpose of this appendix is both architectural and calibrational: it specifies the required arrows and supplies one exact Iwasawa growth recovery witness together with an exact abstract bounded-control theorem. These are recoveries or interface theorems at fixed benchmark strength, not new arithmetic discoveries and not a proof of the Iwasawa main conjecture or of Mazur’s full arithmetic specialization.

MS-B.2. Variance-aware cyclotomic and Iwasawa diagrams

Definition 2.9 (Cyclotomic tower). A *cyclotomic tower* is a declared arithmetic field system

$$K_0 \subset K_1 \subset K_2 \subset \cdots \subset K_\infty$$

together with its chosen transition data and index convention. The field inclusions alone do not determine the variance of every arithmetic object attached to the layers.

Definition 2.10 (Cyclotomic arithmetic datum). A *cyclotomic arithmetic datum* is a family $\mathcal{A}_{\text{cyc}} = \{A_n\}_{n \geq 0}$ attached to the layers, together with every restriction, corestriction, norm, trace, quotient, inflation, localization, completion, or duality map that the proposed Interface will use. Examples include class groups, unit groups, Selmer groups or complexes, Galois cohomology, local conditions, Hecke modules, congruence modules, and finite-layer analytic data.

Definition 2.11 (Arithmetic variance record). An *arithmetic variance record* declares, for each source construction:

- (i) whether the relevant system is direct, inverse, contravariant, covariant, or mixed;
- (ii) the actual source transition maps and their directions;
- (iii) any duality or opposite-category passage;
- (iv) compatibility with composition and identities;
- (v) whether the maps preserve the arithmetic feature named in the purpose profile;
- (vi) the exact theorem or artifact supporting each conversion of variance.

Definition 2.12 (Directed Core profile conversion). A *directed Core profile conversion* is a proved construction that turns the chosen arithmetic diagram into an actual directed system

$$B_{i,w}(F_0) \longrightarrow B_{i,w}(F_1) \longrightarrow B_{i,w}(F_2) \longrightarrow \cdots$$

or its repaired post-threshold analogue, with all maps defined in the declared profile category. Reversing arrows typographically is not such a conversion.

Proposition 2.13 (Field inclusions do not fix module variance). *The inclusions $K_n \subset K_{n+1}$ do not, by themselves, determine a directed Core tower for class, Selmer, cohomological, unit, or dual module data.*

Proof. Different arithmetic constructions use restriction, corestriction, norm, Pontryagin duality, invariants, coinvariants, or derived functors, and these have different variances. The actual maps consumed by a Core colimit must therefore be specified and realized. $\square$

Proposition 2.14 (Inverse systems require a conversion theorem). *An inverse arithmetic system may support the canonical Type IV colimit artifact only after a proved variance, duality, reindexing, or opposite-category construction supplies the actual directed Core profile system.*

Proof. The canonical Type IV source is a colimit of a directed profile system. An inverse limit and a colimit have different universal properties. Identifying or converting them requires additional mathematics and cannot be inferred from common indexing. $\square$

MS-B.2bis. Coefficients, topology, completion, and duality

Definition 2.15 (Iwasawa coefficient record). An *Iwasawa coefficient record* declares the prime p , coefficient ring, Iwasawa algebra Λ , left/right module conventions, topological module category, compact/discrete convention, and every passage to the Core field k . It records whether the passage uses reduction, tensor product, derived tensor, localization, quotient, Pontryagin duality, continuous duality, completion, or a composite route.

Definition 2.16 (Topological realization record). A *topological realization record* specifies which maps are continuous, which limits are taken in a topological category, whether completion is separated and complete, and whether the resulting algebraic object agrees with the object used to construct the Core profile.

Definition 2.17 (Pontryagin-dual variance record). A *Pontryagin-dual variance record* states the source and target categories, the exact contravariance, the topology retained or forgotten, and the theorem transporting transition maps and exactness properties through Pontryagin duality.

Proposition 2.18 (Coefficient reduction is not information preservation). *Passing from a Λ -module, $\mathbb{Z}_p$ -module, Hecke module, or integral complex to a field-valued persistence object is not lossless unless a theorem proves preservation of the declared feature.*

Proof. Reduction and tensor operations may introduce or destroy torsion, while nonflat base change produces Tor terms. Duality and completion may also alter exactness and variance. Purpose-faithfulness must therefore be proved after the actual coefficient passage, not before it. $\square$

Proposition 2.19 (Completion identity is non-scalar). *A missing equality between an algebraic limit, a topological completion, and the declared apex cannot be repaired by metric slack or finite numerical agreement.*

Proof. The issue is object identity and universal property, not a bottleneck discrepancy between already defined profiles. It is a non-scalar obligation. $\square$

MS-B.3. Purpose-faithful realization and tower-axis selection

Definition 2.20 (Cyclotomic realization hypothesis). *A cyclotomic realization hypothesis is a purpose-faithful object-and-morphism assignment*

$$\mathfrak{R}_{\text{cyc}} : \{A_n\}_{n \geq 0} \rightsquigarrow \{F_n\}_{n \geq 0}$$

that declares its variance, coefficient passage, source semantics, filtration orientation, and target category. If functoriality is asserted, its action on all consumed morphisms and its coherence with composition must be proved.

Definition 2.21 (Tower-axis selection). *A tower-axis selection declares the single ordered persistence parameter extracted from the arithmetic data. It records the source parameters, selected direction, normalization, order convention, branch choice, treatment of ties and incomparable indices, skipped layers, discarded information, and any theorem-backed profile discrepancy introduced by the extraction.*

Definition 2.22 (Cyclotomic Core-readable tower). *A cyclotomic datum is Core-readable on a declared scope if its realization supplies:*

- (i) constructible filtered or persistence objects F_n ;
- (ii) actual realized transition morphisms;
- (iii) window-first profiles $B_{i,W}(F_n)$ and repaired evaluations $\text{Eval}_{i,W,\tau}(F_n)$;
- (iv) declared endpoint and cofinal conventions;
- (v) an admissible apex provenance record when an apex is invoked;
- (vi) an actual tower comparison morphism when Type IV is invoked.

Definition 2.23 (Realized tower morphism record). *A realized tower morphism record binds each actual morphism $F_n \rightarrow F_{n+1}$, and each induced profile morphism, to its arithmetic source map, realization theorem, coefficient convention, normalization, software or formal artifact when applicable, and immutable identity.*

Definition 2.24 (Cyclotomic tower defect). *A cyclotomic tower defect, denoted δ^{cyc} , is a local Interface name for an effect introduced by realization, extraction, comparison, truncation, or approximation. Before audit use it must be classified as a supported actual profile discrepancy, a morphism-profile discrepancy, a theorem-controlled nonmetric effect, a non-scalar blocker, advisory data, or unavailable data.*

Proposition 2.25 (Tower index is not automatically a persistence parameter). *The arithmetic layer index is not automatically an admissible real persistence parameter.*

Proof. Constructible one-parameter persistence requires an ordered parameter, actual transition maps, finite critical structure on the monitored scope, and declared endpoint semantics. Arithmetic indexing alone does not establish these properties. $\square$

Proposition 2.26 (Cyclotomic data require realization before diagnostics). *No Type IV value may be assigned to a raw cyclotomic datum before the actual directed Core tower, apex profile, and comparison morphism have been constructed.*

Proof. The diagnostic is defined from the kernel and cokernel of the actual morphism $\phi_{i,W,\tau}$. Raw source data do not supply that morphism. $\square$

Remark 2.27 (Cyclotomic tower does not imply Type IV tower). *The existence of infinitely many arithmetic layers is neither terminal tameness nor apex agreement. These are independent verifier obligations.*

MS-B.3bis. Apex provenance and semantic authority

Definition 2.28 (Apex identity record). An *apex identity record* classifies F_∞ as one of:

realized_source_apex, proved_categorical_apex, proved_completion_apex,
proved_finite_reduction_proxy, formal_placeholder, advisory, undefined.

It records the source object, universal property or completion theorem, coefficient and topology conventions, compatibility maps, repaired profile construction, and immutable artifacts.

Definition 2.29 (Apex vocabulary alignment). The terms limit, inverse limit, direct limit, completion, terminal object, infinite layer, finite proxy, and apex remain distinct. A vocabulary alignment record states which mathematical construction is actually meant and which theorem identifies it with the target of $\phi_{i,W,\tau}$.

Definition 2.30 (Tower semantic-authority record). A *tower semantic-authority record* states whether the realized transition maps and apex maps are proved to represent the source arithmetic relationships named in the purpose profile. Its local outcome is

pass, reject, undefined.

There is no top-level not_invoked state for an Interface whose conclusion actually consumes tower semantics.

Proposition 2.31 (Apex identity is non-scalar). A *missing, ambiguous, or semantically unauthorized apex identity cannot be repaired by ledger reserve, observed stabilization, or favorable Iwasawa invariants*.

Proof. The apex is the codomain object of the comparison morphism. Without it, the morphism and its kernel and cokernel are undefined. $\square$

Proposition 2.32 (Formal apex is scope-bound). A *formal implementation apex may be consumed only within the exact formal model and purpose for which its semantic-authority theorem has been proved*.

Proof. Changing the source meaning, completion, coefficient object, transition maps, or profile extraction changes the represented mathematical object. The old artifact cannot silently authorize the new apex. $\square$

MS-B.4. Iwasawa-module realization and structure-theorem boundaries

Definition 2.33 (Iwasawa module datum). An *Iwasawa module datum* consists of a module M_{Iw} over a declared Iwasawa algebra Λ , its topological and side conventions, and every finiteness, torsion, cotorsion, noetherian, commutative/noncommutative, or local-condition hypothesis consumed by the proposed Interface.

Definition 2.34 (Iwasawa interface). An *Iwasawa interface* is a purpose-faithful realization

$$\mathfrak{R}_{Iw} : M_{Iw} \rightsquigarrow F_{Iw}$$

or a realized tower $F_{\bullet}^{Iw}$, together with the coefficient, variance, filtration, constructibility, and semantic-authority records required by Appendix MS-A.

Definition 2.35 (Iwasawa-layer filtration). An *Iwasawa-layer filtration* is a declared one-parameter filtration. Its parameter encodes one specified arithmetic scale, for example layer, conductor, local-condition depth, augmentation-ideal depth, valuation depth, or congruence depth. Different scales are not interchangeable without a comparison theorem.

Definition 2.36 (Iwasawa Core profile). An *Iwasawa Core profile* is

$$\text{Eval}_{i,W,\tau}(F_{Iw}) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F_{Iw}))$$

for a declared realization, degree, right-open window, threshold, endpoint convention, and coefficient field.

Definition 2.37 (Iwasawa tower realization). An *Iwasawa tower realization* supplies an actual directed Core system

$$F_0^{\text{Iw}} \longrightarrow F_1^{\text{Iw}} \longrightarrow \dots$$

and an authorized apex. If the source arithmetic modules form an inverse system, the record includes the proved conversion producing this directed Core system.

Definition 2.38 (Iwasawa realization burden). The realization burden records whether F_{Iw} is built from the module, a Selmer complex, finite-layer invariants, a characteristic-ideal profile, an elementary-module proxy, local conditions, analytic data, or a mixed construction. It separately records purpose-faithfulness for every source feature used.

Definition 2.39 (Iwasawa realization defect). An *Iwasawa realization defect*, denoted δ^{Iw} , is a local effect name governed by the same type classification as δ^{cyc} . The symbol alone does not create a metric ledger entry.

Remark 2.40 (Iwasawa data are not automatically persistent). The Λ -module topology, algebraic filtration, tower filtration, and AK persistence parameter are four potentially different structures. An Interface must identify the one actually used.

Proposition 2.41 (Iwasawa module structure is not persistence structure). A Λ -module is not, merely by being noetherian, torsion, finitely generated, or compact, a constructible one-parameter persistence module over k .

Proof. The categories, coefficients, topology, indexing, and morphisms differ. A purpose-faithful extraction and constructibility theorem are required. $\square$

Definition 2.42 (Structure-theorem invocation record). A *structure-theorem invocation record* identifies the exact Iwasawa algebra, commutativity and dimension assumptions, module finiteness/torsion hypotheses, pseudo-isomorphism convention, exceptional submodules, and theorem version used to obtain elementary factors or characteristic-ideal data.

Proposition 2.43 (Structure theorem is hypothesis-bound). No elementary decomposition, characteristic ideal, μ -term, or λ -term may be consumed outside the hypotheses of the exact structure theorem invoked.

Proof. Classical one-variable commutative torsion-module structure does not automatically extend to noncommutative, multivariable, nontorsion, nonnoetherian, or topologically altered settings. $\square$

Proposition 2.44 (Characteristic ideal is not a Core defect profile). A characteristic ideal, Fitting ideal, elementary divisor list, or valuation profile is not the kernel or cokernel persistence profile of a Core tower comparison morphism by default.

Proof. The arithmetic invariant is defined from module-theoretic data, often only up to units or pseudo-isomorphism. The Core defect profile is defined from the actual kernel or cokernel of $\phi_{i,W,\tau}$. Identification requires a separate transport theorem. $\square$

MS-B.4bis. Arithmetic invariant namespaces and growth formulas

Definition 2.45 (Arithmetic Iwasawa μ -invariant). The arithmetic Iwasawa invariant is written μ_{Iw} , with its source convention and structure theorem recorded.

Definition 2.46 (Arithmetic Iwasawa λ -invariant). The arithmetic Iwasawa invariant is written λ_{Iw} , with its source convention and structure theorem recorded.

Definition 2.47 (Arithmetic Iwasawa constant term). When a classical finite-layer growth formula uses a constant commonly denoted ν , this appendix writes it $\nu_{\text{Iw}}^{\text{arith}}$. It is not the Collapse cokernel diagnostic.

Definition 2.48 (Arithmetic finite-layer growth profile). A *finite-layer growth profile* is a function

$$g(n) = \text{a declared order, length, rank, valuation, corank, or defect at layer } n,$$

with its exact arithmetic normalization, exceptional range, error term, and theorem status recorded.

Definition 2.49 (Iwasawa invariant profile declaration). An invariant profile declaration classifies μ_{Iw} , λ_{Iw} , ν_{Iw}^{arith} , characteristic ideals, Fitting ideals, ranks, valuations, and finite-layer lengths as advisory features, realization inputs, transport targets, or theorem evidence under an exact source theorem. It does not classify them as Core diagnostics.

Definition 2.50 (Collapse μ -diagnostic). Fix one window–threshold pair (W, τ) , one finite monitored degree set I_{mon} , and actual Type IV morphisms in every monitored degree. Then

$$\mu_{\text{Collapse}}(W, \tau) := \sum_{i \in I_{\text{mon}}} \text{tgdim}_W(\ker \phi_{i, W, \tau}),$$

whenever every summand is defined and terminal-tame.

Definition 2.51 (Collapse ν -diagnostic). For the same fixed (W, τ) and finite I_{mon} ,

$$\nu_{\text{Collapse}}(W, \tau) := \sum_{i \in I_{\text{mon}}} \text{tgdim}_W(\text{coker } \phi_{i, W, \tau}),$$

whenever every summand is defined and terminal-tame. Aggregation across several windows, thresholds, or infinitely many degrees requires a separate typed aggregation or summability theorem.

Theorem 2.52 (Iwasawa μ is not Collapse μ by default). *The arithmetic invariant μ_{Iw} is not identified with μ_{Collapse} without a separate proved transport theorem on the declared realization and scope.*

Proof. The former is defined from arithmetic module structure; the latter from terminal-generic kernel profiles of actual Core morphisms. Their categories, definitions, and failure modes differ. $\square$

Remark 2.53 (Notation collision warning). Undecorated μ and ν are inadmissible in any verifier-side record containing both arithmetic and Collapse quantities. In particular $\nu_{Iw}^{\text{arith}} \neq \nu_{\text{Collapse}}$ by notation and by type unless a theorem proves a scoped relation.

Remark 2.54 (Arithmetic λ and Collapse ν). Neither λ_{Iw} nor ν_{Iw}^{arith} is automatically equal to, controlled by, or equivalent to ν_{Collapse} .

Proposition 2.55 (Finite-layer growth is not Type IV diagnosis). *A finite-layer growth formula does not determine $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}})$ unless a proved theorem identifies its arithmetic terms with the actual kernel and cokernel profile predicates consumed.*

Proof. A growth formula is source-side. Type IV is morphism-side and window-relative. A purpose-faithful identification is an additional Bridge obligation. $\square$

MS-B.4ter. Exact p -primary length-persistence realization

Definition 2.56 (p -power filtration persistence). Let G be a finite abelian p -primary group. Its p -power filtration persistence is the discrete $\mathbb{F}_p$ -persistence module

$$\mathcal{L}_p(G)_j := p^{j-1}G/p^jG \quad (j \geq 1),$$

with transition map induced by multiplication by p , and with value zero outside the finite nonzero range. It is extended to a right-continuous constructible real-parameter module by keeping the value constant on every interval $[j, j+1)$. For a homomorphism $f : G \rightarrow H$, the induced maps on the quotients define $\mathcal{L}_p(f)$.

Definition 2.57 (Total barcode length). For a finite barcode $\mathcal{B}$ with finite bars, define

$$\text{TotLen}(\mathcal{B}) := \sum_{I \in \mathcal{B}} \ell(I).$$

This is a source-facing exact functional in the present realization and is not identified with a Core metric budget.

Theorem 2.58 (Natural lossless realization of finite p -primary size). *The assignment $G \mapsto \mathcal{L}_p(G)$ is functorial from finite abelian p -primary groups to finite constructible persistence modules over $\mathbb{F}_p$. If*

$$G \cong \bigoplus_{r=1}^s \mathbb{Z}/p^{a_r}\mathbb{Z}, \quad a_r \geq 1,$$

then

$$\text{Barcode}(\mathcal{L}_p(G)) \cong \bigsqcup_{r=1}^s \{[1, a_r + 1)\},$$

and consequently

$$\text{TotLen}(\text{Barcode}(\mathcal{L}_p(G))) = \sum_{r=1}^s a_r = \text{length}_{\mathbb{Z}_p}(G) = \log_p \#G.$$

Thus the order of a finite p -primary group is preserved exactly by the complete barcode, not merely by the first graded rank.

Proof. A homomorphism commutes with multiplication by p , so it induces a morphism on every graded quotient and these morphisms commute with the transition maps. For a cyclic summand $\mathbb{Z}/p^a\mathbb{Z}$, every quotient $p^{j-1}G/p^jG$ is one-dimensional for $1 \leq j \leq a$, the transition maps are isomorphisms until the last nonzero degree, and the resulting interval is $[1, a + 1)$. Direct sums give the displayed barcode. Summing bar lengths gives $\sum a_r$, which is both the $\mathbb{Z}_p$ -length and the base- p logarithm of the group order. $\square$

Proposition 2.59 (First graded rank is not size-faithful). *The single quantity $\dim_{\mathbb{F}_p}(G/pG)$ does not determine $\#G$. Hence a realization retaining only the first graded piece is not purpose-faithful for Iwasawa growth or bounded-control cardinalities.*

Proof. The groups $\mathbb{Z}/p\mathbb{Z}$ and $\mathbb{Z}/p^2\mathbb{Z}$ have the same first graded rank and different orders. $\square$

Definition 2.60 (Strict p -filtered exactness record). A short exact sequence of finite p -primary groups is *strict for the p -power filtrations* when the induced filtration on the subobject and quotient agrees with their intrinsic p -power filtrations. A morphism record is strict when both its kernel–image and image–cokernel short exact sequences are strict.

Proposition 2.61 (Strict morphisms preserve kernel and cokernel profiles). *For a strict morphism $f : G \rightarrow H$,*

$$\ker \mathcal{L}_p(f) \cong \mathcal{L}_p(\ker f), \quad \text{coker } \mathcal{L}_p(f) \cong \mathcal{L}_p(\text{coker } f).$$

Without the strictness record, neither identification is asserted.

Proof. Associated graded is exact on strict short exact sequences. Apply this first to kernel–image and then to image–cokernel. $\square$

Proposition 2.62 (Purpose-faithful but not globally fully faithful). *The complete barcode of $\mathcal{L}_p(G)$ determines the isomorphism class of the finite abelian p -primary group G , because its bar lengths recover the invariant factors a_r . However, the functor $\mathcal{L}_p$ is not asserted to be full, faithful on all homomorphism sets, or exact on arbitrary short exact sequences. Kernel and cokernel transport is consumed only under the strictness hypothesis of Proposition 2.61.*

Remark 2.63 (G17 realization witness). Theorem 2.58 is a non-synthetic natural realization theorem on a non-trivial arithmetic class. It supplies a mathematical witness for the natural-realization option of the v20 external-demonstration contract. Canonical release registration remains a separate governance action.

MS-B.5. Actual-morphism Type IV and diagnostic transport

Definition 2.64 (Iwasawa-to-Collapse μ transport statement). An Iwasawa-to-Collapse transport statement is a scoped implication involving μ_{Iw} , λ_{Iw} , ν_{Iw}^{arith} , finite-layer growth, or characteristic-ideal data on one side and an actual Core defect predicate on the other.

Definition 2.65 (Iwasawa-to-Collapse diagnostic transport theorem). Such a statement is a theorem only when it supplies:

- (i) the exact arithmetic source category and hypotheses;
- (ii) coefficient, topology, variance, and one-parameter extraction records;
- (iii) a purpose-faithful realization theorem;
- (iv) actual profile transition maps and apex maps;
- (v) the actual morphisms $\phi_{i,W,\tau}$;
- (vi) terminal-tameness evidence;
- (vii) a theorem identifying the named arithmetic predicate with the stated kernel/cokernel profile predicate;
- (viii) all finite-certificate, reduction, artifact, and Return obligations consumed by the conclusion.

Definition 2.66 (Default status of Iwasawa-to-Collapse transport). Every such transport statement is [Spec] until all obligations in the preceding definition are discharged.

Definition 2.67 (Arithmetic Type IV record). For each invoked (i, W, τ) , an arithmetic Type IV record contains the actual morphism, its source and target profile identities, provenance, semantic authority, kernel and cokernel profiles, terminal-tameness data, and exactly one local status:

not_invoked, undefined, certified_zero, obstructed.

No numerical μ or ν value is attached in the first two states.

Proposition 2.68 (Object agreement is not Type IV zero). *Equal source and target dimensions, isomorphic objects without a specified map, equal barcodes, equal characteristic ideals, or equal finite-layer growth does not prove Type IV zero.*

Proof. Kernel and cokernel are properties of an actual morphism. Object-level data do not determine them. $\square$

Proposition 2.69 (Terminal status is not inferred from finite prefix). *No finite prefix of arithmetic layers determines a terminal-generic Type IV status without a registered finite-to-infinite reduction theorem or independent terminal evidence.*

Proof. Terminal-generic predicates quantify over a cofinal tail. Finite observation does not establish that quantifier. $\square$

MS-B.5bis. Pseudo-isomorphism, pseudo-nullity, and transient defects

Definition 2.70 (Pseudo-isomorphism interface record). A *pseudo-isomorphism interface record* states the source module category, definition of pseudo-null, actual morphism when available, kernel and cokernel classes, codimension convention, and the theorem transporting this source relation to any Core-readable comparison.

Proposition 2.71 (Pseudo-isomorphism is not isomorphism). *An arithmetic pseudo-isomorphism or equality of characteristic ideals is not an isomorphism of realized Core profiles. Equality modulo pseudo-null modules likewise does not prove full Type IV defect vanishing.*

Proof. Pseudo-null kernel or cokernel may be nonzero. Characteristic ideals can forget precisely such subobjects. A Core isomorphism requires zero kernel and cokernel for the actual realized morphism. $\square$

Definition 2.72 (Arithmetic transient-defect record). An arithmetic transient-defect record inspects the full profiles

$$\ker \phi_{i,W,\tau}, \quad \text{coker } \phi_{i,W,\tau},$$

using detector source stage `kernel_defect` or `cokernel_defect`. It declares the defect threshold, component, endpoint convention, family, completeness source, artifact identity, and one of:

$$\text{not_invoked}, \quad \text{undefined}, \quad \text{certified_absent}, \quad \text{certified_present}.$$

Proposition 2.73 (Terminal Type IV zero is not transient zero). *The terminal equality*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

does not exclude finite interior, long-transient, or short defect bars.

Proof. Terminal-generic dimension records tail behavior. Defects supported away from the terminal region may have zero terminal-generic contribution while remaining nonzero as persistence objects. $\square$

Proposition 2.74 (Characteristic-ideal equality may miss transient defects). *Equality of characteristic ideals or elementary factors does not exclude transient Core defects unless a theorem proves completeness for the precise defect family.*

Proof. The arithmetic invariant may forget pseudo-null, finite, or otherwise unsupported information, whereas the Core defect profile retains bar-level support on the monitored window. $\square$

MS-B.6. Congruence interface and comparison modes

Definition 2.75 (Congruence datum). A *congruence datum* consists of arithmetic objects, the coefficient modulus or ideal, congruence depth, normalization, and actual maps or identifications supporting the congruence. It may involve modular forms, Galois representations, Hecke algebras, Selmer data, local conditions, tower layers, or analytic profiles.

Definition 2.76 (Congruence realization hypothesis). A *congruence realization hypothesis* is a purpose-faithful realization

$$\mathfrak{R}_{\text{cong}} : \mathcal{C}_{\text{cong}} \rightsquigarrow \text{FiltCh}(k) \quad \text{or} \quad \text{Pers}_k^{\text{cons}},$$

with coefficient passage, variance, parameter extraction, and semantic-authority records.

Definition 2.77 (Congruence Core profile). A congruence Core profile is a repaired evaluation $\text{Eval}_{i,W,\tau}(F_{\text{cong}})$ produced by the realization, not the source congruence relation itself.

Definition 2.78 (Congruence defect). A *congruence defect*, denoted δ^{cong} , is a local Interface effect name. Its value, units, support, and type must be declared before any audit use.

Definition 2.79 (Congruence comparison mode). A source-side congruence record may be classified as

$$\begin{aligned} &\text{source_congruence}, \quad \text{realized_exact}, \quad \text{metric_candidate}, \\ &\quad \text{advisory}, \quad \text{unavailable}, \quad \text{failed}. \end{aligned}$$

An invoked Core comparison resolves separately to

$$\text{exact}, \quad \text{metric_gap_certified}, \quad \text{not_compared}.$$

The state `not_compared` is permitted only when the relevant claim does not consume the comparison; missing comparison evidence for a comparison-dependent claim yields `undefined`, not `not_compared`.

Definition 2.80 (Congruence-to-metric conversion). A *congruence-to-metric conversion* is a theorem producing a certified bottleneck upper bound on the actual windowed pre-threshold profiles from the declared arithmetic congruence data. It identifies the profile pair, units, normalization, crop effects, window-edge effects, scalarization, and two-sided threshold-gap record.

Definition 2.81 (Congruence ledger entry). A congruence ledger entry binds a supported actual profile discrepancy to its source congruence, conversion theorem, protected profile pair, comparison path, canonical Appendix M slot, value domain, aggregation, and artifact identity. Unsupported congruence depth is not a ledger component.

Proposition 2.82 (Source congruence is not persistence equality). *A congruence modulo p^m , an ideal, or a Hecke congruence module does not imply profile equality, interleaving, or post-threshold equality without a realization and comparison theorem.*

Proof. The relations live in different categories and measure different objects. $\square$

Proposition 2.83 (Congruence defect is not Type IV defect). *The scalar or algebraic quantity δ^{cong} is not the kernel or cokernel profile of $\phi_{i,W,\tau}$ unless a theorem constructs and identifies that profile.*

Proof. Type IV defects are persistence objects attached to a morphism, not arbitrary arithmetic discrepancy numbers. $\square$

Proposition 2.84 (Congruence defect affects Core only through supported routes). *A congruence effect influences a metric Core comparison only through a theorem-supported profile bound in a declared ledger route. A non-scalar congruence failure instead blocks or rejects the relevant obligation and cannot be scalarized by assertion.*

Proof. Appendix M accepts actual profile discrepancy bounds, not source-side labels. Logical failure and metric distance are distinct types. $\square$

Remark 2.85 (Congruence is not equality after realization). Even a canonical source congruence may lose information under coefficient passage, quotienting, duality, or one-parameter extraction. Zero realization defect is a theorem conclusion, not a default.

MS-B.7. Congruence failure, Type IV proxies, and finite detectors

Definition 2.86 (Congruence Type IV proxy). A congruence Type IV proxy is a source anomaly used to select windows, thresholds, degrees, or tower layers for further verification. It is advisory unless a transport theorem identifies it with a Core defect predicate.

Proposition 2.87 (Congruence failure is not Type IV diagnosis). *Failure of a congruence is not equivalent to $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$ by default.*

Proof. Congruence failure is source-side; Type IV obstruction is defined from actual terminal-generic kernel or cokernel profiles. $\square$

Definition 2.88 (MS-B detector record). An MS-B finite detector record declares:

- (i) the exact finite predicate and arithmetic family;
- (ii) source stage, chosen from the canonical four-stage set

$$\begin{array}{ll} \text{windowed_prethreshold,} & \text{repaired_postthreshold,} \\ & \text{kernel_defect,} \quad \text{cokernel_defect;} \end{array}$$

- (iii) nonzero soundness and zero completeness separately;
- (iv) birth/germ or decorated-endpoint exhaustiveness;
- (v) stencil or family-mesh coverage;
- (vi) endpoint and cofinal conventions;
- (vii) family membership and excess-bound provenance;

- (viii) non-adaptivity or an admitted adaptive theorem;
- (ix) immutable input and output artifacts;
- (x) proof-axis status `detector__certified`, `detector__failed`, or `detector__pending`;
- (xi) value-axis status `zero`, `nonzero`, or `undefined`.

Proposition 2.89 (Finite congruence sampling is not proof compression). *Checking finitely many congruence levels, primes, characters, specializations, or thresholds is not a proof of a family-wide zero predicate without a direct theorem or a complete finite detector certificate.*

Proof. Soundness of found anomalies does not imply completeness for absent anomalies. Family coverage is a separate theorem obligation. $\square$

Remark 2.90 (Correct use of congruence proxies). Congruence proxies may prioritize expensive calculations. They do not alter a verifier-side value until they re-enter through an admitted typed record.

MS-B.8. Control-theorem interfaces

Definition 2.91 (Arithmetic control statement). An *arithmetic control statement* compares finite-layer and limiting arithmetic objects, often through invariants, coinvariants, restriction, corestriction, or localization maps. Its theorem hypotheses and exact kernel/cokernel conclusion must be recorded.

Definition 2.92 (Core-readable control statement). A control statement is *Core-readable* only when a proved Interface supplies an actual morphism of the profiles consumed by the intended conclusion and proves how the arithmetic control kernel/cokernel relates to the Core kernel/cokernel.

Definition 2.93 (Control defect). A *control defect*, denoted δ^{ctrl} , is a local name for a control-related effect. A finite, bounded, pseudo-null, torsion, negligible, or asymptotically controlled source defect is not automatically a metric profile discrepancy and is not automatically zero.

Definition 2.94 (Control artifact). A control artifact contains the theorem statement, source and target objects, actual maps, hypotheses, kernel and cokernel conclusions, exceptional layers, local conditions, coefficient conventions, and immutable proof or computation references.

Definition 2.95 (Control-strength classification). A control conclusion is classified as one of

isomorphism, injective, surjective, `finite_kernel`,
`finite_cokernel`, `bounded_defect`, `pseudo_isomorphism`,
asymptotic, unknown.

No weaker class is silently read as a stronger one.

Proposition 2.96 (Arithmetic control does not imply Core tower cleanliness by default). *An arithmetic control theorem does not imply $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ unless its realized form kills the terminal-generic kernel and cokernel profiles of the actual $\phi_{i,W,\tau}$.*

Proof. Finite or bounded arithmetic control defects may realize as nonzero transient or terminal persistence defects. The required identification is not automatic. $\square$

Proposition 2.97 (Finite control defect is not zero defect). *A finite kernel or cokernel in the arithmetic control theorem does not become a zero Core defect merely because it is finite.*

Proof. Finiteness is a size property. Vanishing is an equality property. A finite nonzero object may produce a nonzero bar or defect profile. $\square$

Remark 2.98 (Control theorem status). A classical control theorem may be consumed as source theorem evidence at its exact hypotheses. Its translation into AK profiles remains a separate Interface or Bridge theorem.

MS-B.8bis. Exact abstract bounded-control transfer

Definition 2.99 (Bounded-control diagram). A *bounded-control diagram* is a commutative diagram of finite abelian p -primary groups with exact rows

$$\begin{array}{ccccccc} 0 & \rightarrow & A_n & \rightarrow & B_n & \rightarrow & C_n \rightarrow 0 \\ & & \downarrow a_n & & \downarrow b_n & & \downarrow c_n \\ 0 & \rightarrow & A'_n & \rightarrow & B'_n & \rightarrow & C'_n \rightarrow 0, \end{array}$$

together with uniform bounds on the four outer kernel/cokernel groups and, when Core morphism defects are invoked, a strict p -filtered exactness record.

Theorem 2.100 (Bounded-control transfer by the Snake Lemma). *For every bounded-control diagram there is an exact sequence*

$$0 \rightarrow \ker a_n \rightarrow \ker b_n \rightarrow \ker c_n \rightarrow \operatorname{coker} a_n \rightarrow \operatorname{coker} b_n \rightarrow \operatorname{coker} c_n \rightarrow 0.$$

In particular,

$$\begin{aligned} \operatorname{length}_{\mathbb{Z}_p}(\ker b_n) &\leq \operatorname{length}_{\mathbb{Z}_p}(\ker a_n) + \operatorname{length}_{\mathbb{Z}_p}(\ker c_n), \\ \operatorname{length}_{\mathbb{Z}_p}(\operatorname{coker} b_n) &\leq \operatorname{length}_{\mathbb{Z}_p}(\operatorname{coker} a_n) + \operatorname{length}_{\mathbb{Z}_p}(\operatorname{coker} c_n). \end{aligned}$$

Hence uniform outer bounds imply uniform finiteness and boundedness of the middle kernel and cokernel.

Proof. The exact sequence is the Snake Lemma. For finite p -groups, length is additive on short exact sequences. Taking the appropriate subquotients of the Snake sequence gives the two inequalities. $\square$

Theorem 2.101 (Exact AK realization and Return for bounded control). *Assume the diagram is strict for the p -power filtrations. Applying $\mathcal{L}_p$ gives actual Core-readable morphisms whose kernel and cokernel profiles are the realizations of the arithmetic kernel and cokernel groups. The total-bar-length inequalities are exactly equivalent to the preceding $\mathbb{Z}_p$ -length inequalities. Therefore the bounded-control theorem has a complete non-circular Interface–Core–Return realization at exact strength.*

Proof. Proposition 2.61 identifies the actual profile defects. Theorem 2.58 identifies their total bar lengths with the arithmetic lengths. Theorem 2.100 proves the Core-side bounds, and the same exact identities return those bounds to the arithmetic groups. $\square$

Remark 2.102 (Abstract control is not Mazur specialization). The preceding theorem proves the general bounded-defect mechanism. A Mazur-type arithmetic specialization still requires the Selmer/local-condition diagram, its exactness, the outer cohomological bounds, continuity and variance, and any required strictness or strictness-defect theorem. None of those arithmetic hypotheses is manufactured by the Snake Lemma.

MS-B.9. Finite layers, truncation, and finite-to-infinite reduction

Definition 2.103 (Finite-layer truncation). A finite-layer truncation replaces an infinite arithmetic diagram by the actual subsystem through layer N , with all retained transition maps and omitted scope declared.

Remark 2.104 (Finite-layer data are not automatically apex data). Finite-layer agreement may motivate an apex candidate, but it does not define the apex object, its universal property, its repaired profile, or the comparison morphism to it.

Definition 2.105 (Apex policy). An apex policy states whether the target is a source infinite layer, inverse limit, direct limit, completion, derived limit, homotopy limit/colimit, external object, eventually constant value, or proved finite proxy. It also states the comparison maps and the theorem relating that construction to the repaired apex profile.

Definition 2.106 (Finite-layer truncation defect). A finite-layer truncation effect δ_N^{trunc} is classified before use. Only a supported bound on an actual protected profile pair may enter a metric ledger.

Definition 2.107 (Finite-to-infinite reduction theorem). A finite-to-infinite reduction theorem states the exact infinite predicate, finite witness, system maps, bound or stabilization index, hypotheses, apex construction, compatible cone or cocone, apex-agreement map, and failure route. It distinguishes:

finite observation, finite-prefix agreement, scalar convergence,
metric Cauchy, eventual constancy, categorical limit/colimit existence,
declared apex, apex agreement, external Return.

Definition 2.108 (Apex proxy status). An apex proxy has one status:

genuine_apex, proved_finite_reduction, bounded_approximation,
advisory, undefined, failed.

Only the first two can support an exact apex claim, and only under their recorded hypotheses.

Definition 2.109 (Aggregated truncation defect). If several supported truncation profile bounds are consumed, their aggregate

$$\delta^{\text{trunc}} := \bigoplus_{N \in \mathcal{N}_{\text{used}}} \delta_N^{\text{trunc}}$$

is interpreted in the declared Appendix M value system. The notation does not authorize countable aggregation, change units, or double-count an Appendix MS-A alias.

Proposition 2.110 (Finite-layer truncation does not define the apex by default). *A finite subsystem does not define F_∞ or its apex profile without a proved apex policy and agreement theorem.*

Proof. Finite data have bounded scope; an apex is a separate object with a universal, completion, stabilization, or external identification property. □

Proposition 2.111 (Observed stabilization is not eventual stabilization). *A plateau through finitely many layers does not prove eventual constancy, limit equality, or apex agreement.*

Proof. A later layer may change. The universal tail statement requires a theorem. □

Proposition 2.112 (Terminal diagnostics require terminal evidence). *A terminal-generic Type IV conclusion requires terminal-tameness evidence for source, target, kernel, and cokernel profiles on the declared terminal scope.*

Proof. Without terminal tameness the terminal-generic dimension is undefined, not zero. □

Proposition 2.113 (Inverse limits expose derived-limit obligations). *When an Iwasawa passage uses a nonexact inverse limit, $\varprojlim^1$ -type terms and Mittag-Leffler-type hypotheses are independent obligations.*

Proof. Inverse limit is not exact in general. Finite-stage stabilization plots and metric slack do not prove the vanishing of derived-limit obstructions. □

Remark 2.114 (Truncation must be ledgered only when metric). A theorem-supported truncation discrepancy between actual profiles may be ledgered. Missing convergence, missing apex identity, or missing $\varprojlim^1$ -control is non-scalar and must not be encoded as an arbitrary large or small number.

MS-B.10. p -adic L -profile realization

Definition 2.115 (p -adic L -profile datum). A p -adic L -profile datum consists of a declared p -adic analytic object, branch or component, coefficient ring, variable, interpolation domain, normalization, and selected observable such as valuations, Newton polygons, zero multiplicities, special values, derivatives, or finite-precision coefficients.

Definition 2.116 (p -adic branch policy). A branch policy records characters, signs, ordinary or supersingular choices, plus or minus branches, specializations, exceptional-zero conventions, trivial-zero treatment, periods, Euler factors, and the one-parameter extraction.

Definition 2.117 (*p*-adic precision record). A precision record gives the exact input precision, output precision, certified valuation bounds, truncation order, zero-detection rule, cancellation risk, Newton-polygon certainty, interval or ball arithmetic when used, and failure behavior.

Definition 2.118 (*p*-adic *L*-profile realization). A purpose-faithful *p*-adic realization maps the declared analytic data to a constructible filtered or persistence object and proves which analytic feature is represented by which Core predicate.

Definition 2.119 (*p*-adic *L*-profile defect). The local effect name δ^{pL} may refer to branch, precision, interpolation, truncation, zero-detection, normalization, or comparison effects. Each occurrence must be separately typed and supported.

Remark 2.120 (*p*-adic *L*-data are not Core profiles). A power series, measure, distribution, rigid analytic function, valuation sequence, or Newton polygon is not a persistence module merely because it has an ordered variable.

Proposition 2.121 (*p*-adic *L*-profile compatibility is [Spec] by default). *Every identification of p-adic L-behavior with a repaired persistence, higher obstruction, or Type IV predicate is [Spec] until proved.*

Proof. The source analytic predicate and target Core predicate have different types and require a purpose-faithful transport theorem. $\square$

Proposition 2.122 (Finite-precision *p*-adic evidence is not analytic identity). *Finite precision does not prove equality of analytic functions, exact zero multiplicity, interpolation identity, or an arithmetic transport theorem without an exactness or certified-reduction theorem.*

Proof. Finite precision admits unresolved higher coefficients and cancellation. Exact analytic claims quantify beyond the recorded approximation. $\square$

Proposition 2.123 (Zero detection is not Type IV vanishing). *Detected or absent zeros of a p-adic L-profile do not prove $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ without an actual-morphism transport theorem.*

Proof. Analytic zeros and persistence kernel/cokernel defects are different predicates. $\square$

MS-B.10bis. Main-conjecture and analytic–algebraic bridge discipline

Definition 2.124 (Main-conjecture interface statement). A main-conjecture interface statement records an exact source-domain relation, such as an equality or inclusion of characteristic or Fitting ideals, together with every torsion, local-condition, normalization, period, Euler factor, and branch hypothesis. Its source theorem status is recorded independently of its AK transport status.

Proposition 2.125 (Main-conjecture equality is not Core exactness). *An equality of characteristic ideals or a proved main-conjecture statement is not, by itself, equality or isomorphism of Core profiles, a Type IV zero certificate, or an AK Return theorem.*

Proof. The source theorem concerns arithmetic ideals or modules. Core exactness concerns declared realized profiles and actual morphisms. The Return theorem concerns the opposite direction back to the external arithmetic conclusion. These are separate arrows. $\square$

Definition 2.126 (Analytic–algebraic compatibility record). An analytic–algebraic compatibility record identifies the exact source analytic object, algebraic object, normalizations, exceptional factors, equality strength, and the theorem relating both to the same Core-readable predicate or to mutually comparable Core profiles.

Proposition 2.127 (Two source theorems do not compose through AK automatically). *A proved control theorem and a proved main conjecture do not automatically compose to an AK theorem unless their realized objects, maps, coefficient passages, and conclusion directions are connected by proved typed arrows.*

Proof. Theorems compose only when codomain and domain types, hypotheses, maps, and variance match. Shared arithmetic vocabulary is insufficient. $\square$

MS-B.11. Certificate records and local outcomes

Definition 2.128 (Cyclotomic certificate record). A cyclotomic certificate record is an Appendix MS-A arithmetic Interface record specialized to a cyclotomic source and containing all invoked variance, coefficient, transition-map, apex, detector, Type IV, reduction, and Return fields.

Definition 2.129 (Iwasawa certificate record). An Iwasawa certificate record additionally contains the Iwasawa algebra, topology, module hypotheses, structure-theorem status, invariant namespaces, and duality/completion records actually consumed.

Definition 2.130 (Congruence certificate record). A congruence certificate record additionally contains the congruence modulus or ideal, depth, comparison mode, conversion theorem when metric, and no-double-counting relation to other defect records.

Definition 2.131 (MS-B manifest entry). An MS-B manifest entry records at least:

- (i) source class and immutable source identity;
- (ii) evidence class, invocation role, local claim role, permitted use, and forbidden stronger reading;
- (iii) purpose profile and purpose-faithfulness theorem status;
- (iv) coefficient, topology, duality, and variance conventions;
- (v) one-parameter extraction, order, normalization, constructibility, and endpoint convention;
- (vi) actual realized objects and transition morphisms;
- (vii) windows, thresholds, monitored degrees, and window-first profiles;
- (viii) representatives and standard higher-edge eligibility when invoked;
- (ix) detector records when finite proof compression is invoked;
- (x) apex provenance, semantic authority, actual $\phi_{i,W,\tau}$, and terminal tameness when Type IV is invoked;
- (xi) terminal and transient defect statuses separately;
- (xii) exact, metric-gap-certified, or genuinely unconsumed comparison mode;
- (xiii) ledger value system, supported profile pairs, route, aggregation, units, and no-double-counting relations;
- (xiv) finite-to-infinite reduction theorem and apex agreement when required;
- (xv) typed audit, evidence graph, and manifest reconstruction data;
- (xvi) separately proved Return theorem when an external arithmetic conclusion is claimed;
- (xvii) known-theorem-recovery track identity, immutable benchmark class, exact source theorem locator, source-proof quarantine, non-circularity audit, AK arrow status vector, exact recovered strength, and independent validation result when a recovery is advertised;
- (xviii) portfolio shared infrastructure, cross-track contamination audit, trackwise non-compensation, and dual-recovery closure status when both recovery tracks are declared;
- (xix) local outcomes, failure routes, environment, replay procedure, and staleness conditions;
- (xx) artifact identities and deferred Claim Register promotion status.

Definition 2.132 (MS-B local outcome vector). The following fields are stored separately:

Interface outcome, finite-certificate proof/value statuses,
Core audit outcome, Bridge semantic role/evidence class,
 reduction theorem status, Return theorem status,
 external arithmetic theorem/refutation/unresolved status,
 independent recovery-validation status.

The Interface and Core audit outcomes use pass, reject, or undefined. Detector, Bridge, reduction, Return, external-domain, and validation fields retain their own typed vocabularies and are not collapsed to that three-valued set. A clause-level `not_invoked` is allowed only when the clause is genuinely outside the declared claim scope.

Proposition 2.133 (MS-B records must satisfy Appendix MS-A). *Every MS-B certificate record satisfies the full v20 Appendix MS-A schema and the Minimal Manifest Axiom.*

Proof. MS-B interfaces are special cases of arithmetic interfaces. Specialization adds fields but removes none of the parent obligations. □

Proposition 2.134 (Undefined is not reject or not invoked). *Missing evidence for an invoked clause yields undefined; it is not equivalent to reject, zero, pass, or not_invoked.*

Proof. These states encode different logical information: unavailable evidence, proved falsity, proved vanishing, proved success, and scope exclusion. □

MS-B.12. Mixed interfaces and non-compensation

Definition 2.135 (Mixed cyclotomic–Iwasawa interface). A mixed cyclotomic–Iwasawa interface identifies the exact common source diagram, variance conversions, coefficient passages, and compatible Core targets used by both components.

Definition 2.136 (Mixed Iwasawa–congruence interface). A mixed Iwasawa–congruence interface declares whether congruence is imposed before or after duality, completion, finite-layer passage, and profile realization, and proves every consumed commutation.

Definition 2.137 (Mixed Iwasawa– p -adic L -profile interface). A mixed Iwasawa– p -adic interface declares separate algebraic and analytic source objects and the proved compatibility arrows by which they enter compatible Core-readable profiles.

Definition 2.138 (Mixed compatibility record). A mixed compatibility record states the common target category, coefficient field, parameter, windows, thresholds, degrees, endpoint conventions, artifact identities, comparison modes, synchronization theorem, and failure route.

Definition 2.139 (Mixed arithmetic ledger). The display

$$\delta_{\text{mixed}}^{\text{arith}} := \delta^{\text{cyc}} \oplus \delta^{\text{Iw}} \oplus \delta^{\text{cong}} \oplus \delta^{\text{ctrl}} \oplus \delta^{\text{trunc}} \oplus \delta^{pL}$$

is meaningful only after every included term is a supported metric component in the same declared value system or is converted by a proved conservative map. Non-scalar and advisory terms are excluded from this numeric aggregate.

Proposition 2.140 (Mixed interfaces require compatible realization targets). *A mixed interface is admissible only when the invoked components have type-compatible targets and proved comparison or synchronization maps.*

Proof. Without compatible targets there is no well-typed common predicate, map, or ledger path. □

Proposition 2.141 (Mixed components do not compensate). *A pass or favorable value in one component cannot compensate for an undefined or failed required obligation in another.*

Proof. Required obligations form a typed conjunction. Numerical cancellation and advisory agreement do not prove a missing arrow. $\square$

Proposition 2.142 (Cross-component agreement requires a comparison record). *Visual, numerical, or source-semantic agreement between components is non-verdict until an exact or metric-gap-certified comparison record connects their actual Core profiles.*

Proof. Agreement must be a relation in the target category consumed by the conclusion. $\square$

Remark 2.143 (Mixed does not mean stronger). Mixed architectures often add more non-scalar obligations than they add theorem strength. Their value lies in cross-validation only when the compatibility arrows themselves are proved.

MS-B.13. Threshold sweeps and arithmetic stable bands

Definition 2.144 (Arithmetic advisory stable band). An arithmetic advisory stable band is an interval on which selected source indicators appear stable. It is proposer-side information unless transported.

Definition 2.145 (Arithmetic Core stable band). An arithmetic Core stable band is an interval of thresholds on which, for one artifact-coherent family of actual morphisms and a fixed finite monitored scope, every invoked Type IV record is certified_{zero}.

Definition 2.146 (Arithmetic threshold sweep record). A threshold sweep record declares the interval, sampled or symbolic thresholds, fixed windows and degrees, artifact family, provenance, terminal mode, gap data, threshold crossings, detector records, and any theorem reducing a continuum claim to finitely many checks.

Definition 2.147 (Certified Core stable band). A certified Core stable band additionally requires artifact coherence, immutable source/target identity, unchanged morphism provenance, terminal-tameness throughout the band, and every exact or metric comparison record consumed. Metric comparisons through deletion satisfy the two-sided gap hypotheses at every relevant threshold.

Proposition 2.148 (Arithmetic advisory stability is not Core stability). *Stable source invariants do not certify a Core stable band.*

Proof. The Core band predicate concerns actual morphism defects, not source indicator variation. $\square$

Proposition 2.149 (Sampled stability is not certified stable band). *Finitely or countably sampled threshold values do not prove a universal stable band without a reduction theorem controlling all unsampled crossings and artifact changes.*

Proof. Hard deletion and barcode combinatorics may change between samples. $\square$

Proposition 2.150 (Stable band requires threshold-gap discipline when metric comparisons are used). *Every metric comparison consumed in a certified band satisfies the repaired two-sided threshold-gap and zero-separation conditions on its actual profile pair.*

Proof. Hard threshold deletion is not globally non-expansive. Bandwise use does not remove this obstruction. $\square$

MS-B.14. Ledger semantics and non-scalar obligations

Definition 2.151 (MS-B effect classification). Every local effect name is classified as exactly one of:

metric_profile_discrepancy, morphism_profile_discrepancy,
theorem_controlled_nonmetric, non_scalar_obligation,
advisory, unavailable.

Only the first two may enter a strict Core metric budget, and only through a theorem supporting the actual protected profile pair.

Definition 2.152 (MS-B ledger entry). An MS-B ledger entry records the value system $(V, \preceq, \oplus, 0_V)$, units, scalarization when used, protected profiles, threshold, comparison mode, support theorem, route edge, canonical slot, alias/refinement relations, aggregation rule, total, gap, reserve, artifact identity, and local verdict.

Proposition 2.153 (Arithmetic effects do not enter the ledger by naming). *The symbols δ^{cyc} , δ^{Iw} , δ^{cong} , δ^{ctrl} , δ^{trunc} , and δ^{pL} do not enter a Core budget without a support theorem and canonical placement.*

Proof. A ledger bounds an identified profile comparison path. A namespace symbol is not such a bound. $\square$

Proposition 2.154 (Non-scalar obligations do not admit compensation). *Metric reserve cannot repair missing purpose-faithfulness, coefficient passage, constructibility, transition morphisms, apex identity, semantic authority, detector completeness, higher-edge eligibility, terminal properties, finite-to-infinite reduction, claim permission, artifact identity, or Return.*

Proof. These obligations determine whether the objects, predicates, and inference arrows exist and are authorized. They are not distances between already valid profiles. $\square$

Proposition 2.155 (Countable tower aggregation requires extra structure). *An infinite aggregate of layerwise discrepancy bounds requires the declared quantale or complete ordered structure, summability or join theorem, and the finite-to-infinite identification actually used.*

Proof. A finite commutative ledger monoid does not automatically support countable joins or convergence. $\square$

MS-B.15. Complete cyclotomic/Iwasawa Projection–Return chain

Definition 2.156 (MS-B theorem-bearing chain). A theorem-bearing external arithmetic route has the form

$$\begin{array}{ccccccc}
 X_{\text{arith}} & \xrightarrow{\text{proved typed Interface}} & F_X & \xrightarrow{\text{window-first}} & B_{n,W}(F_X) & \xrightarrow{\text{hard deletion}} & \text{Eval}_{n,W,\tau}(F_X), \\
 & & & & & & \\
 \text{Eval}_{n,W,\tau}(F_X) & \xrightarrow{\text{direct theorem or admitted finite certificate}} & C_{\text{fin}}(X) & \xrightarrow[\text{when required}]{\text{proved reduction}} & C_{\text{AK}}(X), & & \\
 & & & & & & \\
 C_{\text{AK}}(X) & \xrightarrow{\text{typed audit and manifest}} & \text{Verdict}_{\text{AK}}(X) & \xrightarrow[\text{Return theorem}]{\text{separately proved}} & \mathcal{E}_{\text{arith}}(X). & &
 \end{array}$$

Every arrow declares source, target, direction, hypotheses, evidence class, role, artifacts, and failure route.

Definition 2.157 (MS-B Return theorem). An MS-B Return theorem proves that a specified, fully typed AK verdict implies an exact cyclotomic, Iwasawa, congruence, control, or p -adic analytic conclusion on the original source datum. It lists every source hypothesis, realization theorem, finite predicate, reduction theorem, defect exclusion, and non-scalar obligation consumed.

Proposition 2.158 (Core pass is not an arithmetic theorem). *A Core pass, Type IV zero, higher Ext^n -vanishing, stable band, or finite detector certificate does not prove an external arithmetic conclusion without the separately proved Return theorem.*

Proof. Each is a predicate about the realized Core shadow. Return is the soundness arrow back to the source semantics. $\square$

Proposition 2.159 (Reduction and Return are independent). *A finite-to-infinite reduction theorem does not imply a Return theorem, and a Return theorem on an infinite AK predicate does not supply the missing finite-to-infinite reduction.*

Proof. Reduction moves within the AK-side finite/infinite predicate architecture. Return crosses from the AK predicate to the source domain. Their sources and targets differ. $\square$

MS-B.15bis. Dual Iwasawa recovery portfolio and source binding

Definition 2.160 (MS-B dual-recovery portfolio). The MS-B portfolio contains:

$$\text{IW-KTR-GROWTH} \quad \text{and} \quad \text{IW-KTR-CONTROL-MAZUR}.$$

Track G recovers the characteristic-growth clause of Iwasawa's 1959 Theorem 4 at exactly that clause's strength. Track C contains an exact abstract bounded-control transfer and a separately scoped Mazur arithmetic specialization program. The abstract control theorem does not by itself complete the Mazur specialization.

Definition 2.161 (MS-B recovery source locators). The local mathematical source-statement binding for Track G is the immutable artifact

AK_HPDEST_v20_0_0_MS_B_Iwasawa_1959_Source_Statement_Binding.txt

with content identity

SHA256:a5f0a65a73c802492167676aba024b01651cd855bbdc00bc3d3c48d46cd6f0d7.

It binds Kenkichi Iwasawa, *On Γ -extensions of algebraic number fields*, Bull. Amer. Math. Soc. 65 (1959), 183–226, DOI 10.1090/S0002-9904-1959-10317-7, as follows:

- (i) the exact benchmark is the characteristic-function conclusion of printed Theorem 4;
- (ii) the permitted precursor package consists of the strict- Γ -finite definitions in §2.3, the finiteness consequence used from Lemma 5.5, Theorem 3, the exact decomposition immediately preceding Theorem 4, Lemma 3.5, and Lemma 4.10;
- (iii) printed Theorem 4 itself is quarantined from the derivation and is used only for final validation and strength comparison;
- (iv) printed Theorem 11 is the separate class-number specialization boundary.

The Mazur source remains bound bibliographically to Barry Mazur, *Rational points of abelian varieties with values in towers of number fields*, Invent. Math. 18 (1972), 183–266, DOI 10.1007/BF01389815, pending its own exact theorem-excerpt and arithmetic-specialization artifacts. Final repository ingestion and CR-v20 registration remain separate governance actions.

Definition 2.162 (Track G precursor theorem package). Let A be a strictly Γ -finite discrete module and let B, C, M be the canonical divisible, elementary, and maximal regular submodules from the registered source structure theory. The permitted precursor package consists exactly of:

- (i) $m(A) < \infty$;
- (ii) $B + C = M$;
- (iii) the exact identity
$$c(n; A) = c(n; A/M) + c(n; B) + c(n; C);$$
- (iv) eventual constancy of $c(n; A/M)$;
- (v) existence of N_B, u_B such that
$$c(n; B) = l(A)n + u_B \quad (n \geq N_B);$$
- (vi) the elementary contribution
$$c(n; C) = m(A)p^n.$$

These are registered source-side theorem inputs preceding the benchmark conclusion. They are not obtained from the benchmark equality and do not contain its eventual constant or a final stabilization index.

Definition 2.163 (Shared infrastructure without shared conclusions). The tracks share Theorem 2.58, the manifest schema, and the non-compensation rules. They do not share benchmark conclusions, source hypotheses, or Return theorems.

Proposition 2.164 (The two recoveries are logically independent). *Exact recovery of the Iwasawa characteristic-growth clause does not prove the Mazur arithmetic control theorem, and the abstract bounded-control transfer does not prove the Iwasawa growth formula.*

Proof. The growth conclusion concerns the characteristic function of a strictly Γ -finite module. The control conclusion concerns kernels and cokernels of actual maps in a Selmer/local-condition diagram. Their objects, hypotheses, and Return consequents differ. $\square$

Proposition 2.165 (Existential release gate versus dual portfolio). *The exact Track G witness supplies the mathematical content required by an existential Iwasawa known-theorem-recovery gate. The canonical gate is nevertheless only ready_for_registration until the final Claim Register and release registry consume the witness. The portfolio is not dual_recovery_certified while the Mazur specialization remains pending.*

MS-B.15ter. Recovery Track G: exact Iwasawa characteristic-growth recovery

Definition 2.166 (Growth benchmark class). Fix a prime p , a topological group $\Gamma \cong \mathbb{Z}_p$, a topological generator γ , and a strictly Γ -finite discrete p -primary module A in the sense of the bound source statement. Put

$$\omega_n := \gamma^{p^n} - 1, \quad A_n := \ker(\omega_n : A \rightarrow A), \quad c_n := \log_p \#A_n.$$

Strict Γ -finiteness means precisely that every A_n is finite. Changing the generator or the discrete/compact dual convention requires an explicit equivalence record.

Definition 2.167 (Classical growth benchmark statement). The immutable Track G benchmark is the following characteristic-growth theorem: there exist uniquely determined nonnegative integers

$$\lambda_{\text{Iw}}(A) := l(A), \quad \mu_{\text{Iw}}(A) := m(A),$$

a uniquely determined integer ν_A , and at least one, generally nonunique, tail witness N such that

$$c_n = \lambda_{\text{Iw}}(A)n + \mu_{\text{Iw}}(A)p^n + \nu_A \quad (n \geq N).$$

The benchmark is the characteristic-function clause of printed Theorem 4. It does not advertise recovery of every structural assertion appearing elsewhere in that theorem.

Definition 2.168 (Admissible non-circular inputs for Track G). Track G may consume:

- (i) the module A , its actual finite subgroups A_n , and their inclusions;
- (ii) the registered precursor package of Definition 2.162;
- (iii) the source invariants $l(A)$ and $m(A)$ as defined by that precursor structure theory.

It may not consume ν_A , a benchmark-supplied final tail witness N , the benchmark equality, or an equivalent truth table. A tail witness may instead be derived from the independently registered precursor package, for example by taking the maximum of the precursor stabilization bounds. The benchmark theorem itself is used only after the AK derivation for validation.

Remark 2.169 (Exact recovery relative to the precursor package). Track G is an exact-strength recovery of the frozen characteristic-growth clause *relative to* the registered source-side precursor package. It is not advertised as an independent reconstruction of Iwasawa's full module-structure theory from AK Core axioms alone. The AK contribution in this track is the typed, non-circular passage from those precursor theorems to an explicit window-first Core predicate, the bounded-window-to-tail reduction, and the exact arithmetic Return equivalence.

Definition 2.170 (Growth defect sequence). Set

$$d_n := c_{n+1} - c_n - \lambda_{\text{Iw}}(A) - \mu_{\text{Iw}}(A)(p-1)p^n,$$

and

$$d_n^+ := \max(d_n, 0), \quad d_n^- := \max(-d_n, 0).$$

For integers $0 \leq N < R$, define the finite signed defect barcodes

$$\mathcal{D}_{A;N,R}^+ := \bigsqcup_{n=N}^{R-1} \{[n, n+1)\}^{d_n^+}, \quad \mathcal{D}_{A;N,R}^- := \bigsqcup_{n=N}^{R-1} \{[n, n+1)\}^{d_n^-}.$$

Both are finite constructible barcodes on the bounded right-open window $W_{N,R} := [N, R)$.

Definition 2.171 (Filtered-complex realization of a signed defect window). For $q \geq 0$ and an integer n , let $\mathcal{K}(q; n)$ be the finite filtered chain complex over $\mathbb{F}_p$ with:

- (i) q degree-zero generators entering at filtration n ;
- (ii) q degree-one generators entering at filtration $n+1$;
- (iii) differential equal to the identity from the degree-one generators to the corresponding degree-zero generators.

Then

$$\mathbf{P}_0(\mathcal{K}(q; n)) \cong \{[n, n+1)\}^q.$$

Define

$$F_{A;N,R}^\pm := \bigoplus_{n=N}^{R-1} \mathcal{K}(d_n^\pm; n).$$

This is a finite-type filtered object and

$$\mathbf{P}_0(F_{A;N,R}^\pm) \cong \mathcal{D}_{A;N,R}^\pm.$$

Definition 2.172 (Track G repaired Core record). Fix

$$\tau_G := \frac{1}{2}.$$

Every bar in $\mathcal{D}_{A;N,R}^\pm$ has length 1, hence

$$B_{0,W_{N,R}}(F_{A;N,R}^\pm) = W_{W_{N,R}} \mathbf{P}_0(F_{A;N,R}^\pm) \cong \mathcal{D}_{A;N,R}^\pm,$$

and

$$\text{Eval}_{0,W_{N,R},\tau_G}(F_{A;N,R}^\pm) \cong \mathcal{D}_{A;N,R}^\pm.$$

This is an exact window-first route. No metric comparison, threshold-gap transfer, higher Ext, Type IV, or transient-defect clause is invoked.

Theorem 2.173 (Precursor structure implies bounded-window Core vanishing). *For every A in the Growth benchmark class, there exists N such that for every finite $R > N$,*

$$\text{Eval}_{0,W_{N,R},\tau_G}(F_{A;N,R}^+) = 0, \quad \text{Eval}_{0,W_{N,R},\tau_G}(F_{A;N,R}^-) = 0.$$

Proof. By Definition 2.162,

$$c_n = c(n; A/M) + c(n; B) + c(n; C).$$

On a sufficiently deep tail, the first term is constant,

$$c(n; B) = l(A)n + u_B, \quad c(n; C) = m(A)p^n.$$

Taking first differences yields

$$c_{n+1} - c_n = l(A) + m(A)(p-1)p^n$$

for every sufficiently large n . Thus $d_n = 0$ on a tail. For a tail witness N , all multiplicities $d_n^\pm$ vanish for $N \leq n < R$, so both finite barcodes and therefore both repaired evaluations are zero. The printed benchmark equality is not used. $\square$

Theorem 2.174 (Finite-window-to-tail reduction). *For fixed λ, μ , the following are equivalent:*

(i) *there exists N such that, for every finite $R > N$, both signed repaired evaluations on $W_{N,R}$ are zero;*

(ii)

$$c_{n+1} - c_n = \lambda + \mu(p-1)p^n \quad (n \geq N).$$

Proof. The repaired evaluations are exactly the signed unit-bar profiles with multiplicities d_n^+ and d_n^- . Vanishing for every finite $R > N$ is therefore equivalent to $d_n^+ = d_n^- = 0$ for every $n \geq N$, hence to $d_n = 0$ on the tail. $\square$

Theorem 2.175 (Exact Growth Return Theorem). *For any integer sequence c_n and fixed nonnegative integers λ, μ , the following are equivalent:*

(i) *there exists N such that*

$$c_{n+1} - c_n = \lambda + \mu(p-1)p^n \quad (n \geq N);$$

(ii) *there exist N and $v \in \mathbb{Z}$ such that*

$$c_n = \lambda n + \mu p^n + v \quad (n \geq N).$$

For an eventually valid formula, the coefficients λ, μ, v are unique, whereas a valid tail witness N need not be unique or minimal.

Proof. Summing the first-difference identity from N to $n-1$ gives

$$c_n - c_N = \lambda(n - N) + \mu(p^n - p^N),$$

hence

$$v = c_N - \lambda N - \mu p^N.$$

The converse follows by taking first differences. Uniqueness of λ, μ, v follows by comparing two eventual formulas; the p^n, n , and constant terms cannot cancel on an infinite tail unless all coefficient differences vanish. Any larger N remains a witness. $\square$

Definition 2.176 (Track G reduction and Return package). The exact package consists of:

- (i) the content-addressed source-statement and precursor binding of Definition 2.161;
- (ii) the natural finite- p -group realization of Theorem 2.58;
- (iii) the explicit filtered-complex and repaired Core route of Definitions 2.171–2.172;
- (iv) the direct bounded-window Core theorem, Theorem 2.173;
- (v) the finite-window-to-tail reduction, Theorem 2.174;
- (vi) the exact Return equivalence, Theorem 2.175;
- (vii) independent comparison with the quarantined characteristic-growth clause of printed Theorem 4.

Theorem 2.177 (Exact recovery of the Iwasawa growth benchmark). *Track IW-KTR-GROWTH is a non-circular exact known-theorem recovery of the benchmark in Definition 2.167.*

Proof. The precursor package determines the structural invariants and proves the bounded-window Core vanishing without using the benchmark equality, its constant, or a tail witness. The finite-window reduction yields the exact eventual first-difference predicate. The Return Theorem yields precisely the benchmark formula and coefficient uniqueness. The quarantined source theorem is then used only to validate the recovered scope and strength. Thus the Interface, Core, Reduction, Return, and independent-validation arrows are all separately displayed and proved. $\square$

Corollary 2.178 (Conditional class-number specialization). *Let K_∞/K be a $\mathbb{Z}_p$ -extension. If a separately proved arithmetic Interface identifies the p -class-number exponent e_n with the characteristic function c_n of a Track G module up to an eventually constant correction, then*

$$e_n = \mu_{\text{Iw}} p^n + \lambda_{\text{Iw}} n + v_{\text{Iw}}^{\text{arith}} \quad (n \gg 0).$$

Proof. Absorb the eventually constant correction into the Return constant and apply Theorem 2.177. $\square$

Remark 2.179 (Track G current status). The mathematical track status is

mathematically_exact; ready_for_registration.

The recovered benchmark is the characteristic-growth clause of the module theorem. The class-number specialization remains conditional on its separate arithmetic source-to-module Interface and is not needed for this exact module-level witness.

MS-B.15quater. Recovery Track C: exact abstract control and Mazur specialization

Definition 2.180 (Control benchmark class). The exact abstract benchmark class consists of strict bounded-control diagrams from Definition 2.99. The arithmetic specialization class additionally fixes a number field F , an elliptic curve or abelian variety, a prime p , a cyclotomic $\mathbb{Z}_p$ -extension, Selmer local conditions, and every ordinary-reduction and cohomological hypothesis in the selected Mazur source theorem.

Definition 2.181 (Mazur-type control benchmark statement). The Mazur validation target is an exact-sequence or restriction-map statement whose finite error groups have orders bounded independently of the tower layer, at the exact hypotheses of the source locator. This statement is not reduced to isomorphism and is not inferred from the abstract Snake Lemma alone.

Definition 2.182 (Admissible non-circular inputs for Track C). The arithmetic specialization may consume the actual Selmer or cohomological complexes, local-condition maps, Galois actions, and precursor exact sequences. It may not consume the target finiteness or uniform bounds to construct the profile defects whose purpose is to recover those bounds.

Definition 2.183 (Control-map realization package). A complete Mazur specialization supplies an actual commutative control diagram, proves the required exact rows and outer cohomological bounds, realizes the middle map by an actual Core morphism, and supplies either strict p -filtered exactness or a separately controlled strictness defect. Equal ranks, equal coranks, or a source theorem name are insufficient.

Definition 2.184 (Exact recovered strength for Track C). The target strength is

$$\text{finite_kernel} + \text{finite_cokernel} + \text{uniformly_bounded},$$

possibly expressed as exact sequences modulo uniformly bounded finite groups. No isomorphism or full defect-zero conclusion is advertised without a stronger source theorem and Return chain.

Definition 2.185 (Bounded-control Core predicate). The Core predicate consists of the actual kernel and cokernel profiles, their exact total bar lengths, the layerwise bounds, and the strictness/semantic-authority records. Terminal Type IV and transient-defect fields remain separate and are not needed to state finite uniform boundedness.

Definition 2.186 (Track C family and Return package). The abstract package is Theorem 2.101. The Mazur specialization package additionally requires the arithmetic exact diagram and outer bounds, a family-level uniformity theorem, and a Return Theorem identifying the realized middle defects with the original control kernel and cokernel.

Theorem 2.187 (Conditional exact recovery of a Mazur-type control benchmark). *If the complete Mazur specialization package of the preceding definitions is proved without using the benchmark conclusion, then the exact abstract control recovery composes with that package to recover the selected Mazur-type theorem at the recorded bounded-control strength.*

Proof. The arithmetic hypotheses produce a strict bounded-control diagram and uniform outer bounds. Theorem 2.101 gives exact middle profile bounds. The arithmetic Return theorem identifies these bounds with the control kernel and cokernel. Independent comparison then validates the benchmark strength. $\square$

Remark 2.188 (Track C current status). The abstract bounded-control theorem is exactly proved. The Mazur arithmetic specialization remains

bridge_program; arithmetic source arrows pending.

No dual arithmetic recovery status is claimed.

MS-B.15quinquies. Portfolio synthesis and main-conjecture boundary

Definition 2.189 (Dual-recovery synthesis record). The synthesis record is

IW-KTR-GROWTH : characteristic quotients; exact tail-growth recovery;
 arithmetic status: exact module benchmark,
 IW-KTR-CONTROL-MAZUR : Selmer/control diagram; exact abstract transfer;
 arithmetic status: Mazur specialization pending.

No inference arrow between the two external conclusions is present.

Proposition 2.190 (Growth plus control is not the main conjecture). *The exact Growth recovery and the abstract control theorem do not imply an equality of a characteristic ideal with a p -adic L -function, an analytic–algebraic main conjecture, or any unregistered Selmer growth formula.*

Proof. The analytic object, normalizations, Euler factors, characteristic-ideal comparison, and analytic Return theorem are absent. $\square$

Proposition 2.191 (Dual capabilities remain benchmark-local). *The portfolio demonstrates exact size-preserving realization, exact Iwasawa module tail recovery, and exact abstract bounded-control transfer. It proves no universal information-preservation theorem for arbitrary arithmetic towers and no full Mazur specialization.*

Definition 2.192 (MS-B dual-recovery closure status). The current status is

partially__recovered;
 IW-KTR-GROWTH=mathematically__exact;
 abstract__control=exact;
 Mazur__specialization=pending;
 G16__witness=ready__for__registration.

The status becomes dual_recovery__certified only after the Mazur arithmetic track itself is exact and registered.

MS-B.16. Package theorem

Theorem 2.193 (Appendix MS-B v20 interface and recovery package). *Within the Problem Interface layer, Appendix MS-B establishes the following package at exactly the displayed strength.*

- (i) *Arithmetic tower variance, coefficient passage, topology, duality, and one-parameter extraction are independent obligations.*
- (ii) *Cyclotomic and Iwasawa diagrams become Core-readable only through actual realized objects and morphisms.*
- (iii) *Inverse systems require a proved conversion before use in the canonical directed Type IV artifact.*

- (iv) *Apex terminology does not establish apex identity, completion agreement, or semantic authority.*
- (v) *Iwasawa structure theorems are consumed only under their exact hypotheses.*
- (vi) μ_{Iw} , λ_{Iw} , and v_{Iw}^{arith} remain distinct from Collapse diagnostics.
- (vii) *The functor $\mathcal{L}_p$ is a natural, finite, constructible, size-lossless realization of finite abelian p -primary groups.*
- (viii) *General morphisms need a strictness or strictness-defect theorem before source kernel/cokernel groups are identified with profile kernel/cokernel objects.*
- (ix) *Characteristic ideals and pseudo-isomorphisms do not imply Core profile isomorphism or full defect zero.*
- (x) *Type IV consumes actual morphisms and separately verified terminal tameness; terminal zero does not imply transient or full-profile zero.*
- (xi) *Source congruence is not Core exactness or metric-gap certification without a proved conversion.*
- (xii) *Finite arithmetic sampling is not proof compression without a direct theorem or complete detector certificate.*
- (xiii) *The Snake-Lemma bounded-control theorem and its exact $\mathcal{L}_p$ -realization are proved.*
- (xiv) *Finite, bounded, or pseudo-null defect is not zero by terminology.*
- (xv) *Finite prefixes, plateaus, limit existence, apex declaration, and apex agreement remain distinct.*
- (xvi) *Inverse-limit passages retain independent derived-limit and Mittag–Leffler obligations.*
- (xvii) *Finite-precision p -adic data do not prove analytic identities or Core diagnostics.*
- (xviii) *A main-conjecture theorem remains source-side until a separate typed AK Interface and Return chain is proved.*
- (xix) *Mixed components require compatible targets and do not compensate for missing obligations.*
- (xx) *Stable-band claims require artifact coherence and repaired threshold-gap discipline.*
- (xxi) *Only theorem-supported actual profile discrepancies enter the metric ledger; non-scalar obligations remain non-compensable.*
- (xxii) *Interface, finite certificate, Core audit, Bridge, reduction, Return, external conclusion, and recovery validation remain separate.*
- (xxiii) *Track G gives an exact non-circular recovery of the classical Iwasawa characteristic-growth benchmark, relative to its registered precursor package, through the natural p -primary length realization, an explicit signed-defect filtered-complex realization, a proved tail reduction, and an exact Return theorem.*
- (xxiv) *The class-number specialization requires a separate class-field Interface and is not silently imported into the module benchmark.*
- (xxv) *Track C proves the abstract bounded-control mechanism exactly, while the Mazur arithmetic specialization remains a registered Bridge Program.*
- (xxvi) *The Growth witness supplies mathematical G16 content, but canonical G16 pass remains pending final Claim Register/release-registry action.*
- (xxvii) *None of these results implies the Iwasawa main conjecture or a universal information-loss theorem.*

Proof. Items (i)–(vi) follow from Sections MS-B.1–MS-B.4bis. Items (vii)–(viii) are Theorem 2.58 and Proposition 2.61. Items (ix)–(xii) follow from Sections MS-B.5–MS-B.7. Items (xiii)–(xiv) follow from Section MS-B.8bis. Items (xv)–(xxii) follow from Sections MS-B.9–MS-B.15. Items (xxiii)–(xxvii) follow from the recovery and synthesis theorems in Sections MS-B.15ter–MS-B.15quinquies. $\square$

MS-B.17. Explicit exclusions

Appendix MS-B asserts none of the following.

- No Iwasawa main conjecture, modularity theorem, automorphy theorem, new Selmer theorem, or p -adic L -identity is proved.
- No field-tower inclusion automatically determines the variance of attached arithmetic objects.
- No inverse arithmetic system is silently treated as the directed Core system required by Type IV.
- No tensor, reduction, completion, duality, invariants, or coinvariants passage is assumed exact or lossless without proof.
- No raw tower supplies actual Core transition morphisms or an apex comparison morphism.
- No finite stage, completion symbol, inverse limit, direct limit, or formal placeholder is an authorized apex without its record.
- No Iwasawa structure theorem is used outside its exact module hypotheses.
- No equality $\mu_{\text{Iw}} = \mu_{\text{Collapse}}$ or relation between λ_{Iw} , $\nu_{\text{Iw}}^{\text{arith}}$, and ν_{Collapse} is asserted by default.
- No characteristic-ideal equality, Fitting-ideal equality, or pseudo-isomorphism is treated as a Core profile isomorphism.
- No terminal Type IV zero is treated as transient-defect zero, full defect-profile zero, or full comparison isomorphism.
- No source congruence is treated as exact or metric Core comparison without a realized comparison theorem.
- No finite sample of layers, primes, characters, specializations, or thresholds is a complete detector without a completeness theorem.
- No finite kernel or cokernel is identified with zero merely because it is finite.
- No finite prefix, observed plateau, scalar convergence, or Cauchy estimate proves eventual constancy or apex agreement by itself.
- No inverse-limit exactness is assumed without addressing derived-limit obstructions when relevant.
- No finite-precision p -adic computation proves an analytic identity without an exactness theorem.
- No main-conjecture theorem is identified with Core exactness or Return by source prestige.
- No mixed component compensates for another component's undefined or failed obligation.
- No metric reserve repairs a missing morphism, realization, detector completeness, apex identity, reduction, artifact, claim permission, or Return theorem.
- No first graded rank $\dim_{\mathbb{F}_p}(G/pG)$ is treated as the full p -primary order of G .
- No non-strict arithmetic map is assumed to have source kernel/cokernel profiles equal to the kernel/cokernel of $\mathcal{L}_p(f)$.
- No eventual constant or stabilization index is inserted into Track G and then advertised as recovered.
- Track G's exact module benchmark is not silently promoted to the full class-number theorem without the class-field Interface of Corollary 2.178.
- The abstract Snake-Lemma control theorem does not discharge Mazur's local arithmetic hypotheses.

- No Mazur-type finite or uniformly bounded kernel/cokernel conclusion is silently strengthened to isomorphism, full defect zero, or terminal Type IV zero.
- No independent benchmark proof fills a missing AK recovery arrow.
- The exact Growth recovery does not compensate for the pending Mazur specialization.
- No partial portfolio status is written as `dual_recovery_certified`.
- No local exact witness is written as canonical G16=pass before final CR-v20 and release-registry action.
- No dual calibration implies an Iwasawa main conjecture, characteristic-ideal equality, analytic continuation, or arbitrary-tower information preservation.
- No local label is treated as a promoted canonical Claim UID before final CR closure.

MS-B.18. Provenance, conformance, and status

Remark 2.194 (Ledger namespace registration for MS-B). The names

$$\delta^{\text{cyc}}, \quad \delta^{\text{lw}}, \quad \delta^{\text{cong}}, \quad \delta^{\text{ctrl}}, \quad \delta_N^{\text{trunc}}, \quad \delta^{\text{trunc}}, \quad \delta^{pL}$$

are local Interface aliases. Every consumed occurrence declares whether it aliases, refines, or aggregates an Appendix MS-A effect and maps to a canonical Appendix M slot. The same primitive discrepancy is never double-counted.

This appendix depends on:

- (i) Appendix MS-A v20.0.0;
- (ii) Main Chapters 1–8 and Foundation Appendices A–G;
- (iii) Technical Extension Appendices H–N at their frozen, reviewed local strengths, with final CR-v20 promotion pending;
- (iv) CM-v20 and TB-v20 only at their verified local strengths;
- (v) the local content-addressed source registry in Definition 2.161;
- (vi) CR-v20 for deferred canonical promotion and release-gate registration.

Conformance statement for Appendix MS-B. The exact status is

`v20_mathematical_conformant`;
`p-primary_natural_realization_proved`;
`IW-KTR-GROWTH=mathematically_exact`;
`abstract_bounded_control=exact`;
`Mazur_arithmetic_specialization=pending`;
`G16_witness=ready_for_registration`;
`Part IV/CR_promotion=pending`.

No new Iwasawa theorem is claimed; the exact Growth result is a non-circular recovery of a known benchmark. The canonical release headline is not set locally.

Status labels for Appendix MS-B. Theorems 2.58, 2.100, 2.101, 2.173, 2.175, and 2.177 are proved Problem-Interface or recovery theorems at their displayed scopes. Theorem 2.187 remains conditional on the arithmetic specialization package. Theorem 2.193 summarizes these exact distinctions.

Repair ancestry. The v19 MS-B artifact and all prior v20 drafts remain immutable ancestry. This reconstruction supersedes their local recovery-status language without mutating earlier records.

3 Appendix MS-C: Mirror Reflection and Categorical Comparison

MS-C.1. Scope, dependency position, and interface status

This appendix defines the v20.0.0 Problem-Interface discipline for Mirror Reflection, transfer, duality, categorical comparison, and their arithmetic, motivic, geometric, symplectic, sheaf-theoretic, and higher-categorical surroundings. It does not enlarge the theorem-bearing Core of Main Chapters 1–8 or Foundation Appendices A–G.

The central rule is

Mirror and categorical structure may organize a comparison;
verifier authority requires a typed Core-readable projection;
an external conclusion requires a proved Return Theorem.

For every realized object consumed by the Core, the fixed order is

$$B_{i,W}(F) := W_W \mathbf{P}_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Window restriction is performed before hard deletion. No unconditional commutation of these operations, no global exact positive-threshold collapse functor, no positive-threshold Serre localization, and no global strict filtered lift are assumed.

Whenever tower semantics are invoked, the canonical comparison is an actual morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty)$$

in the declared abelian profile category. An object equivalence, common barcode, adjunction, duality slogan, finite-stage agreement, or comparison diagram does not create this morphism.

Remark 3.1 (Interface status). Every problem-specific Mirror, categorical, motivic, derived, Fukaya, Langlands, modularity, or automorphy implication is [Spec] by default. It becomes theorem evidence only after the exact Interface, realization, comparison, finite-certificate or direct-proof, Core, finite-to-infinite, Return, artifact, and claim-use obligations actually consumed by the implication have been proved.

Remark 3.2 (Relation to Appendices L and N). Appendix L supplies the frozen v20 Mirror/Transfer variance, comparison-cell, mate, metric, and ledger calculus. Appendix N supplies the frozen v20 derived, dg, sheaf, stack, descent, multiparameter, and higher-categorical extraction discipline. This appendix specializes those Technical Extension rules to the Problem Interface and may not weaken or silently import them into the Core.

Remark 3.3 (No exact-collapse restoration). All collapse-related comparison is expressed at the barcode/profile level through T_τ^{bar} , or through the repaired evaluation $\text{Eval}_{i,W,\tau}$. The sharp mathematical zero-transport boundary is $2\varepsilon \leq \tau$; $2\varepsilon < \tau$ is an operational reserve. Positive-radius transport through deletion additionally requires the two-sided threshold-gap certificate for the actual windowed pre-threshold profiles.

MS-C.1bis. Reflection-interface sovereignty and claim inflation control

Definition 3.4 (Reflection-interface sovereignty). Reflection-interface sovereignty is the rule that Mirror Reflection, transfer, duality, categorical comparison, derived equivalence, motivic comparison, sheaf or stack transformation, Fukaya-side structure, and higher coherence remain outside the AK Core unless they re-enter through declared verifier-side records. Admissible re-entry objects include, as invoked:

- (i) constructible one-parameter persistence or finite-type filtered objects;
- (ii) exact windowed pre-threshold and repaired post-threshold profiles;

- (iii) detector records at exactly one canonical source stage;
- (iv) admissible representatives and standard degree- n eligibility/edge records;
- (v) actual tower comparison morphisms and status-dependent Type IV records;
- (vi) full kernel/cokernel profiles and transient-defect certificates;
- (vii) theorem-supported metric-ledger entries bound to actual profile pairs;
- (viii) non-scalar obligation, finite-to-infinite, Return, manifest, and immutable-artifact records.

An external equivalence or interpretation is not a re-entry object merely by being mathematically meaningful.

Definition 3.5 (Reflection claim inflation). *Reflection claim inflation* occurs when a mirror candidate, comparison diagram, natural transformation, adjunction, derived or Morita equivalence, span, kernel transform, weak equivalence, or higher coherence cell is consumed at a stronger role than its proof, realization, scope, comparison mode, or artifact identity supports.

Proposition 3.6 (No reflection sovereignty override). *No Mirror, categorical, derived, sheaf, stack, Fukaya, motivic, or higher-categorical structure overrides a missing or failed verifier-side requirement.*

Proof. The Core verdict is a function of typed profiles, proofs, actual morphisms, detectors, ledgers, manifests, and claim permissions. External terminology creates none of these objects. A missing invoked obligation is therefore undefined, while a well-typed evidenced false obligation rejects at its declared route. $\square$

Proposition 3.7 (Diagrammatic elegance is not proof). *A commutative-looking diagram, mirror-symmetric picture, adjunction pattern, naturality square, or coherence sketch is not theorem evidence without the typed arrows, orientation, naturality, coherence, comparison mode, and artifacts used by the conclusion.*

Proof. Diagrammatic shape does not determine the arrows, their domains, invertibility, metric effect, or verifier-side projection. Those data are independent proof obligations. $\square$

Remark 3.8 (Reflection meaning is proposer-side by default). Reflection meaning may guide realization, window, threshold, parameter orientation, or candidate transport. It becomes verifier authority only through a proved Interface or Bridge Theorem and successful claim-use preflight.

MS-C.1ter. Frozen v20 dependency baseline, claim roles, and outcome typing

Definition 3.9 (MS-C v20 dependency baseline). Appendix MS-C is interpreted relative to Main Chapters 1–8, Foundation Appendices A–G, the frozen Technical Extension Appendices H–N, reviewed CM-v20 and TB-v20 statements only at their individually verified strength, Appendix MS-A v20.0.0, and Appendix MS-B v20.0.0. The CR-v20 baseline governs already registered claims but does not pre-promote Part IV. No v19 wording, theorem name, literature prestige, implementation output, or AI-generated analysis strengthens a v20 statement.

Definition 3.10 (MS-C local claim roles). Every cross-layer statement has exactly one local semantic role

interface_definition, interface_theorem, bridge_candidate, bridge_program,
bridge_theorem, toy_bridge_theorem, spec, non_claim.

Evidence status and invocation role are stored separately. A local label is not a canonical Claim UID before Part IV freeze, cross-audit, and final Claim Register extraction.

Definition 3.11 (MS-C outcome vector). The following fields are independent:

source/pre-gate, Interface, comparison, finite certificate/direct proof, Core audit,
Type IV, transient defect, finite-to-infinite reduction, Return, external conclusion.

Interface and Core outcomes use pass, reject, or undefined; subordinate clauses may use not_invoked only under an immutable scope exclusion. Comparison, detector, Bridge, Type IV, Return, and external-domain fields retain their own typed vocabularies.

Proposition 3.12 (No terminology-to-theorem upgrade). *The words mirror, equivalence, derived, stable, coherent, exact, stack, descent, Fukaya, or higher categorical do not promote a statement beyond its registered role, evidence, hypotheses, and artifact closure.*

Proof. The registered role, hypotheses, and artifacts determine permitted use. Terminology is not an inference rule and supplies no missing typed object or proof. $\square$

MS-C.2. Mirror Reflection setting

Definition 3.13 (Mirror Reflection setting). A *Mirror Reflection setting* consists of source and reflected environments $\mathcal{A}$ and $\mathcal{A}^\vee$, a typed comparison datum

$$\text{Mir} : \mathcal{A} \dashrightarrow \mathcal{A}^\vee,$$

its object and morphism domains, variance, parameter action, coefficient action, realization targets, comparison cells, purpose profile, and defect/obligation policy. The dashed arrow asserts no total functoriality.

Definition 3.14 (Mirror Reflection datum). A *Mirror Reflection datum* is

$$\mathfrak{M} = (A, A^\vee, \text{Mir}, \mathfrak{R}, \mathfrak{R}^\vee, \Theta, \Pi),$$

where A , $A^\vee$, and the realization routes are immutably identified; Θ contains the actual comparison cells or profile comparisons; and Π is the purpose-faithfulness, permitted-use, and forbidden-reading record.

Definition 3.15 (Mirror realization pair). A *Mirror realization pair* is

$$F_A := \mathfrak{R}(A), \quad F_{A^\vee} := \mathfrak{R}^\vee(A^\vee),$$

with source semantics, coefficients, filtrations, parameter orientation, constructibility, and artifact identity separately proved for each side. Extension-readable outputs require an additional extraction to the exact Core-readable objects consumed.

Remark 3.16 (Mirror Reflection is not assumed symmetric). The word mirror implies neither involution, equivalence, adjunction, duality, nor information preservation. The datum may be covariant, contravariant, partial, correspondence-valued, approximate, or advisory.

Remark 3.17 (Mirror candidate is not mirror theorem). Declaring $A^\vee$ a mirror candidate creates a comparison target only. A mirror theorem requires its own external hypotheses, proof, realization-faithfulness, comparison, and Return chain.

MS-C.2bis. Mirror directionality, variance, and involution discipline

Definition 3.18 (Mirror directionality record). A *Mirror directionality record* declares whether Mir is covariant, contravariant, adjoint-derived, span-valued, correspondence-valued, partially defined, multi-valued, or advisory; its object and morphism domains; its parameter-domain action; endpoint transport; coefficient action; inverse or pseudo-inverse status; and every failure case.

Definition 3.19 (Mirror involution claim). A *Mirror involution claim* states

$$\text{Mir}^\vee \circ \text{Mir} \simeq \text{id}$$

at one declared strength—strict identity, natural isomorphism, equivalence, controlled cell, or advisory analogy—and on one immutable scope. Units, counits, coherence, and artifact identities are mandatory when consumed.

Proposition 3.20 (Mirror terminology does not imply involution). *Mirror terminology does not imply involution, bidirectional equivalence, full faithfulness, essential surjectivity, conservativity, or information preservation.*

Proof. Those are independent categorical properties requiring inverse data and coherence or reflection theorems. $\square$

Proposition 3.21 (Mirror-side identity is external until realized). *A source-side identity, equivalence, or isomorphism becomes Core exact evidence only after transport to an actual equality or coherent profile isomorphism of the declared Core-readable outputs.*

Proof. The two statements live in different categories and have different scope and artifact identities. $\square$

Remark 3.22 (Mirror defect may be directional). The forward and reverse comparison cells may have different existence, coherence, metric modulus, and cost. No symmetric defect policy is inferred.

MS-C.2ter. Exact, anti-exact, adjunction, and correspondence regimes

Definition 3.23 (Covariant exact-reflection regime). A covariant exact-reflection regime supplies an exact functor between declared abelian realization categories, its parameter and filtration transport, and a theorem identifying the actual profile outputs. An exact- equivalence regime additionally supplies a quasi-inverse and coherent unit/counit isomorphisms.

Definition 3.24 (Contravariant mirror regime). A contravariant mirror regime supplies an exact functor

$$D : C_1^{\text{op}} \longrightarrow C_2$$

and explicit parameter reorientation, endpoint-decoration transport, and limit/colimit variance. An anti- equivalence does not produce a target comparison of the original colimit-to-apex form without a separately typed dual protocol.

Definition 3.25 (Adjunction-only regime). An adjunction-only regime consists of $L : C_1 \rightleftarrows C_2 : R$, its unit and counit, and the exact orientation used. No equivalence, exactness, conservativity, kernel preservation, metric control, or persistence invariance is assumed.

Definition 3.26 (Span or correspondence regime). A span, profunctor, or kernel correspondence records the actual correspondence object and every pullback, pushforward, derived correction, properness, finiteness, orientation, and Beck–Chevalley condition used. The correspondence object alone is not a transfer functor or theorem.

Proposition 3.27 (Adjunction is not equivalence or Type IV transport). *A left adjoint may preserve existing colimits, but an adjunction alone neither preserves kernels and cokernels nor reflects zero objects nor identifies a transformed apex. It therefore does not transport Type IV cleanliness by itself.*

Proof. Colimit preservation is formal for a left adjoint, whereas kernel and cokernel preservation require exactness, zero reflection requires conservativity, and apex identification is external data. $\square$

Proposition 3.28 (Anti-equivalence reverses the tower protocol). *An exact anti-equivalence transports colimits to limits and reverses arrows. A mirror use of such a transport requires a limit-to-coapex or other variance-correct protocol; the original colim $\rightarrow$ apex record cannot be reused by notation.*

Proof. Contravariance reverses arrows and exchanges cocones with cones. Hence the universal property and the orientation of the canonical comparison both change. $\square$

MS-C.3. Categorical comparison setting

Definition 3.29 (Categorical comparison setting). A *categorical comparison setting* consists of declared categories, bicategories, dg or A_∞ -categories, derived or stable categories, sheaf/stack environments, or higher categories C_1, C_2 ; a typed comparison functor, correspondence, span, adjunction, profunctor, kernel, or higher relation; realization and extraction targets; comparison cells; variance; exactness strength; defect semantics; and immutable artifacts.

Definition 3.30 (Categorical comparison datum). A *categorical comparison datum* is

$$\mathfrak{C} = (C_1, C_2, \text{Comp}, X_1, X_2, \Gamma, \Pi),$$

where Γ records actual cells and comparison modes and Π records purpose-faithfulness, semantic authority, permitted use, and forbidden stronger readings.

Definition 3.31 (Core-readable categorical comparison). A categorical comparison is *Core-readable* only when its theorem-backed realization or extraction produces the exact verifier-side profiles, representatives, detector fields, actual morphisms, ledger entries, or manifest components consumed by the claim. Merely possessing a categorical model is insufficient.

Remark 3.32 (Categorical comparison is not categorical equivalence). A comparison may be partial, one-sided, correspondence-valued, derived, or advisory. Equivalence is a separate theorem and must be recorded at its exact strength.

MS-C.3bis. Categorical equivalence and realization-faithfulness burden

Definition 3.33 (Categorical equivalence-strength record). A *categorical equivalence-strength record* declares exactly one proved relation, such as strict isomorphism, equivalence, exact equivalence, anti-equivalence, derived equivalence, Morita equivalence, quasi-equivalence, adjunction, fully faithful functor, conservative functor, span/correspondence, or advisory comparison. It separately records monoidal, triangulated, t -exact, heart-preserving, filtered, enriched, parameter, and higher-coherence properties.

Definition 3.34 (Realization-faithfulness burden). The *realization-faithfulness burden* identifies the external categorical feature, the Core-visible feature, the direction of use, coefficient and parameter conventions, source class, naturality scope, proof or controlled mismatch, and exact failure boundary. It is feature-relative and direction-relative; it is not global categorical faithfulness unless separately proved.

Proposition 3.35 (Categorical equivalence is not profile equality). A *categorical, derived, Morita, or quasi-equivalence does not automatically imply equality, interleaving, threshold-gap certification, representative compatibility, or Type IV transport for realized persistence data.*

Proof. The repaired profile depends on realization, coefficient passage, filtration, parameter orientation, window, and threshold. The equivalence preserves only the structures stated in its strength record. $\square$

Proposition 3.36 (Unproved realization-faithfulness blocks transport). *If realization-faithfulness is speculative, unavailable, failed, or conditionally unresolved for a consumed feature, the corresponding transport or Return route is undefined and cannot be theorem evidence.*

Proof. The cross-layer identification is itself the missing theorem. $\square$

Remark 3.37 (Equivalence strength is feature-relative). A functor may preserve triangulated structure while failing to preserve a chosen filtration, persistence parameter, threshold-gap certificate, degree- n eligibility, heart kernel/cokernel, or tower apex.

MS-C.3ter. Semantic authority and purpose profiles

Definition 3.38 (Reflection purpose profile). A reflection purpose profile records the external feature, the exact Core-visible predicate, direction, monitored degrees, windows, thresholds, parameter orientation, comparison mode, required Return conclusion, source class, and prohibited stronger reading.

Definition 3.39 (Reflection semantic-authority record). A semantic-authority record proves that the realized objects, cells, and morphisms represent the external relationships named in the purpose profile. Its outcome is pass, reject, or undefined. A missing semantic-authority proof cannot be replaced by an object-level equivalence or a small metric cost.

Proposition 3.40 (Purpose declaration is not purpose-faithfulness). *Stating that a mirror or categorical construction is intended to preserve a feature does not prove preservation or reflection of that feature.*

Proof. An intended interpretation contains no proof that the realization preserves or reflects the named predicate. $\square$

MS-C.4. Realization targets

Definition 3.41 (Mirror/Categorical realization target). A Mirror or categorical interface declares each target and its role:

- (i) a finite-type filtered object in $\text{FiltCh}(k)$;
- (ii) a constructible persistence object in $\text{Pers}_k^{\text{cons}}$;
- (iii) a windowed pre-threshold or repaired post-threshold profile;
- (iv) an admitted detector object or certificate;
- (v) an admissible representative $C_{\tau, W}F$;
- (vi) a standard degree- n eligible realized object and edge record;
- (vii) an actual tower comparison morphism and defect profiles;
- (viii) a derived, sheaf, stack, dg, A_∞ , stable, or higher object in a Technical Extension setting;
- (ix) a canonical manifest/certificate component;
- (x) an advisory, Search, visualization, or calibration artifact.

The last class is non-verdict by default.

Definition 3.42 (Core-testable realization target). A target is *Core-testable* only when it supplies the exact typed input of an invoked Core theorem or verifier clause, including its profile identity, scope, coefficient field, parameter convention, and artifact identity.

Definition 3.43 (Extraction route). An *extraction route* is a theorem-backed map, functor, construction, or protocol from an extension- readable object to the exact Core-readable profile, representative, actual morphism, detector record, ledger entry, or manifest field consumed. It records all discarded structure and its effect classification.

Proposition 3.44 (No Core use without Core-testable target). A Mirror or categorical comparison cannot support Core certification without a Core-testable target or a proved extraction route to the consumed verifier-side data.

Proof. Without such a target the verifier has no typed input on which to evaluate the invoked clause. $\square$

Remark 3.45 (Technical-extension target is not automatically Core target). Derived, sheaf, stack, dg, A_∞ , stable, or higher structure remains Technical Extension data until a scope-matched theorem extracts the exact Core-readable record.

MS-C.4bis. Extraction, filtration, and one-parameter reduction

Definition 3.46 (Reflection extraction defect). A reflection extraction effect $\delta^{\text{ref-extract}}$ is classified as exactly one of:

- (i) an actual metric discrepancy between identified Core-readable profiles;
- (ii) a morphism-profile discrepancy requiring its own support theorem;
- (iii) theorem-controlled nonmetric semantic loss;
- (iv) a non-scalar existence, typing, naturality, coherence, or identity obligation;
- (v) advisory information;
- (vi) unavailable control.

Only the first two may enter the strict metric ledger, and only with an Appendix-M-compatible bound on the actual profile pair.

Definition 3.47 (Categorical one-parameter reduction). A *categorical one-parameter reduction* records all source parameters, their orders, the selected persistence direction and orientation, normalization, ties and incomparable values, branches, skipped strata, endpoint conventions, transition morphisms, information discarded, constructibility and pointwise finite- dimensionality, family membership, non-adaptivity, and metric/nonmetric effects.

Proposition 3.48 (Extension-readable does not imply Core-readable). *An extension-readable object is not Core-readable until the exact theorem-backed extraction and every invoked verifier-side obligation are supplied.*

Proof. The external object and the Core input belong to different typed environments. $\square$

Proposition 3.49 (One-parameter reduction is not lossless by default). *A one-parameter reduction is not exact, faithful, conservative, or obstruction-reflecting by default. Discarded gradings, higher cells, automorphisms, stack structure, or A_∞ -operations are non-scalar losses unless a theorem proves irrelevance or constructs an actual metric profile bound.*

Proof. The omitted structures may carry the very obstruction or coherence used by the external conclusion. $\square$

Remark 3.50 (Common reflection extraction risks). Typical risks include forgetting A_∞ -operations, changing filtration conventions, replacing a stack by a coarse moduli object, projecting a multiparameter family to one scale, or identifying energy and barcode filtrations without a comparison theorem.

Proposition 3.51 (Right-unbounded reflection detector boundary). *A right-unbounded extracted parameter range with unbounded possible births has no uniform complete finite detector merely because finitely many categorical objects, charts, thresholds, or tower stages were sampled. A finite certificate requires a finite-birth, stabilization, family, descent, global-bound, or finite-to-infinite theorem defeating the Core no-go boundary.*

Proof. A bar or defect may be born beyond every sampled stage. Completeness therefore requires an additional theorem ruling out such unseen births or reducing them to finite data. $\square$

MS-C.5. Mirror comparison cells

Definition 3.52 (Mirror comparison cell). A *Mirror comparison cell* is an actual natural transformation, pseudo-natural transformation, modification, homotopy, invertible or noninvertible higher cell, weak equivalence, span comparison, kernel comparison, or controlled correspondence

$$\alpha_{\text{Mir}} : P \Longrightarrow Q$$

with declared source and target composites, orientation, scope, naturality, composition law, and artifact identity.

Definition 3.53 (Mirror comparison cost). A *Mirror comparison cost* $\delta(\alpha_{\text{Mir}})$ has a declared codomain and semantics. It becomes a metric-ledger value only after certified conversion to the active quantale and an upper-bound theorem for the actual profile discrepancy consumed.

Definition 3.54 (Mirror comparison status). A Mirror comparison uses exactly one local status

strict, exact_isomorphic, metric_controlled, two_cell_controlled,
advisory_only, not_compared, failed, undefined.

These are local typing tokens, not Core atomic statuses.

Definition 3.55 (Mirror comparison ledger). The symbol

$$\delta^{\text{Mir}} := \bigoplus_{\alpha \in A_{\text{Mir}}} \delta(\alpha)$$

is meaningful only for the finite or theorem-summable family of primitive metric contributions actually traversed, after common-quantale conversion, dependency-aware path aggregation, and no-double-counting checks. Alternative comparison paths are not summed merely because they exist.

Proposition 3.56 (Only admissibly projected Mirror comparisons may support certification). *A Mirror comparison supports an invoked certified route only when its local status projects to Core mode exact or metric_gap_certified with every required equality, coherent isomorphism, metric bound, threshold-gap, and artifact obligation. Advisory, not-compared, failed, and undefined routes do not support the affected claim.*

Proof. The Core consumes the projected comparison evidence, not the local categorical label. $\square$

Remark 3.57 (Advisory Mirror comparison is non-verdict). An advisory comparison may guide Search or model selection but cannot alter a verifier-side value.

MS-C.5bis. Comparison-mode projection, zero reflection, and coherence

Definition 3.58 (Reflection comparison mode). Every consumed reflection comparison resolves to exactly one canonical Core mode

exact, metric_gap_certified, not_compared.

Local strict or exact_isomorphic projects to exact only with the actual equality or coherent profile isomorphism. Local metric_controlled projects to metric_gap_certified only with a certified bottleneck bound on the actual pre-threshold pair and the complete predicate-specific threshold-gap chain. A controlled higher cell has no automatic projection.

Definition 3.59 (Status-mode compatibility for reflection comparisons). The record stores local status, Core mode, comparison orientation, zero-reflection status, coherence status, and metric-support status separately. failed rejects an invoked path; undefined leaves it undefined; not_compared is valid only when the claim genuinely consumes no comparison.

Proposition 3.60 (Strict source comparison is not Core exact mode). *A strict equality, equivalence, or zero-cost cell in the source domain is not Core exact mode until realized as the actual equality or coherent isomorphism of the declared Core-readable outputs.*

Proof. Source and verifier relations have different types and may use different filtrations, parameters, or artifacts. $\square$

Proposition 3.61 (Controlled comparison is not metric certification by name). *A declared cost does not yield metric-gap certification unless it is converted to an actual profile bound and all two-sided threshold-gap and zero-separation hypotheses are proved.*

Proof. A categorical cost may be nonmetric, non-zero-reflecting, or insensitive to the barcode discrepancy consumed. $\square$

Proposition 3.62 (Zero cost is not exactness without zero reflection). *The equation $\delta(\alpha) = 0$ does not imply strict or exact-isomorphic commutation unless the cost semantics is zero-reflecting on the declared cell class and all required coherence is present.*

Proof. A cost map may identify distinct or noninvertible cells, while categorical coherence is not numerical. Zero reflection and coherence are therefore independent hypotheses. $\square$

MS-C.5ter. Adjunction mates and Beck–Chevalley boundaries

Definition 3.63 (Typed reflection adjunction square). For adjunctions $L : \mathcal{A} \rightleftarrows \mathcal{B} : R$ and $L' : \mathcal{A}' \rightleftarrows \mathcal{B}' : R'$, a left-oriented cell $\alpha : L'Q \Rightarrow SL$ and its right mate $\alpha^b : QR \Rightarrow R'S$ are recorded with units, counits, orientation, and scope.

Definition 3.64 (Reflection Beck–Chevalley certificate). A Beck–Chevalley certificate proves invertibility of the mate in the orientation actually consumed, together with naturality, coherence, coefficient, parameter, and artifact records. Invertibility of a differently oriented cell is not a substitute.

Proposition 3.65 (Taking mates does not preserve metric cost by default). *No equality or inequality between the costs of a cell and its mate follows without a mate-compatible cost or Lipschitz theorem. A ledger entry for one orientation cannot be reused silently for the other.*

Proof. Mate correspondence is defined using units and counits in hom-categories. A quantitative cost is extra structure and need not be invariant under those operations. $\square$

Proposition 3.66 (Correspondence is not push–pull theorem). *A span or kernel correspondence does not supply a push–pull, base-change, or transfer theorem until every functor, derived correction, domain, properness/finiteness hypothesis, and consumed mate is defined and proved.*

Proof. The correspondence object alone does not define the required functors, their domains, exactness, or base-change cells. $\square$

MS-C.6. Categorical comparison cells and controlled composition

Definition 3.67 (Categorical comparison cell). *A categorical comparison cell α_{cat} compares two fully typed categorical pipelines, such as $Q' \text{Comp}$ and $\text{Comp}_Q Q$, with every object, arrow, variance convention, whiskering operation, and scope displayed.*

Definition 3.68 (Categorical comparison cost). *A categorical cost $\delta(\alpha_{\text{cat}})$ is a typed value with declared composition and whiskering laws. It is not a metric discrepancy merely because it is numerical.*

Definition 3.69 (Categorical comparison ledger). *The aggregate*

$$\delta^{\text{cat}} := \bigoplus_{\alpha \in A_{\text{cat}}} \delta(\alpha)$$

exists only under the active quantale, certified primitive conversions, composition laws, whiskering moduli, coherence, and dependency graph. Otherwise no verifier-side aggregate is defined.

Proposition 3.70 (Categorical comparison effects use typed routes). *Only theorem-supported actual metric or morphism-profile discrepancies affect the strict Core budget through the ledger. Coherence, existence, naturality, source identity, permission, and Return failures are non-scalar blockers; advisory values remain non-verdict.*

Proof. The metric budget upper-bounds identified profile discrepancies. The other obligations do not have that type. $\square$

Remark 3.71 (Categorical naturality is not free). *Naturality, pseudonaturality, associativity, interchange, and whiskering control are separate obligations. A pasted diagram has no aggregate defect without proved laws covering the paste.*

Theorem 3.72 (Controlled pasting boundary). *A finite pasting of comparison cells has the bound prescribed by the declared vertical/horizontal composition laws, whiskering moduli, and coherence certificates. In the additive scalar special case with non-expansive whiskering, the pasted cost is at most the sum of the primitive costs. Without those hypotheses no aggregate bound is licensed.*

Proof. Apply the certified laws in the order of the pasting diagram. Coherence identifies the parenthesized composites to which each inequality applies. $\square$

MS-C.6bis. Parameter transport and orientation discipline

Theorem 3.73 (Exact affine transport of repaired reflection profiles). *Let $r(t) = at + c$ with $a > 0$. For finite barcodes and right-open windows,*

$$r_* W_W B = W_{r(W)} r_* B, \quad r_* T_{\tau}^{\text{bar}} B = T_{a\tau}^{\text{bar}} r_* B, \quad d_B(r_* B, r_* C) = a d_B(B, C).$$

Hence repaired evaluation, zero transport, and metric reserve are preserved after coherent scaling of windows, thresholds, and radii.

Proof. Increasing affine transport preserves right-open endpoint order and multiplies every finite length, endpoint displacement, and diagonal matching cost by a . $\square$

Proposition 3.74 (Nonlinear reparameterization has no default constant threshold). *For a general increasing nonlinear parameter map, equal source lengths at different births may acquire different target lengths. No universal constant threshold scaling is therefore inferred.*

Proof. For example, under a nonlinear increasing map, two intervals of equal source length at different births can acquire different target lengths. A birth-independent threshold scale cannot represent both. $\square$

Proposition 3.75 (Order reversal requires a dual persistence regime). *An order-reversing mirror parameter map reverses morphisms and changes endpoint decorations. It requires a typed anti-equivalence, dualization, and parameter reorientation rather than ordinary forward-persistence relabeling.*

Proof. Precomposition with an order-reversing map changes covariance and endpoint orientation, so ordinary forward persistence is not preserved by relabeling alone. $\square$

MS-C.7. Mirror/categorical comparison with repaired evaluation

Definition 3.76 (Mirror-evaluation reflection). A *Mirror-evaluation reflection* compares the fully typed composites

$$A \mapsto \text{Mir}(A) \mapsto \text{Eval}_{i,W^\vee,\tau^\vee}(\mathfrak{R}^\vee \text{Mir}(A))$$

and

$$A \mapsto \text{Eval}_{i,W,\tau}(\mathfrak{R}(A)) \mapsto \text{Mir}_{i,W,\tau}(\text{Eval}_{i,W,\tau}(\mathfrak{R}(A))),$$

with parameter, coefficient, degree, endpoint, and threshold transport declared.

Definition 3.77 (Categorical evaluation comparison). A categorical evaluation comparison compares $\text{Eval}(\text{Comp}(X))$ with $\text{Comp}_{\text{Eval}}(\text{Eval}(X))$ through an actual comparison cell or actual profile comparison. Shared notation does not identify the two composites.

Definition 3.78 (Evaluation reflection defect). The effect $\Delta_{\text{Eval},\text{ref}}$ is classified before use as an exact comparison, actual metric profile discrepancy, morphism-profile discrepancy, theorem-controlled nonmetric effect, non-scalar blocker, advisory value, or unavailable control.

Proposition 3.79 (Evaluation reflection is not automatic). *Mirror or categorical comparison does not automatically commute with persistence extraction, window restriction, or hard threshold deletion.*

Proof. The comparison may alter filtrations, parameters, endpoint decorations, bar lengths, or realization targets. $\square$

Proposition 3.80 (Evaluation reflection defect uses its typed route). *A nonzero actual metric evaluation discrepancy enters the declared ledger with a support theorem. A nonmetric or non-scalar evaluation effect follows its own obligation and cannot be scalarized by naming it a defect.*

Proof. Metric discrepancy and semantic/coherence failure have different codomains and failure behavior. $\square$

Remark 3.81 (Preferred Core route). Direct reconstruction of $B_{i,W}$ and $\text{Eval}_{i,W,\tau}$ on the actual Core-readable object is preferred. Every shortcut bears its complete comparison burden.

MS-C.7bis. Window, crop, threshold-gap, and exact-transport discipline

Definition 3.82 (Reflection metric record). A reflection metric record identifies the exact windowed pre-threshold profile pair, window, threshold, bottleneck bound, crop and window-edge effects, scalarization, protected family, two-sided closed-band exclusion, strict reserve, dependency graph, and immutable artifacts.

Proposition 3.83 (Source-level closeness is not profile-level closeness). *A source distance, homotopy size, energy estimate, categorical cost, or analogy score does not imply the bottleneck bound required by the Core unless a theorem or verified computation bounds the actual pre-threshold profile pair.*

Proof. The source quantity may fail to control filtration shifts, crop effects, endpoint changes, or barcode lengths. $\square$

Proposition 3.84 (Exact reflection comparison is not robustness). *An exact comparison at one profile pair does not prove positive-radius robustness. Robustness requires its own metric support and threshold-gap record.*

Proof. Exactness is a radius-zero statement. Positive-radius robustness additionally requires an actual metric bound and exclusion of threshold crossings. $\square$

Remark 3.85 (Reflection gap protection is measured after windowing). Every deletion-crossing gap is computed on the actual windowed pre-threshold profiles, never on source objects, unwindowed outputs, or already deleted profiles.

Theorem 3.86 (Exact realized reflection transport). *Suppose a reflection comparison supplies an actual isomorphism $\beta : B_{i,W}(F_A) \xrightarrow{\sim} B_{i,W}(F_{A^\vee})$ preserving interval decorations and the fixed threshold convention. Then it induces an isomorphism*

$$\text{Eval}_{i,W,\tau}(F_A) \xrightarrow{\sim} \text{Eval}_{i,W,\tau}(F_{A^\vee}).$$

This theorem transports only the displayed repaired profile predicate and no unlisted external semantics.

Proof. Apply barcode-level hard deletion occurrence-wise to the profile isomorphism. $\square$

Theorem 3.87 (Controlled repaired-zero reflection). *Suppose the actual windowed pre-threshold profiles have bottleneck distance at most ε , both avoid the threshold band required by the v20 two-sided gap theorem, and the registered zero-separation hypotheses hold. Then repaired zero transports in the proved direction; in particular the sharp zero boundary uses $2\varepsilon \leq \tau$. No raw detector rank or external categorical invariant is transported.*

Proof. Apply the inherited v20 two-sided threshold-gap transport theorem to the actual windowed pre-threshold pair. The zero-separation hypothesis yields the stated repaired-zero implication; no theorem in that argument controls raw detector ranks or external invariants. $\square$

MS-C.8. Mirror/categorical comparison with barcode-level collapse

Definition 3.88 (Mirror-collapse reflection). A *Mirror-collapse reflection* compares applying T_τ^{bar} before or after the realized Mirror operation, with the exact barcode/profile source, target, threshold transport, and comparison cell specified.

Definition 3.89 (Categorical collapse comparison). A categorical collapse comparison compares $T_\tau^{\text{bar}} \text{Comp}(X)$ and $\text{Comp}_\tau(T_\tau^{\text{bar}} X)$ only after both expressions are typed in the declared profile category.

Definition 3.90 (Collapse reflection defect). The effect $\Delta_{T,\text{ref}}$ is classified as exact, actual metric profile discrepancy, theorem-controlled nonmetric effect, non-scalar blocker, advisory, or unavailable. The symbol alone is not a ledger entry.

Proposition 3.91 (Collapse reflection is not automatic). *Mirror or categorical comparison does not automatically commute with hard barcode deletion.*

Proof. The comparison may change bar lengths, thresholds, endpoint decorations, or realization targets. $\square$

Proposition 3.92 (Collapse reflection defect uses its typed route). *Only a theorem-supported actual metric collapse discrepancy enters the strict ledger. Missing commutation, coherence, typing, or semantic authority remains a non-scalar blocker.*

Proof. A metric ledger bounds an actual profile discrepancy. Missing typing or coherence concerns existence of the comparison and has no scalar metric value by default. $\square$

Remark 3.93 (Collapse comparison is weaker than evaluation comparison). A comparison of deletion alone does not control persistence extraction or window restriction. The full Core-relevant object is $\text{Eval}_{i,W,\tau}$.

MS-C.9. Mirror/categorical comparison with standard higher eligibility

Definition 3.94 (Eligible categorical realization). Fix $n \geq 1$. A Mirror or categorical realization is eligible for the standard v20 degree- n Core edge only when it supplies an admissible representative $C_{\tau, W}F$, an actual degree- n profile isomorphism, a meaningful realized object

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod}),$$

a fixed edge extractor with zero compatibility, an artifact-backed identification of $H^{-n}(A_{n;F,W,\tau})$ with the extracted repaired profile, and all coefficient, naturality, representative-change, realization-change, family, non-adaptivity, and artifact records. The case $n = 1$ specializes to the inherited two-term regime.

Proposition 3.95 (Mirror/categorical comparison does not replace eligibility). *A Mirror, derived, or categorical equivalence does not replace any invoked standard degree- n eligibility obligation.*

Proof. Eligibility concerns a specific representative, amplitude, extractor, and edge identification. An external comparison may help prove these facts but does not contain them by default. $\square$

Remark 3.96 (Comparison may support eligibility, not replace it). A theorem may transport eligibility only when it preserves the actual representative, amplitude, profile class, extractor, edge map, and zero compatibility at the declared scope.

Proposition 3.97 (No higher converse from equivalence alone). *Neither categorical equivalence nor standard degree- n eligibility implies $\text{Ext}^n = 0 \Rightarrow \text{Eval}_n = 0$. A separate natural zero-reflection theorem on the declared profile class is required.*

Proof. The standard edge theorem is one-way. An equivalence of ambient categories does not make the edge extractor zero-reflecting on the realized profile class. $\square$

Remark 3.98 (No synthetic higher bridge). Encoding a profile as a shifted complex may yield a toy identity; it does not create independent mirror, motivic, Fukaya, or arithmetic meaning or a natural external PH-to-Ext bridge.

MS-C.10. Reflection and status-dependent Type IV

Definition 3.99 (Mirror Type IV proxy). A *Mirror Type IV proxy* is a reflection-side signal used to prioritize an actual morphism-level calculation. It has no diagnostic value until a theorem identifies it with the canonical defect profiles.

Definition 3.100 (Categorical Type IV proxy). A *Categorical Type IV proxy* is an object, scalar, fiber, cofiber, cone, equivalence score, descent obstruction, or higher cell suspected to correlate with Type IV. It is advisory until the required identification theorem is proved.

Definition 3.101 (Actual Type IV diagnostic for reflection data). Fix one window–threshold pair (W, τ) , one finite monitored degree set I_{mon} , a declared tower scope, and exactly one status

$$\text{not_invoked}, \quad \text{undefined}, \quad \text{certified_zero}, \quad \text{obstructed}.$$

For `not_invoked`, the record gives a scope rationale and no numerical diagnostics. For `undefined`, it records the missing or invalid obligations and no numerical diagnostics. For `certified_zero` or `obstructed`, it contains:

- (i) valid actual morphisms $\phi_{i,W,\tau}$, source/target identities, provenance, source semantics, semantic authority, and an ambient abelian profile category;
- (ii) terminal mode `bounded_terminal` or `right_unbounded`;
- (iii) terminal-tameness and full kernel/cokernel profile artifacts;
- (iv) degree-wise

$$\mu_{i,W,\tau} = \text{tgdim}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} = \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}),$$

and totals over the fixed finite set I_{mon} ;

- (v) a separate full-interior-defect status `not_checked`, `certified_absent`, or `certified_present`;
- (vi) when transient detection is invoked, a separate record containing $\tau_{\text{def}} > 0$, the component and canonical defect stage, the detector information class, certificate and value statuses, and the aggregate long-transient status. The component is one of `full_profile` or `interior_noncofinal`; the stage is one of `kernel_defect` or `cokernel_defect`.

The terminal status is determined only by the fixed-scope totals $(\mu_{\text{Collapse}}(W, \tau), \nu_{\text{Collapse}}(W, \tau))$. Aggregation across several windows, thresholds, or infinitely many degrees requires a separate theorem.

Proposition 3.102 (Mirror/categorical Type IV proxy is not Type IV diagnosis). *No Mirror or categorical proxy determines terminal Type IV zero or obstruction without the actual comparison morphism, terminal semantics, and kernel/cokernel profiles.*

Proof. Type IV is a property of the actual morphism, not of its source and target objects or an external proxy. $\square$

Remark 3.103 (Correct use of reflection proxies). A proxy may trigger construction of the canonical artifact. It does not supply the resulting status.

Definition 3.104 (Reflection transient-defect record). A reflection transient-defect record inspects the actual full kernel or cokernel profile at one canonical defect stage. Detector proof status, mathematical value, full-interior-defect status, and aggregate long-transient status are stored independently.

Proposition 3.105 (Terminal zero is not transient or full zero). *Terminal `certified_zero` does not imply long-transient absence, short-defect exclusion, full kernel/cokernel vanishing, or comparison-map isomorphism.*

Proof. Terminal-generic dimension ignores defect bars that die before the terminal region. Such interior or short bars may remain nonzero. $\square$

MS-C.10bis. Type IV transport and artifact coherence for reflection interfaces

Definition 3.106 (Reflection Type IV artifact-coherence record). A *Reflection Type IV artifact-coherence record* binds the monitored scope, tower objects, transition maps, apex, comparison morphisms, terminal mode, parameter transport, realization version, and all defect profiles to one content-addressed artifact family.

Proposition 3.107 (Changing reflection artifacts changes the diagnostic claim). *Changing the comparison morphism, apex, terminal mode, realization, parameter map, window, threshold, or monitored scope creates a new Type IV claim and makes the prior artifact stale for that claim.*

Proof. The changed fields alter the identity or semantics of the morphism and its defect profiles. $\square$

Remark 3.108 (Reflection proxy escalation). Proxy evidence may justify the expense of an actual artifact but cannot predetermine its verdict.

Theorem 3.109 (Exact equivalence transport of Type IV). *Let Q be an exact equivalence of declared abelian profile categories. Suppose it transports the complete directed tower, all transition maps, the declared apex, apex agreement, and each actual comparison morphism, and that the transported terminal convention and tameness records agree. Then*

$$Q(\ker \phi_i) \cong \ker Q(\phi_i), \quad Q(\text{coker } \phi_i) \cong \text{coker } Q(\phi_i),$$

and the transported fixed-scope Type IV diagnostics agree. Transient and full-interior statuses transport only when their complete detector/profile records are transported as well.

Proof. An exact equivalence preserves kernels and cokernels up to canonical isomorphism. The stated tower and apex hypotheses identify the transformed comparison with the declared target artifact. Terminal-generic dimension is invariant under compatible profile isomorphism. $\square$

Proposition 3.110 (Object equivalence does not transport morphism defects). *Equivalences of source and target objects do not compare arbitrary morphisms. A conjugacy or coherent square is required before kernels, cokernels, fibers, cofibers, or Type IV records may be transported.*

Proof. Even with identical source and target objects, the identity and zero morphisms have different kernels and cokernels. A coherent square relating the actual maps is therefore necessary. $\square$

Proposition 3.111 (Anti-equivalence requires dual Type IV Return). *An anti-equivalence exchanges kernels with variance-correct cokernels and colimits with limits. Any external claim based on that dual artifact requires a separately stated dual diagnostic and Return protocol; it is not the canonical colimit-to-apex Type IV record by default.*

Proof. Contravariance reverses the comparison and exchanges the relevant universal constructions. The external interpretation of the dual defect consequently needs a separately proved soundness arrow. $\square$

MS-C.10ter. Finite detectors, descent, and finite-to-infinite reflection

Definition 3.112 (Reflection finite-detector record). A finite detector record declares one canonical source stage, soundness, zero-completeness, birth/germ or endpoint exhaustiveness, stencil/family coverage, endpoint/cofinal policy, family membership, excess-bound provenance, non-adaptivity, and separate certificate and value statuses.

Proposition 3.113 (Finite categorical observation is not finite proof). *Finitely many objects, charts, cells, thresholds, tower stages, or numerical comparisons are not a finite proof of a global, family-wide, terminal, or mirror theorem without a direct theorem or complete detector/reduction package.*

Proof. Finiteness of the displayed data does not prove coverage of every unobserved object, birth, overlap, or terminal stage. $\square$

Definition 3.114 (Reflection certificate-descent record). A certificate-descent record contains local verifier records and coherent identifications of scopes, windows, thresholds, profile identities, detectors, representatives, actual morphisms, ledgers, artifacts, and outcome fields on every required overlap and higher intersection.

Proposition 3.115 (Pairwise agreement and local pass do not imply global pass). *Pairwise object isomorphisms do not establish higher cocycle coherence, and a list of locally passing certificates does not prove global artifact identity, global budget closure, descent effectivity, or finite-to-infinite closure.*

Proof. Higher cocycle coherence and global conjunctive obligations are not determined by pairwise isomorphisms or local verdict labels. $\square$

Definition 3.116 (Reflection finite-to-infinite record). Whenever finite reflection evidence is used for an infinite or terminal conclusion, the record distinguishes finite observation, prefix agreement, scalar convergence, metric Cauchy behavior, eventual constancy, limit/colimit existence, declared apex, apex agreement, certificate descent, and external Return. It contains the actual system maps, reduction theorem, compatible cone/cocone, apex comparison, and all derived-limit or Mittag-Leffler obligations.

Proposition 3.117 (Finite prefix or categorical convergence is not Return). *A finite prefix, homotopy colimit, derived limit, Cauchy estimate, effective descent, or observed stabilization proves neither apex agreement nor an external Mirror/categorical conclusion without the separately registered identification and Return arrows.*

Proof. Each listed construction supplies at most an ambient object or finite approximation. The declared apex identity and external semantic implication are separate arrows. $\square$

MS-C.11. Reflection transport and Return statements

Definition 3.118 (Mirror transport statement). A Mirror transport statement is one directional implication between an exactly typed external Mirror predicate and an exactly typed verifier-side predicate. The forward Interface direction and reverse Return direction are different claims.

Definition 3.119 (Categorical transport statement). A categorical transport statement is one directional implication between a declared categorical predicate and a declared Core-visible predicate, with source class, variance, coefficients, parameter scope, comparison mode, and prohibited converse fixed.

Definition 3.120 (Reflection transport theorem). A *Reflection transport theorem* records source and target semantics, exact direction, source class and hypotheses, purpose-faithful realization, parameter/coefficient transport, exact Core predicate and stage, comparison cells and modes, detector/eligibility/Type IV/reduction obligations, metric and nonmetric effects, proof, immutable artifacts, failure routes, and forbidden stronger readings.

Definition 3.121 (Default status of Mirror/categorical transport). An unproved reflection implication remains `bridge_candidate`, `bridge_program`, or `spec`; it is not theorem evidence.

Proposition 3.122 (Mirror/categorical transport does not enlarge the Core). A *proved Problem-Interface transport theorem* remains outside the Core unless a future release imports and reproves it under a conservative-extension and Claim Register action.

Proof. Problem-specific semantics and Return conclusions are not part of the frozen Core by appearance alone. $\square$

Remark 3.123 (Transport directions must be separated). External-to-Core realization and Core-to-external Return have different sources, targets, and soundness burdens. Neither follows from the other.

Definition 3.124 (Reflection Return Theorem). A *Reflection Return Theorem* proves

$$Q_{AK}(X) \implies C_{ref}(X),$$

where the antecedent is the exact audited AK conclusion actually established and the consequent is the exact Mirror/categorical external conclusion. It records every semantic identification, finite-to-infinite scope, comparison, hypothesis, proof, artifact, and failure boundary.

Definition 3.125 (Complete reflection theorem-bearing chain). A complete theorem-bearing route has the form

$$\begin{aligned} X_{ext} &\longrightarrow E_X \longrightarrow F_X \longrightarrow B_{i,W}(F_X) \longrightarrow \text{Eval}_{i,W,\tau}(F_X) \\ &\longrightarrow Q_{fin}(X) \longrightarrow Q_{AK}(X) \longrightarrow \text{Verdict}_{AK}(X) \xrightarrow{\text{Return}} C_{ref}(X). \end{aligned}$$

Here E_X is an optional higher enhancement. Every arrow has a typed theorem, role, status, scope, and artifact. Eligibility, Type IV, transient, descent, or finite-to-infinite branches enter only through their own complete packages.

MS-C.11bis. Transport theorem promotion and external theorem boundary

Definition 3.126 (Core-to-reflection transport theorem). A Core-to-reflection theorem is precisely a Reflection Return Theorem whose antecedent is bound to the canonical manifest and exact audited verdict fields, not a verbal summary such as “Core pass.”

Definition 3.127 (Reflection theorem promotion). Promotion from [Spec] or Bridge Program to theorem evidence requires a complete proof of every invoked arrow, exact claim role and permitted use, dependency and artifact closure, successful claim-use preflight, and final CR-v20 registration after Part IV freeze. Registration without proof does not promote.

Proposition 3.128 (No external theorem by successful reflection certification). A *successful Interface or Core certificate* does not prove mirror symmetry, categorical or derived equivalence, motivic equivalence, Langlands correspondence, modularity, automorphy, or a Fukaya theorem without the exact external Return Theorem.

Proof. The certificate concerns the realized verifier-side shadow; the external theorem has different semantics. $\square$

Proposition 3.129 (Transport hypotheses are non-scalar). *Missing realization, semantic authority, naturality, coherence, descent, finite-to-infinite, permission, or Return hypotheses cannot be repaired by metric slack, zero cost, local pass, or diagrammatic elegance.*

Proof. These are existence and logical obligations, not profile distances. $\square$

Proposition 3.130 (Missing arrow blocks reflection conclusion). *If any required external-realization, extraction, comparison, certificate, Core, reduction, audit, or Return arrow is missing or undefined, the advertised external conclusion remains unlicensed.*

Proof. A composition of implications is defined only when every required arrow exists with matching source, target, and hypotheses. $\square$

MS-C.12. Reflection certificate, manifest, and artifact records

Definition 3.131 (Mirror Reflection certificate record). A Mirror Reflection certificate record is the canonical proof-object projection of one Mirror datum, its purpose-faithful realization, every consumed comparison and verifier-side field, and all immutable artifacts. It does not contain an external theorem merely because its source is a mirror construction.

Definition 3.132 (Categorical comparison certificate record). A categorical comparison certificate record is defined analogously for one categorical comparison datum, with exact variance, equivalence strength, cells, extraction, and purpose profile.

Definition 3.133 (MS-C manifest entry). An MS-C manifest entry records at least:

- (i) source/target objects and categories, semantic authority, content-addressed source identity, role, evidence status, invocation role, permitted use, and forbidden reading;
- (ii) Mirror/comparison operation, variance, domain, coefficients, parameter and endpoint transport, filtration, constructibility, and realization-faithfulness;
- (iii) actual realized objects, extraction routes, windows, thresholds, degrees, pre-threshold and repaired profile identities;
- (iv) comparison cells, mates, units/counits, Beck–Chevalley data, local status, Core mode, zero-reflection, coherence, and metric support;
- (v) detector, representative, degree- n eligibility, and zero-reflection fields when invoked;
- (vi) tower semantics, transition maps, apex, actual comparison morphism, terminal records, diagnostics, and separately typed full-interior/transient fields when invoked;
- (vii) descent, atlas/Restart, finite-to-infinite, derived-limit, and apex-agreement fields when invoked;
- (viii) effect classification, actual profile support, quantale, scalarization, path/dependency graph, threshold clearances, strict reserve, aliases, and no-double-counting records;
- (ix) exact Return claim and theorem status for every advertised external conclusion;
- (x) outcome vector, claim-use preflight, dependency closure, environment, replay, staleness conditions, repair ancestry, deferred CR status, and immutable artifact identities.

Proposition 3.134 (MS-C records must satisfy Minimal Manifest). *Every invoked MS-C record satisfies the v20 canonical manifest and every specialized field required by the consuming theorem.*

Proof. Mirror and categorical semantics do not remove reconstructibility or typing obligations. $\square$

Remark 3.135 (Artifact examples). Artifacts may include category and functor definitions, cells and mates, kernels/correspondences, models, filtration and parameter proofs, exact barcodes and matchings, detectors, actual tower morphisms, descent data, ledgers, formal proofs, environments, and source hashes. Basename, path, title, or citation alone is not immutable identity.

Proposition 3.136 (Stale reflection artifacts are non-consumable). *Changing a source, target, variance convention, parameter map, filtration, threshold, window, mate orientation, cost semantics, model, cover, environment, or realization version makes the prior artifact stale for the changed claim unless a new compatibility theorem and content-addressed record are supplied.*

Proof. The changed field alters the mathematical or computational identity of the comparison. Reuse therefore requires a new compatibility proof and artifact binding. $\square$

MS-C.13. Distinct reflection conclusion levels

Definition 3.137 (Reflection interface pass). A Reflection Interface pass means only that the external comparison has been projected into a complete, typed Core-readable entry at the declared scope.

Definition 3.138 (Core pass for reflection data). A Core pass is the Chapter 7 audit result for the exact invoked repaired-profile, detector, eligibility, Type IV, transient, ledger, atlas, Restart, finite-to-infinite, claim, and manifest clauses. It is not an external theorem.

Definition 3.139 (Mirror/categorical conclusion). A Mirror/categorical conclusion is an external statement about mirror correspondence, categorical or derived equivalence, functorial comparison, motivic relation, geometric duality, descent, or higher structure.

Proposition 3.140 (Core pass is not Mirror/categorical conclusion). *A Core pass does not imply an external reflection conclusion without a Return Theorem whose antecedent is the exact audited AK-side conclusion.*

Proof. The conclusion levels have different semantics and proof obligations. $\square$

Remark 3.141 (Nine-level distinction). The following remain distinct:

external pre-gate, Interface, comparison, finite certificate, Core audit,
Type IV/transient, finite-to-infinite, Return, external theorem.

No success at one level imputes success at another.

MS-C.14. Relation to the Fukaya-categorical successor interface

Remark 3.142 (Fukaya extension deferred). Appendix MS-F is the successor Problem-Interface specialization for Fukaya and Floer data in the fixed v20 Part IV order. MS-C supplies only the generic reflection/categorical contract and does not pre-prove any MS-F theorem or silently import Fukaya-side semantics.

Definition 3.143 (Fukaya-ready comparison). A comparison is *Fukaya-ready* when it declares the symplectic/-Fukaya source, mirror target, variance, A_∞ or dg model, auxiliary choices, comparison cells, realization and extraction routes, persistence parameter and filtration, repaired profiles, actual morphisms when invoked, and ledger alias/refinement relations to MS-C, Appendix L, and Appendix M.

Proposition 3.144 (Fukaya-ready is not Fukaya-proved). *Fukaya-ready status proves no Fukaya-category, Floer, microlocal-sheaf, or homological-mirror-symmetry statement.*

Proof. It is a typed handoff condition, not the required external theorem chain. $\square$

Definition 3.145 (MS-C to MS-F handoff record). The handoff record binds source and target identities, parameter and filtration conventions, the purpose profile, comparison cells, ledger-namespace relations, artifact identities, permitted use, and unresolved obligations. MS-F may refine but may not silently mutate this record.

MS-C.15. Reflection effect classification and adjusted metric budget

Definition 3.146 (Reflection tower-artifact transport defect). An effect $\Delta_{\phi, \text{ref}}$ arises when a reflection route transports, substitutes, or reconstructs an actual tower comparison morphism. It is a metric contribution only when a theorem bounds the actual morphism-derived profile discrepancy. Otherwise it is a nonmetric or non-scalar obligation. Its relation to Appendix L and MS-F names is declared as alias, refinement, aggregate, or disjoint contribution.

Definition 3.147 (Reflection-adjusted budget). Let $(V, \oplus, 0_V, \preceq)$ be the active Appendix-M-compatible quantale. The adjusted metric ledger is the dependency-aware aggregate of exactly the supported metric contributions actually traversed among

$$\Sigma \delta_{\text{base}}, \quad \delta^{\text{Mir}}, \quad \delta^{\text{cat}}, \quad \Delta_{\text{Eval, ref}}, \quad \Delta_{T, \text{ref}}, \quad \Delta_{\phi, \text{ref}}, \quad \delta^{\text{ref-extract}}.$$

Omitted routes contribute the identity; nonmetric and non-scalar effects are excluded from the numerical aggregate and retain their own statuses.

Proposition 3.148 (Reflection evidence must pass the adjusted scalarized budget). *A metric reflection route may support Audit Mode only when the scalarized total upper-bounds the actual composed pre-threshold profile discrepancy and*

$$\text{val}_B(\Sigma\delta_{\text{ref}}) < \text{gap}_\tau(\mathfrak{B}_{\text{ref}})$$

for the protected family, with every two-sided threshold-gap and dependency-path obligation proved.

Proof. This is the inherited v20 metric transport condition applied to the actual reflection path. $\square$

Remark 3.149 (Reflection is not budget-free). Conceptual canonicity, equivalence, or beauty does not prove zero discrepancy. Exactness, zero-reflecting cost, and metric control are separate statements.

MS-C.15bis. Ledger namespace, non-compensation, path dependence, and double-count prevention

Definition 3.150 (Reflection ledger namespace). The symbols

$$\delta^{\text{Mir}}, \delta^{\text{cat}}, \Delta_{\text{Eval,ref}}, \Delta_{T,\text{ref}}, \Delta_{\phi,\text{ref}}, \delta^{\text{ref-extract}}$$

are local aliases, not independent Core ledgers. Every consumed metric occurrence maps to a canonical Appendix M slot, identifies the actual profile pair and route edge it bounds, and declares units, conversion, aggregation, scalarization, and artifact identity.

Definition 3.151 (Reflection primitive discrepancy). A primitive discrepancy is one underlying mismatch before aliases, refinements, or aggregates are assigned. The dependency graph declares whether displayed entries are identical, nested, sequential, alternative, or disjoint.

Proposition 3.152 (No double counting of reflection defects). *One primitive discrepancy may not be counted twice under different reflection, Appendix L, Appendix M, or MS-F names unless a theorem proves that the recorded contributions are disjoint.*

Proof. Otherwise the ledger no longer represents the actual composed path bound. $\square$

Proposition 3.153 (Reflection defects do not compensate for missing artifacts). *No favorable metric value repairs missing realization, extraction, semantic authority, coherence, detector completeness, eligibility, actual morphism, terminal tameness, descent, reduction, Return, permission, or artifact identity.*

Proof. These are non-scalar obligations determining whether the inference exists. $\square$

Remark 3.154 (Ledger-catalog registration). The local namespace remains Problem-Interface notation until final ledger-catalog and CR-v20 extraction after Part IV freeze. No provisional JSON/CSV Claim Register or per-claim promotion artifact is normative during individual appendix revision.

Proposition 3.155 (Alternative reflection paths are not accumulated). *Alternative comparison routes are competing witnesses, not sequential errors. They are not summed unless a separate theorem requires joint traversal.*

Proof. Alternative paths each bound the same endpoint comparison. They are not sequentially traversed edges, so triangle-inequality accumulation does not apply. $\square$

MS-C.16. Appendix MS-C v20 interface package

Theorem 3.156 (Appendix MS-C interface package). *Within the Problem Interface layer, the following package is available at exactly the displayed strength.*

- (i) *Mirror and categorical data remain external until theorem-backed verifier-side re-entry.*
- (ii) *Source/target domains, variance, coefficient and parameter action, realization, purpose profile, comparison cells, and artifact identity are independent obligations.*
- (iii) *Mirror terminology implies no involution, equivalence, symmetry, or information preservation.*

- (iv) *Exact, anti-exact, adjunction-only, and correspondence regimes have different transport consequences.*
- (v) *Equivalence strength is feature-relative; external equivalence does not automatically identify repaired profiles, representatives, actual morphisms, or apexes.*
- (vi) *Extension-readable objects require a typed extraction and one-parameter/constructibility theorem.*
- (vii) *Comparison cells use local statuses whose Core projection is only exact, metric_gap_certified, or genuinely not_compared.*
- (viii) *Zero numerical cost is not exactness without zero reflection and coherence.*
- (ix) *Adjunction mates, Beck–Chevalley conditions, and correspondence push–pull routes require their own orientation-specific proofs.*
- (x) *Controlled pasting requires composition laws, whiskering moduli, coherence, and common-quantale conversion.*
- (xi) *Increasing affine parameter transport gives the exact scaled window/threshold/metric law; nonlinear and order-reversing transport require separate regimes.*
- (xii) *Repaired evaluation and hard deletion do not commute with reflection by default.*
- (xiii) *Exact and metric reflection transport consume the actual windowed pre-threshold profiles and the full v20 threshold-gap discipline.*
- (xiv) *Mirror/categorical comparison does not replace standard degree-n eligibility or natural zero-reflection.*
- (xv) *Type IV requires a status-dependent record, fixed (W, τ) , finite monitored degrees, an actual morphism, terminal semantics, and full defect artifacts.*
- (xvi) *Terminal Type IV, full-interior defect, long-transient defect, and full isomorphism remain separate.*
- (xvii) *Exact equivalence transports Type IV only when the complete tower, apex, comparison morphism, and terminal records are transported; anti-equivalence requires a dual protocol.*
- (xviii) *Finite observation, local pass, pairwise overlap agreement, descent, eventual constancy, apex agreement, and Return remain distinct.*
- (xix) *Every external conclusion requires the complete enhancement–Interface–Core–Reduction–Return chain.*
- (xx) *MS-C manifests retain separate comparison, detector, Core, Type IV, transient, reduction, Return, and external outcome fields and content-addressed artifact identities.*
- (xxi) *Fukaya-ready status is only a typed handoff to MS-F and proves no Fukaya or HMS theorem.*
- (xxii) *Only supported actual metric discrepancies enter the strict ledger; non-scalar obligations are non-compensating and primitive discrepancies are not double-counted.*
- (xxiii) *No MS-C result enlarges the Core or receives canonical CR-v20 promotion before final Part IV freeze and cross-audit.*

Proof. Items (i)–(v) follow from Sections MS-C.1–MS-C.3ter. Item (vi) follows from Section MS-C.4bis. Items (vii)–(x) follow from Sections MS-C.5–MS-C.6. Item (xi) follows from Section MS-C.6bis. Items (xii)–(xiii) follow from Sections MS-C.7–MS-C.8. Item (xiv) follows from Section MS-C.9. Items (xv)–(xvii) follow from Sections MS-C.10–MS-C.10bis. Item (xviii) follows from Section MS-C.10ter. Items (xix)–(xx) follow from Sections MS-C.11–MS-C.13. Item (xxi) follows from Section MS-C.14. Items (xxii)–(xxiii) follow from Sections MS-C.15–MS-C.15bis and the deferred-governance rule. $\square$

MS-C.17. Explicit exclusions

Appendix MS-C asserts none of the following.

- No mirror symmetry, HMS, Fukaya, Floer, categorical/derived/Morita equivalence, motivic equivalence, Langlands, modularity, automorphy, or arithmetic theorem is proved merely by this appendix.
- No mirror operation is automatically total, functorial, involutive, bidirectional, fully faithful, essentially surjective, conservative, monoidal, exact, or information-preserving.
- No adjunction is automatically an equivalence; no correspondence is automatically a push–pull or Beck–Chevalley theorem.
- No source identity, equivalence, naturality cell, mate, coherence cell, or zero cost is automatically a Core exact comparison.
- No categorical cost, energy estimate, analogy score, or homotopy size is automatically a bottleneck bound or metric-ledger component.
- No extension-readable derived, sheaf, stack, dg, A_∞ , stable, Fukaya, or higher object is automatically Core-readable.
- No one-parameter extraction is assumed lossless, exact, faithful, conservative, or non-adaptive.
- No raw external invariant replaces $B_{i,W}$, $\text{Eval}_{i,W,\tau}$, a detector, representative, actual morphism, or manifest field.
- No unconditional commutation of windowing, deletion, realization, Mirror, transfer, or categorical comparison is assumed.
- No global exact positive-threshold collapse functor, Serre localization, or strict filtered lift is restored.
- No metric deletion transport is used without actual pre-threshold profile support and the two-sided gap chain; no exact comparison is positive-radius robustness by itself.
- No nonlinear parameter transport has a default constant threshold scaling; no order reversal is ordinary forward-persistence relabeling.
- No Mirror/categorical comparison replaces standard degree- n eligibility, and no converse $\text{Ext}^n = 0 \Rightarrow \text{Eval}_n = 0$ is asserted without natural zero-reflection.
- No actual Type IV morphism is manufactured from object equivalence, common barcodes, equal Betti data, source/target closeness, fibers/cofibers, cones, or proxies.
- No numerical Type IV values are asserted in `not__invoked` or undefined states.
- No aggregation across several windows, thresholds, or infinitely many degrees is performed without a separate aggregation/summability theorem.
- No terminal Type IV zero implies transient absence, full-interior zero, full defect zero, or isomorphism.
- No anti-equivalence is treated as preserving the canonical colimit-to-apex protocol without a dual limit/-coapex and Return theorem.
- No finite display, finite sample, local pass, pairwise overlap agreement, hyperdescent slogan, finite prefix, Cauchy estimate, or observed stabilization proves a global or infinite conclusion.
- No Interface pass, comparison pass, Core pass, budget pass, Type IV zero, or descent pass is an external Mirror/categorical theorem without Return.
- No metric slack compensates for missing realization, semantic authority, coherence, detector completeness, eligibility, actual morphism, terminal tameness, descent, reduction, Return, permission, or artifact identity.

- No primitive discrepancy is double-counted or silently omitted; no alternative path is summed as if traversed.
- No stale artifact is reused after a source, target, variance, parameter, filtration, threshold, window, model, cover, mate orientation, cost semantics, environment, or realization change.
- No Fukaya-ready handoff proves a Fukaya or homological mirror theorem.
- No Bridge candidate, Program, Spec, Search artifact, AI output, manifest, or Claim Register entry is theorem evidence by itself.
- No current CR-v20 baseline registration is fabricated for Part IV.

MS-C.18. Provenance, conformance, and final local status

This v20 reconstruction depends on:

- (i) Main Chapters 1–3 and Foundation Appendix A for window-first evaluation, hard-deletion repair, exact/-metric comparison, and threshold-gap transport;
- (ii) Chapter 3 and Appendix B for detector and representative discipline;
- (iii) Chapter 4 and Appendix C for standard degree- n eligibility and higher-edge boundaries;
- (iv) Chapter 5 and Appendix D for actual-morphism Type IV, terminal semantics, and transient separation;
- (v) Chapter 6 and Appendix E for coverage, atlas, overlap, Restart, and local-to-global obligations;
- (vi) Chapter 7 and Appendices F–G for atomic status, non-compensation, claim-use preflight, canonical manifests, artifact identity, replay, and failure routing;
- (vii) Chapter 8 and Appendix MS-A for Interface, finite-to-infinite, Return, and external conclusion chains;
- (viii) Appendix L for the frozen Mirror/Transfer variance, 2-cell, mate, parameter, metric, and tower calculus;
- (ix) Appendix M for typed quantale ledger semantics;
- (x) Appendix N for derived, dg, sheaf, stack, descent, multiparameter, and higher-enhancement boundaries;
- (xi) Appendix MS-B only through its frozen v20 interface and artifact conventions, without importing its arithmetic conclusions into MS-C;
- (xii) CR-v20 baseline governance only for already registered claims, with final Part IV promotion deferred.

Its local conformance status is

v20__mathematical__conformant;
Mirror/categorical interface schema hardened;
MSC to MSF handoff defined;
Part IV Claim Register promotion pending.

Local role classification. Definitions are `interface_definition` or operational policy. Propositions and Theorems 3.72, 3.73, 3.86, 3.87, 3.109, and 3.156 are Problem-Interface statements at exactly their displayed hypotheses. They prove no problem-specific Mirror, motivic, Fukaya, Langlands, modularity, automorphy, or arithmetic conclusion. Every unproved cross-layer implication remains `bridge_candidate`, `bridge_program`, `spec`, or `non_claim`.

Deferred governance. Local labels remain stable TeX locators, not canonical v20 Claim UUIDs. No provisional JSON/CSV Claim Register or per-claim promotion receipt is normative. Final UUIDs, dependency closure, ledger-catalog relations, and CR-v20 promotion are generated only after the complete Part IV sequence and Companion are frozen and cross-audited.

Repair ancestry. This text conservatively replaces the v19 MS-C interface architecture. The v19 artifact remains immutable repair ancestry; no prior verdict or source statement is mutated in place.

4 Appendix MS-F: Fukaya-Categorical Extension

MS-F.1. Scope, dependency position, and interface status

This appendix defines the v20.0.0 Problem-Interface discipline for Fukaya-categorical, Floer-theoretic, wrapped, sectorial, microlocal-sheaf-adjacent, and homological-mirror-symmetry-facing data. It is the terminal specialization of the released MS stack and does not enlarge the theorem-bearing Core of Main Chapters 1–8 or Foundation Appendices A–G.

The governing rule is

Fukaya and Floer structures supply external source semantics;
only a typed Core-readable projection and a proved Return Theorem
license an external conclusion.

Every Core-facing persistence route uses the fixed order

$$B_{i,W}(F) := W_W \mathbf{P}_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Window restriction precedes hard deletion. No unconditional commutation, global exact positive-threshold collapse functor, positive-threshold Serre localization, or global strict filtered lift is assumed. Metric transport through deletion requires an actual pre-threshold bottleneck bound, the two-sided closed-band exclusion $[\tau - \varepsilon, \tau + \varepsilon]$ on both compared profiles, and, for exact zero transport, $2\varepsilon \leq \tau$; the operational reserve is $2\varepsilon < \tau$.

Whenever tower semantics are invoked, the canonical Type IV datum is an actual morphism

$$\phi_{i,W,\tau}^{\text{Fuk}} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty)$$

in the declared abelian profile category. A continuation map, wrapped direct-limit symbol, category equivalence, common barcode, sampled stabilization, or mirror diagram does not create this morphism.

Remark 4.1 (Interface status). Every problem-specific Fukaya, Floer, wrapped, sectorial, microlocal-sheaf, or HMS implication is [Spec] by default. It becomes theorem evidence only after every source, realization, coefficient, filtration, comparison, finite-certificate or direct-proof, higher-edge, Type IV, transient, finite-to-infinite, Return, claim-use, and artifact obligation actually consumed by the implication has been proved.

Remark 4.2 (Relation to Appendix MS-C). MS-F accepts the immutable MS-C-to-MS-F handoff record: source and target identities, variance, parameter and filtration conventions, purpose profile, comparison cells, ledger namespace relations, artifact identities, permitted use, and unresolved obligations. MS-F may refine that handoff but may not silently mutate it.

Remark 4.3 (No exact-collapse restoration). All collapse-facing statements use barcode-level deletion or repaired evaluation. Fukaya terminology, filtered A_∞ -structure, action filtrations, continuation maps, or categorical equivalences do not restore any deprecated exact-collapse semantics.

MS-F.1bis. Fukaya interface sovereignty

Definition 4.4 (Fukaya interface sovereignty). Fukaya interface sovereignty is the rule that Fukaya categories, Floer complexes, filtered A_∞ -data, symplectic meanings, brane structures, bounding cochains, continuation data, microlocal models, and mirror artifacts remain outside the AK Core unless they re-enter through declared verifier-side records. Admissible re-entry objects include, as invoked:

- (i) constructible one-parameter persistence or finite-type filtered objects;
- (ii) exact windowed pre-threshold and repaired post-threshold profile records;

- (iii) finite detector records at exactly one canonical source stage;
- (iv) admissible representatives and standard degree- n eligible edge records;
- (v) actual tower comparison morphisms and status-dependent Type IV records;
- (vi) full kernel/cokernel profiles and transient-defect records;
- (vii) path-bound metric-ledger entries supported by actual profile comparisons;
- (viii) non-scalar obligation, finite-to-infinite, descent, Return, manifest, and immutable-artifact records.

Proposition 4.5 (No Fukaya sovereignty override). *No Fukaya-theoretic theorem, Floer construction, categorical equivalence, microlocal model, or HMS assertion overrides a missing or failed verifier-side obligation.*

Proof. The Core consumes typed records, not source-domain prestige. A missing invoked object or proof remains undefined; a well-typed evidenced false condition rejects at its declared route. $\square$

Remark 4.6 (Fukaya meaning is proposer-side by default). Fukaya meaning may guide source selection, filtration, threshold, detector, or comparison design. It becomes verifier authority only through a proved Interface or Bridge Theorem and successful claim-use preflight.

MS-F.1ter. Frozen v20 baseline, handoff acceptance, and typed outcomes

Definition 4.7 (MS-F v20 dependency baseline). MS-F is interpreted relative to Main Chapters 1–8, Foundation Appendices A–G, frozen Technical Extension Appendices H–N, reviewed CM-v20 and TB-v20 only at their exact local strengths, and the final v20 Appendices MS-A, MS-B, and MS-C. CR-v20 governs registered claims but does not pre-promote this appendix. No v19 phrase, literature name, source theorem, software output, or AI-generated text strengthens a v20 claim.

Definition 4.8 (MS-C handoff acceptance record). The acceptance record stores the immutable MS-C handoff identity and classifies every inherited field as accepted, refined, not_invoked, or incompatible. A refinement records the old and new values, a proof of compatibility, and new artifact identity. An incompatible required field leaves the affected route undefined.

Definition 4.9 (MS-F local claim roles). Every cross-layer statement has one local role

interface_definition, interface_theorem, bridge_candidate, bridge_program,
bridge_theorem, toy_bridge_theorem, spec, non_claim.

Evidence status, invocation role, and permitted use are stored separately. Local labels are not canonical Claim UUIDs before Part IV freeze and final Claim Register extraction.

Definition 4.10 (MS-F outcome vector). Source lifecycle, Interface, finite certificate/direct proof, Core audit, higher-edge, Type IV, transient defect, finite-to-infinite, descent, Return, and external theorem outcomes are separate fields. Interface and Core use pass, reject, or undefined; subordinate clauses may be not_invoked only under an immutable scope exclusion. No headline status overwrites the component vector.

MS-F.2. Fukaya-categorical source data

Definition 4.11 (Fukaya-categorical source datum). A *Fukaya-categorical source datum* is a declared symplectic, Floer, wrapped, sectorial, microlocal, or categorical object together with the exact model used. Examples include an exact or monotone symplectic manifold, Liouville sector, Weinstein manifold, brane collection, filtered Floer complex, curved or uncurved A_∞ -category, wrapped or partially wrapped category, Fukaya–Seidel category, or a proposed sheaf model. The name of a geometric source does not determine the category or its filtration.

Definition 4.12 (Fukaya-side category). A *Fukaya-side category* is a declared category, dg category, filtered or curved A_∞ -category, stable enhancement, or higher category $\mathcal{F}(X)$, with object class, morphism complexes, structure maps, curvature convention, units, topology, and filtration stated at the strength consumed by the Interface.

Definition 4.13 (Auxiliary Fukaya data). Auxiliary data include coefficient and Novikov conventions; grading, spin, brane, orientation, and local-system data; perturbations and almost-complex structures; Hamiltonians, wrapping, stops, and sector data; compactness, bubbling, and transversality hypotheses; bounding cochains or bulk deformations; and filtration orientation and normalization. Each field is either supplied, proved unnecessary for the declared source regime, or marked missing.

Definition 4.14 (Fukaya source status). The source lifecycle field uses exactly one of

declared, constructed, source_validated, Core_readable, failed, undefined.

Separate booleans or typed fields record higher eligibility and tower readability. Lifecycle status is not a Core audit verdict and Core_readable does not imply a Core pass.

Remark 4.15 (Auxiliary data are not optional). A certificate consuming a Fukaya model must identify all choices needed to reconstruct that model or cite a proved invariance theorem covering the omitted choice.

Remark 4.16 (Source status is not Core status). Construction, source validity, Core readability, higher eligibility, Type IV readability, and Core pass are logically different layers.

MS-F.2bis. Auxiliary-data audit discipline

Definition 4.17 (Auxiliary data package). The package records every source choice, hypothesis, dependency, proof or computation, permitted variation, invariance theorem, and failure route. Analytic obligations such as compactness, transversality, orientation, and bubbling control are non-scalar unless a separate theorem represents a particular perturbation by an actual profile discrepancy.

Definition 4.18 (Source-status transition record). A transition record binds the old and new lifecycle states, the exact theorem or artifact authorizing the transition, all discharged hypotheses, and immutable input/output identities.

Proposition 4.19 (Source-status transitions are not automatic). *No source lifecycle transition, Core-readability declaration, higher-eligibility declaration, or tower-readability declaration is automatic.*

Proof. Each requires different objects and proof obligations. Source construction does not imply constructible persistence, and constructible persistence does not imply a standard higher-edge or an actual apex comparison. □

Remark 4.20 (Undeclared auxiliary choices are interface failures). A missing reconstructibility-critical choice yields undefined, not an assumed canonical choice.

MS-F.2ter. Filtered A_∞ , curvature, and Novikov discipline

Definition 4.21 (Filtered Fukaya source package). A filtered Fukaya source package declares the coefficient/Novikov object, valuation, filtration orientation, completed tensor products, the filtration degree of every structure map m_k , convergence of the relevant series, curvature m_0 , units, bounding-cochain equation when used, and the exact uncurved or deformed morphism complex extracted for Core use.

Proposition 4.22 (Curved source data do not automatically define a chain complex). *A curved A_∞ -object with nonzero m_0 does not by itself supply a differential squaring to zero on the morphism object consumed by the Core.*

Proof. The curved relations include curvature terms. A valid chain complex requires a declared unobstructed regime, bounding cochain, deformation, quotient, or other theorem producing an actual differential with square zero. □

Proposition 4.23 (Novikov coefficients are not field-valued Core coefficients by default). *A Novikov ring or field, completed module, or valuation-filtered complex is not automatically a finite-dimensional k -linear Core object. Coefficient passage, completion, finite-dimensionality, and information loss require separate records.*

Proof. The source coefficient object may be topological, completed, non-Noetherian, or infinite-dimensional over the chosen field. None of the required Core properties follows without the declared passage and its proof. $\square$

Remark 4.24 (Unobstructedness namespace). Weak, bounding-cochain-relative, bulk-deformed, monotone, exact, and strict unobstructedness are distinct source predicates and must not be merged in a Return statement.

MS-F.3. Fukaya realization targets

Definition 4.25 (Fukaya realization target). A Fukaya Interface declares one or more targets with roles: a filtered Floer complex; a finite-type object in $\text{FiltCh}(k)$; a constructible object in $\text{Pers}_k^{\text{cons}}$; a windowed pre-threshold or repaired profile; an admissible representative; a standard degree- n eligible object; an actual tower comparison morphism; a higher/technical-extension object with a declared Core extraction; a certificate component; or an advisory artifact. Advisory and extension-only targets are non-verdict until re-entry is proved.

Definition 4.26 (Core-readable Fukaya realization). A realization is Core-readable only when it supplies typed constructible/finite-type objects, actual morphisms, coefficient and parameter records, exact profile identities, and all invoked downstream evidence needed to form $B_{i,W}$, $\text{Eval}_{i,W,\tau}$, or a verifier-side certificate field.

Definition 4.27 (Fukaya extraction route). An extraction route is an objectwise construction or functorial map from the declared Fukaya model to an exact Core-readable target. Its variance, action on morphisms, composition law, coefficient passage, filtration shift, constructibility, purpose-faithfulness, and artifacts are explicit. An object assignment is not called a functor without those laws.

Proposition 4.28 (No Core use without Core-readable Fukaya realization). *A Fukaya source cannot support a Core clause without a Core-readable realization or a declared extraction to the exact verifier-side data consumed.*

Proof. Source-side objects do not by themselves instantiate the Core input types or comparison records. $\square$

Remark 4.29 (Fukaya category is not automatically persistence). Filtered morphism spaces and action values may motivate persistence, but constructible one-parameter structure, actual transition maps, and finite-dimensionality still require proof.

MS-F.3bis. One-parameter filtration and normalization discipline

Definition 4.30 (Fukaya one-parameter filtration policy). The policy declares every source parameter, the selected persistence parameter, increasing/decreasing orientation, normalization, endpoint and tie conventions, transition maps, discarded axes, constructibility, family membership, and non-adaptivity. Action, energy, area, valuation, Hamiltonian slope, wrapping depth, stop depth, and categorical degree are not interchangeable by name.

Definition 4.31 (Filtration normalization defect). The local symbol $\delta^{\text{Fuk-norm}}$ denotes a normalization effect only after classification as an actual metric profile discrepancy, theorem-controlled nonmetric change, non-scalar obligation, advisory quantity, or unavailable datum.

Proposition 4.32 (Action or energy is not a Core parameter by name). *A geometrically meaningful action, energy, slope, or wrapping scale is not a Core persistence parameter until its order, maps, constructibility, and purpose-faithfulness are established.*

Proof. The Core consumes a constructible ordered diagram, not merely a scalar attached to generators. $\square$

Remark 4.33 (Filtration reparametrization risk). A change of action normalization can alter lengths, windows, thresholds, and terminal support. Exact transport requires a proved parameter map and corresponding profile isomorphism.

Theorem 4.34 (Positive affine filtration transport). *Let $r(t) = at + c$ with $a > 0$, extended by $r(\pm\infty) = \pm\infty$. On a declared affine-compatible constructible barcode class, reparameterization preserves endpoint decoration and finite/infinite endpoint type, and*

$$r_* W_W B \cong W_{r(W)} r_* B, \quad r_* T_\tau^{\text{bar}} B \cong T_{a\tau}^{\text{bar}} r_* B, \quad d_B(r_* B, r_* C) = a d_B(B, C).$$

Proof. The map sends an interval $[b, d)$ to $[ab+c, ad+c)$, multiplying finite lengths and endpoint displacements by a , while preserving order, decoration, and infinite endpoints. The three identities follow intervalwise. $\square$

Proposition 4.35 (Nonlinear or reversing reparameterization needs a new theorem). *A nonlinear parameter map has no uniform threshold scaling by default, and an order-reversing map does not preserve the canonical right-terminal Type IV convention. Such routes require their own extraction, metric, and dual-terminal protocols.*

Proof. A nonlinear map changes interval lengths nonuniformly, while order reversal exchanges right and left terminal behavior. The affine intervalwise argument therefore does not apply. $\square$

MS-F.4. Persistence extraction from Fukaya data

Definition 4.36 (Fukaya persistence extraction). A Fukaya persistence extraction is a declared objectwise or functorial route

$$\mathfrak{E}_{\text{Fuk}} : \mathcal{F}(X) \rightsquigarrow \text{FiltCh}(k) \quad \text{or} \quad \mathfrak{E}_{\text{Fuk}} : \mathcal{F}(X) \rightsquigarrow \text{Pers}_k^{\text{cons}}.$$

It records the exact source model, selected objects/morphisms, coefficient passage, filtration, morphism shifts, constructibility, and feature-relative purpose-faithfulness.

Definition 4.37 (Action-filtration extraction). An action-filtration extraction uses a declared action functional and orientation, including normalization and continuation-shift conventions.

Definition 4.38 (Energy-filtration extraction). An energy-filtration extraction uses an energy or area scale only after a theorem constructs actual filtered maps and proves the relation to the intended Core predicate.

Definition 4.39 (Microlocal-sheaf extraction). A microlocal-sheaf extraction factors through a declared sheaf/-cosheaf or microlocal model, its equivalence or comparison strength, and a constructible one-parameter extraction. Object descent and certificate descent are separate obligations.

Definition 4.40 (Fukaya extraction defect). The local symbol $\delta^{\text{Fuk-ext}}$ names an extraction effect. It enters the metric ledger only when a theorem bounds the actual bottleneck discrepancy of the exact profiles traversed by the certificate.

Proposition 4.41 (Extraction must preserve constructibility). *A Fukaya persistence extraction is Core-admissible only on scopes where the output is constructible and pointwise finite-dimensional, or where a separately registered admissible replacement theorem applies.*

Proof. Those are the input hypotheses of the repaired persistence calculus. $\square$

Proposition 4.42 (Fukaya extraction defects must be ledgered). *A nonzero extraction effect influences a metric Core route only through a path-bound supported ledger entry. Nonmetric semantic loss and missing functoriality remain non-scalar.*

Proof. Only actual profile discrepancy bounds have the type required by the Appendix-M metric ledger. $\square$

Remark 4.43 (Extraction is an interface hypothesis). An extraction is not certified by naming a Floer complex or a sheaf model; its correctness and directional purpose-faithfulness are theorem obligations.

MS-F.4bis. Continuation, perturbation, model replacement, and metric transport

Definition 4.44 (Continuation defect). A continuation effect $\delta^{\text{Fuk-cont}}$ records the actual filtration shift, source/-target complexes, continuation map, reverse map when used, homotopies, composition identities, and the theorem relating these data to a profile comparison.

Definition 4.45 (Perturbation defect). A perturbation effect $\delta^{\text{Fuk-pert}}$ records the exact change of analytic or auxiliary data and is metric only when it supports an actual profile-level discrepancy bound.

Definition 4.46 (Model-replacement defect). A model-replacement effect $\delta^{\text{Fuk-model}}$ records the source and replacement models, comparison functors or maps, filtration effect, equivalence strength, and verifier-side consequence.

Proposition 4.47 (Continuation map is not tower comparison artifact). *A continuation map is not $\phi_{i,W,\tau}^{\text{Fuk}}$ unless the full directed system, apex, compatible cocone, induced repaired-profile morphism, and semantic authority are supplied.*

Proof. A stage map and a colimit-to-apex comparison have different domains and universal properties. $\square$

Proposition 4.48 (Model equivalence is not persistence-profile equality). *A quasi-equivalence, Morita equivalence, sheaf equivalence, or dg replacement does not imply equality or metric control of repaired profiles unless filtration and parameter transport are proved.*

Proof. Unfiltered categorical equivalence need not preserve the additional filtered structure used by persistence. $\square$

Theorem 4.49 (Filtered continuation interleaving criterion). *Let T_a denote translation of the filtration parameter by $a \geq 0$, not a cohomological shift. Suppose filtered chain maps*

$$f : C \longrightarrow T_a D, \quad g : D \longrightarrow T_b C$$

are supplied, together with filtered chain homotopies

$$(T_a g) \circ f \simeq \eta_C^{a+b}, \quad (T_b f) \circ g \simeq \eta_D^{a+b},$$

where η^{a+b} denotes the canonical filtration-translation structure map and all maps and homotopies respect the displayed shifts on the declared scope. Then the induced persistent-homology modules are (a, b) -interleaved. In the symmetric case $a = b = \varepsilon$, if the resulting modules are q -tame or constructible in the regime where the isometry theorem applies, then

$$d_I \leq \varepsilon, \quad d_B \leq \varepsilon.$$

Proof. Passing the displayed filtered chain maps and homotopies to homology gives the two interleaving morphisms and the required composite identities with the persistence structure maps. The interleaving bound is therefore ε in the symmetric case. The bottleneck bound follows only under the separately declared q -tameness/constructibility hypotheses supporting the persistence isometry theorem. $\square$

Proposition 4.50 (One continuation map is not an interleaving). *A single filtered continuation map or one-sided energy estimate does not by itself produce an interleaving or a bottleneck certificate.*

Proof. An interleaving requires comparison maps in both directions and compatibility of their composites with the structure shifts. A one-sided map supplies only one part of that datum. $\square$

Remark 4.51 (Continuation and perturbation must be ledgered when consumed). Only theorem-supported actual profile discrepancies enter the metric total. Compactness, transversality, coherence, inverse-map existence, and artifact identity remain non-scalar.

MS-F.4ter. Fukaya finite-detector discipline

Definition 4.52 (Fukaya detector record). A detector record chooses exactly one source stage

windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect,

and stores soundness, zero-completeness, birth/germ or stencil coverage, endpoint convention, family membership, excess provenance, non-adaptivity, immutable artifacts, certificate status, and mathematical value.

Proposition 4.53 (Finite Floer computation is not finite proof). *A finite generator list, action window, continuation sample, polygon count, or barcode sample is not a complete certificate without a direct theorem or an admitted complete detector package.*

Proof. Finiteness of a computation does not prove coverage of all retained bars, all relevant operations, or a cofinal tail. $\square$

Proposition 4.54 (Detector stage is not transported by analogy). *A Fukaya/Mirror or model comparison does not change the detector source stage unless a theorem identifies the actual source profiles or defect profiles and transports completeness.*

Proof. The four detector stages have different source objects. A comparison of external models does not identify those objects or their coverage theorems by itself. $\square$

MS-F.5. Fukaya-side obstructions and Core obstructions

Definition 4.55 (Fukaya-side obstruction). A Fukaya-side obstruction is a precisely named source predicate: failure of a differential or A_∞ relation, curvature/bounding-cochain obstruction, bubbling or compactness failure, transversality failure, orientation failure, continuation incoherence, wrapping/stop incompatibility, or mirror comparison failure.

Definition 4.56 (Core obstruction). A Core obstruction is a typed AK predicate such as repaired-profile nonzero, a well-typed nonzero Ext^n -group, an obstructed Type IV record, a certified long-transient defect, a failed metric reserve, or a failed mandatory manifest/claim clause. The exact degree, scope, source stage, and status are recorded.

Proposition 4.57 (Fukaya-side obstruction is not Core obstruction by default). *Neither obstruction namespace implies the other without a directional purpose-faithful transport theorem.*

Proof. The predicates have different source types and failure mechanisms. $\square$

Remark 4.58 (Obstruction transport is [Spec]). Every unproved cross-namespace implication is a Bridge candidate or program, not theorem evidence.

MS-F.5bis. Obstruction namespace separation

Definition 4.59 (Fukaya obstruction namespace). Every obstruction field declares its namespace, source object, logical polarity, scope, evidence, and permitted reading. The namespaces include analytic construction, curved/deformed A_∞ , Floer cohomology, mirror comparison, repaired persistence, higher Ext, Type IV, transient defect, metric ledger, and governance.

Proposition 4.60 (Unobstructedness is not Core vanishing). *Exact, monotone, weak, or bounding-cochain-relative unobstructedness does not imply repaired-profile zero, higher Ext^n -zero, Type IV terminal zero, transient absence, or metric pass by default.*

Proof. Unobstructedness authorizes a source construction; the displayed Core predicates concern a chosen re-alization, threshold, comparison morphism, or metric path. $\square$

Proposition 4.61 (Core obstruction is not Fukaya obstruction). *A Core obstruction may reflect extraction, parameter, threshold, tower, or budget failure and therefore does not imply bubbling, curvature, transversality, or Fukaya nonexistence without Return.*

Proof. The source and target semantics are distinct. $\square$

Remark 4.62 (Obstruction names are not evidence). The words obstruction, vanishing, exact, clean, stable, and unobstructed are overloaded and require decorated names in every theorem-facing record.

MS-F.6. Standard higher Fukaya eligibility for Ext^n -use

Definition 4.63 (Fukaya eligible regime). Fix $n \geq 1$. A Fukaya Interface is in the standard degree- n eligible regime on $(F; W, \tau)$ when:

- (i) an admissible representative $C_{\tau, W}F$ exists;
- (ii) an actual profile isomorphism identifies $\mathbf{P}_n(C_{\tau, W}F) \cong \text{Eval}_{n, W, \tau}(F)$;
- (iii) a realization $A_{n; F, W, \tau} \in D^b(k\text{-mod})$ is independently meaningful;
- (iv) $A_{n; F, W, \tau} \in D^{[-n, 0]}(k\text{-mod})$, or a separately registered admissible replacement holds;
- (v) an edge extractor E_n is fixed, satisfies $E_n(0) = 0$, and is bound to a declared profile class;
- (vi) an artifact-backed identification

$$H^{-n}(A_{n; F, W, \tau}) \cong E_n(\text{Eval}_{n, W, \tau}(F))$$

is supplied with its naturality scope;

- (vii) coefficient, representative-change, realization-change, family, non-adaptivity, claim, and artifact obligations are complete.

The case $n = 1$ is the inherited two-term regime.

Proposition 4.64 (Fukaya category does not replace eligibility). *Fukaya, Floer, or A_∞ -structure does not replace any standard degree- n eligibility obligation.*

Proof. The higher edge is a theorem about the declared realized object and edge certificate, not about source naming. $\square$

Remark 4.65 (Floer grading is not Core amplitude). Maslov or cohomological grading does not by itself produce amplitude $[-n, 0]$ or the required edge identification.

MS-F.6bis. Floer grading and higher-edge hardening

Definition 4.66 (Floer-to-Core amplitude claim). A Floer-to-Core amplitude claim identifies a declared truncation or realization with an object of $D^{[-n, 0]}(k\text{-mod})$, and records compatibility with the repaired degree- n profile, edge extractor, representative changes, and artifacts.

Proposition 4.67 (Floer degree is not the Core Ext^n -edge). *Floer degree, Maslov index, or morphism-space grading is not automatically the residual degree- n edge used by the Core theorem.*

Proof. The Core edge requires the specific artifact-backed H^{-n} -identification above. $\square$

Proposition 4.68 (Unproved amplitude blocks Fukaya Ext^n -use). *Missing amplitude, edge, naturality, coefficient, or artifact evidence leaves an invoked higher categorical clause undefined.*

Proof. These are non-scalar typing and theorem obligations. $\square$

Theorem 4.69 (One-way higher Fukaya discharge). *In the complete standard degree- n eligible regime,*

$$\text{Eval}_{n, W, \tau}(F) = 0 \implies \text{Ext}^n(A_{n; F, W, \tau}, k) = 0.$$

A converse requires a separately proved natural zero-reflection package on the exact profile class.

Proof. Zero compatibility gives $H^{-n}(A) = 0$; apply the field-coefficient higher Ext edge. Eligibility alone does not reflect profile zero. $\square$

MS-F.7. Fukaya/Mirror comparison and MS-C handoff

Definition 4.70 (Fukaya/Mirror comparison datum). A Fukaya/Mirror datum consists of a Fukaya model, mirror-side model, comparison variance and strength, auxiliary choices, parameter action, filtered/dg/ A_∞ comparison cells, realized profile comparisons, purpose profile, and the accepted/refined MS-C handoff identity.

Definition 4.71 (Fukaya/Mirror comparison cost). The local symbol $\delta^{\text{Fuk-Mir}}$ is a metric component only when a theorem bounds an actual profile discrepancy. Higher coherence, equivalence existence, filtration preservation, terminal-support transport, and Return are non-scalar unless separately represented by metric theorems.

Definition 4.72 (Fukaya/Mirror adjusted budget). Let $(V, \oplus, 0_V, \preceq)$ be the active Appendix-M-compatible quantale. The adjusted metric total is the dependency-aware aggregate of exactly the supported metric effects traversed among

$$\Sigma \delta_{\text{base}}, \delta^{\text{Fuk-ext}}, \delta^{\text{Fuk-norm}}, \delta^{\text{Fuk-cont}}, \delta^{\text{Fuk-pert}}, \delta^{\text{Fuk-model}}, \delta^{\text{Fuk-Mir}}, \Delta_{\phi, \text{Fuk}}.$$

Omitted routes contribute 0_V ; nonmetric and non-scalar effects do not enter the numerical total.

Proposition 4.73 (Fukaya/Mirror comparison costs must be ledgered). *Every consumed nonzero metric Fukaya/Mirror discrepancy is included once, with actual profile support and a canonical ledger slot.*

Proof. The metric gate must upper-bound the actual composed discrepancy and cannot ignore a traversed component. $\square$

Proposition 4.74 (Fukaya/Mirror evidence must pass adjusted budget). *A metric comparison route may support Audit Mode only when*

$$\text{val}_B(\Sigma \delta_{\text{Fuk}}) < \text{gap}_\tau(\mathfrak{B}_{\text{Fuk}})$$

and the scalarized total bounds the actual protected pre-threshold profile discrepancy.

Proof. This is the v20 metric ledger gate applied to the declared comparison path. $\square$

Remark 4.75 (No free homological mirror symmetry). A useful or exact Core comparison is not HMS without a separate external theorem and Return chain.

MS-F.7bis. Fukaya/Mirror comparison modes, variance, and terminal support

Definition 4.76 (Fukaya/Mirror comparison mode). The local comparison mode is exactly one of

source_strict, source_equivalent, source_antiequivalent, realized_exact,
realized_metric, ledger_controlled, advisory_only, unavailable,
failed, undefined.

Only realized_exact projects to Core exact; only realized_metric with an actual bottleneck and threshold-gap certificate projects to metric_gap_certified.

Definition 4.77 (Fukaya comparison status–mode compatibility). Source lifecycle, comparison mode, comparison proof status, metric support, zero-reflection, coherence, and Core mode are separate fields. A missing required comparison is undefined, not not_compared; the latter is allowed only when the governing claim genuinely does not consume comparison.

Proposition 4.78 (Source equivalence is not Core exactness). *A Fukaya equivalence, quasi-equivalence, HMS equivalence, or microlocal equivalence is not Core exactness unless it induces an actual isomorphism of the consumed repaired profiles with compatible parameters.*

Proof. The source equivalence need not preserve filtration, windows, thresholds, or profile extraction. $\square$

Proposition 4.79 (Controlled comparison is not metric certification by name). *A controlled categorical or energy estimate is not metric-gap certification until converted into a certified bottleneck bound on the actual windowed pre-threshold profiles.*

Proof. The quantities otherwise live in different metric or categorical types. $\square$

Definition 4.80 (Fukaya terminal-support transport record). A terminal-support transport record proves that a realized equivalence or parameter map preserves the declared terminal mode, endpoint decoration, cofinal/non-cofinal support, and tgdim_W of every transported defect profile. Exact categorical equivalence alone does not supply this record.

Proposition 4.81 (Anti-equivalence requires dual tower protocol). *A contravariant or anti-equivalent Fukaya/Mirror route exchanges diagram variance and generally interchanges colimit and limit semantics. It cannot be read as the canonical directed Type IV artifact without a separately proved dualization, directed conversion, apex agreement, and Return protocol.*

Proof. Contravariance reverses transition maps, so the universal property of the resulting diagram differs from the canonical directed colimit source. The listed conversions are therefore independent obligations. $\square$

Remark 4.82 (Comparison modes delegate to MS-C). MS-F refines the MS-C comparison record; whenever both apply, the stronger collection of obligations governs.

MS-F.8. Fukaya-side towers and status-dependent Type IV

Definition 4.83 (Fukaya-side tower). A Fukaya-side tower is a declared direct, inverse, or mixed diagram of source data. It records actual transition maps, variance, cofinal indexing, auxiliary-data compatibility, and the candidate infinite object. A source diagram is not assumed directed after extraction.

Definition 4.84 (Core-readable Fukaya tower). A tower is Core-readable only when extraction supplies actual directed filtered/profile maps on a fixed scope, a declared apex and semantic-authority theorem, a compatible cocone, and the induced comparison morphisms. Inverse or homotopy-coherent source systems require a proved conversion.

Definition 4.85 (Fukaya tower comparison artifact). The artifact binds a fixed (W, τ) , finite monitored degree set I_{mon} , source and apex profile identities, actual morphisms $\phi_{i,W,\tau}^{\text{Fuk}}$, ambient abelian category, source provenance, semantic authority, terminal mode, terminal tameness, and immutable artifacts.

Definition 4.86 (Fukaya Type IV proxy). A source anomaly, continuation instability, generator growth, mirror mismatch, or sampled nonstabilization is a Fukaya Type IV proxy and is advisory until identified with the actual kernel/cokernel profile predicate.

Definition 4.87 (Actual Fukaya-side Type IV diagnostic). Fix one window–threshold pair (W, τ) and one finite monitored-degree set I_{mon} . For every $i \in I_{\text{mon}}$, a valid record binds an immutable artifact identifier to the actual morphism

$$\phi_{i,W,\tau}^{\text{Fuk}} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty),$$

its source and target profile identities, tower and transition-map provenance, claim-required source semantics, semantic-authority verdict, apex identity and apex-evaluation record, ambient abelian profile category and typing proof, and terminal mode

$$\text{bounded_terminal} \quad \text{or} \quad \text{right_unbounded}.$$

The source, target, kernel, and cokernel profiles are supplied with the terminal-tameness evidence required by the declared mode. Define

$$\mu_{i,W,\tau} := \text{tgdim}_W \ker \phi_{i,W,\tau}^{\text{Fuk}}, \quad \nu_{i,W,\tau} := \text{tgdim}_W \text{coker } \phi_{i,W,\tau}^{\text{Fuk}},$$

and aggregate only over the fixed finite set I_{mon} .

The Type IV record has exactly one status

$$\text{not_invoked}, \quad \text{undefined}, \quad \text{certified_zero}, \quad \text{obstructed}.$$

For `not_invoked` it contains an immutable scope rationale proving that no tower/apex comparison is part of the claim and asserts no numbers. For `undefined` it contains a partial or invalid artifact identifier, or an explicit missing-obligation marker with the missing/ill-typed fields, and asserts no diagnostic numbers. For `certified_zero` or `obstructed` it additionally contains the full kernel and cokernel defect-profile artifacts, all degree-wise and total μ, ν , the separate full-interior-defect status

`not_checked, certified_absent, certified_present.`

Whenever transient detection is invoked, the valid record also contains the complete transient detector package. The terminal status is `certified_zero` exactly when all required inputs are valid and the totals equal $(0, 0)$, and is `obstructed` exactly when all required inputs are valid and the totals are nonzero.

Proposition 4.88 (Fukaya Type IV proxy is not Type IV diagnosis). *No Fukaya proxy determines the Type IV status or numerical diagnostics without the actual artifact and terminal evidence.*

Proof. The diagnostic is defined from a specific morphism's kernel and cokernel profiles. □

Remark 4.89 (Correct use of Fukaya proxies). A proxy may prioritize construction of the canonical artifact; it does not replace it.

MS-F.8bis. Artifact coherence, transient defects, and Type IV transport

Definition 4.90 (Artifact-coherent Fukaya Type IV record). Artifact coherence fixes the tower, transition maps, apex, extraction, parameter orientation, terminal mode, monitored scope, and morphism provenance throughout a claimed diagnostic range. Any change creates a new claim unless an explicit coherence theorem identifies the records.

Definition 4.91 (Fukaya tower-artifact transport defect). The symbol $\Delta_{\phi, \text{Fuk}}$ is metric only when a theorem bounds an actual discrepancy between the morphism-derived defect profiles. Object-level distances do not automatically control it.

Definition 4.92 (Fukaya transient-defect record). The record inspects exactly one defect source,

`kernel_defect or cokernel_defect,`

at a threshold $\tau_{\text{def}} > 0$. It declares component `full_profile` or `interior_noncofinal`, endpoint and terminal conventions, information class, family and coverage, non-adaptivity, immutable artifact, detector proof status, and mathematical value. Its aggregate long-transient status is one of

`not_invoked, undefined, certified_absent, certified_present.`

The separate full-interior field is `not_checked`, `certified_absent`, or `certified_present`.

Proposition 4.93 (Changing Fukaya Type IV artifacts creates a new claim). *Changing any artifact-coherence field changes the diagnostic claim unless a theorem transports the entire record.*

Proof. The kernels, cokernels, terminal support, and monitored scope are properties of the fixed artifact. □

Proposition 4.94 (Terminal zero is not transient or full zero). *Terminal `certified_zero` does not imply long-transient absence, full-interior zero, full defect zero, or isomorphism. Long-transient `certified_absent` excludes only monitored noncofinal bars longer than τ_{def} .*

Proof. Terminal-generic dimension ignores noncofinal support, while a positive transient threshold ignores sufficiently short bars. Neither predicate characterizes the entire defect profile. □

Theorem 4.95 (Exact filtered equivalence transport of Type IV). *Suppose an exact equivalence transports the full directed tower, transition maps, apex cocone, actual comparison morphism, and defect profiles, and a terminal-support transport record identifies `tgdim` on the fixed scopes. Then kernel/cokernel profiles and all terminal Type IV values transport. Full-interior and transient statuses transport only when their complete detector and support data also transport.*

Proof. Exactness transports kernels and cokernels; the additional terminal-support theorem transports the numerical functional. Detector conclusions require their own completeness and artifact identities. $\square$

Remark 4.96 (Terminal-tameness is not optional). Missing terminal-tameness yields undefined, never a fabricated zero.

MS-F.8ter. Fukaya stable-band discipline

Definition 4.97 (Fukaya advisory stable band). An advisory band is a threshold interval on which source indicators or sampled profiles appear stable. It is non-verdict.

Definition 4.98 (Fukaya certified Core stable band). A certified band fixes artifact family, source semantics, terminal mode, parameter/filtration policy, finite monitored degrees, and ledger path; has Type IV status certified_zero at every threshold; and supplies any required continuum-to-finite reduction and two-sided gap evidence. Transient or full-interior claims are additional fields.

Proposition 4.99 (Sampled Fukaya stability is not a certified Core band). *Finite or countable samples do not establish a universal stable band without an artifact-coherent reduction controlling all unsampled threshold crossings and profile changes.*

Proof. Hard deletion and defect support can change between samples. $\square$

MS-F.8quater. Wrapped and sectorial finite-to-infinite discipline

Definition 4.100 (Wrapped finite-to-infinite record). When finite Hamiltonian slopes, wrapping stages, stop removals, or sectors are used for an infinite wrapped or completed target, the record contains the actual cofinal system, continuation maps and coherence, finite witness, finite-to-infinite theorem, categorical colimit or homotopy-colimit construction, completion if any, apex cocone, induced profile comparison, apex agreement, derived-limit or compactness obligations, and artifacts.

Proposition 4.101 (Wrapped category is not automatically the AK apex). *A wrapped Fukaya category, localization, ind-completion, homotopy colimit, or large-slope limit is not the AK apex profile without a theorem identifying the exact construction and comparison morphism.*

Proof. These constructions have different universal properties and may not commute with coefficient passage, persistent homology, windowing, or hard deletion. $\square$

Proposition 4.102 (Cofinal finite computation is not automatic reduction). *Computing finitely many slopes or observing stabilization does not prove the infinite wrapped predicate. A direct uniform theorem or complete reduction package is required.*

Proof. Cofinality concerns an unbounded tail, whereas a finite computation controls only its displayed stages. A later stage may change unless a theorem excludes that possibility. $\square$

MS-F.8quinquies. Sectorial, sheaf, and descent boundaries

Definition 4.103 (Fukaya descent record). A descent record declares the cover/sector system, overlap objects, higher cocycles, object-descent theorem, certificate-descent theorem, global persistence extraction, compatibility with windows and thresholds, and immutable local/global artifacts.

Proposition 4.104 (Pairwise sectorial agreement is not global certificate descent). *Pairwise compatibility of local Fukaya, sectorial, or microlocal models does not prove higher descent, a global Core-readable object, or a global certificate.*

Proof. Higher cocycles, effectivity, extraction compatibility, and certificate assembly are independent obligations. $\square$

MS-F.9. Fukaya certificate records

Definition 4.105 (Fukaya-categorical certificate record). A Fukaya certificate is the canonical v20 verifier-side projection of one declared source route together with all required evidence and typed outcomes. It is not a Fukaya theorem merely because its source is symplectic.

Definition 4.106 (MS-F manifest entry). The manifest records at least:

- (i) source identity, source regime, lifecycle, auxiliary package, and immutable source artifacts;
- (ii) accepted/refined MS-C handoff, local claim role, evidence and invocation roles, permitted use, and forbidden reading;
- (iii) coefficient/Novikov, curvature/bounding-cochain, filtration, parameter, variance, and constructibility records;
- (iv) realization/extraction objects and actual morphisms, including functoriality scope;
- (v) fixed windows, thresholds, monitored degrees, endpoint conventions, pre-threshold and repaired profiles;
- (vi) detector stage, soundness, completeness, coverage, certificate/value statuses, and artifacts when invoked;
- (vii) admissible representative and standard degree- n eligibility/edge records when invoked;
- (viii) continuation, perturbation, model, mirror, and parameter comparison modes and proof artifacts;
- (ix) tower, apex, cocone, actual ϕ , semantic authority, terminal mode/tameness, Type IV status, all degree-wise/-total μ , ν , full-interior and transient records when invoked;
- (x) stable-band, wrapped finite-to-infinite, descent, atlas/Restart, apex-agreement, and derived-limit records when invoked;
- (xi) metric quantale, actual supported profile pairs, path, scalarization, total, gap, reserve, and no-double-counting relations;
- (xii) every non-scalar obligation and atomic status;
- (xiii) concrete bridge-track scope, TrackID when invoked, bridge role/evidence/status vector, exact benchmark and Return target, or an immutable not_invoked rationale;
- (xiv) exact Core verdict, Return claim and theorem status, external conclusion status, claim-use preflight, dependency closure, repair ancestry, deferred CR status, and immutable artifacts.

Proposition 4.107 (Fukaya certificate records must satisfy Minimal Manifest). *Every theorem-facing Fukaya certificate satisfies Appendix G's canonical manifest and reconstructibility requirements.*

Proof. Fukaya source semantics add fields but remove no Core audit obligation. □

Remark 4.108 (Fukaya artifacts). Artifacts support replay and identity; they do not determine truth or pass by their existence alone.

MS-F.10. Fukaya transport, reduction, and Return statements

Definition 4.109 (Fukaya-to-Core transport statement). A Fukaya-to-Core statement is a directional theorem candidate from an exact source predicate to an exact Core-visible predicate, with source class, realization, coefficient, filtration, and failure boundary.

Definition 4.110 (Core-to-Fukaya transport statement). A Core-to-Fukaya statement is a Return candidate whose antecedent is the exact audited AK conclusion and whose consequent is the exact source-domain conclusion.

Definition 4.111 (Fukaya transport theorem). A transport theorem supplies every typed arrow, hypothesis, direction, comparison, finite-to-infinite/descent step, proof, artifact, and forbidden converse used by the implication.

Definition 4.112 (Default status of Fukaya transport). An unproved transport has one of the roles

bridge_candidate, bridge_program, spec.

It is not theorem evidence.

Definition 4.113 (Fukaya Return Theorem). A Fukaya Return Theorem proves

$$Q_{AK}(X) \implies C_{Fuk}(X)$$

on a declared source class, where the antecedent is exactly the audited AK conclusion. It records all realization, preservation/reflection, finite-to-infinite, descent, mismatch, and source-semantic hypotheses.

Definition 4.114 (Complete Fukaya Projection–Return chain). A theorem-bearing route has the form

$$\begin{aligned} X_{Fuk} \longrightarrow E_X \longrightarrow F_X \longrightarrow B_{i,W}(F_X) \longrightarrow \text{Eval}_{i,W,\tau}(F_X) \\ \longrightarrow Q_{fin}(X) \longrightarrow Q_{AK}(X) \longrightarrow \text{Verdict}_{AK}(X) \xrightarrow{\text{Return}} C_{Fuk}(X). \end{aligned}$$

Here E_X is an optional Fukaya/derived enhancement. Every arrow has an exact theorem, role, status, scope, and artifact. Higher-edge, Type IV, transient, descent, or finite-to-infinite branches enter only through their own packages.

Proposition 4.115 (Fukaya transport does not enlarge the Core). *A proved Problem-Interface transport remains outside the Core unless a later release explicitly imports and reproves it under Core governance.*

Proof. External semantics and release-level Core status are distinct. □

Proposition 4.116 (Missing arrow blocks external conclusion). *If any required source, realization, extraction, detector/direct-proof, Core, reduction, descent, audit, or Return arrow is missing or undefined, the external conclusion is unlicensed.*

Proof. The theorem-bearing chain is a typed composition. Removing one required arrow leaves no valid composite from the source object to the advertised external conclusion. □

Remark 4.117 (Transport directions are distinct). Fukaya-to-Core purpose-faithfulness and Core-to-Fukaya Return are separate theorems; neither direction implies the other.

MS-F.10bis. Fukaya theorem promotion schema

Definition 4.118 (Fukaya theorem promotion). Promotion from [Spec], Bridge candidate, or Bridge Program to theorem evidence requires a complete proof of the exact chain, exact claim typing and permitted use, dependency/artifact closure, successful claim-use preflight, and final Claim Register promotion after Part IV freeze.

Proposition 4.119 (No HMS promotion by Core pass). *A Core certificate does not prove HMS, Fukaya equivalence, microlocal equivalence, unobstructedness, compactness, or a Floer theorem without the exact Return theorem.*

Proof. The Core verdict concerns the realized AK shadow, not the full source semantics. □

Proposition 4.120 (Fukaya transport hypotheses are non-scalar). *Metric slack, sampled stability, categorical elegance, or source-side success cannot repair a missing transport, coherence, descent, finite-to-infinite, permission, or artifact obligation.*

Proof. Those obligations determine existence and semantic validity, not metric size. □

MS-F.10ter. Current v20 bridge decision and successor boundary

Definition 4.121 (Concrete Fukaya bridge track). A *concrete Fukaya bridge track* fixes one immutable source class and one exact directional statement of one of the forms

$$\mathcal{P}_{\text{Fuk}}(X) \implies \mathcal{Q}_{\text{AK}}(X), \quad \mathcal{Q}_{\text{AK}}(X) \implies \mathcal{C}_{\text{Fuk}}(X),$$

together with the exact source theorem or construction package, purpose-faithful realization, coefficient and filtration passage, finite/direct proof, reduction or descent when invoked, Return theorem when the target is external, strength comparison, claim status, and immutable artifacts. A generic schema or list of possible Fukaya phenomena is not a concrete track.

Definition 4.122 (v20 Fukaya bridge decision). For AK-HPDST v20.0.0, the concrete problem-facing Fukaya bridge field is

not_invoked.

No theorem is claimed from the existence of a Fukaya category, unobstructedness, continuation, wrapped/sectorial structure, microlocal sheaf structure, or HMS data to a Core verdict. Conversely, no Core verdict is claimed to imply a Fukaya, Floer, symplectic, microlocal, or HMS conclusion.

The present appendix supplies the admissibility, typing, comparison, obstruction, tower, ledger, reduction, descent, and Return *schema* required for a future concrete track. The local continuation-interleaving, affine-parameter, higher-edge, and exact-equivalence transport results are verifier-side Interface theorems under already Core-readable hypotheses; they are not problem-facing Fukaya bridge theorems.

Proposition 4.123 (Non-invocation is not a missing bridge obligation). *Under the declared v20 MS-F scope, the absence of a concrete Fukaya bridge theorem is a justified scope exclusion, not an undefined invoked clause and not a release claim of failure.*

Proof. MS-F advertises a terminal Problem-Interface schema and MS-stack closure, not a concrete Fukaya/HMS theorem or Known-Theorem Recovery. No external Fukaya conclusion is included in the declared v20 claim scope, so no problem-facing bridge arrow is required for the local MS-F package. Any later concrete bridge creates a new track, proof graph, artifact set, and Claim Register entry. The separate release-level Known-Theorem Recovery obligation is managed by the MS-B track and is not duplicated by MS-F. $\square$

Remark 4.124 (Successor priority). A Fukaya bridge may be retained as a high-priority successor candidate, but it is not silently assigned to a specific version. Promotion in v20.1, v21, or a later release depends on selecting a tractable benchmark and proving the complete directional chain without importing the target conclusion into the realization.

MS-F.11. Fukaya Interface, Core, and external outcomes

Definition 4.125 (Fukaya interface pass). Interface pass means only that the source has been projected into a complete typed verifier-side entry at the declared scope. undefined records missing/ill-typed evidence; reject records a well-typed evidenced false invoked Interface condition.

Definition 4.126 (Core pass for Fukaya data). Core pass is the conjunction of the invoked repaired-profile, detector, higher-edge, Type IV, transient, ledger, finite-to-infinite, descent, manifest, and claim clauses. It is not an external theorem.

Definition 4.127 (Fukaya-side theorem). A Fukaya-side theorem is an external source-domain conclusion with exact source class, hypotheses, and proof.

Proposition 4.128 (Core pass is not Fukaya theorem). *Core pass does not imply a Fukaya theorem without a Return Theorem whose antecedent is the exact audited conclusion.*

Proof. The two predicates live in different semantic domains. $\square$

Remark 4.129 (Layer distinction). Source pre-gate, source lifecycle, Interface, finite certificate, Core audit, Type IV/transient, reduction, descent, Return, and external theorem remain distinct.

MS-F.12. Relation to the MS appendices and terminal closure

Definition 4.130 (MS interface closure). MS closure is the typed conjunction of the MS-A arithmetic meta-schema, MS-B arithmetic instances and recovery registrations, MS-C Mirror/categorical handoff, and MS-F Fukaya specialization, all subordinate to window-first evaluation, standard higher eligibility, actual-morphism Type IV, noncompensating ledger semantics, finite-to-infinite discipline, and Return.

Proposition 4.131 (MS-F closes but does not upgrade the MS layer). *MS-F closes the released MS interface stack structurally but does not upgrade an MS statement into a Core or external theorem.*

Proof. Closure assembles and constrains already typed components; it does not create missing proofs or Return arrows. $\square$

Remark 4.132 (End of MS layer). The released order is $\text{MS-A} \rightarrow \text{MS-B} \rightarrow \text{MS-C} \rightarrow \text{MS-F}$. MS-B's registered recovery artifacts remain separate sidecar evidence; MS-F introduces no additional Known-Theorem Recovery claim in this revision.

MS-F.12bis. Fukaya ledger and non-compensation

Definition 4.133 (Fukaya primitive discrepancy). Every local effect is classified as exactly one of

metric_profile_discrepancy, morphism_profile_discrepancy,
theorem_controlled_nonmetric, non_scalar_obligation,
advisory, unavailable.

Local names include extraction, normalization, continuation, perturbation, model, Mirror, and artifact-transport effects. Only the first two classes may enter the strict metric total, and only with actual support theorems.

Proposition 4.134 (No double counting of Fukaya defects). *The same primitive discrepancy may not be counted more than once under MS-F, MS-C, Appendix L, or Appendix M names. Alias, refinement, aggregate, or disjointness relations are mandatory.*

Proof. The ledger must upper-bound the actual path without duplicate or omitted contributions. $\square$

Proposition 4.135 (Fukaya non-scalar obligations do not compensate). *No favorable metric reserve repairs missing source validity, curvature resolution, coefficient passage, constructibility, functoriality, detector completeness, eligibility, actual morphism, terminal tameness, reduction, descent, Return, permission, or artifact identity.*

Proof. These obligations are not elements of the bottleneck value system. $\square$

Remark 4.136 (Ledger-catalog registration for MS-F). Every consumed local name declares its canonical Appendix-M slot and its relation to the MS-C reflection and Appendix-L Mirror/Transfer namespaces. Unregistered names remain Interface-local notation.

MS-F.13. Appendix package theorem

Theorem 4.137 (Appendix MS-F v20 interface package). *Within the Problem Interface layer, Appendix MS-F establishes the following package.*

- (i) *Fukaya, Floer, wrapped, sectorial, microlocal, and HMS structures remain external source semantics.*
- (ii) *The MS-C handoff is accepted or explicitly refined; silent mutation is prohibited.*
- (iii) *Auxiliary, curvature, bounding-cochain, Novikov, coefficient, compactness, and transversality data are typed noninterchangeable obligations.*
- (iv) *A curved A_∞ -source does not automatically supply a chain complex.*
- (v) *Core readability requires a purpose-faithful constructible one-parameter extraction with actual maps.*

- (vi) *Action, energy, slope, valuation, wrapping, stop, and degree parameters are not interchangeable.*
- (vii) *Positive affine reparameterization has the exact window, threshold, and bottleneck scaling stated above; nonlinear and order-reversing routes require separate theorems.*
- (viii) *A filtered continuation interleaving requires two-sided maps and compatible filtered homotopies; one map is insufficient.*
- (ix) *Finite Floer computation becomes proof only through a direct theorem or complete detector package at one canonical source stage.*
- (x) *Fukaya obstructions and Core obstructions remain separate namespaces.*
- (xi) *Higher Ext^n -use requires the complete standard degree- n eligible package and is one-way absent natural zero-reflection.*
- (xii) *Source equivalence, quasi-equivalence, anti-equivalence, and HMS data are not Core exact or metric comparisons without realized profile transport.*
- (xiii) *Type IV uses a fixed (W, τ) , finite monitored degree set, actual comparison morphism, semantic authority, terminal tameness, and status-dependent record.*
- (xiv) *Terminal Type IV, long-transient defect, full-interior defect, full defect zero, and isomorphism remain distinct.*
- (xv) *Exact-equivalence transport of numerical Type IV additionally requires terminal-support transport.*
- (xvi) *Wrapped/sectorial finite-to-infinite claims require cofinal systems, compatible continuation maps, categorical construction, apex agreement, and reduction; a wrapped category name is not an apex proof.*
- (xvii) *Pairwise sector or sheaf agreement is weaker than object descent, and object descent is weaker than certificate descent.*
- (xviii) *Only actual supported metric profile discrepancies enter the quantale ledger; non-scalar obligations are noncompensating and primitive discrepancies are not double counted.*
- (xix) *Interface, Core, reduction, descent, Return, and external theorem outcomes remain distinct.*
- (xx) *Every external Fukaya or HMS conclusion requires the complete Projection–Return chain.*
- (xxi) *The concrete problem-facing Fukaya bridge track is not `_invoked` in v20.0.0; MS-F supplies a bridge-admissibility schema but claims no Fukaya-to-Core, Core-to-Fukaya, or HMS bridge theorem.*
- (xxii) *MS-F closes the released MS stack without adding a new Core theorem or a new Known-Theorem Recovery claim.*

Proof. Items (i)–(iv) follow from Sections MS-F.1–MS-F.2ter. Items (v)–(ix) follow from Sections MS-F.3–MS-F.4ter. Items (x)–(xi) follow from Sections MS-F.5–MS-F.6bis. Item (xii) follows from Section MS-F.7bis. Items (xiii)–(xv) follow from Sections MS-F.8–MS-F.8bis. Items (xvi)–(xvii) follow from Sections MS-F.8quater–MS-F.8quinquies. Item (xviii) follows from Section MS-F.12bis. Items (xix)–(xx) follow from Sections MS-F.9–MS-F.11. Item (xxi) follows from Section MS-F.10ter. Item (xxii) follows from Section MS-F.12. $\square$

MS-F.14. Explicit exclusions

Appendix MS-F asserts none of the following.

- No Fukaya category, Floer complex, compactness, transversality, orientation, unobstructedness, or A_∞ -existence theorem is proved.
- No HMS, Fukaya equivalence, Fukaya–Seidel equivalence, or symplectic classification is proved.

- No microlocal sheaf equivalence is proved.
- No curved A_∞ -object automatically defines a Core chain complex, and no bounding cochain is assumed to exist.
- No Novikov coefficient passage is assumed exact, finite-dimensional, conservative, or lossless.
- No action, energy, valuation, Hamiltonian slope, wrapping depth, stop depth, or categorical degree is automatically the persistence parameter.
- No object assignment is automatically functorial, and no source equivalence automatically preserves filtration or repaired profiles.
- No continuation map alone is an interleaving, bottleneck certificate, tower artifact, or apex agreement.
- No finite generator, polygon, slope, action-window, or barcode computation is a complete detector by finiteness alone.
- No Fukaya unobstructedness is repaired-profile zero, higher Ext zero, Type IV zero, transient absence, or metric pass by default.
- No Floer or Maslov grading is the standard degree- n edge or amplitude by default.
- No reverse implication $\text{Ext}^n = 0 \Rightarrow \text{Eval}_n = 0$ is asserted without natural zero-reflection.
- No source strictness, equivalence, quasi-equivalence, anti-equivalence, Morita equivalence, or HMS statement is Core exactness by terminology.
- No categorical, energy, Hofer-type, or continuation cost is a bottleneck bound without a conversion theorem.
- No metric transport through hard deletion is used outside the two-sided threshold-gap regime.
- No raw source anomaly or proxy replaces an actual Type IV morphism and defect profiles.
- No Type IV value is aggregated across several windows/thresholds or infinitely many degrees without a separate aggregation theorem.
- No terminal Type IV zero implies long-transient absence, full-interior zero, full defect zero, or isomorphism.
- No exact equivalence preserves numerical tgdim without terminal-support transport.
- No wrapped category, localization, ind-completion, homotopy colimit, large-slope limit, or sectorial gluing is automatically the AK apex.
- No finite prefix, sampled stable band, pairwise overlap, or observed stabilization proves a global/infinite theorem, descent, apex agreement, or Return.
- No source-side or higher-categorical object is automatically Core-readable.
- No ledger name enters the metric total without actual profile support, canonical placement, and no-double-counting data.
- No metric slack repairs a non-scalar source, realization, detector, eligibility, morphism, terminal, reduction, descent, Return, claim, or artifact obligation.
- No concrete Fukaya-to-Core, Core-to-Fukaya, unobstructedness, microlocal, or HMS bridge theorem is claimed in v20.0.0; bridge schemas and conditional Interface theorems are not such a bridge.
- No Interface or Core pass is a Fukaya/HMS theorem, and no local label is a promoted Claim UID before final CR closure.

MS-F.15. Provenance, conformance, and final local status

This reconstruction depends on:

- (i) Main Chapters 1–8 and Foundation Appendices A–G for window-first evaluation, finite detectors, standard higher edges, Type IV/transient records, ledger noncompensation, manifests, finite-to-infinite, descent, and Return;
- (ii) frozen Technical Extension Appendices H–N, especially Appendix L comparison, mate, and variance semantics, Appendix M path-bound quantale ledgers, and Appendix N higher, sheaf, stack, and descent boundaries;
- (iii) Appendix MS-A for Problem-Interface sovereignty and complete theorem-bearing chains;
- (iv) Appendix MS-B for arithmetic specialization and the separately managed Iwasawa recovery artifacts;
- (v) Appendix MS-C for the immutable/refinable Mirror-categorical handoff and terminal-support transport boundary;
- (vi) CM-v20 and TB-v20 only at their reviewed immutable strengths;
- (vii) CR-v20 only for current governance and deferred final Part IV promotion.

Its local conformance status is

v20__mathematical__conformant;
Fukaya/Floer interface schema hardened;
MS-C handoff accepted with explicit refinement discipline;
concrete Fukaya bridge track not invoked in v20.0.0;
no external Fukaya/HMS theorem or recovery claimed;
MS problem-interface closure established;
Part IV Claim Register promotion pending.

Local role classification. Definitions are Problem-Interface or operational schema. Propositions and Theorems are local conditional or meta-theoretic statements proved only at their displayed hypotheses. The v20 concrete Fukaya bridge field is not `_invoked`; any future problem-specific Fukaya, Floer, wrapped, microlocal, or HMS implication must be instantiated separately as a `bridge_candidate`, `bridge_program`, `spec`, or `non_claim` before proof and promotion. No local label is a canonical v20 Claim UID before final Part IV freeze, cross-audit, and Claim Register extraction.

Repair ancestry. This text conservatively replaces the v19 MS-F interface architecture. The v19 artifact remains immutable repair ancestry and no earlier verdict is mutated in place.

5 Appendix AGI-N: Resolution-Assisted Normalization Interface

AGI-N.1. Scope and interface status

This appendix defines the v20.0.0 Interface discipline for resolution-assisted normalization. It belongs to the Problem Interface layer and does not enlarge the Core mathematical package of Chapters 1–8.

The central rule is

Normalization may prepare proof objects; it is not itself a proof or Return theorem.

Resolution, geometric normalization, desingularization, smoothing, regularization, reparameterization, chart refinement, model replacement, discretization, and defect isolation may prepare a source datum for Core-readable evaluation. None of them replaces:

$B\text{-Gate}^+$, $\text{Eval}_{i,W,\tau}(F)$, a complete finite detector,
standard degree- n eligibility,

$\phi_{i,W,\tau}$, terminal/transient defect records,
 $\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B})$, finite-to-infinite reduction, Return.

All Core-relevant persistence data use the fixed order

$$B_{i,W}(F) := W_W \mathbf{P}_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

No unconditional crop–collapse commutation, global non-expansiveness, exact functoriality, or Serre localization is assumed.

Whenever Type IV is invoked, the required comparison is an actual declared morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty).$$

Remark 5.1 (Interface status). Every problem-specific implication from normalization to regularity, smoothness, arithmetic validity, PDE well-posedness, geometric resolution, categorical equivalence, or singularity removal is [Spec] by default unless separately proved and registered as an Interface or Bridge Theorem.

Remark 5.2 (Why this appendix precedes Navier–Stokes appendices). AGI-N supplies the normalization and rescaling contract used by downstream PDE interfaces. NS-A may generate exploratory normalized data; NS-B must prove soundness and Return arrows; NS-C may organize a proof-first program. AGI-N itself does not infer Navier–Stokes regularity.

Remark 5.3 (No exact-collapse restoration). This appendix does not restore a global exact collapse functor, positive-threshold Serre localization, global strict filtered lift, or metric stability without the two-sided threshold-gap theorem.

AGI-N.1bis. Normalization interface sovereignty

Definition 5.4 (Normalization interface sovereignty). *Normalization interface sovereignty* is the rule that resolution, normalization, smoothing, regularization, scale alignment, chart refinement, discretization, model replacement, blow-up analysis, and defect isolation remain proposer-side or pre-gate operations unless they re-enter the AK Core through declared verifier-side records. The admissible verifier-side re-entry routes are:

- (i) constructible one-parameter persistence or finite-type filtered objects;
- (ii) window-first repaired evaluation records $\text{Eval}_{i,W,\tau}(F)$;
- (iii) finite detector records at exactly one declared source stage;
- (iv) admissible representatives $C_{\tau,W}F$;
- (v) standard degree- n eligibility records for any invoked Ext^n -clause;
- (vi) declared actual tower comparison artifacts $\phi_{i,W,\tau}$;
- (vii) terminal Type IV and transient-defect records;
- (viii) typed metric-ledger entries with support theorems for actual profile discrepancies;
- (ix) finite-to-infinite reduction records when an infinite target is claimed;
- (x) Return or soundness-out records when a source-domain conclusion is claimed;
- (xi) manifest-readable certificate records with immutable artifact identities.

Definition 5.5 (Normalization verifier-side record). A *normalization verifier-side record* is a record produced after normalization that has one of the Core-readable types listed in Definition 5.4 and is sufficiently documented for independent manifest reconstruction.

Proposition 5.6 (No normalization sovereignty override). *No normalization, resolution, smoothing, regularization, numerical preprocessing, model replacement, or successful implementation overrides the verifier-side Core requirements.*

Proof. The Core verdict is determined by typed verifier-side data: repaired evaluations, detector certificates, representatives, higher eligibility records, actual tower morphisms, diagnostics, ledgers, reductions, Return records, gates, and manifests. Normalization operations may prepare or transform data, but they do not themselves have those types unless the required records are supplied. $\square$

Remark 5.7 (Normalization evidence is proposer-side by default). Resolution charts, smoothing scripts, scale choices, numerical regularization, blow-up sequences, and local model replacements may guide construction. They are not verifier-side evidence until converted into declared Core-readable records or supported metric defects.

AGI-N.2. Normalization source data

Definition 5.8 (Normalization source datum). A *normalization source datum* is an object, diagram, field, dataset, or structured family requiring preprocessing before Core-readable evaluation. Examples include:

- (i) a singular geometric object;
- (ii) a PDE field, weak solution, or weak-solution candidate;
- (iii) a discretized numerical field;
- (iv) a multi-scale filtered object;
- (v) a noisy persistence dataset;
- (vi) an arithmetic family with incompatible local scales;
- (vii) a categorical comparison object with mismatched grading or filtration;
- (viii) a sheaf, derived, or higher-categorical object requiring extraction to persistence data;
- (ix) a tower whose finite-stage, limiting, and apex behavior are not yet aligned;
- (x) a candidate singular region requiring local chart replacement or blow-up analysis.

Definition 5.9 (Resolution-assisted normalization). A *resolution-assisted normalization* is a declared partial operation

$$\mathfrak{N} : X \dashrightarrow X^{\text{nor}}$$

together with an operation-kind record identifying every step used. Permitted operation kinds include geometric normalization, resolution or desingularization, blow-up or blow-down analysis, smoothing, regularization, scale normalization, coordinate normalization, renormalization, chart replacement, filtration alignment, defect isolation, model replacement, finite truncation, apex approximation, and persistence-extraction preparation.

Definition 5.10 (Normalization auxiliary data). *Normalization auxiliary data* are the choices required to make a normalization route audit-readable, including local charts, scale parameters, smoothing parameters, truncation levels, resolution centers, blow-up maps, discretization scales, model-replacement policies, parameter maps, extraction conventions, source and target categories, morphism classes, and artifact references.

Definition 5.11 (Normalization status). A normalization operation has exactly one lifecycle status:

declared, defined, controlled, Core_readable,
advisory, failed, undefined.

The statuses are not ordered automatically. In particular, controlled does not imply Core_readable, and Core_readable does not imply a Core pass.

Remark 5.12 (Dashed arrow). The notation $\dashrightarrow$ indicates that normalization may be partial, conditional, approximate, algorithmic, noninvertible, or dependent on auxiliary choices. It is not assumed to be a strict globally defined functor.

Remark 5.13 (Auxiliary choices are part of the interface). Normalization choices are not informal implementation details. If a choice affects consumed Core-readable data, it must be declared in the manifest and its mathematical effect must be typed.

AGI-N.2ter. Operation taxonomy and model replacement

Definition 5.14 (Normalization operation-kind record). A *normalization operation-kind record* assigns every step in a normalization route exactly one declared kind:

geometric_normalization, resolution, smoothing,
regularization, reparameterization, model_replacement,
truncation, advisory_preprocessing.

Composite routes record their ordered factorization and may not hide one operation kind under another name.

Definition 5.15 (Geometric normalization datum). A *geometric normalization datum* is a morphism

$$\nu : X^\nu \longrightarrow X$$

in a declared geometric category, with X^ν normal in the declared sense and with every claimed birational, finite, integral, proper, or universal property separately stated. This definition does not assert existence in an undeclared category or under undeclared finiteness hypotheses.

Definition 5.16 (Resolution datum). A *resolution datum* is a morphism

$$\pi : \tilde{X} \longrightarrow X$$

in a declared geometric category, with $\tilde{X}$ regular or smooth at the declared strength and with properness, birationality, exceptional locus, and functoriality scope separately recorded. This definition does not assert a general resolution-of-singularities theorem.

Definition 5.17 (Model-replacement datum). A *model-replacement datum* is a typed comparison

$$X \dashrightarrow X'$$

in which the target is used because it is computationally, analytically, categorically, or geometrically better behaved. The record states the comparison strength: equality, isomorphism, weak equivalence, quasi-isomorphism, derived equivalence, profile isomorphism, metric comparison, or advisory replacement.

Proposition 5.18 (Normalization, resolution, and replacement are not interchangeable). *Geometric normalization, resolution, smoothing, regularization, weak equivalence, derived equivalence, and persistence-profile equivalence are not interchangeable operation kinds.*

Proof. They have different source and target categories and preserve different structures. Normality, regularity, smoothness, weak equivalence, derived equivalence, and profile equivalence are distinct predicates. No implication among them is available without a theorem whose hypotheses and target structure are declared. $\square$

Definition 5.19 (Exceptional-information record). An *exceptional-information record* identifies information supported on the non-isomorphism locus of ν or π , including conductor data, branch data, exceptional divisors, multiplicities, discrepancy data, local monodromy, deleted strata, or any source feature intentionally absent from the normalized model.

Proposition 5.20 (Exceptional information is not silently erased). *A normalization or resolution route may not be called lossless for a declared purpose while omitting purpose-relevant exceptional information without a preservation or irrelevance theorem.*

Proof. A lossless claim concerns preservation of the features consumed by the target conclusion. If relevant information is confined to the exceptional or conductor locus and is omitted from the extracted model, the required preservation implication is unproved. The omission is therefore a non-scalar obligation or a declared information-loss record, not an automatic zero defect. $\square$

AGI-N.2quater. Choice dependence, canonicity, and functoriality

Definition 5.21 (Normalization canonicity status). A normalization route has exactly one canonicity status:

canonical, canonical_up_to_declared_equivalence,
choice_dependent_controlled, choice_dependent_advisory,
failed, undefined.

The declared equivalence must name the category and strength at which uniqueness is claimed.

Definition 5.22 (Functorial normalization record). A *functorial normalization record* declares:

- (i) the source category and admissible morphism class;
- (ii) the target category;
- (iii) the object assignment;
- (iv) the morphism assignment;
- (v) composition and identity laws;
- (vi) any natural comparison with the original object;
- (vii) the exact, weak, derived, filtered, or profile-level strength;
- (viii) counterexamples or excluded morphisms.

Proposition 5.23 (Functoriality is scope-relative). *A normalization procedure described as functorial on one category or morphism class is not functorial on a larger category by default and does not automatically induce functoriality of the extracted persistence profiles.*

Proof. Functoriality is a statement about specified objects, morphisms, and composition. Enlarging the morphism class or applying a subsequent extraction changes the required commutative squares. Those additional claims require separate proofs. $\square$

Proposition 5.24 (Choice independence requires comparison artifacts). *Two normalization routes arising from different charts, centers, meshes, smoothing scales, or model choices are not interchangeable in a certificate without an exact or metric comparison of the consumed Core-readable profiles and an artifact identity for that comparison.*

Proof. Different choices may change the realized filtration, window, threshold scale, detector stage, or tower artifact. A certificate may identify them only through a typed comparison at the exact strength consumed. $\square$

AGI-N.2bis. Normalization audit package and status transitions

Definition 5.25 (Normalization audit package). A *normalization audit package* consists of:

- (i) the source datum;
- (ii) the normalization operation;
- (iii) all auxiliary choices;
- (iv) the intended Core-readable target;
- (v) the status of the operation;
- (vi) the declared defect namespace;
- (vii) artifact references sufficient for reconstruction.

Definition 5.26 (Normalization status-transition certificate). A *normalization status-transition certificate* is a record justifying a transition between normalization statuses, for example from *declared* to *defined*, from *defined* to *controlled*, or from *controlled* to *Core-readable*.

Proposition 5.27 (Normalization status transitions are not automatic). *No normalization status transition is automatic.*

Proof. Each status asserts a different typed obligation. A declared operation need not be defined; a defined operation need not have controlled defect; a controlled operation need not produce constructible persistence; and a Core-readable output need not satisfy gates, diagnostics, or budget. Therefore each transition requires its own certificate. $\square$

Proposition 5.28 (Defined normalization is not controlled normalization). *A normalization operation being defined on a source datum does not imply that its Core-relevant defect is zero, bounded, scalarized, or ledger-admissible.*

Proof. Definedness is a domain-of-operation condition. Controlledness is a quantitative or typed audit condition about how the operation changes Core-relevant data. The former does not imply the latter. $\square$

Remark 5.29 (Auxiliary-choice drift). Changing charts, scales, smoothing parameters, truncation levels, or extraction conventions changes the audit package. If the change affects consumed verifier-side data, it must be recorded as a new or updated normalization route rather than silently inherited from the previous route.

AGI-N.3. Normalization as pre-gate

Definition 5.30 (Normalization pre-gate). A *normalization pre-gate* is a declared condition determining whether a source datum X is suitable for conversion into a Core-readable normalized object. It may check definedness, auxiliary-choice closure, constructibility, comparison support, detector applicability, higher eligibility, tower-artifact availability, finite-to-infinite reduction availability, and ledger support.

Definition 5.31 (Normalization pre-gate record). A *normalization pre-gate record* records:

- (i) the source datum and proposed normalization route;
- (ii) the immutable pre-gate scope;
- (iii) each checked clause and its evidence;
- (iv) an interface outcome in

pass, reject, undefined;
- (v) any clause-level not_invoked marker together with an immutable scope-exclusion rationale;
- (vi) typed failure routes and artifact references.

Proposition 5.32 (Normalization pre-gate does not replace Core gate). *A normalization pre-gate may determine whether a normalized object is submitted to Core evaluation, but it does not replace B-Gate⁺, finite detector certification, higher eligibility, monitored Type IV or transient diagnostics, required tower artifacts, finite-to-infinite reduction, or the budget condition.*

Proof. A pre-gate is an Interface preparation condition. The named Core and bridge obligations have different typed inputs and conclusions. Passing the pre-gate supplies none of them automatically. $\square$

Proposition 5.33 (Missing pre-gate evidence is undefined). *A missing required pre-gate artifact or theorem yields undefined, not pass and not not_invoked.*

Proof. The v20 weaker-safe rule distinguishes absent evidence from an immutable scope exclusion. Only the latter licenses not_invoked; otherwise the verifier cannot determine the clause. $\square$

Remark 5.34 (Pre-gate pass is not Core pass). Passing a normalization pre-gate means only that the normalized datum is eligible for the next interface step. It does not imply that the normalized object passes the AK Core.

AGI-N.4. Normalization realization targets

Definition 5.35 (Normalized realization target). A normalization interface must declare the realization target of X^{nor} as exactly one primary target, with any secondary targets declared as derived records:

- (i) a filtered object in $\text{FiltCh}(k)$;
- (ii) a constructible persistence object in $\text{Pers}_k^{\text{cons}}$;
- (iii) a windowed pre-threshold profile $B_{i,W}(F^{\text{nor}})$;
- (iv) a repaired profile $\text{Eval}_{i,W,\tau}(F^{\text{nor}})$;
- (v) an admissible representative $C_{\tau,W}F^{\text{nor}}$;
- (vi) a standard degree- n eligible realized object $A_{n;F^{\text{nor}},W,\tau}$;
- (vii) a declared actual tower comparison artifact $\phi_{i,W,\tau}^{\text{nor}}$;
- (viii) a discretized or measurement-level object with a declared conversion theorem;
- (ix) a certificate-record component;
- (x) an advisory artifact.

Definition 5.36 (Core-readable normalized object). A normalized object X^{nor} is *Core-readable* if it yields a finite-type filtered object or a constructible one-parameter persistence object for which the window-first profile

$$B_{i,W}(F^{\text{nor}}) := \mathbb{W}_W \mathbf{P}_i(F^{\text{nor}})$$

and repaired evaluation

$$\text{Eval}_{i,W,\tau}(F^{\text{nor}}) = T_{\tau}^{\text{bar}}(B_{i,W}(F^{\text{nor}}))$$

are defined on the declared scope.

Definition 5.37 (Normalized extraction route). A *normalized extraction route* is a typed process taking X^{nor} to a declared Core-readable target. It records its domain, codomain, parameter convention, morphism policy, information loss, artifact implementation, and purpose-faithfulness status.

Proposition 5.38 (No Core use without Core-readable normalized object). *A normalized object cannot support Core certification unless it is Core-readable or supplies an independently typed verifier-side certificate component.*

Proof. The Core verifier operates on declared profiles, representatives, eligible realizations, diagnostics, ledgers, tower artifacts, and manifests. A normalized source object without one of these typed outputs supplies no Core input. $\square$

Remark 5.39 (Normalization target is not automatic). Normalization may produce a smoother or better-behaved source object, but this does not automatically make it constructible one-parameter persistence data. The extraction route must be declared and verified.

AGI-N.4bis. One-parameter extraction and constructibility

Definition 5.40 (Normalization one-parameter extraction). A *normalization one-parameter extraction* selects a single ordered persistence parameter from data that may naturally involve several scales, charts, variables, coordinates, discretization levels, or physical units. It records the source parameters, selected parameter, order, units, ties, incomparable cases, discarded information, endpoint convention, and extraction defect.

Definition 5.41 (Normalization extraction-faithfulness status). A normalization extraction-faithfulness claim has exactly one status:

proved, conditional, operational,
spec, failed, undefined.

Only proved or fully discharged conditional status may support theorem evidence.

Proposition 5.42 (Scale or coordinate choice is not a Core parameter by name). *A scale, coordinate, chart index, discretization level, or physical variable is not a Core persistence parameter merely by being chosen during normalization.*

Proof. The Core requires a constructible one-parameter persistence or filtered object with a declared order and window-first evaluation. A named scale supplies no such object until extraction and constructibility are verified. $\square$

Proposition 5.43 (Smoothing or resolution does not imply constructibility). *Smoothing, regularization, desingularization, or chart refinement does not by itself imply that the extracted persistence object is constructible on the declared windows.*

Proof. Constructibility is a property of the extracted persistence object. A regular source may still produce infinite-rank, nonconstructible, multiparameter, or improperly ordered data. $\square$

Remark 5.44 (Multi-scale normalization risk). Common risks include collapsing space, time, scale, amplitude, valuation, and discretization depth into one parameter, or treating a locally convenient chart scale as globally canonical.

AGI-N.4ter. Parameter, window, and threshold reconciliation

Definition 5.45 (Normalization parameter-transport record). *A normalization parameter-transport record declares an order map*

$$r : \mathbb{R} \longrightarrow \mathbb{R},$$

the induced source and target windows W and W^{nor} , the source and target thresholds τ and τ^{nor} , endpoint behavior, and whether the transport is exact, metric-controlled, advisory, failed, or undefined.

Theorem 5.46 (Positive affine normalization transport). *Let $r(t) = at + c$ with $a > 0$, extended by $r(+\infty) = +\infty$, and let r_*B be the barcode obtained by transporting every interval endpoint by r , with endpoint decoration preserved. Then*

$$r_*W_W B = W_{r(W)} r_*B, \quad r_*T_\tau^{\text{bar}} B = T_{a\tau}^{\text{bar}} r_*B.$$

For finite barcode profiles B, C ,

$$d_B(r_*B, r_*C) = a d_B(B, C).$$

Proof. Positive affine maps preserve order, intersections, empty intervals, and finite/infinite endpoint type. Every finite interval length is multiplied by a , giving the threshold identity. Every finite endpoint matching cost and diagonal cost is multiplied by a ; infinite-bar birth costs are also multiplied by a . Taking infima over matchings gives the bottleneck identity. $\square$

Proposition 5.47 (Nonlinear reparameterization has no automatic scalar threshold rule). *For a general increasing nonlinear r , no location-independent rule $\tau \mapsto \tau^{\text{nor}}$ is inferred that makes hard threshold deletion commute on every interval.*

Proof. The transformed length of $[x, x + \ell]$ is $r(x + \ell) - r(x)$, which may depend on x as well as ℓ . Hence source intervals of equal length may lie on opposite sides of any proposed target threshold. $\square$

Proposition 5.48 (Window and threshold reconciliation is non-scalar when missing). *A missing parameter, window, endpoint, or threshold reconciliation theorem is a non-scalar obligation and cannot be repaired by adding metric reserve.*

Proof. Without reconciliation, the profiles being compared are not known to represent the same parameter scope or threshold predicate. Numerical slack cannot identify their types. $\square$

AGI-N.4quater. Normalized finite-detector discipline

Definition 5.49 (Normalized detector record). A normalized detector record declares exactly one source stage:

windowed_prethreshold, repaired_postthreshold,
kernel_defect, cokernel_defect.

It also records the exact source profile, endpoint convention, detector map, family or information class, nonzero soundness, zero completeness, birth/germ or stencil coverage, excess-bound provenance, non-adaptivity, artifact identity, certificate status, and mathematical value.

Proposition 5.50 (Finite normalized observation is not finite proof). *A finite set of normalized chart values, mesh values, ranks, critical scales, solver outputs, or sampled profiles is not a finite proof certificate without the complete detector record governing the claimed predicate.*

Proof. Finiteness of the data does not establish soundness, completeness, endpoint coverage, family membership, or artifact identity. Those are the hypotheses that turn observations into proof. $\square$

Proposition 5.51 (Normalization does not change detector stage silently). *A detector record may not be re-labeled from a topological profile stage to a kernel or cokernel defect stage, or conversely, merely because normalization relates the underlying source objects.*

Proof. The four stages have different source objects and different zero predicates. A stage change requires an actual morphism and a theorem transporting the detector semantics. $\square$

Proposition 5.52 (Right-unbounded normalized scopes retain the finite-detector no-go boundary). *On a right-unbounded normalized scope with unbounded possible birth positions, no uniform complete finite detector is inferred without additional finite-birth, stabilization, global-bound, termination, or finite-to-infinite reduction structure.*

Proof. Normalization changes the presentation of the family but does not by itself bound future birth locations. The v20 right-unbounded no-go boundary therefore remains applicable. $\square$

AGI-N.5. Normalization defects

Definition 5.53 (Normalization defect). A *normalization defect* is a typed discrepancy between source, normalized, and extracted data. It is not numerical by default.

Definition 5.54 (Resolution defect). A *resolution defect* δ^{res} records a discrepancy arising from a resolution, desingularization, blow-up, blow-down, exceptional-locus treatment, or chart replacement.

Definition 5.55 (Scale-normalization defect). A *scale-normalization defect* δ^{scale} records a discrepancy arising from rescaling, renormalization, unit conversion, or parameter normalization.

Definition 5.56 (Filtration-alignment defect). A *filtration-alignment defect* $\delta^{\text{filt-align}}$ records a discrepancy arising when a normalized filtration is aligned with the declared Core parameter.

Definition 5.57 (Normalization extraction defect). A *normalization extraction defect* $\delta^{\text{nor-ext}}$ records a discrepancy introduced when extracting Core-readable filtered or persistence data.

Definition 5.58 (Reverse-transport defect). A *reverse-transport defect* δ^{rev} records a controlled discrepancy in a proved normalized-to-source transport. The existence of the reverse transport theorem itself is non-scalar.

Definition 5.59 (Chart-normalization defect). A *chart-normalization defect* δ^{chart} records profile-level discrepancy arising from chart selection, refinement, transition mismatch, or local-to-global assembly.

Definition 5.60 (Smoothing or regularization defect). A *smoothing or regularization defect* δ^{smooth} records a supported discrepancy introduced by smoothing, mollification, regularization, or denoising.

Definition 5.61 (Normalization truncation defect). A *normalization truncation defect* $\delta^{\text{nor-trunc}}$ records a supported discrepancy introduced by finite-scale, finite-layer, finite-window, finite-rank, or finite-precision truncation.

Definition 5.62 (Normalized tower-artifact transport defect). A *normalized tower-artifact transport defect* $\Delta_{\phi, \text{nor}}$ records a supported morphism-profile discrepancy introduced when a tower comparison artifact is transported or replaced. Object-level closeness does not supply this component.

Definition 5.63 (Normalization effect class). Every normalization effect consumed by a certificate is typed as exactly one of:

metric_profile_discrepancy, morphism_profile_discrepancy,
theorem_controlled_nonmetric_effect, non_scalar_obligation,
advisory, unavailable.

Only the first two classes may enter the bottleneck metric ledger after a support theorem.

Definition 5.64 (Normalization ledger entry). A normalization ledger entry records the primitive discrepancy, effect class, source and target actual windowed pre-threshold profiles or defect profiles, support theorem, MetricUseID, quantale value, scalarization, aggregation path, alias/refinement relation, and immutable artifacts.

Proposition 5.65 (Normalization defects affect Core only through typed records). *A normalization effect affects a metric Core inference only through a supported metric-ledger entry, and affects a nonmetric inference only through its separately discharged theorem or obligation record.*

Proof. Metric discrepancies are consumed through the quantale ledger and scalarization. Semantic preservation, existence, functoriality, completeness, and Return are not metric predicates and retain their own proof obligations. $\square$

Proposition 5.66 (Implementation error is not bottleneck error). *A mesh size, solver residual, chart tolerance, smoothing parameter, or model norm does not enter the metric ledger unless a theorem or verified computation bounds the actual bottleneck discrepancy of the consumed profiles.*

Proof. These quantities live in implementation or source spaces. The metric ledger supports a persistence inference only when they are converted to the actual profile metric used by that inference. $\square$

Remark 5.67 (Normalization is not budget-free). A normalization procedure may simplify analysis, but its supported metric effects must be ledgered and its non-scalar obligations must be discharged separately.

AGI-N.5bis. Defect namespace and double-counting discipline

Definition 5.68 (Primitive normalization discrepancy). A *primitive normalization discrepancy* is an atomic mismatch introduced or exposed by normalization before aggregation.

Definition 5.69 (Normalization discrepancy relation). Whenever an AGI-N defect is used with Appendix M, Appendices H–J, MS, or NS defect names, the relation is declared as exactly one of

alias, refinement, aggregate, disjoint.

Definition 5.70 (Normalization MetricUseID). A *normalization MetricUseID* identifies one actual bottleneck-discrepancy bound and every inference that reuses it, including threshold transport, detector zero transport, overlap transport, and Restart transport.

Proposition 5.71 (No double counting of normalization defects). *The same primitive discrepancy or MetricUseID may not be counted twice in one aggregation path.*

Proof. The ledger must upper-bound the actual discrepancy consumed by a path, not count one bound once for each logical consequence. Alias and reuse relations therefore prevent duplication. $\square$

Proposition 5.72 (Undeclared defect relation blocks consumption). *If two defect symbols may overlap and their relation is undeclared, a certificate may not consume them as independent budget components.*

Proof. The verifier cannot reconstruct whether the proposed aggregate is a sum, maximum, refinement, or duplicate. The budget is therefore undefined. $\square$

Proposition 5.73 (Alternative normalization routes are not automatically aggregated). *Defects on alternative normalization routes are not combined by $\oplus$ unless one proof path actually composes those routes.*

Proof. A ledger aggregate follows a consumed proof path. Alternative candidates are choices, not sequential discrepancies. $\square$

Remark 5.74 (Ledger-catalog registration). All AGI-N metric components must be registered in or mapped to Appendix M before final certificate consumption. Local notation alone does not provide metric authority.

AGI-N.6. Normalized repaired evaluation

Definition 5.75 (Normalized repaired evaluation). Let X^{nor} have a Core-readable realization F^{nor} . Define

$$\begin{aligned} B_{i,W}(F^{\text{nor}}) &:= W_W \mathbf{P}_i(F^{\text{nor}}), \\ \text{Eval}_{i,W,\tau}(F^{\text{nor}}) &:= T_\tau^{\text{bar}}(B_{i,W}(F^{\text{nor}})). \end{aligned}$$

No crop-after-collapse expression is substituted without the exact crop-collapse criterion.

Definition 5.76 (Normalization-adjusted metric budget). Let $(V, \oplus, 0_V, \preceq)$ be the declared metric-ledger quantale. The normalization-adjusted metric aggregate $\mathbf{d}_{\text{nor}}^{\text{met}} \in V$ contains only supported metric-profile and morphism-profile discrepancies on the consumed proof path. Its real scalarization is

$$\varepsilon_{\text{nor}} := \text{val}_B(\mathbf{d}_{\text{nor}}^{\text{met}}).$$

Non-scalar obligations are not terms of this aggregate.

Proposition 5.77 (Normalized evidence must pass adjusted budget). *A metric inference relying on normalized data may pass only if the record proves that*

$$d_B(B, C) \leq \varepsilon_{\text{nor}}$$

for every actual profile pair consumed by that inference and

$$\varepsilon_{\text{nor}} < \text{gap}_\tau(\mathfrak{B}_{\text{nor}}).$$

A repaired-zero transfer additionally consumes the v20 post-threshold separation condition, operationally $2\varepsilon_{\text{nor}} < \tau$.

Proof. The first inequality gives semantic support to the scalarized ledger value. The strict gap comparison licenses threshold classification transport, and the additional separation condition licenses zero transport. $\square$

Remark 5.78 (Normalized pass is not source pass). A Core pass for a normalized object does not automatically imply any pass or theorem about the original source object.

Remark 5.79 (Evaluation record is still required). A normalization ledger does not replace the exact repaired evaluation or the complete detector record.

AGI-N.6bis. Normalization comparison modes and gap safety

Definition 5.80 (Normalization comparison). A *normalization comparison* compares source data, normalization routes, extracted pre-threshold profiles, repaired profiles, defect profiles, or tower artifacts at a declared type.

Definition 5.81 (Normalization comparison mode). Every local normalization comparison has exactly one mode:

source_exact, source_equivalent, realized_exact,
realized_metric, ledger_controlled, advisory_only,
not_compared, failed, undefined.

Only realized_exact projects to Core mode exact; only realized_metric with the complete profile-level metric record projects to metric_gap_certified. The mode not_compared cannot discharge an invoked comparison obligation.

Proposition 5.82 (Source exactness is not normalized Core exactness). *An equality, isomorphism, proper birational morphism, weak equivalence, quasi-isomorphism, or derived equivalence in the source domain is not a Core exact comparison of normalized profiles unless transported to an actual profile isomorphism.*

Proof. The source relations preserve structures in their own categories. A Core exact comparison is a relation between the declared realized profiles. The required type-changing theorem is independent. $\square$

Proposition 5.83 (Normalization closeness is not metric-gap certification). *A small source-side or implementation-side error is not a metric-gap-certified comparison unless it bounds the actual windowed pre-threshold profiles and both profiles carry the two-sided threshold-gap record.*

Proof. Hard deletion is not globally non-expansive. The repaired metric theorem therefore requires the actual profile pair and the two-sided threshold band. $\square$

Proposition 5.84 (Object closeness does not control normalized defect morphisms). *Closeness of normalized source and apex objects does not automatically bound the kernel or cokernel defect profiles of a normalized tower comparison morphism.*

Proof. Kernel and cokernel depend on an actual morphism, not only on its source and target objects. A separate single-morphism or defect-profile comparison theorem is required. $\square$

Remark 5.85 (Gap is measured after realization and windowing). Threshold clearance is measured on the realized windowed pre-threshold profiles, not on source scales, unwindowed data, or after-deletion profiles.

AGI-N.6ter. Higher eligible realization after normalization

Definition 5.86 (Normalized standard degree- n eligible regime). Fix $n \geq 1$. A normalized interface is standard degree- n eligible on $(F^{\text{nor}}, W, \tau)$ when it contains:

- (i) a degree- n compatible admissible representative;
- (ii) an independently meaningful realization $A_{n;F^{\text{nor}},W,\tau} \in D^{[-n,0]}(k\text{-mod})$;
- (iii) a fixed edge extractor E_n with $E_n(0) = 0$;
- (iv) an artifact-backed isomorphism

$$H^{-n}(A_{n;F^{\text{nor}},W,\tau}) \cong E_n(\text{Eval}_{n,W,\tau}(F^{\text{nor}}));$$

- (v) representative-change, realization-change, coefficient, naturality, claim, and artifact records.

Theorem 5.87 (Normalized higher obstruction discharge). *In the regime of Definition 5.86,*

$$\text{Eval}_{n,W,\tau}(F^{\text{nor}}) = 0 \implies \text{Ext}^n(A_{n;F^{\text{nor}},W,\tau}, k) = 0.$$

Proof. The edge extractor sends the zero profile to zero, so the artifact-backed identification gives $H^{-n}(A_{n;F^{\text{nor}},W,\tau}) = 0$. Over a field, the canonical higher edge identifies $\text{Ext}^n(A, k)$ with the dual of $H^{-n}(A)$. $\square$

Proposition 5.88 (Normalization does not supply reverse higher-edge reflection). *The reverse implication from $\text{Ext}^n = 0$ to repaired profile zero is unavailable unless a separate natural zero-reflection theorem is supplied for the declared extractor and profile class.*

Proof. Normalization and amplitude do not make the edge extractor faithful or zero-reflecting. $\square$

AGI-N.7. Source-to-normalized transport

Definition 5.89 (Source-to-normalized transport statement). A source-to-normalized transport statement has direction

source structure $\implies$ specified normalized or Core-visible behavior.

Definition 5.90 (Normalized-to-source transport statement). A normalized-to-source transport statement has direction

specified normalized or Core-visible behavior $\implies$ source-domain conclusion.

Definition 5.91 (Normalization transport theorem). A normalization transport theorem supplies the exact source and target semantics, operation-kind factorization, auxiliary choices, parameter reconciliation, purpose-faithful realization, repaired profiles, detector records, higher eligibility when invoked, tower artifacts and defect records when invoked, finite-to-infinite reduction when required, metric and non-scalar obligations, artifacts, and a proof of the stated direction.

Definition 5.92 (Default status of normalization transport). Every source-to-normalized or normalized-to-source transport statement is [Spec] by default unless separately proved and registered.

Proposition 5.93 (Normalized Core pass does not imply source conclusion). *A Core pass for X^{nor} does not imply a conclusion about X unless a normalized-to-source transport or Return theorem proves that implication.*

Proof. The normalization may discard, alter, approximate, or reparameterize information. The audited Core conclusion concerns the realized normalized object, not the original source semantics. $\square$

Remark 5.94 (Two transport directions). The forward and reverse directions are distinct. The reverse direction is usually the one required for an external theorem and therefore carries the stronger information-preservation burden.

AGI-N.7bis. Losslessness and theorem-promotion discipline

Definition 5.95 (Lossless normalization statement). A *lossless normalization statement* asserts preservation of every feature needed by one exact declared conclusion, including exceptional-locus information, parameter semantics, detector predicate, eligible edge, tower artifact, reduction target, and Return semantics when those are invoked.

Definition 5.96 (Normalization theorem promotion). Normalization theorem promotion changes a normalization-assisted implication from [Spec] or Bridge-candidate status to theorem status. It requires a proof, immutable source and target bindings, claim-use preflight, and Claim Register promotion.

Proposition 5.97 (Normalization is not lossless by declaration). *Declaring a normalization canonical, standard, birational, weakly equivalent, derived equivalent, or regularizing does not establish losslessness for the declared Core or source conclusion.*

Proof. Each adjective controls a different structure. None automatically preserves every arrow from projection through Return. $\square$

Proposition 5.98 (Reverse-transport hypotheses are non-scalar). *Missing hypotheses for normalized-to-source Return cannot be repaired by metric slack, advisory evidence, or a successful pre-gate.*

Proof. They determine semantic implication, not numerical approximation. $\square$

Remark 5.99 (The reverse direction is the dangerous direction). The reverse direction is where exceptional information, noninvertibility, chart dependence, truncation, and model replacement can invalidate an otherwise clean normalized Core pass.

AGI-N.7ter. Complete normalization Bridge and Return chain

Definition 5.100 (Normalization Projection–Return chain). A complete normalization-assisted problem chain is

$$\begin{aligned} X &\longrightarrow X^{\text{nor}} \longrightarrow F^{\text{nor}}, \\ F^{\text{nor}} &\longrightarrow B_{i,W}(F^{\text{nor}}) \longrightarrow \text{Eval}_{i,W,\tau}(F^{\text{nor}}), \\ \text{Eval}_{i,W,\tau}(F^{\text{nor}}) &\longrightarrow Q_{\text{fin}} \longrightarrow Q_{\text{AK}}, \\ Q_{\text{AK}} &\longrightarrow \text{Verdict}_{\text{AK}} \longrightarrow C_X. \end{aligned}$$

where every arrow is separately typed as Interface, finite certification, finite-to-infinite reduction, higher-obstruction transport, or Return.

Definition 5.101 (Normalization outcome vector). A normalization-assisted record stores separate outcomes for:

interface, finite_certificate, Core,
higher_edge, TypeIV, transient,
reduction, Return, external.

No component is inferred from another by status alone.

Proposition 5.102 (No outcome-layer collapse). *Interface pass does not imply Core pass; finite-certificate pass does not imply an infinite conclusion; terminal Type IV zero does not imply full defect zero; and Core pass does not imply a source theorem.*

Proof. Each implication omits a separately typed verifier, reduction, defect, or Return theorem. □

Definition 5.103 (AGI-N v20 bridge status). Appendix AGI-N itself declares no concrete external-domain normalization Bridge in v20.0.0. Its local bridge status is

schema_available; concrete_bridge_not_invoked.

Downstream appendices, including NS-B or NS-C, may invoke a concrete track only by supplying the complete Projection–Return chain.

AGI-N.7quater. Resolution towers and finite-to-infinite reduction

Definition 5.104 (Resolution or refinement tower). A *resolution or refinement tower* is an actual directed diagram

$$X_0^{\text{nor}} \longrightarrow X_1^{\text{nor}} \longrightarrow X_2^{\text{nor}} \longrightarrow \dots$$

with declared transition maps, extraction-compatible realized tower, and a separately declared apex policy. A list of models without morphisms is not a tower.

Definition 5.105 (Termination or coverage certificate). A *termination or coverage certificate* proves either:

- (i) finite termination at a stage satisfying the declared normalization objective;
- (ii) exhaustive finite coverage of all retained features on the declared scope; or
- (iii) applicability of a separately proved finite-to-infinite reduction theorem.

Proposition 5.106 (Finite prefix is not normalized apex). *A finite prefix, observed plateau, repeated barcode, or numerically stable refinement sequence does not identify the infinite normalized object or declared apex.*

Proof. Future stages may introduce new features, maps may fail to stabilize, and the declared apex may involve a colimit, homotopy colimit, completion, localization, or external limit. A reduction theorem is required. □

Proposition 5.107 (Resolution tower finite proof requires termination or reduction). *A finite proof claim over an infinite resolution or refinement tower requires a termination, coverage, or finite-to-infinite reduction certificate; finite computation alone is insufficient.*

Proof. The target statement quantifies beyond the computed prefix. The certificate must therefore provide the mathematical step that closes the unobserved tail. □

AGI-N.8. Resolution and singularity claims

Definition 5.108 (Singularity proxy). A *singularity proxy* is an interface-level signal suggesting the presence of a singular, unstable, or unresolved region. Examples include:

- (i) blow-up of a numerical or analytic observable;
- (ii) concentration of persistence mass;
- (iii) nonzero residual repaired degree-1 profile

$$\text{Eval}_{1,w,\tau}(F) \neq 0;$$

- (iv) unstable Betti integral behavior;
- (v) anomalous tower defect;
- (vi) failure of normalization pre-gate;
- (vii) local scale collapse;
- (viii) chart incompatibility;
- (ix) failure of finite-layer-to-apex comparison.

Definition 5.109 (Resolution success claim). A *resolution success claim* is a statement asserting that a resolution-assisted normalization has removed, controlled, isolated, or localized a singularity or defect.

Proposition 5.110 (Resolution success is not Core success). A *resolution success claim does not imply Core pass unless the resulting normalized object satisfies Core gate, diagnostic, artifact, and budget conditions*.

Proof. Resolution success is an interface-level or domain-level statement. Core pass requires B-Gate⁺, monitored Type IV cleanliness, required artifacts, and budget admissibility. The former does not imply the latter without further verification. $\square$

Proposition 5.111 (Resolution success is not problem-specific regularity). A *resolution success claim does not imply problem-specific regularity, smoothness, or non-blow-up unless a separate transport theorem proves that implication*.

Proof. Resolution-assisted normalization may alter the object, isolate a defect, or replace local data. A problem-specific regularity statement concerns the original source problem. A transport theorem is required to connect the normalized behavior back to the original problem. $\square$

Remark 5.112 (Singularity proxy is not singularity theorem). A singularity proxy may indicate where to inspect. It is not itself a theorem asserting singularity formation or singularity absence.

AGI-N.8bis. Local normalization atlases and descent

Definition 5.113 (Normalization atlas). A *normalization atlas* is a declared cover $\{U_a\}$ with local normalization routes, extracted local profiles, overlap comparisons, and immutable chart artifacts.

Definition 5.114 (Normalization object descent). Object descent is the existence of a global normalized object or profile reconstructed from local objects, actual transition morphisms, and the required higher cocycle or hyperdescent data.

Definition 5.115 (Normalization certificate descent). Certificate descent is the stronger statement that local verifier records glue to a global record with coherent scope, artifacts, detector stages, thresholds, ledgers, Type IV data, and Return obligations.

Proposition 5.116 (Pairwise chart agreement is not global descent). *Pairwise agreement of local normalized charts or profiles does not imply object descent or certificate descent*.

Proof. Triple and higher overlaps may fail coherence, and even object descent does not automatically glue detector completeness, artifact identity, ledgers, tower morphisms, or Return semantics. $\square$

Proposition 5.117 (Local Core passes do not imply global Core pass). *Core passes on every local normalized chart do not imply a global Core pass without certificate descent and the relevant atlas coverage theorem.*

Proof. A global verdict may fail through uncovered regions, incoherent overlaps, inconsistent thresholds, duplicated ledger entries, or missing higher descent data. $\square$

AGI-N.9. Normalization and monitored Type IV

Definition 5.118 (Normalized tower). A *normalized tower* is an actual directed system

$$F_0^{\text{nor}} \longrightarrow F_1^{\text{nor}} \longrightarrow \dots$$

with declared transition morphisms and a separately declared apex object F_∞^{nor} .

Definition 5.119 (Normalized tower comparison artifact). Fix one right-open window W , one threshold $\tau > 0$, and a finite monitored degree set I_{mon} . For each $i \in I_{\text{mon}}$, a normalized tower comparison artifact is an actual morphism in the declared abelian persistence category

$$\phi_{i,W,\tau}^{\text{nor}} : \text{ColimProfile}_{i,W,\tau}(F_\bullet^{\text{nor}}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty^{\text{nor}}).$$

Definition 5.120 (Normalization Type IV proxy). A Normalization Type IV proxy is a proposer-side signal suggesting possible tower pathology. It is not a diagnostic value or comparison artifact.

Definition 5.121 (Normalized Type IV status). The normalized Type IV status is exactly one of

not_invoked, undefined, certified_zero, obstructed.

No numerical pair is imputed in the first two states.

Definition 5.122 (Actual normalized Type IV diagnostic). The *actual normalized Type IV diagnostic* is the degree-wise and aggregated kernel/cokernel terminal-generic data computed from the complete normalized Type IV record of Definition 5.123. It is undefined unless that record supplies the actual morphisms, terminal semantics, tameness, and defect profiles.

Definition 5.123 (Complete normalized Type IV record). A complete normalized Type IV record contains:

- (i) immutable tower, transition, extraction, apex, and artifact identities;
- (ii) source provenance, source semantics, and semantic-authority status for the claimed use;
- (iii) the fixed (W, τ) and finite I_{mon} ;
- (iv) the ambient abelian profile category and actual morphisms $\phi_{i,W,\tau}^{\text{nor}}$;
- (v) terminal mode bounded_terminal or right_unbounded;
- (vi) terminal-tameness of source, target, kernel, and cokernel;
- (vii) full kernel and cokernel profiles;
- (viii) degree-wise values

$$\begin{aligned} \mu_i^{\text{nor}} &= \text{tgdim}_W \ker \phi_{i,W,\tau}^{\text{nor}}, \\ \nu_i^{\text{nor}} &= \text{tgdim}_W \text{coker } \phi_{i,W,\tau}^{\text{nor}}; \end{aligned}$$

- (ix) total values

$$\begin{aligned} \mu_{\text{Collapse}}^{\text{nor}} &= \sum_{i \in I_{\text{mon}}} \mu_i^{\text{nor}}, \\ \nu_{\text{Collapse}}^{\text{nor}} &= \sum_{i \in I_{\text{mon}}} \nu_i^{\text{nor}}. \end{aligned}$$

- (x) full-interior-defect and transient-defect records;
- (xi) typed status and failure route.

The status is `certified_zero` exactly when all required inputs are valid and both totals vanish; it is obstructed when all required inputs are valid and at least one total is nonzero.

Proposition 5.124 (Normalization Type IV proxy is not Type IV diagnosis). *A normalization proxy does not determine the normalized or source Type IV status.*

Proof. The diagnostic is defined from kernels and cokernels of the actual comparison morphisms in the complete record. A proxy supplies neither those morphisms nor their defect profiles. $\square$

Remark 5.125 (Decorated normalized diagnostics). The pair $(\mu_{\text{Collapse}}^{\text{nor}}, \nu_{\text{Collapse}}^{\text{nor}})$ belongs to the declared normalized tower and artifact. It is not the source diagnostic without a transport theorem.

Remark 5.126 (Correct use of normalization proxies). Normalization proxies may prioritize an actual Type IV computation; they do not replace it.

AGI-N.9bis. Type IV coherence and transient defects

Definition 5.127 (Artifact-coherent normalized Type IV record). An artifact-coherent normalized Type IV claim fixes the normalization route, extraction route, tower, transition maps, apex, (W, τ) , I_{mon} , terminal mode, and comparison morphisms throughout.

Definition 5.128 (Normalized transient-defect record). A normalized transient-defect record declares exactly one source stage

`kernel_defect` or `cokernel_defect`,

a defect threshold $\tau_{\text{def}} > 0$, full-profile or interior-noncofinal component, endpoint and cofinal convention, family and information class, detector soundness and completeness, non-adaptivity, immutable artifacts, certificate status, mathematical value, and aggregate long-transient status.

Definition 5.129 (Normalized full-interior-defect status). The full-interior-defect status is exactly one of

`not_invoked`, `undefined`, `certified_zero`, `obstructed`.

It is independent of the terminal Type IV status.

Proposition 5.130 (Changing normalized artifacts creates a new claim). *Changing any artifact-coherence field creates a new Type IV claim unless an exact transport theorem identifies the records.*

Proof. The diagnosed morphism and its terminal semantics have changed. $\square$

Proposition 5.131 (Undefined normalized Type IV is non-verdict). *A missing artifact, authority record, actual morphism, terminal-tameness certificate, or defect profile yields undefined, not $(0, 0)$.*

Proof. The numerical diagnostic is not defined until every required typed input exists. $\square$

Proposition 5.132 (Terminal zero is not full defect zero). *The status `certified_zero` for terminal Type IV does not imply absence of finite interior or long-transient defect bars and does not imply that $\phi_{i, W, \tau}^{\text{nor}}$ is an isomorphism.*

Proof. Terminal-generic dimension ignores noncofinal finite defects. Their absence requires the separate full or transient detector records. $\square$

Remark 5.133 (Terminal-tameness is required). Finite-window visual stability does not supply terminal-generic dimension.

AGI-N.10. Normalization certificate records

Definition 5.134 (Normalization certificate record). A normalization certificate record is a certificate whose source has passed through a declared normalization route and whose verifier-side data are extracted from the normalized object.

Definition 5.135 (AGI-N manifest entry). An AGI-N manifest entry records, at minimum:

- (i) source identity, source semantics, and intended external conclusion;
- (ii) operation-kind factorization, normalization route, status, and normalized object;
- (iii) auxiliary choices, canonicity status, functoriality scope, and exceptional-information record;
- (iv) realization target, extraction route, purpose-faithfulness, and constructibility;
- (v) parameter, window, endpoint, and threshold reconciliation;
- (vi) exact pre-threshold and repaired profiles;
- (vii) detector stage and complete finite-detector record when invoked;
- (viii) representative and standard degree- n eligibility records when invoked;
- (ix) normalization comparison modes and all actual profile comparisons;
- (x) metric-ledger quantale, support theorems, MetricUseIDs, scalarization, and protected families;
- (xi) all non-scalar obligations and typed failure routes;
- (xii) resolution/refinement tower, transitions, apex policy, and reduction record when invoked;
- (xiii) complete terminal Type IV, full-interior, and transient-defect records when invoked;
- (xiv) normalization atlas, higher coherence, and certificate descent when local data are glued;
- (xv) Interface, finite-certificate, Core, higher-edge, reduction, Return, and external outcomes;
- (xvi) concrete bridge TrackID or an immutable not_invoked rationale;
- (xvii) claim references, repair ancestry, artifacts, versions, and hashes.

Proposition 5.136 (Normalization certificate records must satisfy Minimal Manifest). *Every normalization certificate record must satisfy Appendix G and the additional AGI-N fields required by the clauses it invokes.*

Proof. Normalization adds choices, transformations, and possible information loss; it does not weaken the v20 audit contract. □

Remark 5.137 (Normalization artifacts). Artifacts may include resolution charts, exceptional-locus data, blow-up maps, scale maps, smoothing parameters, model replacements, filtered objects, extraction code, detector records, repaired profiles, tower morphisms, error estimates, descent data, reduction proofs, Return proofs, and ledger records. Artifacts support reconstruction but do not create proof by storage.

AGI-N.11. Normalization pass, Core pass, and problem conclusion

Definition 5.138 (Normalization interface pass). A normalization interface pass means that the source has been converted into a complete, consistent, Core-readable entry under the declared Interface contract.

Definition 5.139 (Core pass for normalized data). A Core pass for normalized data is the exact Chapter 7 verdict on the normalized verifier-side record. It may concern repaired zero, higher Ext discharge, terminal or transient defects, or another typed predicate.

Definition 5.140 (Problem-specific conclusion). A problem-specific conclusion is a statement in the original source semantics, such as regularity, smoothness, non-blow-up, desingularization, equivalence, finiteness, or well-posedness.

Proposition 5.141 (Core pass for normalized data is not problem conclusion). *A Core pass for normalized data does not imply a problem-specific conclusion without the exact reduction and Return arrows required by that conclusion.*

Proof. The antecedent and consequent inhabit different typed conclusion levels. □

Remark 5.142 (Three-layer distinction). At minimum, normalization Interface pass, Core pass, and source theorem remain distinct; in v20 the finite certificate, reduction, higher-edge, transient, and Return outcomes are also stored separately.

AGI-N.12. Relation to Navier–Stokes interface

Remark 5.143 (Use in Navier–Stokes appendices). In the Navier–Stokes appendices, resolution-assisted normalization may be used to prepare velocity, vorticity, energy, pressure, enstrophy, or localized scale profiles for Core-readable persistence extraction. However, it remains a pre-gate interface. It does not prove Navier–Stokes regularity, non-blow-up, uniqueness, or smoothness.

Definition 5.144 (PDE normalization pre-gate). A *PDE normalization pre-gate* is a normalization pre-gate applied to PDE-derived data, such as:

- (i) local energy profiles;
- (ii) vorticity concentration profiles;
- (iii) pressure fields;
- (iv) scale-invariant quantities;
- (v) weak-solution approximants;
- (vi) discretized flow fields;
- (vii) candidate singular neighborhoods;
- (viii) blow-up rescaling windows;
- (ix) local regularity charts.

Definition 5.145 (PDE normalized profile). A *PDE normalized profile* is a normalized PDE-derived datum realized as a filtered or persistence object and evaluated through

$$\text{Eval}_{i,w,\tau}(F_{\text{PDE}}^{\text{nor}}).$$

Proposition 5.146 (PDE normalization pre-gate does not imply PDE regularity). *Passing a PDE normalization pre-gate does not imply PDE regularity.*

Proof. A PDE normalization pre-gate only prepares PDE-derived data for further interface processing or Core-readable evaluation. PDE regularity is a problem-specific conclusion about solutions of a PDE. The former does not imply the latter without a transport theorem. □

Remark 5.147 (PDE profile is not PDE theorem). A repaired evaluation record derived from a PDE normalized profile may be useful evidence inside the AK interface. It is not, by itself, a theorem about Navier–Stokes or any other PDE.

AGI-N.12bis. PDE, discretization, and rescaling hardening

Definition 5.148 (PDE normalization evidence status). A PDE normalization artifact has exactly one status:

sample, calibration, bounded_approximation,
certified_Core_record, transport_theorem,
failed, undefined.

Only a complete certified record or proved transport theorem may support the corresponding verifier-side use.

Definition 5.149 (PDE rescaling defect). A PDE rescaling defect $\delta^{\text{PDE-scale}}$ is a supported profile discrepancy introduced by blow-up rescaling, parabolic scaling, localization, renormalization, or chart-dependent scale selection. The existence of a rescaled solution, limit, or reverse theorem remains non-scalar.

Definition 5.150 (PDE rescaling contract). A PDE rescaling contract records the equation, solution class, source and rescaled domains, time intervals, boundary conditions, scaling exponents, normalization center, convergence mode, extraction route, discretization and measurement records, and the exact source-to-rescaled and rescaled-to-source claims.

Proposition 5.151 (Numerical normalized profile is not regularity theorem). *A numerical or sampled normalized PDE profile is not a regularity, smoothness, uniqueness, or non-blow-up theorem.*

Proof. A numerical profile has finite modeling and computational scope; the listed conclusions quantify over the source equation and solution class. □

Proposition 5.152 (Blow-up rescaling profile is not singularity exclusion). *A clean or zero rescaled profile does not exclude source singularity formation without a proved reverse transport theorem for the exact rescaling contract.*

Proof. Rescaling changes coordinates, scale, domain, and potentially the limiting object. □

Proposition 5.153 (Sampled stabilization is not certified regularity). *Observed stabilization across sampled normalized windows is not regularity unless upgraded through a complete detector, finite-to-infinite reduction, and normalized-to-source Return chain.*

Proof. A finite sample supplies no universal continuum or source-domain quantifier. □

Proposition 5.154 (Weak-solution normalization does not create strong-solution semantics). *Normalizing, smoothing, or discretizing a weak solution or candidate does not promote it to a strong or classical solution of the original PDE.*

Proof. Solution-class membership is an analytic theorem about the original equation and regularity, not a property created by preprocessing. □

Remark 5.155 (Discretization and measurement remain externally typed). Mesh, timestep, truncation, precision, sensor error, and solver residuals are governed by Appendices I and J and enter AGI-N only through declared conversion and alias records.

AGI-N.12ter. Handoff contract to NS-A, NS-B, and NS-C

Definition 5.156 (AGI-N to NS handoff record). An AGI-N to NS handoff record contains:

- (i) the PDE, domain, boundary conditions, time interval, and solution class;
- (ii) the source field or candidate and its evidence status;
- (iii) the normalization or rescaling route and all auxiliary choices;
- (iv) discretization and measurement dependencies from Appendices I and J;

- (v) the one-parameter extraction, parameter map, windows, thresholds, and monitored degrees;
- (vi) pre-threshold, repaired, detector, higher-edge, tower, and ledger records actually available;
- (vii) the exact unresolved analytic, reduction, and Return obligations;
- (viii) immutable artifacts and failure routes.

Proposition 5.157 (Normalization handoff is not NS soundness). *A complete AGI-N to NS handoff record does not prove that the normalized AK predicate is sound or complete for Navier–Stokes regularity, blow-up, uniqueness, or smoothness.*

Proof. The handoff records typed inputs and open obligations. The analytic identification between those inputs and the PDE conclusion belongs to NS-B or a separately proved Bridge Theorem. $\square$

Remark 5.158 (Downstream responsibility). NS-A consumes the handoff for exploration only; NS-B must prove source-to-profile and profile-to-source soundness at the exact claimed strength; NS-C must not count a proposed proof route as a completed bridge.

AGI-N.13. Summary of resolution-assisted normalization interface

Theorem 5.159 (Appendix AGI-N interface package). *Within the Problem Interface layer, the following package is available.*

- (i) *Resolution-assisted normalization is a pre-gate Interface, not a Core proof.*
- (ii) *Normalization, resolution, smoothing, regularization, reparameterization, and model replacement are separately typed.*
- (iii) *Geometric normalization and resolution data declare all finiteness, properness, birationality, regularity, exceptional-locus, and functoriality hypotheses rather than assuming them by name.*
- (iv) *Choice dependence, canonicity, and functoriality are scope-relative and artifact-backed.*
- (v) *Exceptional information is preserved, proved irrelevant, or explicitly recorded as omitted.*
- (vi) *Core readability requires a declared constructible one-parameter realization and window-first repaired evaluation.*
- (vii) *Positive affine parameter changes transport windows, thresholds, and bottleneck distance exactly at the displayed scale; nonlinear changes have no automatic scalar threshold rule.*
- (viii) *Finite normalized observations become proof only through complete stage-typed detector records.*
- (ix) *Standard degree-n eligible normalization supports the one-way higher Ext discharge and no unproved reverse implication.*
- (x) *Source exactness, weak or derived equivalence, and implementation error are not Core exactness or metric-gap certification by name.*
- (xi) *Only supported actual profile discrepancies enter the metric ledger; semantic and existence obligations remain non-scalar.*
- (xii) *Primitive discrepancies and MetricUseIDs obey alias, path, and double-counting discipline.*
- (xiii) *A complete normalization problem demonstration separates Interface, finite certification, Core, higher-edge, reduction, Return, and external outcomes.*
- (xiv) *AGI-N declares the normalization Bridge schema but invokes no concrete external Bridge in this appendix.*
- (xv) *Infinite resolution or refinement towers require actual transition morphisms, a declared apex, and a termination, coverage, or finite-to-infinite reduction theorem.*

- (xvi) *Pairwise local chart agreement is neither object descent nor certificate descent.*
- (xvii) *Normalized Type IV diagnostics use one fixed (W, τ) , a finite degree set, actual morphisms, complete terminal records, and separate interior/transient records.*
- (xviii) *Terminal Type IV zero is not full defect zero or map isomorphism.*
- (xix) *A Core pass for normalized data is not a source theorem without a proved Return chain.*
- (xx) *PDE rescaling, numerical profiles, weak-solution preprocessing, and sampled stabilization do not imply regularity or non-blow-up.*
- (xxi) *The AGI-N to NS handoff records exact available data and unresolved obligations but is not an NS soundness theorem.*
- (xxii) *All invoked clauses are serialized through the expanded AGI-N manifest and Claim Register governance.*

Proof. The items collect the definitions and propositions of Sections AGI-N.1–AGI-N.12bis. Each item is either a typed Interface definition, a proved internal transport theorem with displayed hypotheses, or an explicit non-implication preserving the v20 conservative-extension boundary. $\square$

AGI-N.14. Explicit exclusions

The following are not asserted in Appendix AGI-N.

- No general normalization, resolution, desingularization, smoothing, or regularization existence theorem is proved.
- No geometric normalization is identified with a resolution.
- No proper birational morphism, weak equivalence, quasi-isomorphism, derived equivalence, or model replacement is identified with persistence-profile equality by default.
- No normalization route is assumed invertible, canonical, functorial, or lossless outside its declared category and morphism class.
- No exceptional-locus, conductor, branch, multiplicity, or discrepancy information is declared irrelevant without proof.
- No scale, coordinate, chart, mesh, or physical variable is automatically a Core persistence parameter.
- No nonlinear reparameterization receives an automatic globally valid threshold scaling rule.
- No finite normalized observation or numerical plateau is a finite proof certificate.
- No finite prefix or repeated profile identifies an infinite normalized apex.
- No pairwise chart agreement implies higher descent or global certificate pass.
- No pre-gate replaces B-Gate⁺, repaired zero, detector completeness, higher eligibility, Type IV, transient defects, finite-to-infinite reduction, Return, or the metric budget.
- No normalization creates a reverse Extⁿ-to-profile implication.
- No object-level closeness controls kernel or cokernel defect profiles without a morphism theorem.
- No terminal Type IV zero implies full defect zero or comparison-map isomorphism.
- No undefined diagnostic is interpreted as zero.
- No metric slack repairs missing realization, functoriality, constructibility, detector completeness, apex agreement, descent, or Return.

- No Core pass for normalized data is a theorem about the original source object.
- No PDE normalization, blow-up rescaling, smoothing, weak-solution processing, or simulation proves Navier–Stokes regularity, uniqueness, smoothness, or non-blow-up.
- No concrete external normalization Bridge is claimed by AGI-N in v20.0.0.
- No global exactness, global non-expansiveness, Serre localization, or global strict filtered lift for hard threshold deletion is assumed.

AGI-N.15. Provenance and status

The material in this appendix depends on:

- (i) Chapters 1–8, especially the finite-certificate, higher-edge, Type IV, atlas, reduction, and Return contracts;
- (ii) Foundation Appendices A–G for metric repair, detector and representative discipline, higher eligibility, Type IV/transient calculus, overlap and Restart, non-compensation, and manifest semantics;
- (iii) Appendices H–J for advisory, discretization, and measurement boundaries;
- (iv) Appendices L–N for transport cells, quantale ledger semantics, and higher-categorical/descent boundaries;
- (v) Appendices MS-A–MS-F for purpose-faithfulness, comparison, and Problem Interface sovereignty;
- (vi) Appendix CR-v20 for final claim-use governance after Part IV is frozen.

v20 conformance status.

v20 mathematical conformant.
 Normalization, resolution, and model-replacement taxonomy hardened.
 Finite detector, higher Ext, Type IV/transient, descent, reduction, and Return boundaries integrated.
 Concrete external normalization bridge not invoked in AGI-N.
 Navier–Stokes normalization handoff ready.
 Part IV Claim Register promotion pending.

Status labels for Appendix AGI-N. All inherited labels remain local stable statement identifiers. The new v20 definitions, propositions, and theorems are Interface definitions, Interface theorems, operational policies, or explicit non-claims at their displayed strength. No local label or successful build constitutes global Claim Register promotion.

Registration timing. Canonical Claim UUIDs, compact CR entries, and final artifact hashes are generated after the complete Problem Interface sequence and Companion are frozen and cross-audited. This deferred registration changes governance timing, not mathematical strength.

6 Appendix NS-A: Navier–Stokes Exploration Protocol

NS-A.1. Scope and exploration status

This appendix defines an exploration protocol for Navier–Stokes-related data in the AK framework. It belongs to the Problem Interface layer. It does not enlarge the Core mathematical package of Chapters 1–8.

The central rule of this appendix is:

Navier–Stokes exploration is not Navier–Stokes proof.

The protocol may generate candidate windows, candidate filtrations, advisory indicators, persistence profiles, normalized objects, or tower candidates for further Core evaluation. However, none of these exploratory outputs replaces:

$$\begin{aligned} B\text{-Gate}^+, \quad \text{Eval}_{1,W,\tau}(F) &= 0, \\ (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) &= (0, 0), \quad \phi_{i,W,\tau}, \\ \text{val}_B(\Sigma\delta) &< \text{gap}_\tau(\mathfrak{B}). \end{aligned}$$

All Core-relevant persistence data produced from Navier–Stokes source data must be evaluated through the repaired windowed profile:

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Whenever tower diagnostics are invoked, the required comparison is a declared tower comparison artifact:

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty).$$

Remark 6.1 (Interface status). The material in this appendix is interface-level and exploration-level material. Any implication from observed Navier–Stokes behavior to regularity, smoothness, uniqueness, continuation, non-blow-up, blow-up, or singularity exclusion is [Spec] by default unless separately proved under the theoretical soundness layer and proof-first program.

Remark 6.2 (Relation to Appendix AGI-N). Appendix AGI-N supplies the resolution-assisted normalization interface. The present appendix may use such normalization as a pre-gate for Navier–Stokes-derived data, but normalization remains separate from proof. A normalized PDE profile is still not a PDE theorem.

Remark 6.3 (No exact-collapse restoration). This appendix does not restore the old exact-collapse framework. In particular, it does not assume a global exact collapse functor, a positive-threshold Serre localization, or a global strict filtered lift. All Core-relevant NS-derived evaluations must be expressed through

$$\text{Eval}_{i,W,\tau}$$

and barcode-level threshold deletion

$$T_\tau^{\text{bar}}.$$

NS-A.2. Navier–Stokes source data

Definition 6.4 (Navier–Stokes source datum). A *Navier–Stokes source datum* is a mathematical, numerical, or observational object associated with a Navier–Stokes problem. Examples include:

- (i) a velocity field $u(x, t)$;
- (ii) a pressure field $p(x, t)$;
- (iii) vorticity

$$\omega = \nabla \times u;$$

- (iv) local energy density;
- (v) enstrophy density;
- (vi) strain tensor or deformation tensor data;
- (vii) weak-solution approximants;
- (viii) Leray–Hopf type data;
- (ix) numerical simulation snapshots;
- (x) discretized flow fields;
- (xi) scale-localized flow profiles;

- (xii) candidate singular neighborhoods;
- (xiii) blow-up rescaling profiles;
- (xiv) localized parabolic cylinder data.

Definition 6.5 (Exploration input). An *exploration input* is a Navier–Stokes source datum together with:

- (i) a declared spatial domain;
- (ii) a declared time interval;
- (iii) boundary or decay conditions, if used;
- (iv) viscosity parameter;
- (v) sampling or discretization scale, if applicable;
- (vi) normalization policy, if applicable;
- (vii) observable family;
- (viii) window, scale, and threshold policies;
- (ix) artifact references.

Definition 6.6 (NS source status). A Navier–Stokes source datum has one of the following statuses:

- (i) *raw*: source data are declared but not yet processed;
- (ii) *observable-extracted*: PDE observables have been extracted;
- (iii) *normalized*: resolution-assisted normalization has been applied;
- (iv) *Core-readable*: filtered or persistence realization is declared;
- (v) *tower-readable*: a Core-readable tower and tower comparison artifact are declared;
- (vi) *advisory*: used only for exploration;
- (vii) *failed*: required preprocessing, realization, constructibility, artifact, or budget conditions fail.

Remark 6.7 (Source data are not Core data). A velocity field, vorticity field, pressure field, simulation output, or weak-solution approximant is not automatically a Core object. It becomes Core-readable only after a declared filtered or persistence realization route is provided.

Remark 6.8 (Source status is not proof status). The status of source data records where they sit in the exploration pipeline. It does not assert a Navier–Stokes theorem, nor does it determine Core pass.

NS-A.3. PDE observables

Definition 6.9 (PDE observable). A *PDE observable* is a scalar, vector, tensor, or derived quantity extracted from Navier–Stokes source data for exploration, normalization, realization, or advisory analysis. Typical examples include:

- (i) $|u|$, the velocity magnitude;
- (ii) $|\omega|$, the vorticity magnitude;
- (iii) $|\nabla u|$, the strain or gradient magnitude;
- (iv) local kinetic energy density;
- (v) local enstrophy density;

- (vi) pressure magnitude or pressure oscillation;
- (vii) scale-invariant local energy quantities;
- (viii) local intermittency indicators;
- (ix) parabolic cylinder averages;
- (x) blow-up candidate scores;
- (xi) localized dissipation proxies;
- (xii) spatial or temporal concentration profiles.

Definition 6.10 (Observable family). An *observable family* is a declared collection

$$\mathcal{O}_{\text{NS}} = \{O_\lambda\}_{\lambda \in \Lambda}$$

of PDE observables used for exploration, filtration construction, normalization, advisory analysis, or Core-readable realization.

Definition 6.11 (Observable status). An observable has one of the following statuses:

- (i) *advisory*: used only for exploration or prioritization;
- (ii) *pre-gate*: used to select or normalize data before Core evaluation;
- (iii) *realization*: used to construct filtered or persistence objects;
- (iv) *ledgered*: used with declared error contribution;
- (v) *verifier-side*: used directly in a Core-readable certificate record.

Definition 6.12 (Observable extraction defect). An *observable extraction defect* is a declared discrepancy

$$\delta^{\text{obs}}$$

arising from computing, approximating, interpolating, sampling, smoothing, or otherwise extracting an observable from Navier–Stokes source data.

Remark 6.13 (Observable status must be declared). The same observable may be advisory in one run and realization-relevant in another. Its status must be declared in the certificate or exploration record.

Remark 6.14 (Observable is not theorem). An observable may correlate with regularity, concentration, or instability. Such correlation is not a proof of a Navier–Stokes conclusion unless a separate theoretical transport theorem is supplied.

NS-A.4. Filtered realization of Navier–Stokes data

Definition 6.15 (Navier–Stokes filtered realization hypothesis). A *Navier–Stokes filtered realization hypothesis* is a declared map or procedure

$$\mathfrak{R}_{\text{NS}} : \mathcal{D}_{\text{NS}} \longrightarrow \text{FiltCh}(k) \quad \text{or} \quad \mathfrak{R}_{\text{NS}} : \mathcal{D}_{\text{NS}} \longrightarrow \text{Pers}_k^{\text{cons}},$$

where $\mathcal{D}_{\text{NS}}$ is a declared class of Navier–Stokes source data, normalized data, or observables.

Definition 6.16 (PDE persistence extraction). A *PDE persistence extraction* is a procedure that maps PDE observables to constructible persistence data. Examples include:

- (i) sublevel-set persistence of an observable;
- (ii) superlevel-set persistence of a concentration field;

- (iii) cubical persistence from discretized flow data;
- (iv) time-windowed persistence profiles;
- (v) scale-parameter persistence from localized quantities;
- (vi) persistence of thresholded vorticity or enstrophy regions;
- (vii) persistence of normalized blow-up rescaling profiles;
- (viii) persistence of parabolic cylinder aggregate fields.

Definition 6.17 (Core-readable Navier–Stokes object). A Navier–Stokes source datum is *Core-readable* if the realization produces:

$$F_{\text{NS}} \in \text{FiltCh}(k) \quad \text{or} \quad P_{\text{NS}} \in \text{Pers}_k^{\text{cons}}$$

such that, for declared i, W, τ , the repaired evaluation

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}}) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F_{\text{NS}}))$$

or the direct persistence expression

$$T_{\tau}^{\text{bar}}(W_W P_{\text{NS}})$$

is defined in the constructible monitored regime.

Definition 6.18 (NS realization defect). An *NS realization defect* is a declared discrepancy

$$\delta^{\text{NS-real}}$$

arising from mapping PDE source data or observables into filtered or persistence data.

Proposition 6.19 (No Core use without PDE realization). *Navier–Stokes source data cannot support Core certification unless they are realized as Core-readable filtered or persistence data, or as verifier-side certificate data.*

Proof. The Core verifier evaluates repaired persistence profiles, admissible representatives, eligible realized objects, diagnostics, ledgers, tower artifacts, and gate verdicts. Raw PDE fields, numerical observables, or simulation outputs are not automatically such objects. Therefore a declared realization is required. $\square$

Remark 6.20 (Purpose-faithfulness). A PDE realization must be purpose-faithful. If the exploration concerns vorticity concentration, then the realization must preserve the relevant concentration structure. If it concerns energy localization, then the realization must preserve the relevant energy-localization structure. Purpose-faithfulness does not mean that every PDE property is preserved.

NS-A.5. Window, scale, and threshold policies

Definition 6.21 (Navier–Stokes window). A *Navier–Stokes window* is a declared right-open interval

$$W = [a, a')$$

in a chosen exploration parameter, such as:

- (i) physical time;
- (ii) spatial scale;
- (iii) parabolic scale;
- (iv) observable magnitude;
- (v) filtration parameter;

- (vi) simulation step;
- (vii) normalized concentration level;
- (viii) blow-up rescaling depth.

Definition 6.22 (Scale policy). A *scale policy* specifies how spatial, temporal, and parabolic scales are selected and related. It may include:

$$r, \quad r^2, \quad t_0 - r^2 < t < t_0, \quad B_r(x_0),$$

or other declared local-scale structures.

Definition 6.23 (Threshold policy for NS exploration). A *threshold policy for NS exploration* specifies the collapse threshold τ , threshold sweep, stable band candidate, or threshold-selection rule used for repaired persistence evaluation.

Definition 6.24 (NS evaluation policy). An *NS evaluation policy* declares:

- (i) monitored degrees i ;
- (ii) windows W ;
- (iii) thresholds τ ;
- (iv) observable family;
- (v) realization route;
- (vi) whether the resulting profile is advisory, pre-gate, or verifier-side;
- (vii) ledger components associated with the evaluation.

Remark 6.25 (Scale is not proof). Scale selection may focus attention on candidate singular or regular regions. It does not prove regularity or blow-up. Scale policies are exploration and realization policies, not PDE theorems.

Remark 6.26 (Threshold stability is not PDE regularity). A stable band in repaired persistence data may be useful for Core analysis. It is not, by itself, Navier–Stokes regularity. A transport theorem is required to convert Core-visible stability into a PDE conclusion.

NS-A.6. Advisory indicators

Definition 6.27 (Navier–Stokes advisory indicator). A *Navier–Stokes advisory indicator* is an exploratory quantity used to rank, flag, or prioritize regions for further Core evaluation. Examples include:

- (i) vorticity concentration score;
- (ii) enstrophy growth score;
- (iii) local energy concentration score;
- (iv) scale-invariant warning score;
- (v) spectral instability indicator;
- (vi) Betti integral of a thresholded observable;
- (vii) persistence mass concentration;
- (viii) normalization Type IV proxy;
- (ix) candidate blow-up score Φ_{NS} ;
- (x) finite-resolution instability score;

(xi) parabolic cylinder anomaly score.

Definition 6.28 (Navier–Stokes potential). A *Navier–Stokes potential* is an advisory navigation quantity

$$\Phi_{\text{NS}} : \mathcal{X}_{\text{NS}} \longrightarrow \mathbb{R} \cup \{\infty\}$$

on a declared exploration space $\mathcal{X}_{\text{NS}}$, used to prioritize candidate regions, scales, windows, or normalization targets.

Definition 6.29 (NS advisory-to-budget conversion). An *NS advisory-to-budget conversion* is a declared rule assigning a ledger component

$$\delta^{\text{NS-adv}}$$

to an advisory NS indicator when that indicator is used in a certificate-relevant way.

Proposition 6.30 (NS advisory indicators do not determine Core verdicts). *No Navier–Stokes advisory indicator determines Core pass or Core reject.*

Proof. By Appendix H and Chapter 6, advisory quantities may guide inspection but do not determine Core verdicts. Core verdicts are determined by repaired evaluation records, gate verdicts, monitored diagnostics, required artifacts, and budget comparison. $\square$

Remark 6.31 (Correct use of Φ_{NS}). The potential

$$\Phi_{\text{NS}}$$

may identify promising or dangerous regions. It is not a proof of regularity, blow-up, or Core admissibility.

NS-A.7. Exploration pre-gates

Definition 6.32 (Navier–Stokes exploration pre-gate). A *Navier–Stokes exploration pre-gate* is a condition used to decide whether a candidate region or dataset should be submitted to Core-readable realization. Examples include:

- (i) sufficient data resolution;
- (ii) bounded measurement noise;
- (iii) successful normalization;
- (iv) admissible scale policy;
- (v) constructibility of extracted persistence;
- (vi) acceptable discretization error;
- (vii) nonpathological sampling pattern;
- (viii) declared boundary condition compatibility;
- (ix) availability of repaired evaluation records;
- (x) availability of tower comparison artifacts when tower diagnostics are invoked.

Definition 6.33 (NS pre-gate record). An *NS pre-gate record* records:

- (i) the candidate region, scale, or dataset;
- (ii) the pre-gate condition;
- (iii) pass/fail/not-applicable status;
- (iv) the reason for failure, if any;

(v) artifact references supporting the decision.

Proposition 6.34 (NS pre-gates do not replace Core gates). *A Navier–Stokes exploration pre-gate may select candidates for Core evaluation, but it does not replace B-Gate⁺, monitored Type IV diagnostics, required tower comparison artifacts, or the budget condition.*

Proof. Pre-gates are interface-level filters. Core gates are verifier-side conditions. Passing a pre-gate does not imply passing the Core. $\square$

Remark 6.35 (Pre-gate failure). Failure of an exploration pre-gate means that the candidate is not suitable for the declared exploration route. It is not a proof of singularity, non-regularity, or Core rejection.

NS-A.8. Discretization and measurement discipline

Definition 6.36 (NS discretization error). An *NS discretization error* is a declared error bound

$$\delta^{\text{NS-disc}}$$

arising from grid resolution, time stepping, mesh approximation, numerical truncation, or discrete persistence computation.

Definition 6.37 (NS measurement error). An *NS measurement error* is a declared error bound

$$\delta^{\text{NS-meas}}$$

arising from observing, sampling, interpolating, or estimating Navier–Stokes source data or observables.

Definition 6.38 (NS algorithmic error). An *NS algorithmic error* is a declared error bound

$$\delta^{\text{NS-alg}}$$

arising from numerical solvers, persistence algorithms, integration schemes, interpolation routines, or implementation choices.

Definition 6.39 (NS exploration ledger). The *NS exploration ledger* aggregates exploration-level errors:

$$\Sigma \delta_{\text{NS}} := \delta^{\text{NS-disc}} \oplus \delta^{\text{NS-meas}} \oplus \delta^{\text{NS-alg}} \oplus \delta^{\text{NS-real}} \oplus \delta^{\text{NS-adv}} \oplus \delta^{\text{NS-param}} \oplus \delta^{\text{obs}} \oplus \delta^{\text{nor}} \oplus \delta^{\text{filt-align}},$$

with omitted components set to the identity element of the declared ledger. If an exploration record introduces additional registered NS-interface defects, the aggregation must be extended by declaration rather than consumed silently.

Proposition 6.40 (Exploratory evidence must be ledgered before Core use). *If numerical or measured Navier–Stokes evidence is used in a Core certificate record, its discretization, measurement, normalization, realization, observable-extraction, and algorithmic errors must be included in the declared δ -ledger.*

Proof. By Appendices I, J, AGI-N, and M, all approximation, measurement, normalization, realization, extraction, and implementation discrepancies used by a certificate must enter the budget ledger. Therefore exploratory PDE evidence must be ledgered before Core use. $\square$

Remark 6.41 (Simulation is not proof). A numerical simulation may generate valuable exploratory evidence. It is not a proof of Navier–Stokes regularity or non-regularity. If simulation-derived data are used in a certificate, their errors must be declared and ledgered.

NS-A.9. Exploration-to-Core re-entry

Definition 6.42 (Exploration-to-Core re-entry). *Exploration-to-Core re-entry* is the process by which exploratory Navier–Stokes data are converted into a Core certificate record. It requires:

- (i) a Core-readable realization;
- (ii) declared windows;
- (iii) threshold policy;
- (iv) monitored degrees;
- (v) repaired evaluation records

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}});$$
- (vi) admissible representative data

$$C_{\tau,W}F_{\text{NS}}$$
 if representatives are invoked;
- (vii) eligibility data, if Ext^n is invoked;
- (viii) tower comparison artifacts, if Type IV diagnostics are invoked;
- (ix) monitored Type IV diagnostics;
- (x) ledgered errors;
- (xi) gate verdicts;
- (xii) artifact references.

Definition 6.43 (NS Core candidate). An *NS Core candidate* is a realized Navier–Stokes-derived object prepared for Core evaluation but not yet certified.

Definition 6.44 (NS Core certificate record). An *NS Core certificate record* is a certificate record obtained from an NS Core candidate after all required repaired evaluation records, admissible representative records when invoked, eligibility records when invoked, tower artifacts when invoked, Core gates, diagnostics, ledgers, and artifact references have been supplied.

Proposition 6.45 (Exploration output is not certificate). *An exploration output is not a Core certificate unless it satisfies the Minimal Manifest Axiom and all Core verifier conditions.*

Proof. Exploration output may contain candidate windows, indicators, plots, persistence summaries, normalized objects, or candidate towers. A Core certificate additionally requires declared objects, windows, threshold and collapse policies, repaired evaluation records, representative and eligibility records when invoked, ledger policy, diagnostics, tower artifacts when invoked, gate verdicts, artifact references, and final status. Thus exploration output is not a certificate by itself. $\square$

Remark 6.46 (Re-entry is the critical boundary). The boundary between exploration and proof is crossed only at Core re-entry. Before re-entry, the objects are advisory or candidate data. After re-entry, they are evaluated by the ordinary Core verifier. Even after Core pass, a PDE conclusion still requires a separate theoretical transport theorem.

NS-A.10. NS Type IV proxies

Definition 6.47 (NS-derived tower). An *NS-derived tower* is a directed system

$$F_0^{\text{NS}} \longrightarrow F_1^{\text{NS}} \longrightarrow F_2^{\text{NS}} \longrightarrow \dots$$

obtained from Navier–Stokes-derived data, such as grid refinements, scale refinements, time-window refinements, normalized blow-up profiles, or finite-resolution approximants.

Definition 6.48 (NS tower comparison artifact). An *NS tower comparison artifact* is a declared comparison

$$\phi_{i,W,\tau}^{\text{NS}} : \text{ColimProfile}_{i,W,\tau}(F_{\bullet}^{\text{NS}}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_{\infty}^{\text{NS}}).$$

Definition 6.49 (NS Type IV proxy). An *NS Type IV proxy* is an exploratory signal suggesting possible monitored generic Type IV obstruction in a realized Navier–Stokes-derived tower. Examples include:

- (i) finite-scale collapse followed by limiting reappearance;
- (ii) discrepancy between finite-resolution and continuum-limit persistence;
- (iii) anomalous tower behavior under grid refinement;
- (iv) persistence features appearing only at a limiting scale;
- (v) normalization failure in a limiting regime;
- (vi) mismatch between local finite windows and global limiting behavior;
- (vii) candidate defect visible only after parabolic rescaling.

Definition 6.50 (Actual NS Type IV diagnostic). Fix one right-open window W , one threshold $\tau > 0$, and a finite monitored degree set

$$I_{\text{mon}}^{\text{NS}} \subset \mathbb{Z}.$$

For every $i \in I_{\text{mon}}^{\text{NS}}$, an *actual NS Type IV diagnostic datum* consists of:

- (i) an actual repaired profile tower

$$M_{0;i,W,\tau}^{\text{NS}} \longrightarrow M_{1;i,W,\tau}^{\text{NS}} \longrightarrow \dots$$

with declared transition morphisms satisfying identity and composition;

- (ii) a declared source provenance for the colimit profile, chosen from

genuine_colimit, verified_stabilization, implementation_artifact.

- (iii) a declared apex profile $N_{i,W,\tau}^{\text{NS}}$, its source semantics, and an apex-agreement record;

- (iv) an actual comparison morphism

$$\phi_{i,W,\tau}^{\text{NS}} : \text{colim}_m M_{m;i,W,\tau}^{\text{NS}} \longrightarrow N_{i,W,\tau}^{\text{NS}};$$

- (v) the terminal mode bounded_terminal or right_unbounded;
- (vi) terminal-tameness proofs for the source, target, kernel, and cokernel profiles;
- (vii) immutable artifact identity and claim-use status.

When all listed data are valid, define

$$\mu_{i,W,\tau}^{\text{NS}} := \text{tgdim}_W \ker(\phi_{i,W,\tau}^{\text{NS}}), \quad \nu_{i,W,\tau}^{\text{NS}} := \text{tgdim}_W \text{coker}(\phi_{i,W,\tau}^{\text{NS}}),$$

and

$$\mu_{\text{Collapse}}^{\text{NS}} := \sum_{i \in I_{\text{mon}}^{\text{NS}}} \mu_{i,W,\tau}^{\text{NS}}, \quad \nu_{\text{Collapse}}^{\text{NS}} := \sum_{i \in I_{\text{mon}}^{\text{NS}}} \nu_{i,W,\tau}^{\text{NS}}.$$

The specialized Type IV status is exactly one of

$$\text{not_invoked}, \quad \text{undefined}, \quad \text{certified_zero}, \quad \text{obstructed}.$$

No numerical pair is imputed in the first two cases. The status is `certified_zero` precisely when the complete datum is valid and both totals vanish; it is `obstructed` when the complete datum is valid and at least one total is nonzero.

Proposition 6.51 (NS Type IV proxy is not Type IV diagnosis). *An NS Type IV proxy does not determine whether*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

or

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0).$$

Proof. Monitored Type IV diagnostics are computed from window-relative terminal-generic dimensions tgdim_W of kernel and cokernel defect profiles of declared tower comparison artifacts. An NS proxy may suggest where such diagnostics should be computed, but it is not the diagnostic itself. $\square$

Remark 6.52 (Correct use of NS Type IV proxies). NS Type IV proxies may guide tower construction, refinement, or rechecking. They may not replace the computation of

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}).$$

NS-A.11. Exploration artifacts

Definition 6.53 (NS exploration artifact). An *NS exploration artifact* is any artifact supporting an exploration record or Core re-entry record. Examples include:

- (i) PDE source data;
- (ii) velocity or vorticity fields;
- (iii) simulation files;
- (iv) grid or mesh specifications;
- (v) time-step records;
- (vi) observable definitions;
- (vii) normalization artifacts;
- (viii) persistence extraction scripts;
- (ix) repaired evaluation records;
- (x) barcode files;
- (xi) Betti integral tables;
- (xii) advisory indicator logs;
- (xiii) discretization and measurement error estimates;

- (xiv) NS tower comparison artifacts;
- (xv) diagnostic tables;
- (xvi) certificate manifests.

Definition 6.54 (NS exploration manifest entry). *An NS exploration manifest entry records:*

- (i) the Navier–Stokes source datum;
- (ii) the source status;
- (iii) the observable family;
- (iv) the realization hypothesis;
- (v) window, scale, and threshold policies;
- (vi) advisory indicators;
- (vii) pre-gate results;
- (viii) normalization policy, if used;
- (ix) discretization, measurement, realization, observable-extraction, and algorithmic errors;
- (x) repaired evaluation records, if available;
- (xi) tower comparison artifacts, if available;
- (xii) artifact references;
- (xiii) re-entry status.

Proposition 6.55 (Artifacts are required for re-entry audit). *An NS exploration output cannot be used for Core re-entry unless the required artifact references are supplied.*

Proof. Core re-entry requires audit-readable reconstruction. Without artifacts, an auditor cannot reconstruct the source data, observables, realization route, repaired evaluation records, error estimates, diagnostics, tower artifacts, or ledger entries. Thus artifact references are required. $\square$

NS-A.12. Exploration pass, Core pass, and PDE conclusion

Definition 6.56 (NS exploration pass). *An NS exploration pass means that a candidate datum has passed the declared exploration pre-gates and is suitable for Core re-entry.*

Definition 6.57 (Core pass for NS-derived data). *A Core pass for NS-derived data means that the corresponding NS Core certificate record passes B-Gate⁺, monitored Type IV diagnostics, required artifact checks, and the budget condition.*

Definition 6.58 (Navier–Stokes conclusion). *A Navier–Stokes conclusion is a problem-specific statement about existence, smoothness, regularity, uniqueness, continuation, blow-up, non-blow-up, or singularity exclusion of Navier–Stokes solutions.*

Definition 6.59 (NS theoretical transport theorem). *An NS theoretical transport theorem is a theorem proving that a declared Core-visible condition or Core certificate behavior implies a declared Navier–Stokes conclusion. It must specify:*

- (i) the PDE class;
- (ii) the domain and boundary conditions;
- (iii) the solution class;

- (iv) the realization route;
- (v) the Core-visible condition used;
- (vi) all approximation, normalization, and extraction assumptions;
- (vii) the exact Navier–Stokes conclusion;
- (viii) proof references or formal proof artifacts.

Proposition 6.60 (Core pass is not NS conclusion). *A Core pass for NS-derived data does not imply a Navier–Stokes conclusion unless a separate theoretical transport theorem proves that implication.*

Proof. A Core pass certifies Core-visible persistence, diagnostic, artifact, and budget conditions. A Navier–Stokes conclusion concerns PDE solutions. A theoretical transport theorem is required to connect them. $\square$

Remark 6.61 (Three-layer distinction). The following must remain distinct:

NS exploration pass, Core pass, Navier–Stokes theorem.

Confusing these layers would violate the non-claim discipline of Chapter 7.

NS-A.12bis. v20 exploration sovereignty and diagnostic coherence

Definition 6.62 (NS exploration sovereignty). *NS exploration sovereignty* is the rule that exploratory PDE data, advisory indicators, simulations, normalized profiles, finite resolution observations, and potential values cannot override Core verifier requirements. Every theorem-relevant use must re-enter through a Core-readable realization, repaired evaluation records, declared artifacts, a ledger, and the Minimal Manifest Axiom.

Proposition 6.63 (No exploration sovereignty override). *No NS exploration output can function as verifier-side evidence unless it has been converted into a declared Core-readable object, a verifier-side certificate record, or a ledgered component.*

Proof. Exploration produces candidate data and priorities. Core verification uses typed records: repaired evaluations, representatives, eligibility records, tower artifacts, diagnostics, ledgers, and manifests. Since these are different layers, no exploration output overrides the Core layer. $\square$

Definition 6.64 (NS re-entry mode). An NS exploration artifact has one of the following re-entry modes:

- (i) *advisory-only*;
- (ii) *pre-gate*;
- (iii) *realization-candidate*;
- (iv) *ledgered-support*;
- (v) *verifier-side-record*;
- (vi) *failed*;
- (vii) *undefined*.

Proposition 6.65 (Re-entry mode is not automatic). *An advisory or pre-gate NS artifact does not become a verifier-side record merely because it is numerically stable, visually convincing, or repeatedly observed.*

Proof. Verifier-side status requires declared realization, artifacts, and ledger semantics. Repeated observation may support exploration, but it does not supply those records. $\square$

Definition 6.66 (NS one-parameter extraction). An *NS one-parameter extraction* is a declared procedure selecting one Core persistence parameter from PDE variables such as physical time, parabolic scale, spatial scale, observable magnitude, rescaling depth, grid resolution, or normalized concentration level.

Definition 6.67 (NS parameter-extraction defect). An *NS parameter-extraction defect* is a declared discrepancy

$$\delta^{\text{NS-param}}$$

arising from selecting, collapsing, reparameterizing, discretizing, or comparing PDE-side scales as a one-parameter persistence variable.

Proposition 6.68 (PDE scale is not automatically a Core parameter). *Physical time, parabolic scale, spatial radius, grid resolution, and observable magnitude are not automatically the Core persistence parameter.*

Proof. A Core persistence parameter must be selected by a declared one-parameter extraction and must yield constructible persistence data on the declared windows. PDE variables may be candidates for this role, but the identification is not automatic. $\square$

Definition 6.69 (NS diagnostic coherence record). An *NS diagnostic coherence record* records the monitored scope, tower artifact, apex object, terminal mode, terminal-tameness proof, and provenance of each Type IV diagnostic used for NS-derived data.

Proposition 6.70 (Changing diagnostic artifacts creates a new NS claim). *Changing the monitored scope, apex policy, terminal mode, tower comparison artifact, realization route, or extraction route of an NS diagnostic creates a new diagnostic claim.*

Proof. The diagnostic is defined from a specific artifact family and terminal-generic profile data. Changing those data changes the mathematical object being diagnosed. $\square$

Definition 6.71 (NS sampled stability). *NS sampled stability* means stability observed over a finite sample of grids, scales, windows, simulations, or thresholds.

Definition 6.72 (NS certified Core stable band). An *NS certified Core stable band* is an artifact-coherent stable band, in the sense of Appendix D, for which the relevant NS-derived repaired diagnostics are certified across the declared band.

Proposition 6.73 (Sampled NS stability is not certified band stability). *NS sampled stability does not imply an NS certified Core stable band without a band-level artifact-coherent proof and the required two-sided threshold-gap records.*

Proof. A finite sample may miss threshold-adjacent or limiting changes. Certified band stability is a uniform artifact-coherent statement, not a sampled observation. $\square$

Remark 6.74 (Exploration remains proposer-side until re-entry). NS exploration, including simulation, visualization, advisory scoring, and candidate-window discovery, remains proposer-side until the data are promoted into an audit-readable Core record.

Remark 6.75 (NS ledger namespace registration). The symbols

$$\delta^{\text{NS-disc}}, \quad \delta^{\text{NS-meas}}, \quad \delta^{\text{NS-alg}}, \quad \delta^{\text{NS-real}}, \quad \delta^{\text{NS-adv}}, \quad \delta^{\text{obs}}, \quad \delta^{\text{NS-param}}$$

must be registered in the ledger catalog before being consumed in a certificate. If any symbol aliases, refines, aggregates, or is disjoint from a normalization defect of Appendix AGI-N, that relation must be declared to prevent double counting.

NS-A.12ter. v20 finite-certificate, higher-edge, reduction, and bridge boundary

Definition 6.76 (NS source-semantics record). An *NS source-semantics record* fixes the equation, spatial dimension, domain, time interval, boundary or decay conditions, viscosity, forcing convention, incompressibility convention, initial-data class, solution class, equality or inequality sense, and the exact source object consumed by the realization. Two records differing in any theorem-relevant field define different interface claims.

Definition 6.77 (NS purpose-faithfulness burden and status). The purpose-faithfulness burden requires a target-relative proof that the PDE feature used by the intended bridge survives observable extraction, normalization, one-parameter reduction, filtration, persistence, windowing, and any comparison step. Its status is exactly one of

proved, conditional, operational, spec, failed, undefined.

Only proved, or conditional with every condition discharged, may support theorem-facing transport.

Definition 6.78 (NS finite detector record). An NS finite detector record declares exactly one source stage among

windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect.

It separately records nonzero soundness, zero completeness, endpoint convention, birth/germ or stencil coverage, family membership, excess-bound provenance, non-adaptivity, immutable artifact identity, certificate status, and mathematical value. The certificate status and value are independent axes.

Proposition 6.79 (Zero-looking finite NS data are not finite proof). *A finite collection of zero ranks, empty sampled complexes, stable simulation values, or absent sampled bars proves repaired-profile zero only when the governing zero-completeness theorem and every applicability field of Definition 6.78 are discharged.*

Proof. Finite observation supplies values only on the sampled information class. Universal repaired-profile vanishing requires a theorem reducing the full declared family to that information. Soundness of nonzero witnesses does not supply completeness of zero observations. $\square$

Definition 6.80 (Adaptive exploration quarantine). An *adaptive exploration quarantine* records every data-dependent choice of observable, chart, window, threshold, mesh, detector, stopping rule, or candidate region made after inspecting the same data later used as evidence. The selected record remains proposer-side unless a theorem covers the adaptive selection rule or independent verification data are supplied.

Proposition 6.81 (Adaptive choice does not inherit fixed-family completeness). *A completeness theorem proved for a fixed non-adaptive detector family does not automatically apply to a detector, window, or threshold selected adaptively from the tested NS data.*

Proof. The adaptive selection changes the information class and the quantified family. A fixed-family theorem has no conclusion on the enlarged selection protocol until the selection rule is included in its hypotheses. $\square$

Definition 6.82 (NS degree- n higher-edge re-entry record). For $n \geq 1$, an NS higher-edge re-entry record is the complete standard degree- n eligibility package: repaired degree- n profile, admissible representative, actual profile isomorphism, independent realization in $D^{[-n,0]}(k\text{-mod})$, fixed zero-compatible extractor, artifact-backed H^{-n} -identification, naturality, non-circularity, change-of-representative data, claim status, and artifact identity.

Proposition 6.83 (No NS higher discharge from synthetic shift). *Placing an NS detector value into cohomological degree $-n$ does not by itself prove a natural NS PH_n -to- Ext^n bridge or satisfy Definition 6.82.*

Proof. A synthetic shift supplies an engineered algebraic identity but not a purpose-faithful source realization, representative-independent naturality, or an artifact-backed edge theorem. $\square$

Definition 6.84 (NS repaired tower-profile system). An *NS repaired tower-profile system* is an actual directed system in the declared persistence category, at fixed (W, τ) , with objects and transition morphisms satisfying identity and composition. A tower of source fields, meshes, barcodes, or labels does not automatically induce this system because hard threshold deletion is not a global functor on arbitrary morphisms.

Definition 6.85 (NS full-interior and transient-defect record). For a valid Type IV artifact, the full-interior status is exactly one of

not_checked, certified_absent, certified_present.

A long-transient detector separately uses source stage kernel_defect or cokernel_defect, a threshold $\tau_{\text{def}} > 0$, a complete detector certificate, and value status

not_invoked, undefined, certified_absent, certified_present.

Proposition 6.86 (Terminal cleanliness does not erase interior defects). *A valid terminal Type IV status certified_zero does not imply full-interior absence, long-transient absence, zero kernel and cokernel profiles, or that the comparison morphism is an isomorphism.*

Proof. Terminal-generic dimension records only terminal-cofinal behavior under the declared terminal convention. Finite noncofinal defect bars may remain invisible to that diagnostic. $\square$

Definition 6.87 (NS finite-to-infinite reduction record). A finite NS certificate may support an infinite tower, continuum, or apex predicate only through a record containing the exact finite witness, a proved reduction theorem, actual directed-system morphisms, the limit or stabilization semantics, a compatible apex cocone, apex agreement, the induced infinite predicate, and every subsequent Return target.

Proposition 6.88 (Finite NS plateau is not apex agreement). *Agreement over finitely many grids, scales, or rescaling stages does not imply eventual constancy, convergence, colimit identification, apex agreement, or a continuum PDE conclusion.*

Proof. Each conclusion contains a universal or limiting assertion absent from a finite prefix. A separate reduction theorem is therefore necessary. $\square$

Definition 6.89 (NS ledger effect class). Every NS discrepancy consumed by a certificate is classified as one of

metric_profile_discrepancy, morphism_profile_discrepancy,
theorem_controlled_nonmetric, non_scalar_obligation,
advisory, unavailable.

Only the first two classes enter the metric ledger, and only with evidence bounding the actual bottleneck discrepancy of the exact profile pair consumed by the proof.

Proposition 6.90 (Numerical PDE error is not automatically bottleneck error). *A mesh width, solver residual, truncation norm, measurement tolerance, or normalization parameter is not a metric-ledger value until a theorem or verified artifact bounds the actual bottleneck discrepancy of the relevant windowed pre-threshold or defect profiles.*

Proof. The quantities have different types and codomains. A support theorem is required to convert the numerical estimate into the discrepancy used by the threshold-gap proof. $\square$

Definition 6.91 (NS-A bridge invocation status). Appendix NS-A has

exploration_schema = available, exploration_derived_PDE_bridge = not_invoked.

This is a scope decision, not a failed regularity proof and not a PDE theorem. The exact criterion-recovery track NS-KTR-SERRIN-L6 registered in Appendix NS-B is independent of simulation, adaptive exploration, sampled stability, and Type IV proxies: it consumes an analytic source-norm record directly. Any bridge built from NS-A exploration artifacts still requires a separately typed NS-B/NS-C route with complete source, reduction, Return, and artifact records.

Definition 6.92 (AGI-N to NS handoff receipt). An AGI-N to NS handoff receipt preserves source identity, normalization operation, auxiliary choices, exceptional-information record, parameter convention, realization target, detector stage, ledger aliases, reduction obligations, Return target, and artifact identity. NS-A may refine these fields but may not mutate them silently.

Theorem 6.93 (NS-A v20 completion package). *The NS exploration layer is v20-conformant only when exploration sovereignty, adaptive-selection quarantine, complete detector typing, standard higher-edge re-entry, actual repaired tower morphisms, terminal/transient separation, finite-to-infinite reduction, typed ledger support, and the explicit non-invocation of a concrete PDE bridge are all respected.*

Proof. Each listed item is a non-compensating requirement established by Definitions 6.78–6.92 and Propositions 6.79–6.90. Omitting any invoked item removes a required route edge and yields an undefined stronger conclusion. $\square$

NS-A.13. Summary of Navier–Stokes exploration protocol

Theorem 6.94 (Appendix NS-A exploration package). *Within the Problem Interface layer, the following package is available.*

- (i) *Navier–Stokes source data are not Core objects by default.*
- (ii) *PDE observables must have declared status: advisory, pre-gate, realization, ledgered, or verifier-side.*
- (iii) *Core use requires a declared filtered or persistence realization.*
- (iv) *Core-readable NS-derived objects are evaluated through*

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}}) = T_{\tau}^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F_{\text{NS}})).$$

- (v) *Window, scale, threshold, and evaluation policies must be declared.*
- (vi) *NS advisory indicators do not determine Core verdicts.*
- (vii) *NS exploration pre-gates do not replace Core gates.*
- (viii) *Numerical, measured, normalized, realized, observable-extracted, advisory-conversion, parameter-extraction, or algorithmic errors must be ledgered before Core use.*
- (ix) *Exploration-to-Core re-entry requires the Minimal Manifest data and ordinary Core verifier conditions.*
- (x) *NS Type IV proxies do not replace monitored Type IV diagnostics.*
- (xi) *NS tower diagnostics require declared tower comparison artifacts*

$$\phi_{i,W,\tau}^{\text{NS}}.$$

- (xii) *Exploration artifacts are required for audit-readable Core re-entry.*
- (xiii) *A Core pass for NS-derived data does not imply a Navier–Stokes conclusion without a separate theoretical transport theorem.*
- (xiv) *NS exploration sovereignty prevents advisory, numerical, or pre-gate artifacts from overriding Core verifier requirements.*
- (xv) *NS re-entry modes distinguish advisory-only material from verifier-side records.*
- (xvi) *PDE-side scale variables require a declared one-parameter extraction before they become Core persistence parameters.*
- (xvii) *NS Type IV diagnostics require artifact-coherent diagnostic records; changing the artifact data creates a new claim.*
- (xviii) *Sampled NS stability is not certified Core stable-band evidence without Appendix D band-level coherence.*

Proof. Items (i)–(iii) follow from the definitions of NS source data, observables, and filtered realization. Item (iv) follows from the definition of Core-readable NS object. Item (v) follows from the window, scale, threshold, and evaluation policy definitions. Item (vi) is Proposition 6.30. Item (vii) is Proposition 6.34. Item (viii) is Proposition 6.40. Item (ix) follows from the exploration-to-Core re-entry definition and Proposition 6.45. Item (x) is Proposition 6.51. Item (xi) follows from Definition 6.48. Item (xii) is Proposition 6.55. Item (xiii) is Proposition 6.60. Items (xiv)–(xviii) follow from Definition 6.62, Proposition 6.63, Definition 6.64, Proposition 6.65, Definition 6.66, Proposition 6.68, Definition 6.69, Proposition 6.70, and Proposition 6.73. $\square$

NS-A.14. Explicit exclusions

The following are not asserted in Appendix NS-A.

- No Navier–Stokes existence theorem is proved.
- No Navier–Stokes uniqueness theorem is proved.
- No Navier–Stokes regularity theorem is proved.
- No Navier–Stokes smoothness theorem is proved.
- No non-blow-up theorem is proved.
- No blow-up theorem is proved.
- No singularity exclusion theorem is proved.
- No numerical simulation is treated as proof.
- No PDE observable is treated as a Core object without realization.
- No advisory indicator is treated as a Core verdict.
- No Φ_{NS} value is treated as a proof.
- No exploration pre-gate replaces B-Gate⁺.
- No exploration pre-gate replaces the repaired topological condition

$$\text{Eval}_{1,W,\tau}(F) = 0.$$

- No exploration pre-gate replaces eligibility.
- No exploration pre-gate replaces monitored Type IV diagnostics.
- No exploration pre-gate replaces the tower comparison artifact

$$\phi_{i,W,\tau}.$$

- No NS Type IV proxy replaces

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}).$$

- No Core pass for NS-derived data is treated as a Navier–Stokes theorem.
- No normalization success is treated as PDE regularity.
- No discretized zero diagnostic is treated as continuum zero diagnostic without diagnostic convergence.
- No exploration artifact supports Core acceptance by itself.
- No global exactness of T_{τ}^{bar} is assumed.
- No positive-threshold Serre localization is assumed.
- No global strict filtered lift C_{τ} is assumed.

NS-A.15. Provenance and status

The material in this appendix depends on:

- (i) the interface protocol of Chapter 8;
- (ii) the non-claim discipline of Chapter 7;
- (iii) the repaired barcode-level collapse and constructible persistence framework of Appendix A;
- (iv) the admissible representative discipline of Appendix B;
- (v) the eligible realization regime of Appendix C;
- (vi) the monitored Type IV diagnostic calculus and tower comparison artifact discipline of Appendix D;
- (vii) the manifest schema of Appendix G;
- (viii) the advisory-invariant discipline of Appendix H;
- (ix) the grid-to-continuum and discretization discipline of Appendix I;
- (x) the measurement-level stability discipline of Appendix J;
- (xi) the ledger catalog of Appendix M;
- (xii) the normalization interface of Appendix AGI-N.

Its role is to define a disciplined exploration protocol for Navier–Stokes-derived data without allowing exploratory evidence to become proof.

Status labels for Appendix NS-A. *Problem-interface definitions:* Definitions [6.4](#), [6.5](#), [6.6](#), [6.9](#), [6.10](#), [6.11](#), [6.12](#), [6.15](#), [6.16](#), [6.17](#), [6.18](#), [6.21](#), [6.22](#), [6.23](#), [6.24](#), [6.27](#), [6.28](#), [6.29](#), [6.32](#), [6.33](#), [6.36](#), [6.37](#), [6.38](#), [6.39](#), [6.42](#), [6.43](#), [6.44](#), [6.47](#), [6.48](#), [6.49](#), [6.50](#), [6.53](#), [6.54](#), [6.56](#), [6.57](#), [6.58](#), [6.59](#).

Problem-interface propositions/theorems: Propositions [6.19](#), [6.30](#), [6.34](#), [6.40](#), [6.45](#), [6.51](#), [6.55](#), [6.60](#), Theorem [6.94](#).

Operational cautions: Remarks [6.1](#), [6.2](#), [6.3](#), [6.7](#), [6.8](#), [6.13](#), [6.14](#), [6.20](#), [6.25](#), [6.26](#), [6.31](#), [6.35](#), [6.41](#), [6.46](#), [6.52](#), [6.61](#).

Additional v20.0 conservative hardening definitions: Definitions [6.62](#), [6.64](#), [6.66](#), [6.67](#), [6.69](#), [6.71](#), [6.72](#).

Additional v20.0 conservative hardening propositions: Propositions [6.63](#), [6.65](#), [6.68](#), [6.70](#), [6.73](#).

Additional v20.0 conservative hardening remarks: Remarks [6.74](#), [6.75](#).

Additional v20 completion definitions: Definitions [6.76](#), [6.77](#), [6.78](#), [6.80](#), [6.82](#), [6.84](#), [6.85](#), [6.87](#), [6.89](#), [6.91](#), [6.92](#).

Additional v20 completion propositions/theorem: Propositions [6.79](#), [6.81](#), [6.83](#), [6.86](#), [6.88](#), [6.90](#), and Theorem [6.93](#).

Explicit exclusions: Section NS-A.14.

7 Appendix NS-B: Navier–Stokes Theoretical Soundness Layer

NS-B.1. Scope and theoretical status

This appendix defines the theoretical soundness layer for Navier–Stokes interfaces in the AK framework. It belongs to the Problem Interface layer. It does not enlarge the Core mathematical package of Chapters 1–8.

The central rule of this appendix is:

A Core pass for NS-derived data is not a PDE theorem without a proved soundness bridge.

The purpose of this appendix is to state what must be supplied before exploration outputs, normalized objects, discretized data, measured observables, tower diagnostics, or Core certificates may be interpreted as evidence for a Navier–Stokes statement.

All Core-relevant NS-derived persistence data remain governed by the repaired windowed evaluation:

$$\text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Whenever Type IV diagnostics are invoked, the relevant comparison must be a declared tower comparison artifact:

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_{\bullet}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_{\infty}).$$

Remark 7.1 (Interface status). All implications from Core-visible data to Navier–Stokes regularity, smoothness, uniqueness, continuation, non-blow-up, blow-up, or singularity exclusion are [Spec] by default. They become theorem-level statements only after the required soundness bridge and transport theorem are proved.

Remark 7.2 (Relation to Appendix NS-A). Appendix NS-A defined the exploration protocol. The present appendix does not repeat exploration. It determines which additional theoretical conditions are needed for exploration-derived or Core-derived data to have PDE meaning.

Remark 7.3 (No exact-collapse restoration). This appendix does not restore the old exact-collapse framework. In particular, it does not assume a global exact collapse functor, a positive-threshold Serre localization, or a global strict filtered lift. All soundness claims must be formulated relative to repaired evaluation, admissible representatives, declared tower artifacts, and ledgered defects.

NS-B.2. Navier–Stokes continuum target

Definition 7.4 (Navier–Stokes continuum target). A *Navier–Stokes continuum target* is a declared PDE object

$$(u, p)$$

on a declared spacetime domain

$$\Omega \times I,$$

where u is a velocity field and p is a pressure field satisfying the Navier–Stokes equations in a declared sense.

Definition 7.5 (Solution class). A *solution class* is the declared analytic class in which the continuum target is interpreted. Examples include:

- (i) smooth classical solutions;
- (ii) strong solutions;
- (iii) mild solutions;
- (iv) weak solutions;
- (v) Leray–Hopf weak solutions;
- (vi) local-in-time regular solutions;
- (vii) discretized approximants with a declared continuum limit.

Definition 7.6 (PDE target statement). A *PDE target statement* is the Navier–Stokes conclusion one intends to support or prove. Examples include:

- (i) regularity on a spacetime cylinder;
- (ii) smooth continuation beyond a time T ;
- (iii) exclusion of blow-up on a region;
- (iv) boundedness of a critical norm;
- (v) vanishing or control of a singular set;

- (vi) uniqueness in a declared solution class;
- (vii) stability under a declared approximation or normalization regime.

Definition 7.7 (PDE target record). A *PDE target record* contains:

- (i) the equation and domain;
- (ii) the boundary or decay conditions;
- (iii) the solution class;
- (iv) the intended PDE target statement;
- (v) the local or global region to which the statement applies;
- (vi) the analytic criteria intended to support the conclusion;
- (vii) artifact references.

Remark 7.8 (PDE target must be declared). Without a declared solution class and target statement, it is impossible to determine what theoretical soundness would mean. A Core certificate alone does not determine the PDE theorem being claimed.

NS-B.3. Continuum realization hypothesis

Definition 7.9 (Continuum realization hypothesis). A *continuum realization hypothesis* is a declared map or procedure

$$\mathfrak{R}_{\text{NS}}^{\text{cont}} : (u, p) \longrightarrow F_{\text{NS}}$$

or

$$\mathfrak{R}_{\text{NS}}^{\text{cont}} : (u, p) \longrightarrow P_{\text{NS}},$$

where

$$F_{\text{NS}} \in \text{FiltCh}(k), \quad P_{\text{NS}} \in \text{Pers}_k^{\text{cons}},$$

and the realized object is compatible with the repaired AK Core protocol.

Definition 7.10 (Purpose-faithful PDE realization). A continuum realization is *purpose-faithful* for a PDE target statement if the Core-visible features it preserves are sufficient for the intended PDE soundness bridge. For example:

- (i) a vorticity-based realization must preserve the relevant vorticity concentration behavior;
- (ii) an energy-based realization must preserve the relevant local energy behavior;
- (iii) a scale-critical realization must preserve the relevant scaling-sensitive quantities;
- (iv) a singular-set realization must preserve the relevant local defect structure;
- (v) a pressure-sensitive realization must preserve the pressure behavior required by the target statement.

Definition 7.11 (Core-visible NS feature). A *Core-visible NS feature* is a repaired persistence, representative, diagnostic, artifact, or ledger feature of a realized NS object, such as:

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}}), \quad C_{\tau,W}F_{\text{NS}}, \quad \phi_{i,W,\tau}^{\text{NS}}, \quad (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}), \quad \Sigma\delta.$$

Proposition 7.12 (No PDE interpretation without purpose-faithfulness). A *Core certificate for NS-derived data cannot be interpreted as evidence for a PDE target statement unless the realization is purpose-faithful for that target*.

Proof. A Core certificate concerns the realized persistence or filtered object. A PDE target statement concerns the original continuum solution. If the realization does not preserve the PDE features relevant to the target, no implication from Core behavior to the PDE target is justified. $\square$

Remark 7.13 (Purpose-faithful is target-relative). A realization may be purpose-faithful for detecting vorticity concentration but not for proving pressure regularity. It may be suitable for local exploration but not for global continuation. Thus purpose-faithfulness must be stated relative to the target theorem.

NS-B.4. Grid-to-continuum soundness

Definition 7.14 (NS discretized target). An *NS discretized target* is a grid, mesh, numerical field, simulation output, or finite-dimensional approximation

$$(u_h, p_h)$$

intended to approximate a continuum target

$$(u, p).$$

Definition 7.15 (NS grid-to-continuum soundness). A discretized Navier–Stokes realization satisfies *grid-to-continuum soundness* if:

- (i) the discretized targets (u_h, p_h) converge to a declared continuum target (u, p) in a stated topology or weak sense;
- (ii) the persistence realizations of (u_h, p_h) converge to the persistence realization of (u, p) in the Core-relevant sense;
- (iii) the repaired evaluations converge or are controlled:

$$\text{Eval}_{i,W,\tau}(F_{\text{NS},h}) \longrightarrow \text{Eval}_{i,W,\tau}(F_{\text{NS}});$$

- (iv) the discretization errors are bounded by declared ledger components

$$\delta_h^{\text{NS-disc}};$$

- (v) any diagnostic transfer from discretized to continuum data is justified.

Definition 7.16 (NS diagnostic convergence). *NS diagnostic convergence* holds if the monitored Type IV diagnostics computed from discretized NS towers converge to, or eventually agree with, the diagnostics of the declared continuum tower:

$$(\mu_{\text{Collapse}}^{(h)}, \nu_{\text{Collapse}}^{(h)}) \longrightarrow (\mu_{\text{Collapse}}, \nu_{\text{Collapse}})$$

in the declared diagnostic sense, with both sides computed from declared tower comparison artifacts.

Definition 7.17 (Continuum tower comparison artifact). A *continuum tower comparison artifact* is a declared comparison

$$\phi_{i,W,\tau}^{\text{cont}} : \text{ColimProfile}_{i,W,\tau}(F_{\bullet}^{\text{cont}}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_{\infty}^{\text{cont}})$$

associated to a continuum NS-derived tower.

Definition 7.18 (Discretized tower comparison artifact). A *discretized tower comparison artifact* is a declared comparison

$$\phi_{i,W,\tau}^{(h)} : \text{ColimProfile}_{i,W,\tau}(F_{\bullet}^{(h)}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_{\infty}^{(h)})$$

associated to a discretized NS-derived tower at grid or resolution scale h .

Proposition 7.19 (Discretized zero diagnostics do not imply continuum zero diagnostics). *At a fixed grid scale h , the condition*

$$(\mu_{\text{Collapse}}^{(h)}, \nu_{\text{Collapse}}^{(h)}) = (0, 0)$$

does not imply

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

for the continuum target without NS diagnostic convergence.

Proof. A finite grid may fail to resolve limiting generic kernel or cokernel defects. Without a convergence theorem connecting discretized defect profiles to continuum defect profiles through declared tower comparison artifacts, no implication from discretized diagnostics to continuum diagnostics is valid. $\square$

Remark 7.20 (Relation to Appendix I). Appendix I established the general grid-to-continuum discipline. This section specializes that discipline to Navier–Stokes-derived data.

NS-B.5. Measurement and observable soundness

Definition 7.21 (Observable soundness). A PDE observable $O(u, p)$ is *sound for a target statement* if control of the observable in the declared sense is sufficient, through a proved analytic theorem, to support the target statement.

Definition 7.22 (Measurement soundness). A measured observable

$$\hat{O}$$

is *measurement-sound* if:

- (i) it approximates the ideal observable $O(u, p)$;
- (ii) the measurement error is bounded by a declared ledger component

$$\delta^{\text{NS-meas}};$$

- (iii) the error bound is small enough relative to the safety margin;
- (iv) the measurement artifacts are available for audit.

Definition 7.23 (Observable-to-regularity bridge). An *observable-to-regularity bridge* is a theorem or declared [Spec] statement of the form:

control of selected PDE observables $\implies$ regularity or continuation in the declared solution class.

Definition 7.24 (Persistence-observable bridge). A *persistence-observable bridge* is a theorem or declared [Spec] statement connecting repaired persistence behavior,

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}}),$$

to control of a PDE observable relevant to the target statement.

Proposition 7.25 (Measured observable control is not PDE regularity by default). *Measured control of a PDE observable does not imply PDE regularity unless an observable-to-regularity bridge is proved and measurement error is ledgered.*

Proof. Measured control concerns approximate observable values. PDE regularity is an analytic property of the continuum solution. A theorem connecting the observable to regularity is required, and measurement errors must be accounted for. $\square$

Proposition 7.26 (Persistence control is not observable control by default). *Control of*

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}})$$

does not imply control of a PDE observable unless a persistence-observable bridge is supplied.

Proof. Repaired persistence behavior is Core-visible topological or filtered data. A PDE observable is an analytic quantity of the continuum solution. The two are not identical by definition. A bridge theorem is required to transport control from one to the other. $\square$

Remark 7.27 (Betti integral caution). A Betti integral or persistence summary derived from a PDE observable is a measurement-level or Core-readable quantity only after realization. It is not a regularity criterion unless a separate observable-to-regularity bridge proves that role.

NS-B.6. Energy, vorticity, pressure, and scale compatibility

Definition 7.28 (Energy-compatible realization). A realization is *energy-compatible* if the filtered or persistence structure preserves the local energy behavior relevant to the declared PDE target statement.

Definition 7.29 (Vorticity-compatible realization). A realization is *vorticity-compatible* if the filtered or persistence structure preserves the vorticity concentration behavior relevant to the declared PDE target statement.

Definition 7.30 (Pressure-compatible realization). A realization is *pressure-compatible* if the filtered or persistence structure preserves the pressure behavior relevant to the declared PDE target statement.

Definition 7.31 (Scale-compatible realization). A realization is *scale-compatible* if it respects the scaling relation required by the Navier–Stokes analysis being used, such as parabolic scaling or critical-norm scaling.

Definition 7.32 (Criticality-compatible realization). A realization is *criticality-compatible* if the Core-visible data preserve the distinction between subcritical, critical, and supercritical behavior relevant to the declared analytic criterion.

Proposition 7.33 (Scale compatibility is not automatic). *A filtered or persistence realization of NS data is not scale-compatible by default.*

Proof. Persistence parameters, spatial scales, temporal windows, and collapse thresholds may transform differently under Navier–Stokes scaling. Without an explicit scale policy and proof of compatibility, the realization need not preserve scale-critical structure. $\square$

Remark 7.34 (Compatibility is target-dependent). Energy compatibility, vorticity compatibility, pressure compatibility, scale compatibility, and criticality compatibility are not universal. Each must be stated relative to a target theorem and a chosen realization.

NS-B.7. The NS soundness bridge

Definition 7.35 (NS soundness bridge). An *NS soundness bridge* is a theorem or declared [Spec] statement connecting Core-visible behavior of a realized NS object to a PDE target statement. It has the schematic form:

$$\text{Core certificate behavior of } F_{\text{NS}} \implies \text{PDE target statement for } (u, p).$$

Definition 7.36 (Admissible NS soundness bridge). An NS soundness bridge is *admissible* only if it declares:

- (i) the solution class;
- (ii) the PDE target statement;
- (iii) the realization hypothesis;
- (iv) purpose-faithfulness of the realization;
- (v) repaired evaluation records used;
- (vi) admissible representative and eligibility data, if Ext^n is invoked;
- (vii) declared tower comparison artifacts, if Type IV diagnostics are used;
- (viii) grid-to-continuum soundness, if discretized data are used;
- (ix) measurement soundness, if measured observables are used;
- (x) normalization transport, if normalized data are used;
- (xi) analytic criteria connecting the realized features to the PDE target;
- (xii) all ledgered defects;

(xiii) the final logical implication.

Definition 7.37 (Bridge invocation and evidence status). An NS soundness bridge carries two independent status axes. Its invocation status is exactly one of

not_invoked, invoked.

Its evidence status is exactly one of

proved, conditional, spec, rejected, unknown.

Theorem evidence requires proved, or conditional with every registered hypothesis discharged. The token not_invoked is available only under a declared scope-exclusion policy; missing bridge evidence for an invoked PDE conclusion is unknown and yields an undefined external conclusion, not a pass.

Remark 7.38 (Bridge status vocabulary alignment). In cross-appendix comparisons, the bridge-status token *speculative* is aligned with [Spec], and *undeclared* is aligned with *undefined*. These alignments are vocabulary identifications only; they do not promote a bridge to theorem-level status.

Proposition 7.39 (No NS conclusion without soundness bridge). *No Navier–Stokes conclusion follows from a Core pass for NS-derived data without an admissible NS soundness bridge.*

Proof. A Core pass concerns the realized persistence or filtered object. A Navier–Stokes conclusion concerns the continuum PDE solution. The soundness bridge is exactly the theorem needed to connect these two levels. Without it, the implication is not established. $\square$

Remark 7.40 (Bridge is the bottleneck). The main theoretical difficulty is not producing exploratory indicators. It is proving that Core-visible behavior of the realized object implies a PDE target statement. This is the soundness bottleneck.

NS-B.8. Standard higher Ext^n eligibility

Definition 7.41 (NS degree- n eligible regime). Fix $n \geq 1$. A Navier–Stokes interface is in the *standard NS degree- n eligible regime* on $(F; W, \tau)$ only when the record supplies:

- (i) the exact repaired degree- n profile $\text{Eval}_{n,W,\tau}(F)$, together with either a direct proof of its value or a complete detector record at its declared stage;
- (ii) an admissible representative $C_{\tau,W}F$;
- (iii) an actual profile isomorphism

$$\rho_n : \mathbf{P}_n(C_{\tau,W}F) \xrightarrow{\sim} \text{Eval}_{n,W,\tau}(F);$$

- (iv) a genuine scope-level realization functor producing

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod});$$

- (v) a fixed edge extractor E_n with $E_n(0) = 0$;
- (vi) an artifact-backed isomorphism

$$H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} E_n(\text{Eval}_{n,W,\tau}(F));$$

- (vii) naturality, non-circularity, representative-change, realization-change, claim-status, and artifact-identity records.

Synthetic shifts of detector values do not satisfy this definition without an independent natural realization theorem.

Proposition 7.42 (NS Ext^n -use requires complete eligibility). *For $n \geq 1$, a Navier–Stokes interface may invoke the Core one-way discharge*

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n;F,W,\tau}, k) = 0$$

only in the standard NS degree- n eligible regime.

Proof. The v20 higher obstruction edge is conditional on the complete degree- n representative, realization, extractor, edge-identification, naturality, and artifact package. Navier–Stokes origin supplies none of those hypotheses automatically. $\square$

Remark 7.43 (PDE obstruction is not Core Ext^n). A PDE-derived cohomological obstruction, weak-solution defect, compactness defect, or analytic obstruction is not automatically

$$\text{Ext}^n(A_{n;F,W,\tau}, k).$$

Any identification requires its own purpose-faithful Interface theorem. The reverse implication from $\text{Ext}^n = 0$ to repaired-profile zero additionally requires a separately proved natural zero-reflection theorem.

NS-B.9. Normalization soundness

Definition 7.44 (NS normalization soundness). A resolution-assisted normalization used in a Navier–Stokes interface is *sound* if:

- (i) the normalization operation is defined on the declared source data;
- (ii) all auxiliary choices are declared;
- (iii) the normalization defects are ledgered;
- (iv) the normalized object is Core-readable;
- (v) the normalized repaired evaluation

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}}^{\text{nor}})$$

is declared when used;

- (vi) a normalized-to-source transport statement is supplied when conclusions about the original PDE solution are intended.

Definition 7.45 (Normalized-to-source NS bridge). A *normalized-to-source NS bridge* is a theorem or declared [Spec] statement of the form:

$$\text{Core-visible behavior of normalized NS data} \implies \text{PDE conclusion for the original NS target.}$$

Proposition 7.46 (Normalization soundness does not imply PDE regularity). *Even sound normalization does not imply Navier–Stokes regularity unless a normalized-to-source regularity bridge is proved.*

Proof. Sound normalization prepares a normalized Core-readable object with ledgered defects. Regularity is a statement about the original continuum solution. A theorem transferring normalized behavior back to the original solution is required. $\square$

Remark 7.47 (Relation to Appendix AGI-N). Appendix AGI-N gives the general normalization interface. This section states the additional soundness requirement when normalization is used for Navier–Stokes conclusions.

NS-B.10. Type IV soundness for NS towers

Definition 7.48 (NS continuum tower). An *NS continuum tower* is a directed tower of Core-readable objects derived from continuum Navier–Stokes data, such as scale towers, time-window towers, local-cylinder towers, normalized refinement towers, or blow-up rescaling towers.

Definition 7.49 (NS Type IV soundness). An NS tower satisfies *Type IV soundness* if the monitored Type IV diagnostics of the realized tower correspond, through a declared soundness bridge, to the limiting PDE behavior being studied.

Definition 7.50 (Type IV-to-PDE bridge). A *Type IV-to-PDE bridge* is a theorem or declared [Spec] statement of the form:

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0) \implies \text{declared PDE target statement.}$$

Proposition 7.51 (Type IV cleanliness is not PDE regularity by default). *The condition*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

for an NS-derived tower does not imply PDE regularity unless Type IV soundness and an appropriate regularity bridge are proved.

Proof. Type IV cleanliness is a Core diagnostic statement about declared tower comparison artifacts. PDE regularity is an analytic statement about Navier–Stokes solutions. The former implies the latter only through a theorem connecting the diagnostic to the PDE target. $\square$

Remark 7.52 (Type IV may be useful but not sufficient). Type IV diagnostics may be valuable for detecting limiting mismatch between finite-stage and limiting structures. However, their PDE meaning must be established separately.

NS-B.11. Theoretical soundness record

Definition 7.53 (NS theoretical soundness record). An *NS theoretical soundness record* is a manifest-level record containing:

- (i) the continuum target (u, p) ;
- (ii) the solution class;
- (iii) the PDE target statement;
- (iv) the realization hypothesis;
- (v) purpose-faithfulness evidence;
- (vi) repaired evaluation records used;
- (vii) representative and eligibility data, if Ext^n is invoked;
- (viii) tower comparison artifacts, if Type IV diagnostics are used;
- (ix) grid-to-continuum soundness data, if applicable;
- (x) measurement soundness data, if applicable;
- (xi) normalization soundness data, if applicable;
- (xii) Type IV soundness data, if tower diagnostics are used;
- (xiii) the declared NS soundness bridge;
- (xiv) bridge status;

- (xv) ledger entries;
- (xvi) artifact references;
- (xvii) status labels for all problem-specific implications.

Proposition 7.54 (Soundness record is required for PDE interpretation). *A Core certificate for NS-derived data cannot be interpreted as a Navier–Stokes theorem without an NS theoretical soundness record.*

Proof. The soundness record supplies the continuum target, solution class, realization, purpose-faithfulness, error accounting, and bridge theorem needed to interpret Core behavior as a PDE statement. Without it, the interpretation is not audit-readable. $\square$

Remark 7.55 (Soundness record is not proof by itself). A soundness record may contain [Spec] statements. Only those bridge components that are actually proved can support theorem-level conclusions.

NS-B.12. Core pass, soundness pass, and PDE theorem

Definition 7.56 (NS soundness pass). *An NS soundness pass means that the theoretical soundness record supplies all required hypotheses and proved bridge statements needed to interpret a Core certificate as a PDE conclusion.*

Definition 7.57 (NS theorem-level pass). *An NS theorem-level pass means:*

- (i) the NS Core certificate passes;
- (ii) the NS theoretical soundness record passes;
- (iii) the relevant NS soundness bridge is proved;
- (iv) the target PDE statement follows in the declared solution class.

Definition 7.58 (NS theorem-level outcome). The theorem-level outcome for a declared NS target is exactly one of

not_invoked, undefined, theorem_level, refuted_or_failed.

It is not_invoked only when the target is outside the declared run scope. It is undefined whenever required Core, reduction, bridge, analytic, artifact, or claim evidence is missing or remains speculative. It is theorem_level only when the complete typed route is proved. It is refuted_or_failed only when an invoked required condition is evidenced false. Missing evidence is never converted into failure, pass, or a numbered obstruction.

Proposition 7.59 (Core pass plus unproved soundness bridge is not theorem-level pass). *A Core pass together with an unproved [Spec] soundness bridge does not constitute an NS theorem-level pass.*

Proof. A theorem-level pass requires a proved bridge. A [Spec] bridge is a specification or conjectural interface statement. Therefore it cannot supply theorem-level implication. $\square$

Remark 7.60 (Layer separation). The following must remain distinct:

exploration pass, Core pass, soundness pass, PDE theorem.

This appendix concerns the third layer.

NS-B.12bis. v20 conservative hardening: soundness sovereignty and bridge promotion

Definition 7.61 (NS soundness sovereignty). *NS soundness sovereignty* is the rule that PDE interpretation is governed by proved soundness bridges, not by Core pass, exploration success, numerical convergence, or terminology.

Proposition 7.62 (No soundness sovereignty override). *No Core-visible NS feature can be interpreted as a PDE theorem unless the declared soundness bridge proves the corresponding implication in the declared solution class.*

Proof. Core-visible features and PDE theorems live in different semantic layers. The soundness bridge is the required logical map between them. $\square$

Definition 7.63 (NS bridge promotion record). An *NS bridge promotion record* records a proposed promotion of a bridge from [Spec] or conditional status to theorem-level status, including the hypotheses, proof artifacts, external analytic theorems, and verification of all ledgered defects.

Proposition 7.64 (Bridge names do not promote bridge status). *Calling a statement a regularity bridge, diagnostic bridge, Type IV-to-PDE bridge, or observable bridge does not promote it beyond [Spec] status.*

Proof. Status depends on proof, hypotheses, and artifact records, not on nomenclature. $\square$

Definition 7.65 (NS diagnostic convergence mode). An NS diagnostic comparison between discretized and continuum towers has one of the following modes:

- (i) *not-compared*;
- (ii) *sampled*;
- (iii) *bounded-approximation*;
- (iv) *eventual-agreement*;
- (v) *proved-convergence*;
- (vi) *failed*;
- (vii) *undefined*.

Proposition 7.66 (Sampled diagnostic agreement is not diagnostic convergence). *Agreement of Type IV diagnostics at finitely many grid scales is not NS diagnostic convergence unless an eventual-agreement or convergence theorem is supplied.*

Proof. Finite agreement does not control unresolved limiting kernel or cokernel defect profiles. Diagnostic convergence is a tower-level statement involving declared artifacts. $\square$

Definition 7.67 (NS compatibility burden). The *NS compatibility burden* is the obligation to prove, relative to the target theorem, that energy, vorticity, pressure, scale, and criticality information survive the realization route used in the Core certificate.

Proposition 7.68 (Compatibility is non-scalar). *Missing energy, vorticity, pressure, scale, or criticality compatibility cannot be repaired by a small scalar ledger margin.*

Proof. Compatibility is a typed semantic condition connecting PDE quantities to realized Core-visible features. A scalar budget can bound declared discrepancies, but it cannot supply a missing semantic theorem. $\square$

Definition 7.69 (NS theorem-promotion schema). An *NS theorem-promotion schema* consists of a Core certificate, a soundness record, and a proved admissible bridge. It also contains every required continuum, grid, measurement, and normalization soundness clause, followed by a final derivation of the PDE target statement.

Proposition 7.70 (Core pass plus incomplete soundness is non-theorem). *A Core pass together with incomplete, conditional-with-unmet-hypotheses, speculative, failed, or undeclared soundness data is not a theorem-level NS result.*

Proof. Theorem-level status requires a complete proved implication from the Core certificate to the PDE target statement. Incomplete soundness leaves the implication unproved. $\square$

Remark 7.71 (Soundness bridge remains the bottleneck). The v20 interface can discipline exploration and certification, but it does not remove the analytic burden of proving the soundness bridge.

Remark 7.72 (NS-B ledger namespace registration). Defects used by NS-B, including discretization, measurement, realization, normalization, observable, diagnostic-convergence, and bridge-related defects, must be registered in Appendix M or explicitly related to the AGI-N and NS-A namespaces by alias, refinement, aggregation, or disjointness declarations.

NS-B.12ter. v20 exact bridge logic, target separation, and recovery boundary

Definition 7.73 (Typed NS bridge chain). A theorem-facing NS bridge chain consists of distinct arrows

$$\begin{aligned} \text{continuum source} &\longrightarrow \text{purpose-faithful realization} \longrightarrow \text{Core-readable profile} \\ &\longrightarrow \text{finite Core certificate} \longrightarrow \text{finite-to-infinite reduction} \\ &\longrightarrow \text{analytic criterion} \longrightarrow \text{PDE Return conclusion.} \end{aligned}$$

Every arrow records source and target semantics, hypotheses, comparison mode, defects, artifacts, evidence status, permitted use, and failure route.

Theorem 7.74 (Conditional NS bridge-composition theorem). *Let S be a declared NS source class, Q a Core predicate, A an analytic criterion, and C a PDE conclusion. Suppose:*

- (i) *a purpose-faithful source-to-Core Interface theorem proves $s \in S \Rightarrow Q(s)$ at the exact strength consumed;*
- (ii) *every finite-to-infinite step used in Q is proved;*
- (iii) *an Interface/Return theorem proves $Q(s) \Rightarrow A(s)$;*
- (iv) *a cited or separately proved analytic theorem proves $A(s) \Rightarrow C(s)$;*
- (v) *all hypotheses, artifacts, defects, and claim-use preflights are discharged.*

Then $Q(s) \Rightarrow C(s)$ on the declared source class.

Proof. This is composition of the displayed proved implications. The theorem introduces no new analytic content beyond the registered component theorems and has no conclusion when any component arrow is missing. $\square$

Theorem 7.75 (Target-separation no-go). *Let $\mathfrak{R} : S \rightarrow C$ be an NS realization and let $P : S \rightarrow \{0, 1\}$ be the truth value of a declared PDE target. If there exist $x, y \in S$ with*

$$\mathfrak{R}(x) \cong \mathfrak{R}(y) \quad \text{but} \quad P(x) \neq P(y),$$

then no predicate q depending only on the realized Core object can satisfy $P = q \circ \mathfrak{R}$ on all of S .

Proof. If such q existed, the isomorphic realized objects would receive the same value under q , forcing $P(x) = P(y)$, a contradiction. $\square$

Remark 7.76 (Meaning of target separation). A realization used to decide a PDE target must separate target-positive from target-negative source objects on the declared family, or a weaker one-way statement must be proved with its exact direction. Topological or persistence compression that forgets this distinction cannot be repaired by a later Core calculation.

Definition 7.77 (Metric-safe grid-to-continuum zero transfer). A metric grid-to-continuum zero transfer records actual windowed pre-threshold profiles B_h, B , a certified bound $d_B(B_h, B) \leq \varepsilon_h$, two-sided avoidance of $[\tau - \varepsilon_h, \tau + \varepsilon_h]$ by both profiles, the zero-separation condition $2\varepsilon_h \leq \tau$ (strictly $<$ in the operational reserve), and detector completeness on every profile whose zero predicate is inferred.

Proposition 7.78 (Raw diagnostic numbers are not metric-stable). *A bottleneck bound between source or target persistence profiles does not by itself control Type IV kernel or cokernel profiles, terminal-generic dimensions, raw detector ranks, or certificate sizes.*

Proof. Those quantities depend on actual morphisms, defect-profile constructions, detector stages, and completeness theorems not supplied by an object-level metric bound. $\square$

Definition 7.79 (Three NS defect-soundness levels). An NS soundness route treats separately:

- (i) terminal-generic Type IV status;
- (ii) long-transient defect status;
- (iii) full defect-profile status.

A PDE Return theorem must name exactly which one it consumes. No implication among the three is available by terminology.

Proposition 7.80 (Terminal Type IV zero is insufficient for full PDE limiting control). *A Type IV-to-PDE bridge that requires absence of all limiting defects cannot consume only terminal-generic zero unless it separately proves that finite noncofinal and transient defects are irrelevant or absent.*

Proof. Terminal-generic zero leaves finite interior defect profiles unconstrained. A PDE statement sensitive to those profiles therefore lacks a required premise. $\square$

NS-B.12quater. Exact small-success recovery: a fixed-exponent Serrin criterion

Definition 7.81 (Frozen Serrin source regime). Fix a finite time horizon $T > 0$. Let u be a Leray–Hopf weak solution of the unforced three-dimensional incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$, with viscosity $\nu_{\text{visc}} > 0$ and divergence-free initial datum in $L^2(\mathbb{R}^3)$. The frozen benchmark is the strict-subcritical fixed-exponent Serrin implication

$$u \in L^6(0, T; L^6(\mathbb{R}^3)) \implies u \text{ is interior regular on } \mathbb{R}^3 \times (0, T].$$

This is the fixed-exponent specialization used from James Serrin, *On the interior regularity of weak solutions of the Navier–Stokes equations*, Arch. Rational Mech. Anal. 9 (1962), 187–195, DOI 10.1007/BF00253344. Its authoritative article locator, exact frozen benchmark wording, solution-class scope, and prohibited stronger readings are carried by the source-binding artifact SRC-NS-SERRIN-L6. The analytic conclusion is quarantined from the realization and is used only at the Return step.

Definition 7.82 (Serrin norm and excess residual). Set

$$\mathcal{N}_6(u) := \|u\|_{L^6(0, T; L^6(\mathbb{R}^3))} \in [0, +\infty].$$

For a rational witness $M \in \mathbb{Q}_{>0}$, define

$$r_M(u) := \max\{0, \mathcal{N}_6(u) - M\} \in [0, +\infty],$$

with $r_M(u) = +\infty$ when $\mathcal{N}_6(u) = +\infty$.

Definition 7.83 (Exact single-bar Serrin realization). Fix

$$W_S := [0, +\infty), \quad \tau_S := 1.$$

For each rational $M > 0$, define the constructible one-bar persistence object

$$P_M^S(u) := \begin{cases} \mathbf{k}_{[0, 1+r_M(u))}, & r_M(u) < +\infty, \\ \mathbf{k}_{[0, +\infty)}, & r_M(u) = +\infty. \end{cases}$$

Its Core predicate is

$$Q_M^S(u) := T_1^{\text{bar}}(W_{W_S} P_M^S(u)) = 0.$$

The comparison mode is exact. No metric ledger, detector, Type IV artifact, higher edge, or finite-to-infinite reduction is invoked.

Theorem 7.84 (Exact zero-reflection of the Serrin bound). *For every source object u in Definition 7.81 and every $M \in \mathbb{Q}_{>0}$,*

$$Q_M^S(u) \iff N_6(u) \leq M.$$

Proof. Windowing by W_S does not change the displayed interval. If $r_M(u) = 0$, the unique finite bar has length exactly $1 = \tau_S$ and is deleted. If $0 < r_M(u) < +\infty$, its length is strictly greater than 1 and it is retained. If $r_M(u) = +\infty$, the infinite bar is retained. Hence the repaired profile is zero exactly when $r_M(u) = 0$, equivalently when $N_6(u) \leq M$. $\square$

Theorem 7.85 (Finite rational witness for the Serrin hypothesis). *For every declared source solution u ,*

$$N_6(u) < +\infty \iff \exists M \in \mathbb{Q}_{>0} Q_M^S(u).$$

The certificate payload is the rational witness M , an artifact-backed analytic proof of the exact source-norm bound $N_6(u) \leq M$, the one-bar realization identity, and the exact repaired-zero computation. The numeral M alone is not theorem evidence.

Proof. If $N_6(u) < +\infty$, choose a rational $M \geq N_6(u)$; then Theorem 7.84 gives $Q_M^S(u)$. Conversely, such a witness implies $N_6(u) \leq M < +\infty$. The statement is finite at proof-object level once the analytic norm bound is itself proved; it does not claim that sampled numerical data compute the exact norm bound. $\square$

Theorem 7.86 (Exact known-theorem recovery of the fixed-exponent Serrin implication). *Assume the frozen source theorem SRC-NS-SERRIN-L6. For a source solution u , if there exists $M \in \mathbb{Q}_{>0}$ with*

$$Q_M^S(u),$$

then u is interior regular on $\mathbb{R}^3 \times (0, T]$. Conversely, the analytic hypothesis $u \in L^6(0, T; L^6(\mathbb{R}^3))$ supplies such a finite rational Core witness. Thus the complete chain

$$\begin{aligned} u &\mapsto N_6(u) \mapsto r_M(u) \mapsto P_M^S(u) \mapsto Q_M^S(u) \\ &\longrightarrow N_6(u) < +\infty \longrightarrow \text{Serrin interior regularity} \end{aligned}$$

recovers the frozen criterion at exactly its one-way strength.

Proof. The first Return arrow is Theorem 7.84; the existence of a rational witness is Theorem 7.85; the final analytic implication is the frozen Serrin source theorem. The realization uses only the source norm and its rational bound, never the regularity conclusion, so the chain is non-circular. $\square$

Remark 7.87 (Meaning and limitation of the small success). This is a genuine continuum NS Interface/Return bridge and an exact Known-Theorem Recovery, but it contributes no new analytic proof of Serrin's theorem. It certifies the criterion predicate by a one-bar Core object and returns the already known conditional regularity conclusion. It does not decide regularity for solutions outside $L_t^6 L_x^6$, does not prove that regularity implies the criterion, and does not approach unconditional global regularity by itself.

Remark 7.88 (One-way separation and the no-go theorem). The realization is exact for the sufficient predicate $N_6(u) < +\infty$; it is not claimed to separate every regular solution from every singular candidate. This is the weaker one-way regime allowed by Remark 7.76, not a violation of Theorem 7.75.

Definition 7.89 (NS known-theorem-recovery status). For v20.0.0, the NS stack has

known_theorem_recovery = exact_small_success,
recovery_track = NS-KTR-SERRIN-L6,
criterion_level_bridge = proved_relative_to_source_theorem,
unconditional_3D_regularity_bridge = not_invoked.

The local mathematical witness is ready for final Claim Register and release-registry promotion. Registration remains a separate governance act.

Definition 7.90 (Criterion-recovery successor package). Track NS-KTR-SERRIN-L6 completes the first criterion-recovery package at deliberately narrow strength. Successor work should replace or enrich the scalar one-bar realization by a more geometrically informative target-separating realization, recover a sharper criterion, or close a local-to-global route. No successor step is inferred from the present recovery.

Theorem 7.91 (NS-B v20 soundness closure). *NS-B closes one complete criterion-level source-to-PDE route, namely Track NS-KTR-SERRIN-L6, and closes the general soundness schema. Every other external NS conclusion remains undefined unless its own complete typed bridge chain and Return theorem are proved. Neither the small-success recovery nor a complete manifest supplies unconditional three-dimensional regularity.*

Proof. Theorems 7.84, 7.85, and 7.86 close the frozen criterion route. The conditional composition theorem still requires all arrows for any other target; the target-separation no-go excludes realizations that erase the consumed predicate; and Definition 7.89 prevents promotion from criterion-level recovery to unconditional regularity. $\square$

NS-B.13. Summary of theoretical soundness layer

Theorem 7.92 (NS-B soundness package). *Within the Problem Interface layer, the following package is available.*

- (i) *A Navier–Stokes continuum target, solution class, and target statement must be declared before PDE interpretation.*
- (ii) *A continuum realization must be purpose-faithful for the declared PDE target.*
- (iii) *Core-visible NS features must be expressed through repaired evaluation, admissible representatives when invoked, declared tower artifacts when used, diagnostics, and ledger records.*
- (iv) *Discretized data require grid-to-continuum soundness.*
- (v) *Discretized zero diagnostics do not imply continuum zero diagnostics without diagnostic convergence.*
- (vi) *Measured observable control does not imply PDE regularity without an observable-to-regularity bridge and ledgered measurement error.*
- (vii) *Persistence control does not imply observable control without a persistence-observable bridge.*
- (viii) *Energy, vorticity, pressure, scale, and criticality compatibility are target-dependent and must be declared.*
- (ix) *No Navier–Stokes conclusion follows from Core pass without an admissible NS soundness bridge.*
- (x) *The Core Extⁿ-clause may be invoked only in the NS eligible regime.*
- (xi) *Sound normalization does not imply PDE regularity without a normalized-to-source regularity bridge.*
- (xii) *Type IV cleanliness is not PDE regularity by default.*
- (xiii) *An NS theoretical soundness record is required for PDE interpretation.*
- (xiv) *Core pass plus unproved [Spec] soundness bridge is not theorem-level pass.*

- (xv) *NS soundness sovereignty prevents Core pass, exploration success, and diagnostic terminology from becoming PDE interpretation without a proved bridge.*
- (xvi) *Bridge-promotion requires an explicit promotion record and proof artifacts.*
- (xvii) *Sampled diagnostic agreement is not diagnostic convergence.*
- (xviii) *Compatibility with energy, vorticity, pressure, scale, and criticality is a non-scalar obligation.*
- (xix) *Theorem-level promotion requires the full NS theorem-promotion schema.*
- (xx) *Track NS-KTR-SERRIN-L6 gives an exact one-bar recovery of the fixed-exponent Serrin criterion, relative to its frozen analytic source theorem.*
- (xxi) *This criterion-level recovery is not an unconditional three-dimensional regularity theorem and supplies no new analytic proof of the source criterion.*

Proof. Items (i)–(ii) follow from the definitions of continuum target and purpose-faithful realization. Item (iii) follows from the definition of Core-visible NS feature. Items (iv)–(v) follow from grid-to-continuum soundness and Proposition 7.19. Item (vi) is Proposition 7.25. Item (vii) is Proposition 7.26. Item (viii) follows from the compatibility definitions. Item (ix) is Proposition 7.39. Item (x) is Proposition 7.42. Item (xi) is Proposition 7.46. Item (xii) is Proposition 7.51. Item (xiii) is Proposition 7.54. Item (xiv) is Proposition 7.59. Items (xv)–(xix) follow from Definition 7.61, Proposition 7.62, Definition 7.63, Proposition 7.64, Definition 7.65, Proposition 7.66, Definition 7.67, Proposition 7.68, Definition 7.69, and Proposition 7.70. Items (xx)–(xxi) follow from Theorem 7.86 and Remark 7.87. $\square$

NS-B.14. Explicit exclusions

The following are not asserted in Appendix NS-B.

- No unconditional three-dimensional Navier–Stokes regularity theorem is proved.
- No new analytic proof of a regularity or smoothness theorem is supplied; Track NS-KTR-SERRIN-L6 only recovers the frozen conditional Serrin implication.
- No uniqueness theorem is proved.
- No existence theorem is proved.
- No continuation theorem is proved.
- No non-blow-up theorem is proved.
- No blow-up theorem is proved.
- No singularity exclusion theorem is proved.
- No Core pass is interpreted as a PDE theorem without a proved soundness bridge.
- No discretized diagnostic is treated as continuum diagnostic without diagnostic convergence.
- No numerical convergence is assumed without a grid-to-continuum statement.
- No measured observable is treated as exact without measurement ledger.
- No vorticity, energy, pressure, scale, or criticality observable is treated as a regularity criterion without an analytic bridge.
- No repaired persistence control is treated as observable control without a persistence-observable bridge.
- No normalization success is treated as regularity.

- No Type IV cleanliness is treated as PDE regularity by default.
- No PDE-derived cohomological obstruction is automatically identified with Core Extⁿ.
- No [Spec] soundness bridge is treated as a theorem.
- No exploration artifact supports theorem-level PDE conclusion by itself.
- No global exactness of T_{τ}^{bar} is assumed.
- No positive-threshold Serre localization is assumed.
- No global strict filtered lift C_{τ} is assumed.

NS-B.15. Provenance and status

The material in this appendix depends on:

- (i) the Navier–Stokes exploration protocol of Appendix NS-A;
- (ii) the normalization interface of Appendix AGI-N;
- (iii) the repaired barcode-level collapse and constructible persistence framework of Appendix A;
- (iv) the admissible representative discipline of Appendix B;
- (v) the eligible realization regime of Appendix C;
- (vi) the monitored Type IV diagnostic calculus and tower comparison artifact discipline of Appendix D;
- (vii) the manifest schema of Appendix G;
- (viii) the grid-to-continuum discipline of Appendix I;
- (ix) the measurement-level stability discipline of Appendix J;
- (x) the ledger catalog of Appendix M;
- (xi) the interface protocol of Chapter 8;
- (xii) the non-claim discipline of Chapter 7.

Its role is to state the theoretical conditions required before NS-derived Core certificates may be interpreted as PDE-relevant statements.

Status labels for Appendix NS-B. *Problem-interface definitions:* Definitions 7.4, 7.5, 7.6, 7.7, 7.9, 7.10, 7.11, 7.14, 7.15, 7.16, 7.17, 7.18, 7.21, 7.22, 7.23, 7.24, 7.28, 7.29, 7.30, 7.31, 7.32, 7.35, 7.36, 7.37, 7.41, 7.44, 7.45, 7.48, 7.49, 7.50, 7.53, 7.56, 7.57, 7.58.

Problem-interface propositions/theorems: Propositions 7.12, 7.19, 7.25, 7.26, 7.33, 7.39, 7.42, 7.46, 7.51, 7.54, 7.59, Theorem 7.92.

Operational cautions: Remarks 7.1, 7.2, 7.3, 7.8, 7.13, 7.20, 7.27, 7.34, 7.40, 7.43, 7.47, 7.52, 7.55, 7.60.

Additional v20.0 conservative hardening definitions: Definitions 7.61, 7.63, 7.65, 7.67, 7.69.

Additional v20.0 conservative hardening propositions: Propositions 7.62, 7.64, 7.66, 7.68, 7.70.

Additional v20.0 conservative hardening remarks: Remarks 7.71, 7.72, 7.38.

Additional v20 completion definitions: Definitions 7.73, 7.81, 7.82, 7.83, 7.77, 7.79, 7.89, and 7.90.

Additional v20 completion propositions/theorems: Propositions 7.78 and 7.80; Theorems 7.74, 7.75, 7.84, 7.85, 7.86, and 7.91.

Additional v20 completion remark: Remarks 7.76, 7.87, and 7.88.

Explicit exclusions: Section NS-B.14.

8 Appendix NS-C: Navier–Stokes Proof-First Program

NS-C.1. Scope and proof-first status

This appendix defines a proof-first program for Navier–Stokes interfaces in the AK framework. It belongs to the Problem Interface layer. It does not enlarge the Core mathematical package of Chapters 1–8.

The central rule of this appendix is:

A proof-first program is a proof architecture, not a completed proof.

The purpose of this appendix is to state, in an auditable way, what must be proved before an AK-based Navier–Stokes interface can support a theorem-level Navier–Stokes conclusion.

All Core-side proof obligations must be expressed through the repaired Core vocabulary:

$$\text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)),$$

admissible representatives

$$C_{\tau,W}F,$$

declared tower comparison artifacts

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_{\bullet}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_{\infty}),$$

and the audit budget condition

$$\text{val}_B(\Sigma\delta) < \text{gap}_{\tau}(\mathfrak{B}).$$

Remark 8.1 (Proof-first status). This appendix does not prove Navier–Stokes regularity, smoothness, uniqueness, continuation, non-blow-up, blow-up, or singularity exclusion. It defines the proof obligations required before such claims may be made.

Remark 8.2 (Relation to Appendices NS-A and NS-B). Appendix NS-A defines exploration. Appendix NS-B defines theoretical soundness. The present appendix reverses the priority: rather than beginning from exploratory outputs, it begins from the target theorem and lists the required bridges, diagnostics, Core certificates, and proof objects.

Remark 8.3 (No exact-collapse restoration). This appendix does not restore the old exact-collapse framework. No global exact collapse functor, positive-threshold Serre localization, or global strict filtered lift C_{τ} is assumed. All proof-first obligations must be stated in terms of repaired evaluation, admissible representatives, declared tower artifacts, and ledgered defects.

NS-C.2. Proof-first principle

Definition 8.4 (Proof-first principle). The *proof-first principle* states that an AK-based Navier–Stokes program must begin by declaring:

- (i) the target PDE theorem;
- (ii) the solution class;
- (iii) the PDE assumptions;
- (iv) the Core-readable realization route;
- (v) the repaired evaluation records required;
- (vi) the admissible representative data required, if any;
- (vii) the declared tower comparison artifacts required, if any;
- (viii) the soundness bridge;

- (ix) the required AK Core certificate;
- (x) all external analytic lemmas;
- (xi) all ledgered defects;
- (xii) all failure conditions;
- (xiii) the proof object schema.

Exploration may support this program, but it may not determine the theorem.

Definition 8.5 (Proof-first record). A *proof-first record* is a manifest-level record that lists:

- (i) the target theorem;
- (ii) required analytic hypotheses;
- (iii) required realization bridge;
- (iv) required repaired evaluation records;
- (v) required Core certificates;
- (vi) required soundness bridges;
- (vii) required transport statements;
- (viii) required tower comparison artifacts;
- (ix) required ledger conditions;
- (x) unresolved gaps;
- (xi) status labels.

Proposition 8.6 (Exploration cannot replace proof-first obligations). *No exploration output, advisory indicator, numerical result, normalization success, Type IV proxy, or Core candidate replaces the proof obligations declared in a proof-first record.*

Proof. Exploration outputs may identify candidates or suggest structures. Proof obligations are logical requirements for theorem-level implication. Candidate generation does not discharge those requirements. $\square$

Remark 8.7 (Why proof-first is required). For a Clay-level PDE problem, exploratory agreement is insufficient. The architecture must specify the theorem, hypotheses, analytic bridges, Core-side certificate conditions, and proof-object requirements before any theorem-level conclusion is claimed.

NS-C.3. Target theorem templates

Definition 8.8 (Navier–Stokes target theorem template). A *Navier–Stokes target theorem template* is a formal statement of the intended PDE conclusion. It must specify:

- (i) the spatial dimension;
- (ii) the domain;
- (iii) the boundary or decay conditions;
- (iv) the viscosity parameter;
- (v) the initial data class;
- (vi) the solution class;

- (vii) the time interval;
- (viii) the exact conclusion to be proved.

Definition 8.9 (Local regularity target). A *local regularity target* is a theorem template asserting regularity or smoothness of a solution on a declared spacetime cylinder

$$Q_r(x_0, t_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

Definition 8.10 (Continuation target). A *continuation target* is a theorem template asserting that a solution can be continued beyond a time T under declared conditions.

Definition 8.11 (Non-blow-up target). A *non-blow-up target* is a theorem template asserting that no singularity forms in a declared region or time interval.

Definition 8.12 (Global regularity target). A *global regularity target* is a theorem template asserting global smoothness or regularity for all time in a declared solution class.

Definition 8.13 (Clay-level target). A *Clay-level target* is a theorem template matching the global existence and smoothness problem for the three-dimensional incompressible Navier–Stokes equations in a standard Clay-compatible formulation.

Remark 8.14 (Global targets are strongest). A global regularity target is substantially stronger than a local regularity target. The proof-first program should generally treat local, continuation, non-blow-up, global, and Clay-level targets as distinct stages.

NS-C.4. Core certificate target

Definition 8.15 (NS Core certificate target). An *NS Core certificate target* is the conjunction of exactly those Core clauses invoked by the declared theorem route. Its record fixes the monitored degrees, one right-open window W , one threshold $\tau > 0$, the comparison modes, detector stages, protected profile family, and all non-scalar obligations. Schematically,

$$C_{\text{NS}} \models B\text{-Gate}^+ \wedge \mathcal{H}_{\text{higher}} \wedge \mathcal{D}_{\text{TypeIV}} \wedge \mathcal{D}_{\text{transient}} \wedge (\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B})),$$

where every optional clause is either invoked with its complete evidence or marked `not_invoked` under a declared scope exclusion. Type IV zero is not required when Type IV is outside scope, and a ledger inequality cannot compensate for a missing representative, detector theorem, actual morphism, reduction, Return theorem, or artifact.

Definition 8.16 (NS Core object of proof). The *NS Core object of proof* is the filtered or persistence object

$$F_{\text{NS}} \quad \text{or} \quad P_{\text{NS}}$$

obtained from the continuum Navier–Stokes target through the declared realization route and used in the Core certificate target.

Definition 8.17 (NS repaired evaluation proof target). An *NS repaired evaluation proof target* is a required verifier-side statement of the form:

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}}) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F_{\text{NS}})),$$

together with the required vanishing or comparison condition in monitored degrees.

Definition 8.18 (NS tower proof target). An *NS tower proof target* is a required verifier-side tower statement involving a declared artifact

$$\phi_{i,W,\tau}^{\text{NS}} : \text{ColimProfile}_{i,W,\tau}(F_\bullet^{\text{NS}}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty^{\text{NS}})$$

and the diagnostic condition

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

Definition 8.19 (Core certificate sufficiency claim). A *Core certificate sufficiency claim* is a statement of the form:

$$C_{\text{NS}} \text{ passes Core} \implies \text{PDE target theorem.}$$

Proposition 8.20 (Core certificate sufficiency is [Spec] by default). *Every Core certificate sufficiency claim for Navier–Stokes is [Spec] by default unless proved by an admissible NS soundness bridge.*

Proof. A Core certificate is a statement about repaired persistence, admissible representatives, diagnostics, artifacts, and budget. A Navier–Stokes theorem is a statement about PDE solutions. The implication requires an analytic soundness bridge. Without that bridge, it remains [Spec]. $\square$

Remark 8.21 (Core pass is necessary only if declared). The proof program may declare Core pass as a sufficient condition, a necessary condition, or an auxiliary certificate. These logical roles must not be confused.

NS-C.5. Required bridge lemmas

Definition 8.22 (Bridge lemma). A *bridge lemma* is a theorem connecting one layer of the proof program to another. For Navier–Stokes, bridge lemmas may connect:

$$\text{PDE observables} \leftrightarrow \text{persistence realization} \leftrightarrow \text{Core certificate} \leftrightarrow \text{PDE target conclusion.}$$

Definition 8.23 (Realization bridge). A *realization bridge* proves purpose-faithfulness of the chosen PDE-to-persistence realization. The proof is relative to the declared target theorem.

Definition 8.24 (Evaluation bridge). An *evaluation bridge* proves that the repaired evaluation records

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}})$$

capture the PDE-relevant features required by the proof route.

Definition 8.25 (Collapse bridge). A *collapse bridge* proves that the barcode-level after-threshold Core data preserve or detect the PDE-relevant obstruction or regularity feature needed by the proof.

Definition 8.26 (Diagnostic bridge). A *diagnostic bridge* proves that monitored Type IV cleanliness or failure, computed from the declared artifact

$$\phi_{i,W,\tau}^{\text{NS}},$$

corresponds to the intended PDE limiting behavior.

Definition 8.27 (Regularity bridge). A *regularity bridge* proves that the verified Core-visible condition implies the declared Navier–Stokes regularity, continuation, or non-blow-up target.

Definition 8.28 (Compactness bridge). A *compactness bridge* proves that limiting behavior of realized or normalized objects is compatible with the required continuum PDE compactness, convergence, or weak-to-strong passage.

Definition 8.29 (Global assembly bridge). A *global assembly bridge* proves that local or windowed Core certificates assemble into the global PDE conclusion required by a global or Clay-level target.

Remark 8.30 (Bridge lemmas are the main mathematical burden). The AK Core can certify its own internal statements. The hard PDE work lies in proving the bridge lemmas that connect Core-visible statements to Navier–Stokes regularity.

NS-C.6. Required analytic hypotheses

Definition 8.31 (Analytic hypothesis set). An *analytic hypothesis set* is the collection of PDE-side assumptions required by the proof-first program. It may include:

- (i) energy inequality;
- (ii) local energy inequality;
- (iii) weak compactness;
- (iv) pressure decomposition;
- (v) vorticity equation control;
- (vi) scale-critical norm bounds;
- (vii) interpolation inequalities;
- (viii) ε -regularity criterion;
- (ix) backward uniqueness;
- (x) continuation criterion;
- (xi) uniqueness or weak-strong uniqueness assumptions;
- (xii) local-to-global assembly assumptions.

Definition 8.32 (External analytic theorem). An *external analytic theorem* is a known or separately proved PDE theorem used by the proof-first program. It must be cited, stated, or imported with its hypotheses.

Definition 8.33 (Analytic gap). An *analytic gap* is a missing or unproved PDE-side statement required to complete the proof-first program.

Definition 8.34 (Core-side gap). A *Core-side gap* is a missing or unverified AK-side requirement, such as missing repaired evaluation records, missing admissible representatives, missing eligibility, missing tower artifacts, failed diagnostics, or failed budget comparison.

Definition 8.35 (Bridge gap). A *bridge gap* is a missing or unproved implication between PDE-side data and Core-side data, or between Core-side pass and a PDE target conclusion.

Proposition 8.36 (Analytic gaps prevent theorem-level pass). *If the proof-first program contains an unresolved analytic gap that is necessary for the target theorem, then theorem-level pass is not obtained.*

Proof. A theorem-level pass requires all required implications to be proved. An unresolved necessary analytic gap means at least one implication is missing. Therefore the theorem-level conclusion does not follow. $\square$

Proposition 8.37 (Bridge gaps prevent theorem-level pass). *If the proof-first program contains an unresolved bridge gap necessary for the target theorem, then theorem-level pass is not obtained.*

Proof. A bridge gap means that one of the required cross-layer implications is missing. Since theorem-level status requires a complete chain from PDE assumptions through Core certification to the PDE conclusion, the target theorem does not follow. $\square$

Remark 8.38 (Gaps should be explicit). A proof-first record should expose analytic, Core-side, and bridge gaps rather than hide them. A visible gap is preferable to an unstated theorem-level claim.

NS-C.7. Continuum, discretization, and proof status

Definition 8.39 (Continuum proof mode). A proof-first program is in *continuum proof mode* if all Core-readable objects and diagnostics are derived from continuum Navier–Stokes data through a proved continuum realization hypothesis.

Definition 8.40 (Discretized proof-support mode). A proof-first program is in *discretized proof-support mode* if discretized or numerical data support exploration, conjecture formation, candidate selection, or empirical checking, but theorem-level claims still require independent continuum soundness.

Definition 8.41 (Discretized theorem mode). A proof-first program is in *discretized theorem mode* only if grid-to-continuum convergence, repaired evaluation convergence, diagnostic convergence, measurement soundness, and all analytic transport bridges are proved sufficiently to support theorem-level claims.

Definition 8.42 (Hybrid proof-support mode). A proof-first program is in *hybrid proof-support mode* if continuum arguments provide the main theorem route while discretized or numerical artifacts provide auxiliary verification, debugging, or candidate selection.

Proposition 8.43 (Discretized proof-support mode is not theorem mode). *Discretized proof-support mode does not yield theorem-level Navier–Stokes conclusions.*

Proof. Discretized proof-support mode allows numerical or finite-resolution data to support exploration. Theorem-level conclusions require continuum statements and proved grid-to-continuum and soundness bridges. Therefore proof-support mode is not theorem mode. $\square$

Remark 8.44 (Preferred theorem route). The preferred route for a theorem-level claim is continuum proof mode. Discretized data may remain valuable for exploration, but the proof should ultimately close at the continuum level or include a fully proved grid-to-continuum bridge.

NS-C.8. Failure modes

Definition 8.45 (Proof-first failure mode). A *proof-first failure mode* is a declared way in which the Navier–Stokes proof-first program may fail to establish its target theorem.

Definition 8.46 (Realization failure). *Realization failure* occurs when the PDE source data cannot be realized as a purpose-faithful Core-readable object.

Definition 8.47 (Evaluation failure). *Evaluation failure* occurs when repaired evaluation records

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}})$$

are missing, nonconstructible, unstable beyond budget, or fail the required Core-side condition.

Definition 8.48 (Eligibility failure). *Eligibility failure* occurs when an invoked degree- n higher-edge record is present and a required standard n -eligibility condition is evidenced false. Missing eligibility evidence yields *undefined*, not failure and not Ext ^{n} -vanishing.

Definition 8.49 (Tower artifact failure). *Tower artifact failure* occurs when Type IV diagnostics are invoked but the required tower comparison artifact

$$\phi_{i,W,\tau}^{\text{NS}}$$

is missing, ill-defined, or not audit-readable.

Definition 8.50 (Diagnostic failure). *Diagnostic failure* occurs only when a valid invoked Type IV record, with actual comparison morphism and terminal-tameness evidence on the fixed monitored scope, has status obstructed, equivalently at least one valid terminal-generic diagnostic is nonzero. Missing tower, apex, morphism, source semantics, or terminal evidence yields undefined, not diagnostic failure and not diagnostic zero.

Definition 8.51 (Budget failure). *Budget failure* occurs when a valid scalarized metric-ledger record proves that the required strict comparison does not hold:

$$\text{val}_B(\mathbf{d}_{\text{met}}) \not\leq \text{gap}_\tau(\mathfrak{B}).$$

A missing scalarization, missing actual bottleneck bound, or missing non-scalar obligation yields *undefined*; it is not silently renamed budget failure.

Definition 8.52 (Soundness failure). *Soundness failure* occurs when the Core certificate passes internally but no proved soundness bridge connects it to the PDE target theorem.

Definition 8.53 (Transport failure). *Transport failure* occurs when a normalized, discretized, localized, or transformed object passes a relevant check, but the required implication back to the original continuum PDE target is missing or false.

Proposition 8.54 (Soundness failure blocks PDE conclusion). *If soundness failure occurs, then no Navier–Stokes theorem-level conclusion follows, even if the Core certificate passes.*

Proof. Soundness failure means that the implication from Core behavior to the PDE target theorem is not proved. Therefore the Core pass remains internal to the AK certificate and does not yield the PDE theorem. $\square$

Remark 8.55 (Failure classification is constructive). A failure mode is not merely negative. It identifies which proof obligation must be repaired: realization, repaired evaluation, eligibility, tower artifact, diagnostics, budget, soundness, or transport.

NS-C.9. Proof object requirements

Definition 8.56 (NS proof object). An *NS proof object* is a structured record intended to support a theorem-level Navier–Stokes conclusion through the AK interface. It must include both Core certificate data and PDE-side soundness data.

Definition 8.57 (NS proof object schema). An NS proof object must contain:

- (i) target theorem template;
- (ii) solution class;
- (iii) continuum source data;
- (iv) realization bridge;
- (v) Core-readable object;
- (vi) declared windows, thresholds, and monitored degrees;
- (vii) repaired evaluation records

$$\text{Eval}_{i,W,\tau}(F_{\text{NS}});$$
- (viii) admissible representative data

$$C_{\tau,W}F_{\text{NS}}$$
 if representatives are invoked;
- (ix) eligibility data, if Ext^n is invoked;
- (x) tower comparison artifacts

$$\phi_{i,W,\tau}^{\text{NS}}$$
 if Type IV diagnostics are invoked;
- (xi) Core certificate target;

- (xii) monitored Type IV diagnostics;
- (xiii) budget ledger;
- (xiv) Core gate verdicts;
- (xv) regularity bridge;
- (xvi) compactness bridge, if needed;
- (xvii) global assembly bridge, if needed;
- (xviii) normalization or discretization transport, if used;
- (xix) artifact references;
- (xx) status labels for every bridge and claim.

Definition 8.58 (Theorem-level NS proof object). An NS proof object is *theorem-level* if every required bridge, transport statement, diagnostic, ledger condition, Core gate condition, artifact requirement, and PDE-side analytic condition is proved or verified under the declared hypotheses.

Definition 8.59 (Proof object status). An NS proof object has one of the following statuses:

- (i) *schema-only*: the required fields are declared but not filled;
- (ii) *partial*: some proof fields are filled and some gaps remain;
- (iii) *Core-complete*: Core fields are complete but PDE-side soundness is incomplete;
- (iv) *soundness-complete*: soundness bridges are proved but Core fields remain incomplete;
- (v) *theorem-level*: all required components are complete and proved;
- (vi) *failed*: at least one required component fails.

Proposition 8.60 (Incomplete NS proof object is not theorem-level). *An incomplete NS proof object is not theorem-level.*

Proof. A theorem-level proof object requires all necessary logical components. If any required bridge, diagnostic, ledger, artifact, or analytic component is missing, the proof object cannot establish the target theorem. $\square$

Remark 8.61 (Status labels are mandatory). Every component of an NS proof object should be labeled as Core theorem, interface theorem, operational policy, [Spec], search artifact, or non-claim. This prevents conjectural bridges from being read as proved theorems.

NS-C.10. Minimal theorem route

Definition 8.62 (Minimal theorem route). A *minimal theorem route* is the shortest chain of proved implications from the target theorem template assumptions to the PDE conclusion through AK Core certification. It has the schematic form:

$$\begin{aligned}
 &\text{PDE assumptions} \Rightarrow \text{purpose-faithful Core realization} \\
 &\quad \Rightarrow \text{repaired Core certificate pass} \\
 &\quad \Rightarrow \text{proved soundness bridge} \Rightarrow \text{PDE conclusion.}
 \end{aligned}$$

Definition 8.63 (Core route segment). A *Core route segment* is the part of a minimal theorem route that proves the required repaired Core certificate:

$$\text{Eval}_{1,W,\tau}(F_{\text{NS}}) = 0, \quad (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0), \quad \text{val}_B(\Sigma\delta) < \text{gap}_\tau(\mathfrak{B}),$$

together with any required representative, eligibility, and artifact conditions.

Definition 8.64 (PDE route segment). A *PDE route segment* is the part of a minimal theorem route that proves the PDE-side analytic implication from the Core certificate or Core-visible condition to the target PDE conclusion.

Definition 8.65 (Route redundancy). A proof route has *redundancy* if it includes advisory indicators, numerical evidence, alternative normalizations, categorical comparisons, or search artifacts not logically required for the theorem-level implication.

Proposition 8.66 (Minimal route should separate proof from support). *A minimal theorem route should separate logically necessary proof components from supportive exploration artifacts.*

Proof. Theorem-level validity depends only on necessary proved implications. Exploration artifacts may motivate or support discovery but should not be mixed with the proof chain unless they are converted into verified proof components. $\square$

Remark 8.67 (Usefulness of redundant support). Redundant support may improve confidence, debugging, and discovery. It should be recorded as advisory or supporting material, not as part of the minimal proof chain.

NS-C.11. Candidate proof program

Definition 8.68 (Candidate AK–NS proof program). A *candidate AK–NS proof program* consists of the following tasks:

- (i) choose a target theorem template;
- (ii) specify the solution class and PDE assumptions;
- (iii) define a purpose-faithful realization from PDE data to Core-readable persistence or filtered data;
- (iv) prove constructibility and repaired evaluation compatibility;
- (v) establish the relevant Core certificate target;
- (vi) prove admissible representative and eligibility conditions when Ext^n is invoked;
- (vii) construct required tower comparison artifacts when Type IV diagnostics are invoked;
- (viii) prove diagnostic soundness for monitored Type IV;
- (ix) prove budget admissibility under all approximation, normalization, and measurement defects;
- (x) prove a regularity, continuation, or non-blow-up bridge;
- (xi) assemble the NS proof object;
- (xii) classify all remaining gaps.

Definition 8.69 (Program status). A candidate AK–NS proof program has one of the following statuses:

- (i) *exploratory*: dominated by search or advisory evidence;
- (ii) *interface-specified*: target, realization, and bridge requirements are stated;
- (iii) *partially proved*: some bridge lemmas or Core components are proved;
- (iv) *Core-certified*: Core certificate is established but PDE soundness is incomplete;
- (v) *soundness-certified*: PDE soundness bridge is proved but Core certificate is incomplete or separate;
- (vi) *theorem-level*: all required components are proved and assembled.

Proposition 8.70 (Core-certified is not theorem-level by default). *A Core-certified AK–NS proof program is not theorem-level unless the PDE soundness bridge is also proved.*

Proof. Core certification establishes AK-internal certificate conditions. A theorem-level Navier–Stokes conclusion requires an additional bridge. That bridge must connect the Core conditions to the declared PDE target. $\square$

Remark 8.71 (Program status should be honest). A proof program should report its status conservatively. Calling an exploratory or Core-certified program theorem-level without a proved soundness bridge violates the non-claim discipline.

NS-C.12. Relation to Clay-level formulation

Proposition 8.72 (Clay-level target requires global bridge). *A Clay-level target cannot be obtained from local or windowed Core certificates unless a global regularity bridge assembles the local certificates into the global PDE conclusion.*

Proof. Local or windowed certificates concern restricted spacetime or parameter domains. A Clay-level target is global in time and space under the declared formulation. A theorem assembling local certificate behavior into global existence and smoothness is therefore required. $\square$

Definition 8.73 (Clay-level proof obligation). *A Clay-level proof obligation is a required component for upgrading a local, continuation, or windowed AK–NS proof route to a Clay-level global result. It includes:*

- (i) global-in-time solution class compatibility;
- (ii) global local-to-global assembly;
- (iii) control of all possible singularity scenarios;
- (iv) compatibility with the standard Clay formulation;
- (v) proof that the AK-side certificate route covers every required PDE case;
- (vi) elimination of all analytic, Core-side, and bridge gaps.

Remark 8.74 (No Clay claim). This appendix does not claim to solve the Clay Navier–Stokes problem. It states what would be needed for an AK-based proof program to reach that level.

NS-C.13. Closure of the Navier–Stokes appendices

Definition 8.75 (NS appendix closure). *The NS appendix closure is the requirement that every Navier–Stokes-facing claim remain classified into one of the following layers:*

exploration, Core certificate, soundness bridge,
proof-first theorem route, PDE theorem.

Proposition 8.76 (NS appendices close without unconditional Navier–Stokes regularity). *Appendices NS-A–NS-C close the Navier–Stokes interface layer and recover the fixed-exponent Serrin criterion through Track NS-KTR-SERRIN-L6, but they do not prove unconditional three-dimensional regularity or a Clay-level theorem.*

Proof. Appendix NS-A defines exploration. Appendix NS-B defines theoretical soundness and closes the narrow criterion-recovery route. Appendix NS-C defines proof-first architecture and the remaining obligations. The completed route assumes the Serrin integrability condition; no route proves that condition for arbitrary Leray–Hopf solutions or assembles it into unconditional global regularity. Thus the NS layer has a genuine small success but is not globally theorem-complete. $\square$

Remark 8.77 (Meaning of closure). Closure here means that the interface architecture is internally disciplined: exploration cannot masquerade as proof, Core pass cannot masquerade as PDE theorem, and proof-first obligations cannot be discharged by terminology. It does not mean that the Navier–Stokes problem is solved.

NS-C.13bis. v20 conservative hardening: proof-object sovereignty and typed theorem routing

Definition 8.78 (NS proof-object sovereignty). *NS proof-object sovereignty* is the rule that theorem-level status is determined by a complete proof object, not by exploration success, Core certification alone, partial bridge diagrams, or terminology.

Proposition 8.79 (No proof-object sovereignty override). *No advisory, numerical, Core-certified, or partially bridged NS program is theorem-level unless the theorem-level proof object is complete.*

Proof. Theorem-level status requires all required Core, soundness, transport, analytic, artifact, and ledger components. Partial completion does not discharge the missing components. $\square$

Definition 8.80 (Typed NS route edge). A *typed NS route edge* is a declared implication in the proof route whose source layer, target layer, hypotheses, defects, artifacts, and status are recorded.

Definition 8.81 (NS route dependency graph). An *NS route dependency graph* is the directed graph of typed route edges required for a theorem-level proof object. It must distinguish PDE-side analytic implications, Core-side certificate implications, soundness bridges, normalization transports, discretization transports, and global assembly bridges.

Proposition 8.82 (Untyped route edges are non-verdict). *An untyped or unstated implication in an NS proof route cannot contribute to theorem-level status.*

Proof. Without source layer, target layer, hypotheses, and status, the implication is not audit-readable and cannot be checked as part of a proof object. $\square$

Definition 8.83 (NS proof evidence mode). An NS evidence item has one of the following modes:

- (i) *theorem*;
- (ii) *conditional-theorem*;
- (iii) *Core-certificate*;
- (iv) *interface-theorem*;
- (v) *Spec*;
- (vi) *advisory*;
- (vii) *search-artifact*;
- (viii) *failed*;
- (ix) *undefined*.

Proposition 8.84 (Evidence modes do not collapse). *A [Spec], advisory, search, or Core-certificate evidence item does not become a PDE theorem merely by appearing in a proof-first record.*

Proof. The evidence mode records the logical status of the item. The proof-first record aggregates obligations; it does not promote their status. $\square$

Definition 8.85 (NS typed budget and non-scalar outcome). The metric-budget outcome is exactly one of

not_invoked, undefined, pass, reject.

It is computed only from a declared metric-ledger element $\mathfrak{d}_{\text{met}}$, its scalarization, a theorem or verified artifact proving that the scalarized value bounds the actual bottleneck discrepancy, and the protected family clearance. Representative existence, higher eligibility, detector completeness, actual comparison morphisms, terminal tameness, finite-to-infinite reduction, Return, source binding, and artifact identity are separate non-scalar obligations. Their absence yields an undefined route and cannot be repaired by metric slack.

Proposition 8.86 (Scalar margin does not repair bridge gaps). *A small scalar budget margin does not repair a missing realization bridge, soundness bridge, compactness bridge, global assembly bridge, or Clay-level bridge.*

Proof. Bridge gaps are typed logical gaps. Scalar ledger bounds control declared discrepancies but do not supply missing implications. $\square$

Definition 8.87 (NS release-level closure). *NS release-level closure* means that NS-A, NS-B, and NS-C close the released Navier–Stokes interface stack in this revision by classifying exploration, Core certification, soundness, proof-first routing, and non-claims. It does not assert theorem-completeness.

Proposition 8.88 (Release-level closure is not theorem-completeness). *NS release-level closure, even with the exact criterion recovery of Track NS-KTR-SERRIN-L6, does not prove unconditional three-dimensional regularity and does not eliminate the analytic, Core-side, or bridge gaps of stronger target levels.*

Proof. Closure is a governance and interface-architecture property. Theorem-completeness requires a completed proof object and proved bridges. $\square$

Remark 8.89 (Proof-first records must expose gaps). A proof-first record is stronger when it explicitly exposes unresolved gaps, because it prevents architectural completeness from being misread as mathematical proof.

Remark 8.90 (NS-C budget notation). Whenever a budget comparison is consumed as a theorem-route condition, the typed form

$$\text{val}_B(\Sigma\delta) < \text{gap}_\tau(\mathfrak{B})$$

should be used, with the protection family and all non-scalar obligations declared.

NS-C.13ter. v20 dependency DAG, residual obstruction cut, and successor program

Definition 8.91 (Canonical NS proof-route DAG). A *canonical NS proof-route DAG* is a finite or locally finite directed acyclic graph whose nodes are typed statements and whose edges are typed implications. The graph distinguishes source assumptions, realization, repaired profiles, detector certificates, higher edges, Type IV/transient diagnostics, finite-to-infinite reduction, analytic criteria, Return, and the external PDE conclusion. Each edge records evidence status, dependencies, artifacts, defects, permitted use, and prohibited stronger readings.

Definition 8.92 (Complete theorem route). A path from the source-assumption node to the PDE-conclusion node is *complete* when every edge is theorem evidence at the consumed strength, every conditional hypothesis is discharged, all non-scalar obligations are satisfied, every metric comparison is certified for the actual profile pair, and all artifacts are immutable and available.

Theorem 8.93 (Route completeness criterion). *A proof object assembled from the canonical route DAG licenses the declared PDE conclusion only if it contains at least one complete theorem route from the source assumptions to that conclusion. A collection of individually valid edges that does not form such a path is insufficient.*

Proof. The PDE conclusion is obtained only by composing implications whose target matches the next source. Disconnected or type-incompatible edges cannot be composed into the required implication. $\square$

Definition 8.94 (Minimal residual obstruction set). Let G be the canonical route DAG after every currently proved edge is marked closed. A *residual obstruction set* is a set of unresolved or rejected edges meeting every source-to-conclusion route. It is *minimal* when no proper subset has this property. Equivalently, it is a minimal unresolved edge cut separating the source assumptions from the target conclusion.

Theorem 8.95 (Residual-cut obstruction theorem). *If a residual obstruction set remains, no theorem-level proof object exists in the declared route DAG. Conversely, closing one complete source-to-conclusion route is sufficient at the architectural level, subject to the truth and exact typing of its component theorems.*

Proof. Every source-to-conclusion path intersects the residual cut, so every possible route contains an unresolved or rejected implication. Conversely, a complete path composes to the target implication by Theorem 8.93. $\square$

Remark 8.96 (Residual set is not a scalar score). The residual obstruction set is a typed cut, not an optimization score. A small number of unresolved edges can still contain the entire Clay-level difficulty, and many closed auxiliary edges cannot compensate for one missing Return theorem.

Definition 8.97 (NS target ladder). The proof program keeps distinct target levels:

local_criterion, local_regularity, continuation,
scenario_exclusion, global_regularity, Clay_level.

Success at one level does not promote the next without a typed assembly edge.

Definition 8.98 (Finite-to-infinite route edge). A finite-to-infinite route edge records the finite witness, theorem reducing it to an infinite or continuum predicate, actual tower or covering maps, limit semantics, apex agreement, and the exact target predicate. It is a separate edge from the Core certificate and from the PDE Return theorem.

Proposition 8.99 (Reduction and Return are noninterchangeable). *A finite-to-infinite reduction can establish an infinite Core predicate without giving it PDE meaning, while a PDE Return theorem cannot be applied to a finite observation that has not been reduced to its required infinite predicate.*

Proof. The two implications have different source and target types. Neither supplies the domain condition required by the other. $\square$

Definition 8.100 (NS bridge-engineering successor program). The retained successor program is ordered as follows:

- (i) freeze one named local or continuation criterion and its authoritative source statement;
- (ii) prove a target-separating, purpose-faithful continuum realization for that criterion;
- (iii) construct its Core-readable profile and finite detector certificate;
- (iv) prove any required finite-to-infinite reduction;
- (v) prove the criterion Return theorem and independently validate exact recovered strength;
- (vi) only then attempt local-to-global or Clay-level assembly.

Items (i)–(v) are completed in v20 for the narrow Track NS-KTR-SERRIN-L6; the same program remains a successor-release candidate for sharper, more geometric, continuation, global, and Clay-level targets.

Definition 8.101 (Completed NS small-success route). Track NS-KTR-SERRIN-L6 is the following complete path at target level local_criterion:

$$\begin{aligned} \text{Leray–Hopf source solution} &\longrightarrow \mathcal{N}_6(u) \longrightarrow r_M(u) \longrightarrow P_M^S(u) \\ &\longrightarrow Q_M^S(u) = 0 \longrightarrow \mathcal{N}_6(u) < +\infty \longrightarrow \text{Serrin interior regularity.} \end{aligned}$$

The realization and exact zero-reflection are proved in Appendix NS-B, and the final edge consumes the frozen source theorem SRC-NS-SERRIN-L6. No detector, Type IV diagnostic, higher edge, metric comparison, or finite-to-infinite reduction is part of this route.

Proposition 8.102 (Small success does not close higher target levels). *Completion of Track NS-KTR-SERRIN-L6 closes a criterion-level known-theorem-recovery path but does not close continuation, scenario-exclusion, global-regularity, or Clay-level routes.*

Proof. The completed path assumes the strict-subcritical $L_t^6 L_x^6$ criterion. Higher target levels require additional edges proving that the criterion holds for the relevant source family or assembling it into an unconditional conclusion. Those edges are absent. $\square$

Definition 8.103 (NS v20 bridge status). The v20 status is

$$\begin{aligned} \text{NS_interface_stack} &= \text{closed}, & \text{NS_known_theorem_recovery} &= \text{exact_small_success}, \\ \text{NS_recovery_track} &= \text{NS-KTR-SERRIN-L6}, & \text{NS_criterion_route} &= \text{complete}, \\ \text{NS_unconditional_regularity_bridge} &= \text{not_invoked}, & \text{NS_Clay_level_proof} &= \text{not_claimed}. \end{aligned}$$

The remaining obligations concern stronger target levels; they are not hidden completed lemmas.

Theorem 8.104 (NS-C v20 proof-first closure). *Appendices NS-A–NS-C provide a complete v20 schema for exploration, Core re-entry, soundness, reduction, Return, and proof-route auditing, and they close one exact criterion-level source-to-PDE route through Track NS-KTR-SERRIN-L6. They do not close any unconditional three-dimensional global-regularity or Clay-level route.*

Proof. Definition 8.101 and Appendix NS-B supply one complete path in the route DAG. Proposition 8.102 shows that this path terminates at the frozen criterion-level conclusion. The residual-cut calculus still blocks every stronger target route because the edges establishing the criterion unconditionally or assembling it globally remain unresolved. $\square$

NS-C.14. Summary of proof-first program

Theorem 8.105 (NS-C proof-first package). *Within the Problem Interface layer, the following package is available.*

- (i) *A proof-first program begins from a declared target theorem, not from exploratory indicators.*
- (ii) *A Core certificate target must be distinguished from a PDE theorem target.*
- (iii) *Core certificate sufficiency for Navier–Stokes is [Spec] by default.*
- (iv) *Core-side proof obligations must be expressed through*

$$\text{Eval}_{i,W,\tau}, \quad C_{\tau,W}F, \quad \phi_{i,W,\tau}, \quad \text{val}_B(\Sigma\delta) < \text{gap}_\tau(\mathfrak{B}).$$
- (v) *The main proof obligations are realization, evaluation, collapse, diagnostic, compactness, global assembly, and regularity bridges.*
- (vi) *Analytic gaps and bridge gaps prevent theorem-level pass.*
- (vii) *Discretized proof-support mode is not theorem mode.*
- (viii) *Failure modes must be classified as realization, evaluation, eligibility, tower artifact, diagnostic, budget, soundness, or transport failures.*
- (ix) *An NS proof object must contain both Core certificate data and PDE-side soundness data.*
- (x) *Incomplete NS proof objects are not theorem-level.*
- (xi) *A minimal theorem route must separate proof components from supportive exploration artifacts.*
- (xii) *Core-certified status is not theorem-level by default.*
- (xiii) *Clay-level targets require a global regularity bridge and are not claimed here.*
- (xiv) *Appendices NS-A–NS-C close the interface layer and one criterion-level recovery route without proving unconditional three-dimensional regularity.*
- (xv) *Theorem-level status is governed by proof-object sovereignty.*
- (xvi) *The proof route must be represented by typed route edges and an audit-readable dependency graph.*

- (xvii) *Evidence modes do not collapse: [Spec], advisory, search, and Core-certificate modes remain distinct.*
- (xviii) *Typed budget failure includes scalar budget failure and missing non-scalar obligations.*
- (xix) *Release-level NS closure is not theorem-completeness.*
- (xx) *Track NS-KTR-SERRIN-L6 closes one exact criterion-level route and constitutes the NS small-success recovery in v20.0.0.*
- (xxi) *The completed criterion route does not close continuation, global-regularity, or Clay-level targets.*

Proof. Items (i)–(ii) follow from the proof-first and target theorem definitions. Item (iii) is Proposition 8.20. Item (iv) follows from the repaired Core vocabulary requirements in Sections NS-C.1 and NS-C.4. Item (v) follows from the bridge lemma definitions. Item (vi) follows from Propositions 8.36 and 8.37. Item (vii) is Proposition 8.43. Item (viii) follows from the failure mode definitions. Items (ix)–(x) follow from the NS proof object schema and Proposition 8.60. Item (xi) is Proposition 8.66. Item (xii) is Proposition 8.70. Item (xiii) is Proposition 8.72. Item (xiv) is Proposition 8.76. Items (xv)–(xix) follow from Definition 8.78, Proposition 8.79, Definition 8.80, Definition 8.81, Proposition 8.82, Definition 8.83, Proposition 8.84, Definition 8.85, Proposition 8.86, Definition 8.87, and Proposition 8.88. Items (xx)–(xxi) follow from Definition 8.101 and Proposition 8.102. □

NS-C.15. Explicit exclusions

The following are not asserted in Appendix NS-C.

- No unconditional three-dimensional Navier–Stokes regularity theorem is proved.
- No new analytic regularity proof is supplied; the only theorem-facing PDE result is the exact recovery of the frozen conditional Serrin criterion.
- No global smoothness theorem is proved.
- No Clay Millennium problem is solved.
- No existence theorem is proved.
- No uniqueness theorem is proved.
- No continuation theorem is proved.
- No non-blow-up theorem is proved.
- No blow-up theorem is proved.
- No singularity exclusion theorem is proved.
- No Core certificate is asserted to imply PDE regularity without a proved soundness bridge.
- No exploratory indicator is treated as proof.
- No numerical evidence is treated as proof.
- No discretized proof-support mode is treated as theorem mode.
- No [Spec] bridge is treated as a theorem.
- No local certificate is treated as global regularity without a global bridge.
- No incomplete proof object is treated as theorem-level.
- No analytic gap is hidden or discharged by terminology.

- No bridge gap is hidden or discharged by terminology.
- No Core-side gap is hidden or discharged by terminology.
- No Type IV proxy replaces

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}).$$

- No Type IV diagnostic is invoked without a declared tower comparison artifact

$$\phi_{i,W,\tau}.$$

- No admissible representative

$$C_{\tau,W}F$$

is replaced by informal filtered-lift terminology.

- No repaired evaluation record

$$\text{Eval}_{i,W,\tau}(F)$$

is replaced by old collapse notation.

- No AK terminology alone is used to claim a Navier–Stokes theorem.
- No global exactness of T_{τ}^{bar} is assumed.
- No positive-threshold Serre localization is assumed.
- No global strict filtered lift C_{τ} is assumed.

NS-C.16. Provenance and status

The material in this appendix depends on:

- (i) the Navier–Stokes exploration protocol of Appendix NS-A;
- (ii) the theoretical soundness layer of Appendix NS-B;
- (iii) the normalization interface of Appendix AGI-N;
- (iv) the repaired barcode-level collapse and constructible persistence framework of Appendix A;
- (v) the admissible representative discipline of Appendix B;
- (vi) the eligible realization regime of Appendix C;
- (vii) the monitored Type IV diagnostic calculus and tower comparison artifact discipline of Appendix D;
- (viii) the manifest requirements of Appendix G;
- (ix) the ledger catalog of Appendix M;
- (x) the non-claim discipline of Chapter 7;
- (xi) the interface protocol of Chapter 8.

Its role is to define a proof-first architecture for a possible AK-based Navier–Stokes program without asserting that the program has been completed.

Status labels for Appendix NS-C. *Problem-interface definitions:* Definitions [8.4](#), [8.5](#), [8.8](#), [8.9](#), [8.10](#), [8.11](#), [8.12](#), [8.13](#), [8.15](#), [8.16](#), [8.17](#), [8.18](#), [8.19](#), [8.22](#), [8.23](#), [8.24](#), [8.25](#), [8.26](#), [8.27](#), [8.28](#), [8.29](#), [8.31](#), [8.32](#), [8.33](#), [8.34](#), [8.35](#), [8.39](#), [8.40](#), [8.41](#), [8.42](#), [8.45](#), [8.46](#), [8.47](#), [8.48](#), [8.49](#), [8.50](#), [8.51](#), [8.52](#), [8.53](#), [8.56](#), [8.57](#), [8.58](#), [8.59](#), [8.62](#), [8.63](#), [8.64](#), [8.65](#), [8.68](#), [8.69](#), [8.73](#), [8.75](#).

Problem-interface propositions/theorems: Propositions [8.6](#), [8.20](#), [8.36](#), [8.37](#), [8.43](#), [8.54](#), [8.60](#), [8.66](#), [8.70](#), [8.72](#), [8.76](#), Theorem [8.105](#).

Operational cautions: Remarks [8.1](#), [8.2](#), [8.3](#), [8.7](#), [8.14](#), [8.21](#), [8.30](#), [8.38](#), [8.44](#), [8.55](#), [8.61](#), [8.67](#), [8.71](#), [8.74](#), [8.77](#).

Additional v20.0 conservative hardening definitions: Definitions [8.78](#), [8.80](#), [8.81](#), [8.83](#), [8.85](#), [8.87](#).

Additional v20.0 conservative hardening propositions: Propositions [8.79](#), [8.82](#), [8.84](#), [8.86](#), [8.88](#), and [8.102](#).

Additional v20.0 conservative hardening remarks: Remarks [8.89](#), [8.90](#).

Additional v20 completion definitions: Definitions [8.91](#), [8.92](#), [8.94](#), [8.97](#), [8.98](#), [8.100](#), [8.101](#), and [8.103](#).

Additional v20 completion proposition/theorems: Proposition [8.99](#); Theorems [8.93](#), [8.95](#), and [8.104](#).

Additional v20 completion remark: Remark [8.96](#).

Explicit exclusions: Section NS-C.15.

NS layer closure: Definition [8.75](#), Definition [8.87](#), Propositions [8.76](#), [8.88](#), Theorem [8.105](#).